

REDEFINING MEDICAL EDUCATION
VIDEO COMPANION WITH EACH CHAPTER

NEXT
GENERATION
CONCEPT BOOK
PHARMACOLOGY



ACTIVE RECALL BASED
INTEGRATED EDITION



CONCEPT BASED LEARNING

Video Companion on Each Chapter

NEXT GENERATION



COMPREHENSIVE REVIEW SERIES

“PHARMACOLOGY”

ACTIVE RECALL BASED

Integrated Edition





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CONCEPTS

- ⇒ **Concept 1.1 General Introduction**
- ⇒ **Concept 1.2 Pharmacokinetics
(drug absorption, distribution,
metabolism and excretion)**
- ⇒ **Concept 1.3 Pharmacodynamics**
- ⇒ **Concept 1.4 Drug development**
- ⇒ **Concept 1.5 Adverse drug reactions**

Concept 1.1 : Introduction

Learning objectives: To know the scientists name who contributed invention related pharmacology
 To study the components of prescription
 To know the different nomenclature of drug name

Time Needed

1 st reading	15 mins
2 nd reading	5 mins

Contributions of Scientists:

1	Oswald Schmiedeberg	Father of Pharmacology. Propounded some of the fundamental concepts of pharmacology.
2	Colonel Ram Nath Chopra	Father of Indian Pharmacology. Established the first center for study and research in pharmacology in India, at the Calcutta School of Tropical Medicine.
3	Sir James Black	Father of Modern Pharmacology
4	Langley	Observed the mutual antagonism of atropine and pilocarpine in 1878, and proposed that both acted on the same 'receptive substance' or 'receptor' on the cell.
5	Clarke	Propounded receptor occupation theory in 1937.
6	Rudolph Buchheim	Founded the first institute for research in pharmacology in Germany in 1847.
7	Otto Loewi	transmission across nerve junctions to be mediated by neurotransmitters
8	Ahlquist	Classified adrenergic receptors into α and β types.

Components of a Prescription:

Part of prescription	Refers to
Superscription	Rx. It means 'take thou'. It is a tribute to Lord Jupiter
Inscription	Main body of the prescription
Subscription	Specific instructions to the pharmacist
Signatura / Transcription	Specific instructions to the patient

POOJA CLINIC
Dr. j. Thirunavukkarasu, M.D.,
308, 1st cross st
Pattabiram
Chennai-72

NAME _____ AGE _____
 ADDRESS _____ DATE _____

Prescriber information

Patient information

Superscription ← **R**

Inscription ← Tab. Metformin 500mg 1---0---1

Subscription ← Any instruction to pharmacist

Transcription ← Any instruction to patient

(SIGNATURE) _____

☐ LABEL
 REFILL 0 1 2 3 4 5 PRN

Fig.1.1

Different Names for Drugs:

	Name of drug	Refers to
1	Chemical name	Structure of the drug
2	Generic name	Non-proprietary name of the drug
3	Brand / Trade name	Proprietary name

For example:

- **Chemical name-** Acetyl salicylic acid
- **Generic name-** Aspirin
- **Brand name-** Ecosprin

Definitions:

1. **Drug** – Any substance or product that is used or is intended to be used to modify or explore physiological systems or pathological states for the benefit of the recipient (WHO, 1966).
ORPHAN DRUGS:-used for diagnosis/treatment/prevention of a rare disease or condition(Fomepizole, digibind, liothironine)
Essential drugs- **Drugs that meet health needs of the majority of population, it should be affordable by all people and it should be available in all time, most of the essential drugs prepared as single compound.**
2. **Pharmacology** – The science of drugs that deals with the interaction of exogenously administered chemical molecules with living systems.
3. **Pharmacy** – The art and science of compounding and dispensing drugs or preparing suitable dosage forms for administration of drugs to man or animals. It includes collection, identification, purification, isolation, synthesis, standardization and quality control of medicinal substances.
4. **Pharmaceutics** – The large scale manufacture of drugs.
5. **Pharmacotherapeutics** – The application of pharmacological information together with the knowledge of the disease for its prevention, mitigation or cure.
6. **Clinical pharmacology** – The scientific study of drugs (both old and new) in man.
7. **Toxicology** – The study of poisonous effect of drugs and other chemicals (household, environmental pollutant, industrial, agricultural, homicidal) with emphasis on detection, prevention and treatment of poisonings.
8. **Pharmacovigilance (PV or PhV), also known as drug safety, is the pharmacological science relating to the collection, detection, assessment, monitoring, and prevention of adverse effects with pharmaceutical products.**

ONE-LINERS:

Greek word: **Pharmacon** – Drug

French word: Drogue – **Dry herb**

The first 'Model List of Essential Drugs' was brought out by the WHO in – 1977

Current Model List of Essential Drugs – 17th (2011)

National Essential Drugs List released by India in – 1996

Current national list of essential medicines in India contains – 348 drugs

Concept 1.2 : Pharmacokinetics:

Learning objectives:

- To know the components of pharmacokinetics
- To know the factors modifying absorption of drug
- To know the different methods of drug transport
- To know the various factors that modify drug distribution
- To know the clinical importance protein bind of drugs
- To know the different types drug metabolism
- To know th clinical importance of microsomal enzyme
- To know orders of kinetic of excretion
- To know the importance of half life of the drug

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

Definition: The quantitative study of drug movement, in, through and out of the body.

- Includes – (Mnemonic: ADME)
- Absorption
- Distribution
- Metabolism
- Excretion

Note: Some scientists prefer to include **dissolution of drugs** also in pharmacokinetics (DADME)

(A) Absorption:

The two major factors that are responsible for drug absorption include lipid solubility and Non-Ionisation.

Non ionisation of drug is depends upon pH of the environment (acidic drug become non ionised in acidic medium, basic drugs become non ionised in basic medium).

strongest acidic drugs and strongest basic drugs always seen in ionised form.

Site of absorption for orally administered drugs:

Weakly acidic drugs like aspirin, barbiturates arenonionised in acidic media, so they are well absorbed from stomach.

- Weakly basic drugs---morphine, diazepam --- non ionised in alkaline pH-----absorbed from duodenum.
- Most common site of drug absorption: **Upper duodenum (because intestine has large surface area and thin mucosa than stomah).**

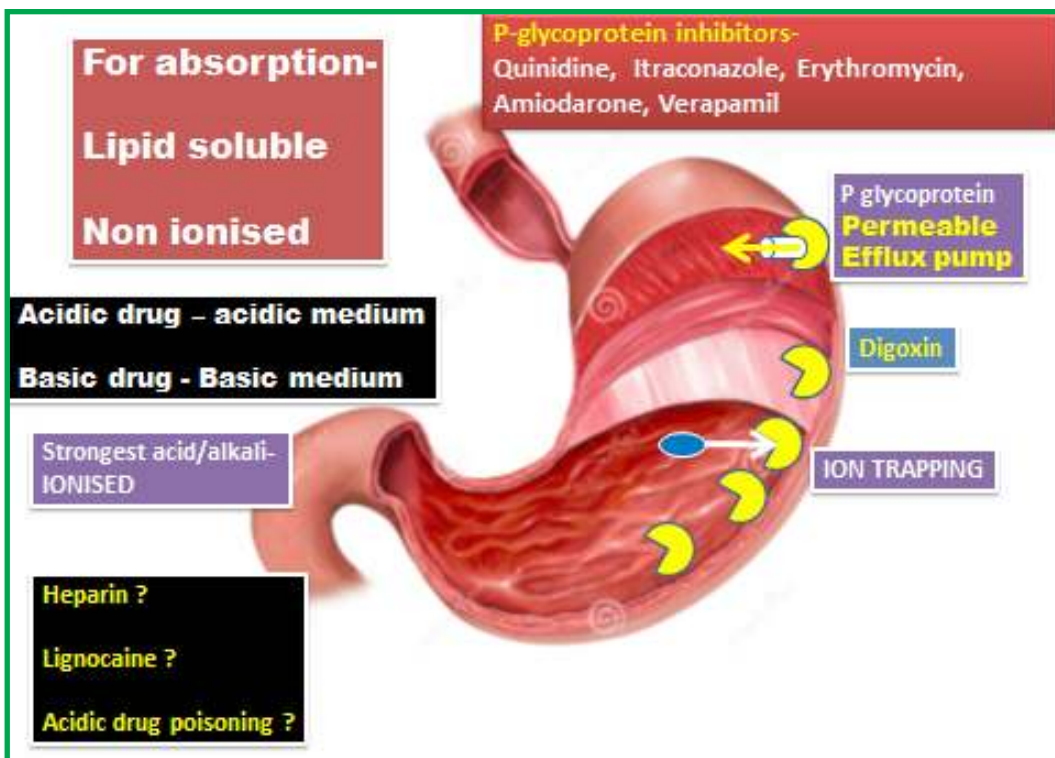


Fig. 1.2

- Conversely, Alkalinisation of urine is done for excretion of acidic drugs.
- Acidification of urine can be done for excretion of basic drugs, but not practically advised due to increased risk of renal stone formation.

E.g. of acidic drugs	E.g. of basic drugs
Sulfonamides	Morphine
Barbiturates	Atropine
NSAIDs	Amphetamines
Penicillin V	Ephedrine
	Chloroquine
	Polymyxin B
	Vancomycin

Drug Transport:

1. Simple passive diffusion(no need of energy or carrier)
2. Facilitated passive diffusion(carrier dependent, no need of energy)
3. Active transport (energy and carrier dependent)
4. Pinocytosis (cell drinking process)
 - Most common method of drug absorption: Simple diffusion
 - Simple diffusion is quantified using: Fick's law

Fick's law: $R = (\Delta C \times A \times P) / T$, where

- R = Rate of simple diffusion across a membrane
- ΔC = Concentration gradient across the membrane
- A = Surface area available for diffusion
- P = Permeability coefficient of the drug (indicates its lipophilicity)
- T = Thickness of the membrane

Drugs which are absorbed by active Transport:

- Levodopa
- α -Methyldopa

ONE-LINERS:

- **pH independent transport is seen for – Ethanol and Diethyl ether**
- Alcohol absorption is from – **Small intestine > Stomach**

Bioavailability(F):

- Percentage of administered drug that enters the systemic circulation in unchanged form.
- Expressed as percentage (It has no units).
- Denotes – **Rate and extent of drug absorption.**
- Depends upon
 - Route of administration (Most important).
 - Local binding to proteins or other structures.
 - First pass metabolism.
- Route with maximum bioavailability – Intravenous (100% or 1 as entire drug is administered directly into systemic circulation).
- Formula: $F_{\text{route}} = \text{AUC}_{\text{route}} / \text{AUC}_{\text{IV}}$

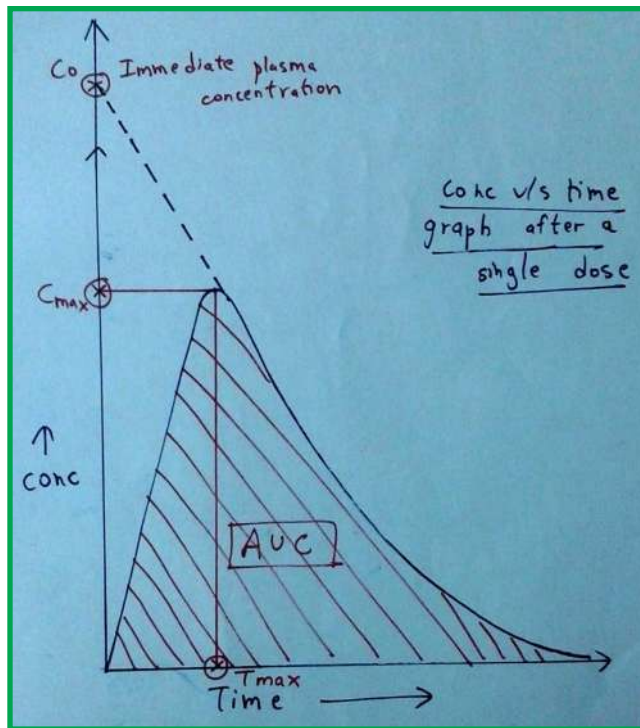


Fig. 1.3: Plasma concentration versus time curve of a drug after a single dose

C_{max} - Maximum plasma concentration obtained.

T_{MAX} - is the time to reach C_{max} (rate of absorption).

AUC- bioavailability(extend of absorption).

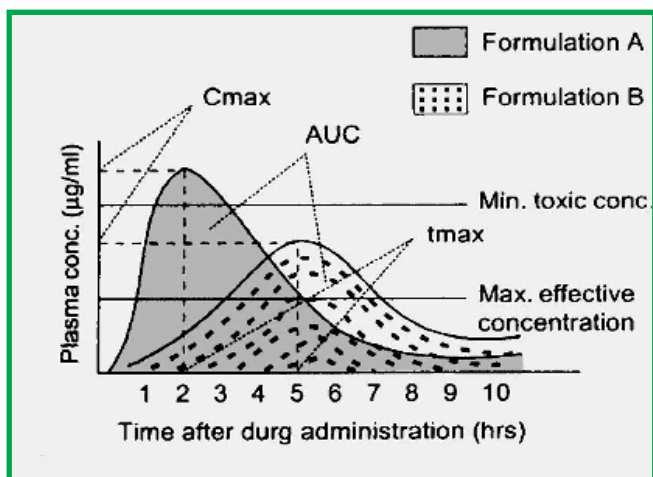


Fig. 1.4

Same drug, same dose, same dosage forms-on comparison not having more than 20%- difference in bio availability means it is called as Bioequivalent.

Factors decreasing oral bioavailability of drugs:

1. Decreased oral absorption
2. High first pass metabolism

1. Decreased oral absorption. This could be due to –

- Polar nature of drugs (**E.g: Aminoglycosides**).
- Cleavage in stomach by acid (e.g: Penicillin G, Erythromycin, PPI).
- Metabolism by enzymes in GIT (e.g: Insulin, Vasopressin).
- Presence of other drugs (e.g: Calcium decreases absorption of tetracyclines; antacids decrease absorption of digoxin).
- Presence of food (exception- **lithium, Halofantrine, griseofulvin, fibrates and some more drugs(look at the table)- these drugs will be more absorbed in the presence of food**).

Food decreases absorption of	Food increases absorption of
• Ampicillin • Captopril • Digoxin • Isoniazid • Levodopa • Rifampicin • Tetracycline	• Carbamazepine • Griseofulvin (fatty food) • Lithium • Lumefantrine • Nitrofurantoin • Riboflavin • Fibrates • Erlotinib

2. High first pass metabolism:

Extent of first pass metabolism:

Low	Intermediate	High	
		Not given orally	Given orally in high doses
Phenobarbitone	Aspirin	Hydrocortisone	Propranolol
Phenylbutazone	Quinidine	Isoprenaline	Alprenolol
Pindolol	Desipramine	Lignocaine	Salbutamol
Tolbutamide	Metoprolol	Testosterone	Verapamil
Theophylline	Nortriptyline		Nitroglycerine
Isosorbidedimonitrate	Chlorpromazine		Morphine
	Pentazocine		Pethidine
			Methyltestosterone
			Propoxyfene

ONE-LINERS:

- Route with highest first pass metabolism – **Oral**
- Most common site of first pass metabolism on oral administration – **Liver**
- First pass metabolism in small intestine occurs with – **Levodopa**
- Rectally given drug absorbed through External hemorrhoidal vein mean – **no first metabolism**

HENDERSON HESSELBACH EQUATION- $pK_a = pH + \log\left(\frac{\text{ionized } A^-}{\text{un ionized } HA}\right)$

$pK_a = pH$ means what is the inference? - 50% of drug is in the ionized form & 50% non ionized form

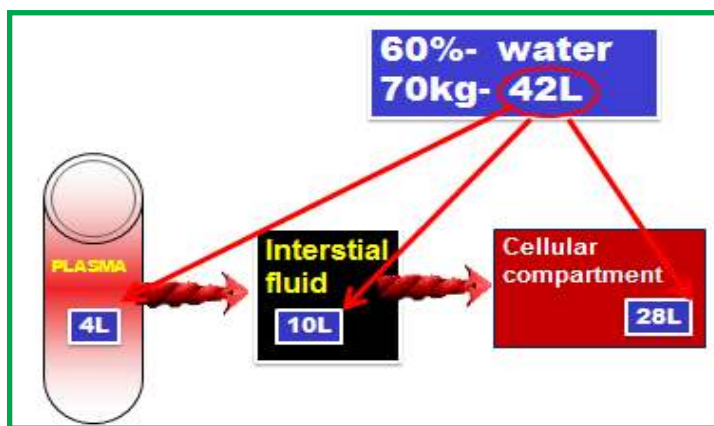
(B) Distribution:

Fig. 1.5

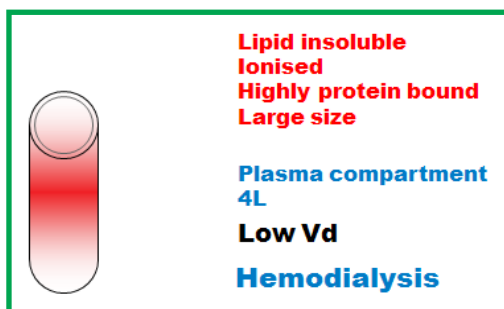
Low volume of distribution

Fig. 1.6

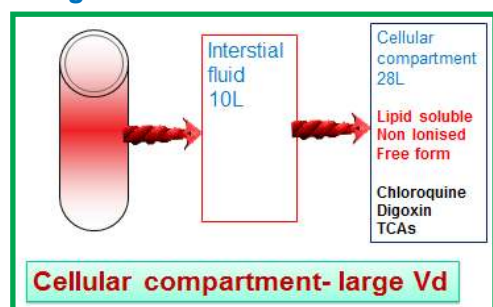
Large volume of distribution

Fig. 1.7

Drugs having large volume of distribution for rapid action needs loading dose (loading dose is depend upon Vd)

There is no role for hemodialysis for drugs having large Vd

Volume of distribution (Vd):

- **Formula – Dose administered / Immediate plasma concentration**
- Expressed in litres or litres/kg body weight
- **Indicates – Extent of tissue penetration of drug**
- Greater the volume of distribution, more is the tissue penetration

Scenario	Volume of distribution	Examples	
Drug is highly bound to plasma proteins	V_d = Plasma volume (3L or 0.15L/kg)	Warfarin Diclofenac	LowVd
Drug is restricted to intravascular compartment	V_d = Blood volume (5L)	Heparin Streptokinase	Low V_d Removed by hemodialysis
Drug is distributed in the extracellular fluid	V_d = ECF volume (15L or 0.25L/kg)	Aminoglycosides	
Drug penetrates into the tissues	$V_d >$ ECF volume (>15L or >0.25L/kg)	Digoxin (6L/kg) Propranolol (4L/kg) Morphine (3.5L/kg)	High V_d No role for dialysis

Factors governing volume of distribution of drugs:

- Lipid:water partition coefficient of the drug (Lipophilicity)
- pKa value of the drug
- Degree of plasma protein binding
- Affinity for different tissues
- Fat:lean body mass ratio (varies with age, sex, obesity)
- Systemic diseases (CHF, uremia, cirrhosis of liver)

Plasma protein binding of drugs:

Acidic drugs binds with albumin.

Basic drugs binds with alpha1 acid glycoprotein.

Each molecule of albumin has two binding site (can able to accomadate two acidic drug)

In hypo albuminemia (nephrotic syndrome, liver failiure), free level of acidic drug will be high.

- Usually prolongs the half-life due to delayed metabolism and excretion.
- **Free form is responsible for therapeutic effect and toxicity.**
- Bound form acts as a reservoir.

Drugs highly bound to plasma proteins:

To albumin	To α_1 -acid glycoprotein
Barbiturates Benzodiazepines NSAIDs Penicillins Phenytoin Sulfonamides Tetracyclines Tolbutamide Valproate Warfarin	β -blockers Bupivacaine Disopyramide Imipramine Lignocaine Methadone Prazosin Quinidine Verapamil

Distribution / Accumulation of drugs in various tissues:

Organ	Drugs getting concentrated
Skeletal muscle, heart	• Digoxin • Emetine
Liver	• Chloroquine • Digoxin • Emetine • Tetracyclines
Kidney	• Chloroquine • Digoxin • Emetine
Thyroid	• Iodine
Brain	• Acetazolamide • Chlorpromazine • Isoniazid
Retina	• Chloroquine (bound to nucleoproteins)
Iris	• Atropine (bound to melanin) • Ephedrine
Bones and teeth	• Bisphosphonates (bound to hydroxyapatite) • Heavy metals (bound to mucopolysaccharides of connective tissue) • Tetracyclines

Adipose tissue	• DDT • Ether • Minocycline • Phenoxybenzamine • Thiopentone
Bone	• Tetracycline • Lead • Arsenic • Radium

Redistribution:

- **Seen with highly lipid soluble drugs.**
- Get distributed immediately to organs of high blood flow followed by redistribution to organs of lower blood flow.
- Rapid onset and rapid termination of action of drug.
- E.g: **thiopentone sodium.**
- Drugs that can be eliminated by hemodialysis.

- | | |
|---|---|
| <ul style="list-style-type: none"> • Carbamazepine • Lithium • Phenobarbitone • Theophylline • Valproate | <ul style="list-style-type: none"> • Ethylene glycol • Methyl alcohol (Methanol) • Salicylates |
|---|---|

Drug that cannot be removed by dialysis:

A- amphetamine

V- Verapamil

O- Opioids, OPC

I- imipramine (TCA)

D- digoxin

Dialysis-

Diazepam(BZD)

Blood Brain Barrier absent -- Pituitary, Pineal, Area Prostrrema CTZ, Median Eminence(example for drugs which never cross BBB-Streptomycin, neostigmine, glycopyrolate, dopamine)

(C) Metabolism:

Conversion of lipophilic moiety into hydrophilic one, so that excretion is possible.

Fate of drug during metabolism:

1. Active drug converted to inactive metabolite (Most common)	
2. Active drug converted to active metabolite	
Active drug	Active metabolite
Allopurinol	Alloxanthine (Oxypurinol)
Amitriptyline	Nortriptyline
Cefotaxime	Desacetylcefotaxime
Chloral hydrate	Trichloroethanol
Codeine	Morphine
Diazepam	Desmethyl diazepam, Oxazepam
Digitoxin	Digoxin
Imipramine	Desipramine
Losartan	E3174
Morphine	Morphine-6-glucuronide
Primidone	Phenobarbitone, Phenylethylmalonamide
Procainamide	N-acetylprocainamide
Spironolactone	Canrenone
3. Inactive drug (Prodrug) converted to active metabolite	
Prodrug	Active metabolite
Acyclovir, Valacyclovir	Acyclovir triphosphate
Famciclovir	Penicyclovir
Bacampicillin	Ampicillin
Cyclophosphamide	Aldophosphamide, Phosphoramidate mustard, Acrolein
Dipivefrine	Epinephrine (Adrenaline)
Enalapril	Enalaprilat
5-Fluorouracil	Fluorouridine monophosphate
Levodopa	Dopamine
6-Mercaptopurine	Methylmercaptopurineribonucleotide
α -methyl dopa	α -methylnorepinephrine

Prednisone	Prednisolone
Proguanil	Cycloguanil
Sulfasalazine	5-aminosalicylic acid
Sulindac	Sulindac sulphide metabolite

Metabolism of xenobiotics:

Phase 1 reactions	Phase 2 reactions
Non-synthetic in nature	Synthetic in nature
Metabolites of drugs are formed	Conjugates of drug metabolites are formed (hence, also called as conjugation reactions)
Products are usually lipophilic or hydrophilic	Products are usually hydrophilic
Occur in microsomes	Occur in cytoplasm (except glucuronidation)
E.g.: • Oxidation (most common Phase I reaction) • Reduction • Cyclization • Decyclization • Hydrolysis	E.g.: • Acetylation • Methylation • Sulfation • Glucuronidation (most common Phase II reaction) • Glutathione conjugation • Glycine conjugation

ONE-LINERS:

Endogenous metabolite undergoing xenobiotic metabolism – **Bilirubin**

Examples of drugs undergoing acetylation (These drugs also cause **lupus-like reaction as a side effect**) – (Mnemonic: **SHIPP-C** or **CHIPPS**)

- Sulfonamides + Dapsone
- Hydralazine
- Isoniazid
- Procainamide
- Para-amino salicylic acid (PAS)
- Clonazepam
- Hoffman's elimination- Non enzymatic self degradation without the help of liver or kidney
- Example- ATRACURIUM (safe in renal and hepatic failure)

Nomenclature of microsomal enzymes:

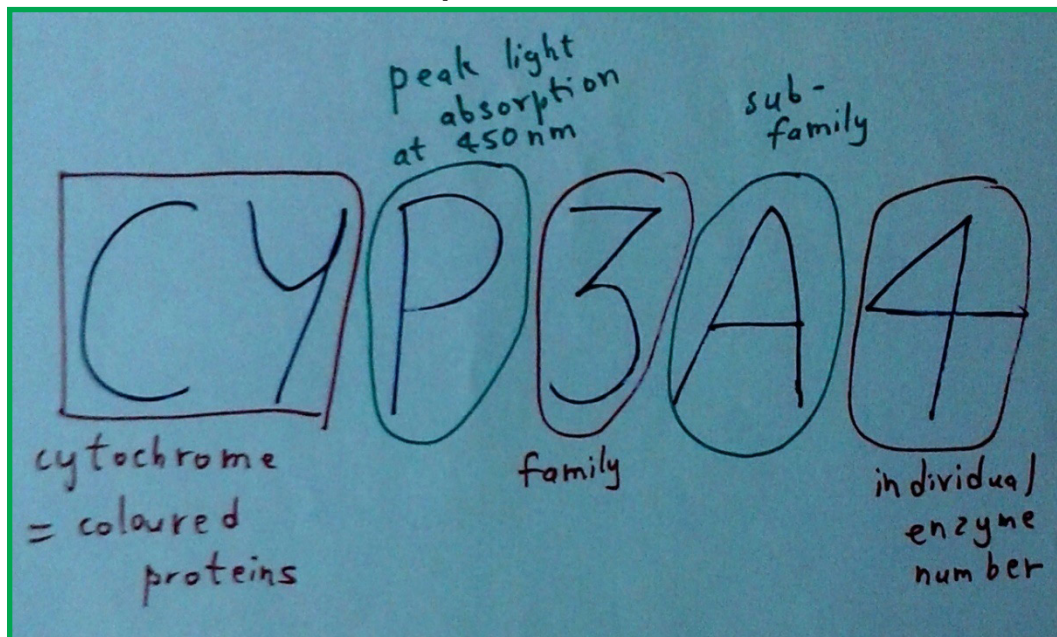


Fig 1.8: Nomenclature of microsomal enzymes

The enzymes are named based on –

- Cytochrome – meaning coloured components of the cell (proteins are coloured due to the presence of heme).
- P450 – Peak light absorption at 450 nm wavelength.
- Family number.
- Sub-family number.
- Individual enzyme number.

CYP450 enzymes primarily involved in drug metabolism and drugs metabolized by them:

Enzyme	Drugs metabolized by the enzyme
CYP3A4	Most of the drugs
CYP2D6	• Neuroleptics • Tricyclic antidepressants (TCAs) • Selective serotonin reuptake inhibitors (SSRIs) • Antiarrhythmics • β-blockers • Codeine converted into morphine by CYP2D6 • Tamoxifen activated by CYP2D6 (Fluoxetine inhibits CYP2D6), fluoxetine interfere with tamoxifen activation

CYP2C8/9	• Phenytoin • Warfarin • Carbamazepine • Ibuprofen • Tolbutamide • Repaglinide • Celecoxib • Losartan
CYP2C19	• Omeprazole • Clopidogrel(pro drug) activated with the help of CYP2C19(omeprazole interfere the activation of clopidogrel, Pantaprazole, Rabeprazole, do not have drug interaction with clopidogrel) • Voriconazole • Phenytoin • Diazepam • Propranolol
CYP1A1/2	• Procarcinogens to carcinogens • Theophylline
CYP2E1	• Paracetamol to NABQI (N-acetyl-p-benzoquinonimine- hepatotoxigenic metabolite) • Alcohol will induce CYP2E1, so chronic alcoholics are more prone for hepatotoxicity with paracetamol

Examples of CYP enzyme inducers and inhibitors –

CYP inducers	CYP inhibitors
• Rifampicin • Griseofulvin • Phenytoin • Phenobarbitone • Carbamazepine • Cigarette smoke (CYP1A1/2) • Chronic alcoholism • St John's wort (natural antidepressant)	• Ketoconazole • Itraconazole • Erythromycin • Clarithromycin • Ciprofloxacin • Verapamil • Diltiazem • Cimetidine • Ranitidine • Valproate • HIV protease inhibitors (most potent is RITONAVIR) • Grapefruit juice • Isoniazid (for all CYP enzymes except CYP2E1)

ONE-LINERS:

- PPI prevent activation of – **clopidogrel** (as both are metabolized by CYP2C19)
- Drugs precipitating paracetamol toxicity – **Isoniazid, zidovudine and Alcohol**

(D) Excretion:

Most important organ for drug excretion – **Kidney**

First-order kinetics	Zero-order kinetics = Saturation kinetics
A constant fraction of the drug is excreted in unit time	A constant amount of the drug is excreted in unit time
Rate of excretion is directly proportional to the plasma concentration; also varies in the same direction with changes in drug dose	Rate of excretion is constant and independent of plasma concentration
$T_{1/2}$ is constant	$T_{1/2}$ increases with increase in dose of the drug
Clearance is constant	Clearance decreases with increase in dose of the drug
Fall in plasma concentration is exponential	Fall in plasma concentration is linear

Examples of drugs following zero-order kinetics:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Alcohol, Aspirin • Theophylline • Warfarin | <ul style="list-style-type: none"> • Phenytoin • Tolbutamide |
|--|--|

One-liners:

- Alcohol is the only drug that follows zero-order kinetics **in all doses**.
- Other drugs **only follow at high doses** – hence, also called as ‘pseudo-zero order’ kinetics.

Clearance of a Drug:

- Volume of plasma that is completely cleared of the drug in unit time.
- Expressed in – volume per unit time (conventionally mL/min).
- Denotes the – **ability of the human body to excrete the drug**.
- Formula for renal clearance –

$$CL = UV/P$$

U = Urinary concentration of drug
P = Plasma concentration of drug
V = urine flow rate

Drugs Excreted Partly via Saliva and Sweat:

- | | |
|---|--|
| <ul style="list-style-type: none"> • Lithium • Rifampicin | <ul style="list-style-type: none"> • Potassium iodide • Heavy metals |
|---|--|

HALF-LIFE / $T_{1/2}$ (also known as elimination half-life):

- Time taken for the plasma concentration of the drug to decrease to half.
- Measured in units of – time (minutes, hours, days, etc).
- Denotes the – **rate of drug elimination**.

- Also denotes the duration of action of the drug.
- In first order kinetic 97% drug is eliminated after 5 half life.
- Regarding absorption, after 5 half lives – the drug reaches steady state plasma level 97%.
- Formula –
 - $T_{1/2} = (0.693 \times V_d) / CL$.

Hit and run drugs – have short half-lives but long durations of action.

E.g.:

- MAO inhibitors
- PPIs(half life 2-4hrs, but acting for 24 hrs, due to irreversible inhibition of proton pump)
- Guanethidine
- Reserpine
- Anti-platelet effect of aspirin

Percentage of drug eliminated in successive half-lives:

Half-life	Percent of drug eliminated
1	50
2	75
3	87.5
4	93.75
5	96.875

ONE-LINERS:

Almost complete elimination of drug occurs in – **5 $T_{1/2}$ OR 4-5 $T_{1/2}$**

Steady-state Concentration of Drug (C_{pss}):

- Concentration at which fluctuation of drug levels in plasma is considered negligible.
- Also known as Target C_{pss} as clinicians aim to attain this drug concentration in plasma.
- Attained after – 5 $T_{1/2}$ OR 4-5 $T_{1/2}$.

Formulae:

- Loading dose formula = $(C_{pss} \times V_d)$.
- Maintenance dose formula = $(C_{pss} \times CL)$.

ONE-LINERS:

- Loading dose depends up on – **Volume of distribution**
- Maintenance dose depends up on – **Clearance**

Concept 1.3 : Pharmacodynamics

Learning objectives: To study the effect of drug in the body
 To know the clinical importance of log dose response curve
 To study the different types of receptor
 To know the importance of therapeutic index

Time Needed

1 st reading	25 mins
2 nd reading	10 mins

Definition – The study of drug effects on the human body.

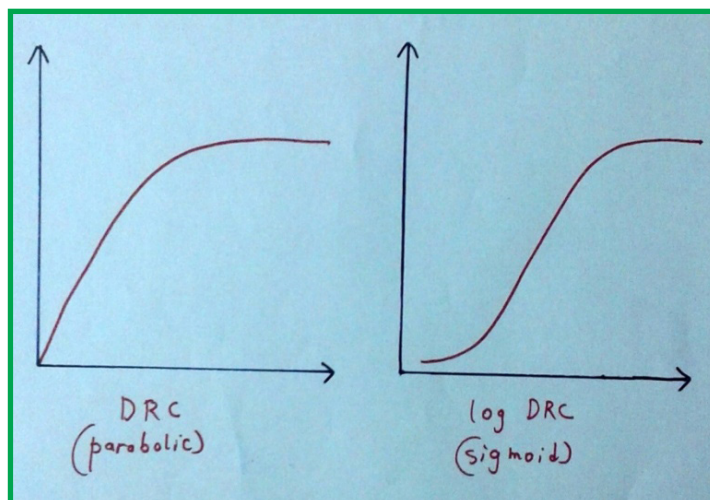


Fig 1.9: Dose response curve and log dose response curve

Dose Response Curve (DRC):

- Magnitude of effect produced by the drug (Y-axis) is plotted against the dose of the drug (X-axis).
- Shape – **Parabolic.**

Log Dose Response Curve (Log DRC):

- Magnitude of effect produced by the drug (Y-axis) is plotted against the logarithm of the dose of the drug (X-axis).
- Shape – **Sigmoid.**

Advantages of Log DRC Over DRC:

1. Central portion of the graph is a straight line; therefore, the response increases linearly with the logarithm of the dose.
2. A wider range of drug doses can be plotted.
3. Comparison between agonists and antagonists becomes simple.

Pharmacodynamic Parameters Defined From Log DRC:

- **Potency**– The dose of the drug at which the response starts appearing.
Whichever drug produces response at a lower dose is said to be more potent.
- 2. **Efficacy**–The maximum response produced by a drug.
Whichever drug produces a greater maximum response is said to be more efficacious.

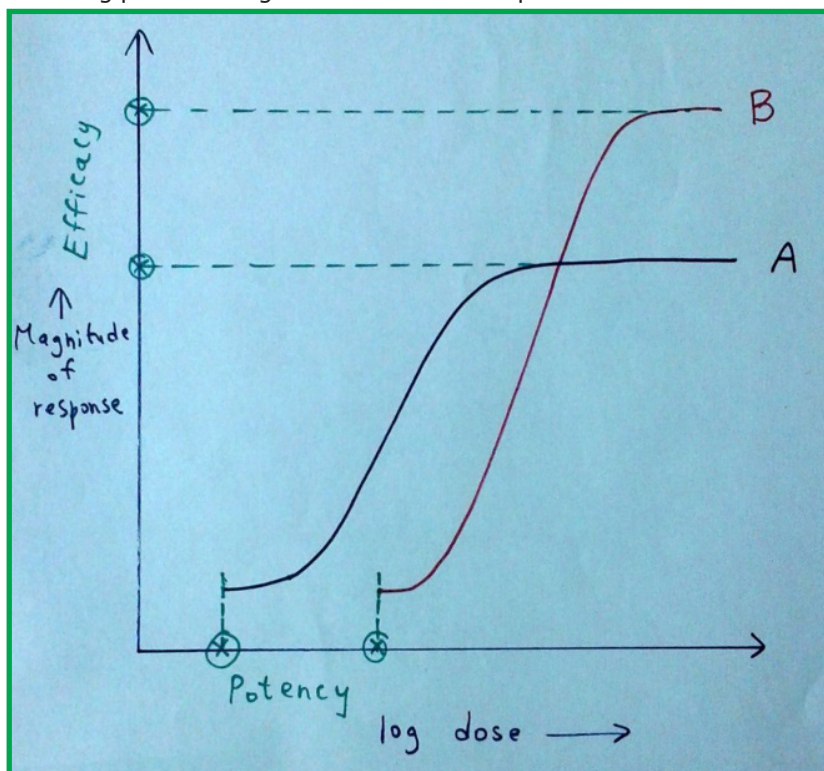


Fig 1.10: Potency and efficacy of drugs

In the above Figure, Drug A is more potent while Drug B is more efficacious.

Quantal dose response curve – The percentage of population showing a response is plotted against the logarithm of the dose of the drug.

Pharmacodynamic Parameters Defined From Quantal DRC:

1. Effective dose 50 (ED50) – Dose which produces therapeutic response in half (50%) of the population.
Indicates – Potency of the drug.
2. Lethal dose 50 (LD50) – Dose which produces toxicity (toxic effect) in half (50%) of the population.
Indicates – Safety of the drug.
Maximum response (height of the curve) - **efficacy**.

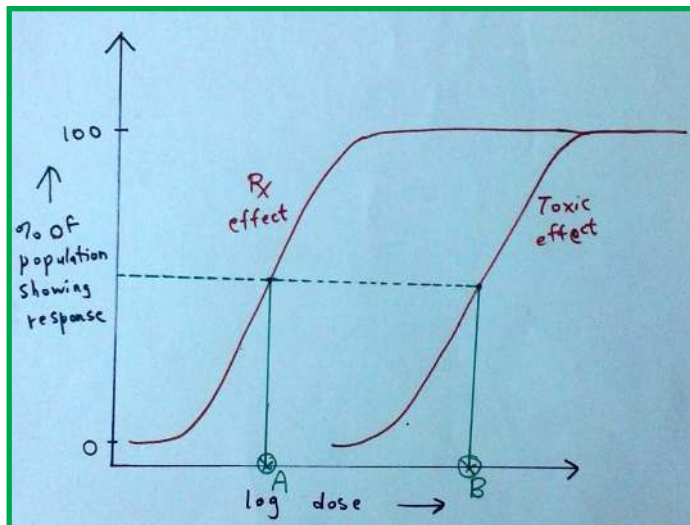


Fig 1.11: Quantal dose response curves for therapeutic and toxic effects of a drug

In the above figure, point A represents ED50 while point B represents LD50.

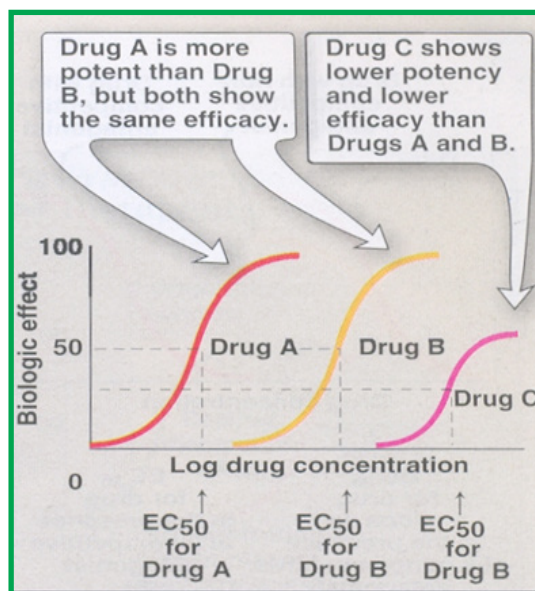


Fig. 1.12

Therapeutic Index (TI):

- Formula: LD_{50}/ED_{50} .
- **Best indicator of safety of drug.**

E.g. of drugs with narrow therapeutic index: (Mnemonic: DAT-LAAT).

- Digoxin
- Antiarrhythmics
- Theophylline
- Lithium
- Aminoglycosides
- Anti-epileptics
- Tricyclic antidepressants

Action of Drugs in Receptors:

Drug	Effect	Intrinsic activity
Agonist	Induces active conformation	1
Partial agonist	Induces partially active conformation	0 to 1
Antagonist	No effect on receptor	0
Inverse agonist	Induces inactive conformation	Negative (0 to -1)

Classification of receptors on the basis of location:

Intranuclear (located in nucleus)	Intracytoplasmic (located in cytoplasm)	Located on the cell surface
• Vitamin A • Vitamin D • Thyroxine • Estrogen • PPAR	• All steroid hormones except estrogen	• Rest all

ONE-LINERS:

- Testosterone and progesterone receptors were originally thought to be cytoplasmic in location; now proposed to be nuclear in location (not convincingly proven).
- Steroid hormone receptors are called nuclear receptors as – They act at the level of nucleus to modulate transcription.
- All amine hormones act on cell membrane except thyroid hormone (acts on nucleus).

Classification of cell surface receptors on the basis of second messenger/s:

1. G-protein coupled receptors (GPCRs) (**serpentine in shape**).
2. Ion channels.
3. Tyrosine kinase coupled receptors.
4. JAK-STAT coupled receptors.

Examples of receptors that act as ion channels:

- Nicotinic cholinergic (N_M and N_N)
- $GABA_A$
- Glycine
- Excitatory glutamate (Kainate, NMDA, AMPA)
- $5-HT_3$

Example for GPCR (G protein coupled receptor).

Muscarinic receptors (M_2 , M_4 - G_i pathway, M_1 , M_3 , M_5 – G_q pathway).

Beta receptors (G_s pathway).

Alpha 2 receptor (G_i pathway).

Alpha 1 receptor (G_q pathway).

Enzyme linked receptor (Tyrosine Kinase).

Insulin.

Examples of molecules that act through JAK/STAT receptors:

- Growth hormone
- Prolactin
- Interferons
- Many cytokines

Examples of chemical antagonism:

- Chelating agents complex toxic heavy metals.
- $KMnO_4$ oxidizes alkaloids during gastric lavage in case of alkaloid poisoning.
- Tannins form insoluble alkaloid altannates with alkaloids in case of alkaloid poisoning.
- Methemoglobin reacts with cyanide to form cyanmethemoglobin.

Example of physiological antagonism.

Adrenaline antagonism on histamine action in anaphylactic shock.

Drugs that may react when mixed in the same syringe or infusion bottle:

- Thiopentone sodium + Succinylcholine chloride
- Penicillin G sodium + Succinylcholine chloride
- Heparin + Penicillin / Tetracyclines / Streptomycin / Hydrocortisone

Therapeutic drug monitoring (TDM): Monitoring for the plasma levels of the drug at regular intervals.

Indications:

1. Drugs with narrow therapeutic index.
2. Drugs with a high inter-individual variation in plasma concentrations.

3. Drugs whose therapeutic effect cannot be determined clinically.
4. Nephrotoxic drugs are to be administered in patients with renal failure.
5. Monitoring of patient compliance (best method).
6. Diagnosis of toxicity.

TDM is not required in case of :

1. Drugs whose therapeutic effect can be determined clinically (e.g.: Warfarin – PT-INR can determine its therapeutic effect).
2. Pro-drugs.
3. Drugs with irreversible action.
4. Drugs with hit-and-run effect.

E.g. of drugs requiring TDM: (Mnemonic: DAT-LAAT-MC)

- Digoxin
- Antiarrhythmics
- Theophylline
- Lithium
- Aminoglycosides

- Anti-epileptics
- Tricyclic antidepressants
- Methotrexate
- Calcineurin inhibitors

Concept 1.4 : Drug development

Learning objectives: To study the various phases of clinical trial

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Clinical Trials:

Mandatory / Compulsory phases of clinical trials :

Phase	Also known as	Approximate number of participants
1	Human pharmacology and safety	20 – 80
2	Therapeutic exploration and dose ranging	100 – 500
3	Therapeutic confirmation / comparison	500 – 3000
4	Post-marketing surveillance	General population

Mandatory / Compulsory phases of clinical trials :

Phase	Conducted on	To determine
1	Small group of healthy individuals	Safety Pharmacokinetics Pharmacodynamics
2	Small group of patients	Efficacy Safety
3	Large group of patients	Efficacy Safety
4	Post-marketing surveillance	Rare adverse effects Chronic adverse effects Effect of the drug on special populations

ONE-LINERS:

- Best phase for determining drug efficacy – **Phase 3** (due to large sample size).
- Best phase for determining drug safety – **Phase 3** (due to large sample size).
- Minimum mandatory duration of Phase 4 studies – **2 years**.

**Optional Phases of Clinical Trials:**

Phase	Conducted on	To determine
Pre-clinical	Laboratory animals	Effect of the drug on animals
0	Small group of healthy volunteers (Microdosing – a sub-therapeutic dose is administered)	Pharmacokinetics – specifically bioavailability
5	Effectiveness research	Active reporting of effectiveness of drug in different populations

ONE-LINERS:

- Investigational new drug (IND) – **Application to conduct a human study for a new drug.**
- IND is submitted between – **Pre-clinical studies and Phase 1.**
- New drug application (NDA) – **Application for marketing a new drug.**
- NDA is submitted between – **Phase 3 and Phase 4.**

Regulation of Clinical Trials:

Good clinical practices (GCP)	Set of guidelines to be followed while conducting a clinical trial	Not applicable for – Pre-clinical studies
Good laboratory practices (GLP)	Set of guidelines to be followed while working in a laboratory	Not applicable for – Phase 4

CPCSEA (Committee for the Purpose of Control and Supervision on Experiments in Animals) GUIDELINES FOR PRECLINICAL STUDIES-

Concept 1.5 : Adverse drug reactions

Learning objectives: To study the classification of adverse drug reaction
To know about pharmacogenomic reaction
To study about teratogenicity

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Classification of ADR:

TYPE A- augmented)reactions:Eg.Hypoglycemia with insulin.

Constipation with morphine.

TYPE B (bizarre) reactions:Eg.anaphylaxis- penicillin allergy.

TYPE C(chronic) reactions: Eg.tardive dyskinesia with Neuroleptics.

Analgesic nephropathy.

TYPE D (delayed) reactions: Eg.teratogenesis, carcinogenesis:

Diethylstilbestrol, thalidomide

TYP E (end of use) reactions: Eg.adrenocortical insufficiency due to sudden withdrawal of corticosteroids.

TYPE F (failure of drug action)

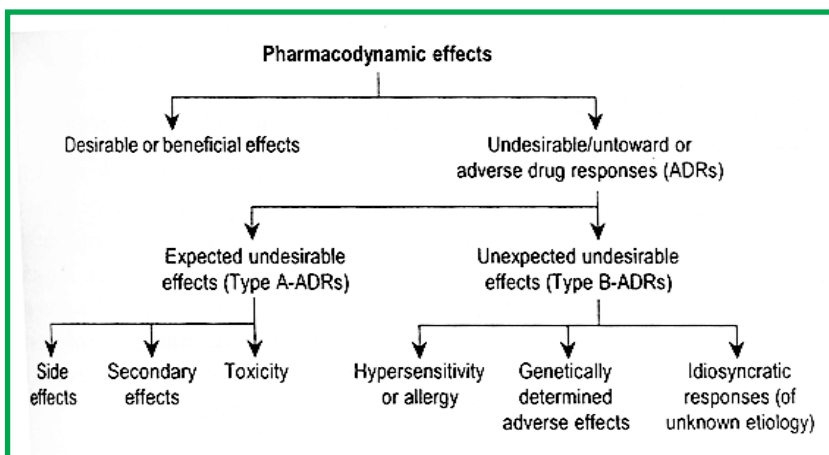


Fig. 1.13

ONE-LINERS:

- In India, pharmacovigilance programme is conducted by – **Central Drugs Standards Control Organization (CDSCO)**.
- International collaboration centre for reporting of ADRs – **Uppsala Monitoring Centre in Sweden**.
- Central Drugs Laboratory in India situated in – **Kolkata**.

Phototoxicity and Photoallergy:

Phototoxicity	Photoallergy
Drug or its metabolite accumulates in the skin ↓ Absorbs light and undergoes a photochemical reaction ↓ Damage to surrounding skin	Drug or its metabolite induces a T-cell mediated immune response ↓ Exposure to light ↓ Papular or eczematous contact dermatitis-like lesions
Occurs on exposure to UV-B (290 – 320 nm)	Occurs on exposure to UV-A (320 – 400 nm)
More common	Less common
Short duration after exposure	Long duration after exposure
E.g : • Amiodarone • Dapsone • Sulfonamides • Fluoroquinolones • Nalidixic acid • Phenothiazines • Thiazides • Tetracyclines • Tar products	E.g : • Chloroquine • Chlorpromazine • Carbamazepine • Griseofulvin • Sulfonamides • Sulfonolyureas

Examples of Pharmacogenetic Considerations:

1. Atypical pseudocholinesterase or pseudocholinesterase deficiency can cause prolonged apnea due to succinylcholine.
2. G6PD deficiency can cause hemolysis due to certain drugs.
3. Low activity of CYP2C9 leads to reduced warfarin metabolism and increased risk of bleeding.
4. Thiopurinemethyltransferase (TPMT) deficiency increases risk of severe bone marrow toxicity due to 6-mercaptopurine and azathioprine.
5. UGT1A1*28 allele of UDP-glucuronosyl transferase increases the risk of diarrhea due to irinotecan.
6. Severe 5-fluorouracil toxicity occurs in patients with dihydropyrimidine dehydrogenase (DPD) deficiency.
7. Over-expression of p-glycoprotein results in tumour resistance to many cancer chemotherapeutic agents.
8. Polymorphism of N-acetyltransferase-2 (NAT-2) results in rapid and slow acetylator status of isoniazid.
9. Acute intermittent porphyria is precipitated by barbiturates due to genetic defects in repression of porphyrin synthesis.
10. CYP2D6 abnormality causes poor metoprolol / debrisoquin metabolizer status.
11. Malignant hyperthermia after halothane is due to abnormal calcium release channel (ryanodine receptor) in the sarcoplasmic reticulum of skeletal muscles.
12. Genetic inability to hydroxylate phenytoin can lead to toxicity at usual doses.

13. Resistance to coumarin anticoagulants is due to an abnormal VKOR1 that is less sensitive to the coumarins.
14. Attack of angle closure glaucoma is precipitated by mydriatics in individuals with narrow iridocorneal angle.
15. HLAB5701 in this alle patient - abacavir causing SJS.

Drugs Causing Hemolysis in G6PD Deficiency:

Group	Definite risk	Possible risk	Doubtful risk
Antimalarials	Primaquine, Dapsone / Chlorprogaunil	Chloroquine	Quinine
Sulfonamides	Sulfomethoxazole, Other sulphonamides, Dapsone	Sulfasalazine, Sulfadimidine	Sulfisoxazole, Sulfadiazine
Antibacterials	Cotrimoxazole, Nalidixic acid, Nitrofurantoin, Niridazole	Ciprofloxacin, Norfloxacin	Chloramphenicol, p-amino salicylic acid (PAS)
Analgesics / Antipyretics	Acetanilide, Phenazopyridine	Acetylsalicylic acid (Aspirin) >3g/d	Acetylsalicylic acid (Aspirin) <3g/d, Acetaminophen (Paracetamol), Phenacetin
Others	Naphthalene, Methylene blue, Rasburicase	Vitamin K analogues, Ascorbic acid >1 g	Doxorubicin, Probenecid

US FDA Risk category of drugs during pregnancy:

Category	Animal studies	Human studies	Examples
A (No risk to humans)	Not teratogenic OR Not available	Not teratogenic	• Magnesium sulfate • Thyroxine
B (No evidence of risk to humans)	Not teratogenic	Not available	• Penicillin V • Amoxicillin • Cefaclor • Erythromycin • Paracetamol • Lidocaine
	Teratogenic	Not teratogenic	

C (Risk cannot be ruled out)	Not available	Not available	• Morphine • Codeine • Atropine • Corticosteroids • Adrenaline • Thiopentone • Bupivacaine
	Teratogenic	Not available	
D (Benefit may outweigh potential risk)	Teratogenic	Teratogenic; but use of the drug is acceptable despite the potential risk	• Aspirin • Phenytoin • Carbamazepine • Valproate • Lorazepam • Methotrexate
X (Risk may outweigh potential benefit)	Teratogenic	Teratogenic; use of the drug is not acceptable due to potential risk	• Estrogens • Isotretinoin • Ergometrine • Thalidomide

Human Teratogenic Drugs:

Thalidomide phocomelia, multiple defects of internal organs.

Anticancer drugs cleft palate, hydrocephalus, multiple defects, foetal death (methotrexate).

Androgens: virilization; limb, esophageal, cardiac defects

Progestins: virilization of female foetus.

Stilboestrol: vaginal carcinoma in teenage female offspring.

Tetracyclines: discoloured and deformed teeth, retarded bone growth.

Warfarin: depressed nose; eye and hand defects, growth retardation

Phenytoin hypoplastic phalanges, cleft lip/palate, microcephaly.

Phenobarbitone: various malformations

Carbamazepine: neural tube defects, assorted abnormalities.

Valproate sod: spina bifida and other neural tube defects, heart and limb Abnormalities.

Alcohol: low IQ baby, growth retardation, foetal alcohol syndrome.

ACE inhibitors: hypoplasia of organs, growth retardation, foetal loss.

Lithium: Epstein anomaly, foetal goiter, cardiac and other abnormalities.

Antithyroid drugs: foetal goiter and hypothyroidism.

Indomethacin / aspirin: premature closure of ductus arteriosus.

Isotretinoin: craniofacial, heart and CNS defects, hydrocephalus.

Miscellaneous

Examples of Enantiomers and their Importance:

Enantiomer	Advantage claimed
S-atenolol	Half dose, better tolerated
S-metoprolol	Half dose
S-amlodipine	Half dose, less peripheral edema
S-omeprazole (Esmoprazole)	Better oral bioavailability
S-pantoprazole	More potent
R-salbutamol	More active, S- may antagonize R-
S-citalopram (Escitalopram)	Lower dose, less side effects
S-naproxen	Less burden on kidney (but inversion occurs in vivo)
Cisatracurium	4x more potent, less histamine release
Levofloxacin	More active, slower elimination
Levocetirizine	Half dose, only R-form active
Desloratidine	Half dose

Drugs Available as Transdermal Patch:

In India	Outside India
- Nitroglycerine - Fentanyl - Nicotine - Estradiol	- Isosorbidedinitrate - Hyoscine - Clonidine

Drugs Available as Liposomal Preparations:

- Amphotericin B
- Cytarabine
- Doxorubicin
- Daunorubicin
- Morphine
- Vincristine
- Verteporfin (Age-related ARMD)

- Verteporfin is approved for – **Age-related macular degeneration (ARMD)**
- Size of liposomes – **60 – 80 nM.**

Young's formula: for dosing in children:

$$\text{Child dose} = (\text{Age} / \text{Age} + 12) \times \text{adult dose}$$

Dilling's formula: for dosing in children:

$$\text{Child dose} = (\text{Age} / 20) \times \text{adult dose}$$

Worksheet

- MCQ OF "GENERAL PHARMACOLOGY" FROM DQB



Active recall-

Two important property needed for a drug to get better absorptions are	
Low volume of distribution means the drugs are seen mainly in which compartment	
Mention the areas in brain where blood brain barrier is absent	
Thiopentone sodium has very short half life because of	
What is the active metabolite of spironolactone	
Most common phase II biotransformation reaction is	
Example for drug undergoing Hoffmann degradation is	



Examples for drug undergoing zero order kinetic of excretion are	
Examples for ligand gated receptors	
Location of thyroid receptor	
Therapeutic index indicates	
In the presence of competitive antagonism what will happen to log dose response curve	
Phase IV clinical trial indicates	
Malignant hyperthermia due to succinyl choline is example for what type of adverse effect	
Moebius syndrome is the teratogenicity caused by	

CONCEPTS

- ⇒ **Concept 2.1 Cholinergic neurotransmission**
- ⇒ **Concept 2.2 Muscarinic Receptor Agonists and Antagonists**
- ⇒ **Concept 2.3 Nicotinic receptors agonist and antagonist**
- ⇒ **Concept 2.4 Anticholinesterase Agents**
- ⇒ **Concept 2.5 Anti cholinergic agents**
- ⇒ **Concept 2.6 Adrenergic neurotransmission & classification of adrenergic**
- ⇒ **Concept 2.7 Adrenergic receptors**
- ⇒ **Concept 2.8 Alpha receptor agonist and antagonist**
- ⇒ **Concept 2.9 Beta receptor agonist and antagonist**

Concept 1.1: Cholinergic neurotransmission

Learning Objectives:

- Know the mechanisms of synthesis, storage, release, and destruction of Ach
- Know the sites of release of Ach as neurotransmitter

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Synthesis, Storage, Release & Metabolism Of Acetylcholine(Ach):

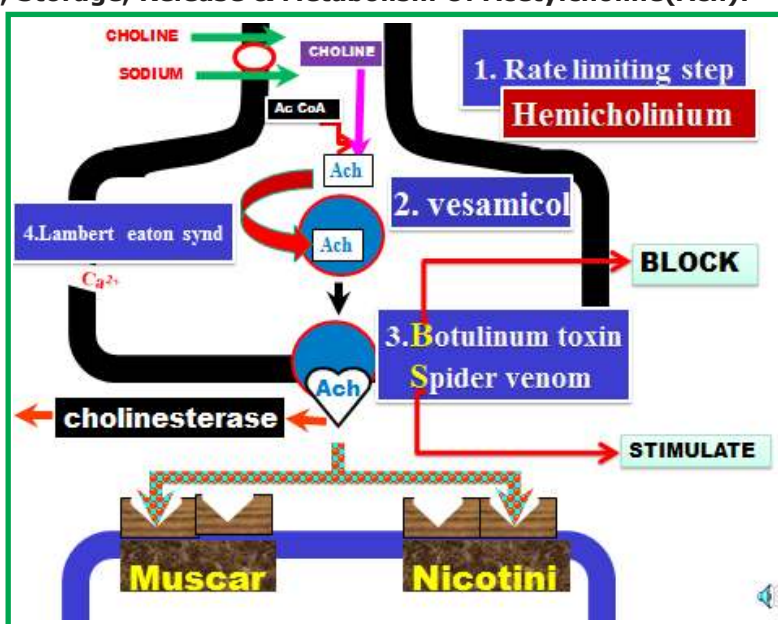


Fig. 2.1

Steps		Modified by
Choline uptake (rate limiting step)		Blocked by hemicholinium
Vesicular uptake of ACh		Blocked by vesamicol
Release of ACh		Blocked by botulinum toxin, Release Increase by spider venom
Lambert Eaton syndrome	Defect in the presynaptic calcium channel result in poor release of ACh	Treated by 3, 4 di aminopyridine (dalfampridine). this drug is also useful in multiple sclerosis to improve walking ability

Common drug metabolized by pseudocholinesterase

Drug	Half Life	Property
Ach	Few seconds	Not used clinically
Sch	3- 5 minutes	Fastest & shortest acting SMR Preferred in treacheal intubation
Mivacurium	15- 20 minutes	Day care SMR
Remifantanyl	3- 5 minutes	Day care analgesic
Esmolol	~9 Min	Ultra short acting beta blocker

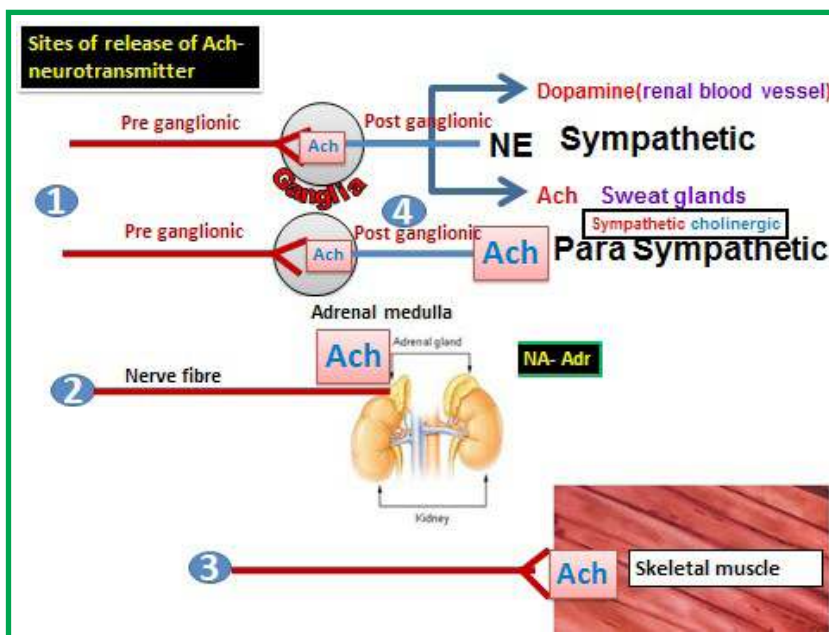


Fig. 2.2

Sites of release of ACh

Fibre	Transmitter
All the preganglionic fibres of sympathetic and parasympathetic fibres at the ganglia	Ach
A nerve fibre supplying adrenal medulla	Ach
At neuromuscular junction (NMJ).	Ach
Post ganglionic parasympathetic fibre	Ach
Post ganglionic sympathetic fibres supplying sweat gland	Ach

Concept 2.2 : Muscarinic Receptor Agonists and Antagonists

Learning objectives:

- Know the location, functions and drugs acting on various muscarinic receptors
- Know the uses and toxicities of muscarinic receptor agonists and antagonists.

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Muscarinic receptors

Subtypes M₁, M₂, M₃, M₄ and M₅ (clinically important are M₁, M₂ and M₃), all are G protein coupled receptors.

M ₁	M ₂	M ₃	M ₄	M ₅
G _q	G _i	G _q	G _i	G _q

Receptor	Location	Function	
Muscarinic Receptors			Agonist
M ₁	Central nervous system (post-synaptic neurons)	Anterograde transmission of the cholinergic impulse	Oxotremorine
	Gastric glands	Increased gastric acid secretion	
M ₂	Heart	Decreased heart rate (-ve chronotropic effect) Decreased conduction (-ve dromotropic effect)	Methacholine
	Smooth muscles	Contraction	
	Pre-synaptic neurons	Inhibition of acetylcholine release	
M ₃	Exocrine glands	Increased glandular secretions	Pilocarpine, CEVIMELINE
	Smooth muscles	Contraction	Bethanechol
	Eye	Miosis	Pilocarpine

M1

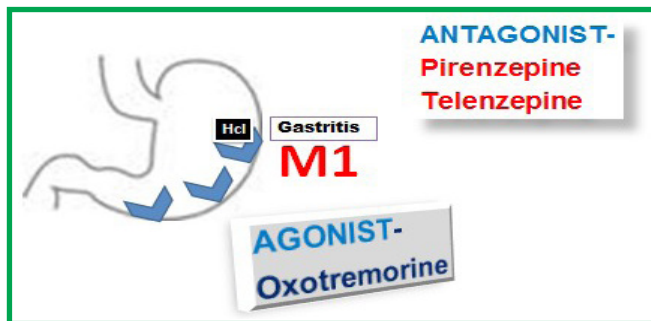


Fig. 2.3

M2

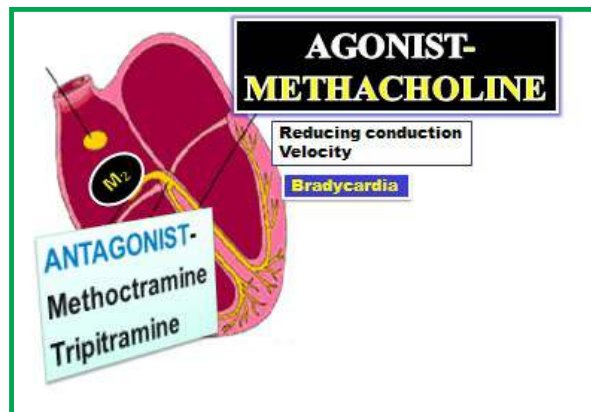


Fig. 2.4

M3

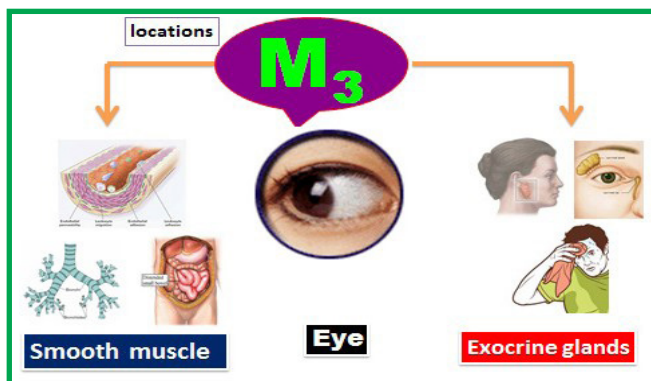


Fig. 2.5

SMOOTH MUSCLE

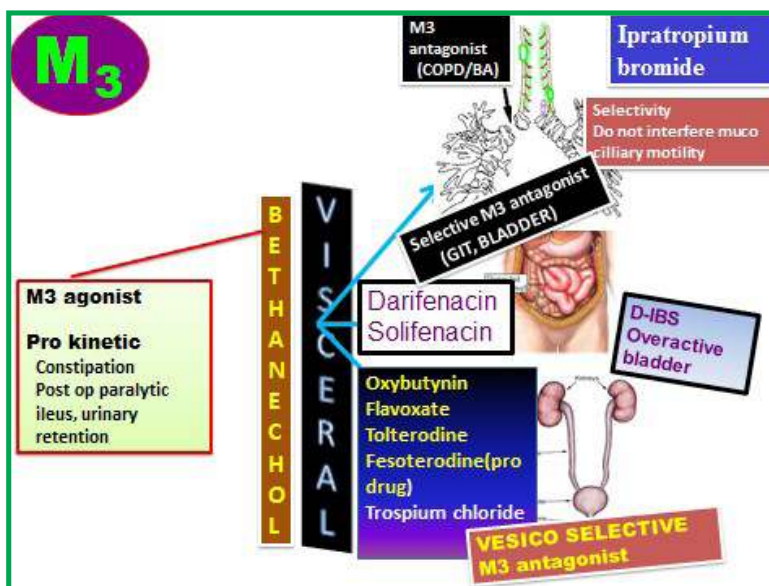


Fig. 2.6

EYE

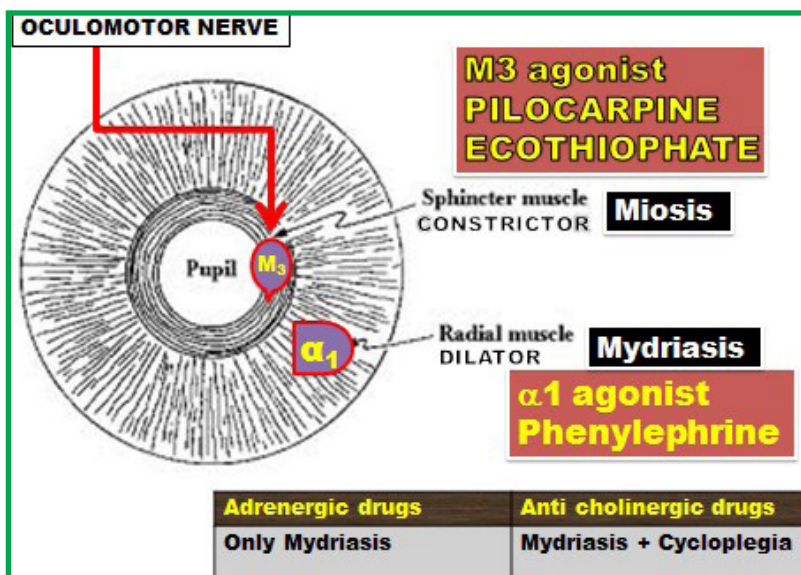
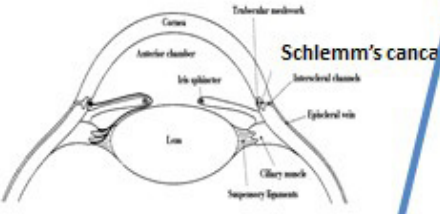


Fig. 2.7

GLUACOMA

PILOCARPINE
ECOTHIOPHATE-



M₃

Antagonist

Atropine ointment-1wk
Homatropine
Cyclopentolate
Tropicamide(3-6hrs)
Fastest & shortest

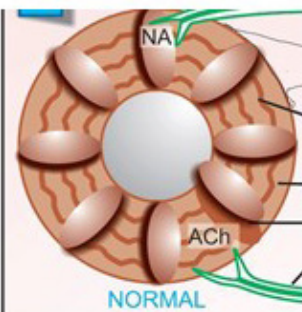
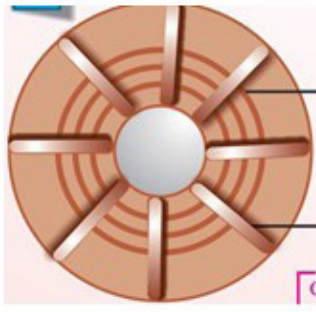
C/I- GLUACOMA

Only for fundus exam-
Mydriasis enough
Phenylephrine- preferred/
Or Tropicamide

Error of refraction
mydriasis & cycloplegia
Tropicamide- DOC
Child <5yr
Atropine oint 1%

Fig. 2.8

EFFECT OF TEST DRUG ON RABBIT EYE

NORMAL

> Light Reflex & Corneal reflex-PRESENT
 > Increased secretion & congestion
 > Intraocular pressure reduced.

Cholinergic- Pilocarpine

Fig. 2.9

EFFECT OF TEST DRUG ON RABBIT EYE

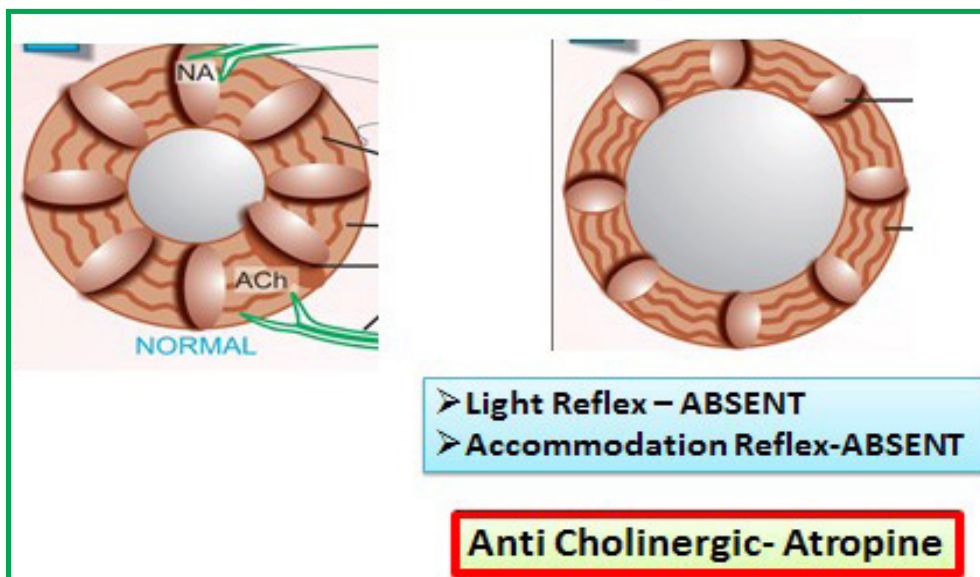


Fig. 2.10

EFFECT OF TEST DRUG ON RABBIT EYE

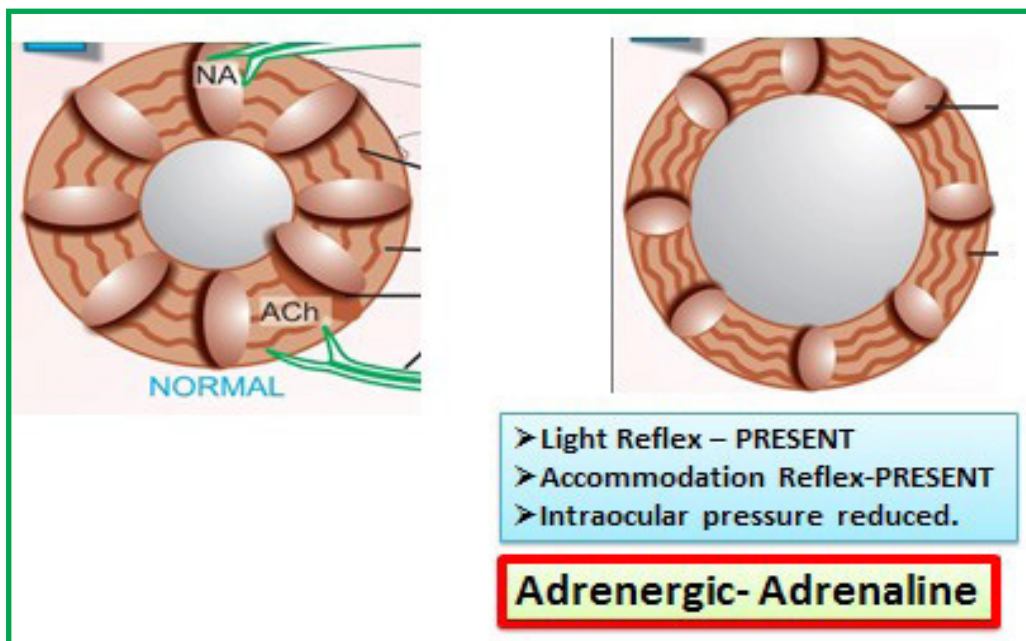


Fig. 2.11

Smooth muscles in which M₃ receptors cause contraction:

	Smooth muscle	Effect due to contraction
1	Sphincter pupillae	Miosis
2	Bronchial smooth muscle	Bronchoconstriction
3	GI smooth muscle	Increased motility and peristalsis
4	Detrusor (urinary bladder)	Voiding of urine

GLANDS

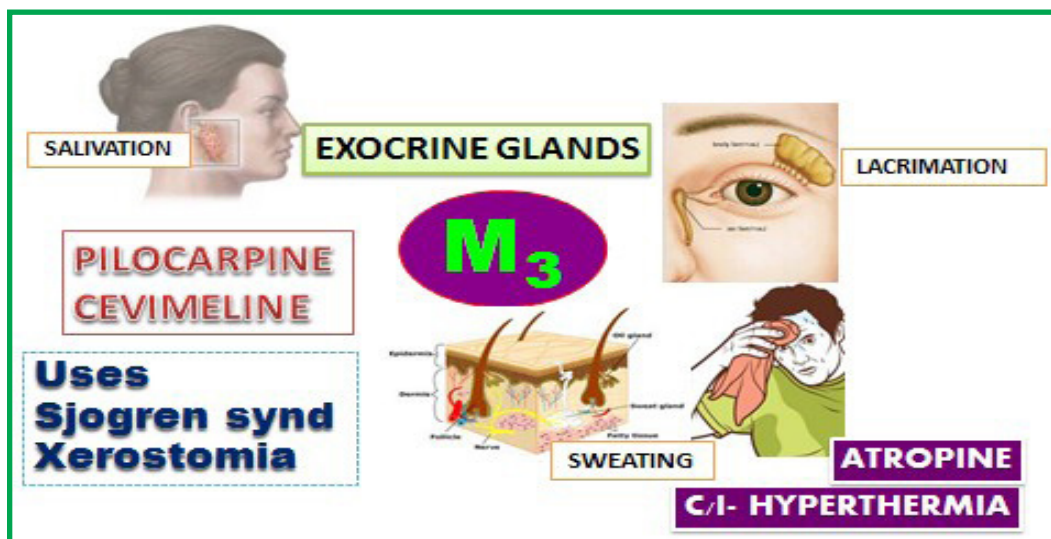


Fig. 2.12

Relatively Selective Agonists and Antagonists:

Receptor	Relatively selective agonist	Relatively selective antagonist
M ₁	MCN-343A Oxotremorine	Pirenzepine Telenzepine
M ₂	Methacholine	Methoctramine Tripitramine
M ₃	Bethanechol	Darifenacin Solefenacin



Worksheet

	Location	Selective agonist	Selective antagonist
M1			
M2			
M3	GI & Bladder smooth muscle		
M3	EYE		
M3	Exocrine glands		

Concept 2.3 : Nicotinic receptor agonist and antagonist

Learning objectives:

- Know the location, functions and drugs acting on various nicotinic receptors

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Nicotinic receptors (Nm & Nn)

Nm receptor-

Receptor	Nm
Type	Ligand gated
Location	Neuro muscular junction (NMJ)
Function	Contraction of skeletal muscle
Agonist	Phenyl trimethyl ammonium (PTMA) Nicotine
Antagonist	Tubocurarine α -Bungarotoxin

Nn receptor

Receptor	Nn
Type	Ligand gated
Location & function	Autonomic ganglia-Anterograde transmission of impulse Adrenal medulla Secretion of catecholamines Central nervous system Excitation or inhibition
Agonist	Dimethyl phenyl piperazinium (DMPP) Nicotine
Antagonist	Hexamethonium Trimethaphan Mecamylamine

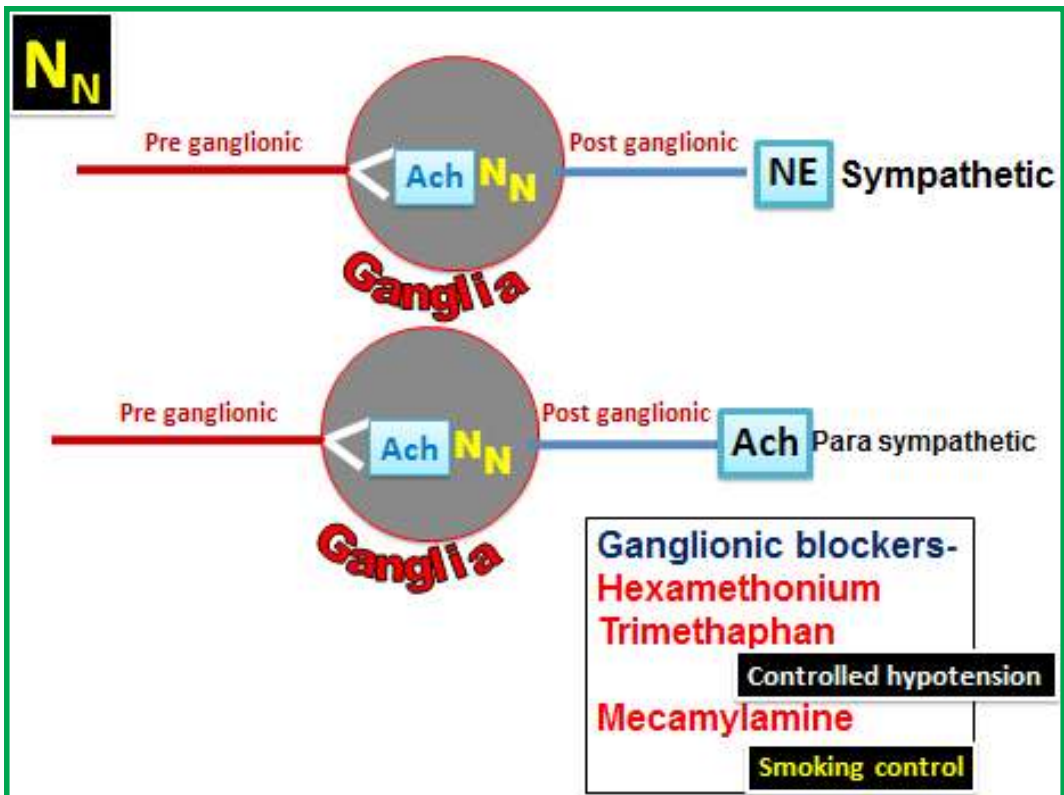


Fig. 2.13

Worksheet

To do

No depolarizing skeletal muscle relaxant acts by blocking which receptor

Name the ganglionic blocker useful to control smoking habbit

Name the ganglionic blocker useful to produce controlled hypotension

Concept 2.4 : Anticholinesterase agents

Learning objectives:

- Know the classification of cholinergic agents
- Know the uses and toxicities o drugs and agents that inhibit AChE.

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Classification of Cholinomimetics:

Direct cholinomimetics : Directly activate the cholinergic receptors.

Indirect cholinomimetics : Increase acetylcholine levels in the synapse by decreasing acetylcholinesterase activity

Examples of direct cholinomimetics:	
Choline esters – • Acetylcholine • Methacholine • Carbachol • Bethanechol	Alkaloids – (suffix INE) • Muscarine • Pilocarpine • Arecoline • Lobeline • Cevemeline
Examples of indirect cholinomimetics:	
REVERSIBLE – Carbamates – • Physostigmine (Eserine) • Neostigmine • Pyridostigmine • Edrophonium • Rivastigmine • Donepezil • Galantamine • Acridine – • Tacrine	IRREVERSIBLE – Organophosphates – • Dyflos (Di-isopropyl fluoro phosphate or DFP) • Echothiophate • Malathion • Diazinon • Tabun • Sarin • Soman • Carbamates – • Carbaryl • Propuxur

Examples of organophosphate nerve gases:

• Tabun • Sarin • Soman

Insecticides:

• Malathion • Diazinon • Parathion
--

OPC Poisoning:

- OPC inhibits cholinesterase enzyme by binding with esteritic site only.
- First line drug of choice for OPC- Atropine (dose depends upon signs and symptoms of atropinisation).
- Oxime are cholinesterase reactivators (oxime binds with anionic site of cholinesterase enzyme).
- Pralidoxime given 1=2gm slow IV infusion for about 15-30 minutes.
- Obidoxime is the most potent oxime.
- Diacetyl mono oxime is highly lipid soluble.

Oxime will not be useful for treating carbamate poisoning (because carbamate binds both anionic as well as esteritic site of cholinesterase enzyme)

Chronic OPC poisoning may cause neurological damage due to inhibition of neuropathy target esterase enzyme

Properties of Direct Cholinomimetics:

Drug	Hydrolysis by		Actions	
	AChE	BuChE	Muscarinic	Nicotinic
Acetylcholine	++	+	+	+
Methacholine	+	-	+	±
Carbachol	-	-	+	++
Bethanechol	-	-	+	-

Uses of anticholinesterase

Condition	Anti cholinesterase
Alzheimer's disease –	• Tacrine (causes hepatotoxicity) • Rivastigmine • Donepezil • Galantamin
Atropine and Datura poisonings	Physostigmine (lipid soluble crosses BBB)
Post-operative atony	Neostigmine
Myasthenia gravis	• Pyridostigmine • Neostigmine Cobra bite • Neostigmine
Reversal of prolonged post-operative neuro-muscular blockade	Neostigmine
Cobra bite	Neostigmine + atropine



Worksheet

To do

Condition	Drug
Preferred anticholinesterase for myasthenia gravis	
Anticholinesterase useful for diagnosis of myasthenia gravis	
Drug of choice for atropine poisoning	
Examples for nerve gases	

Concept 2.5 : Anticholinergic agents

Learning objective: To know the anticholinergic drugs and their uses

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Classification of Anticholinergics:

- Direct anticholinergics – Directly antagonize the cholinergic receptors.
- Indirect anticholinergics – Decrease acetylcholine levels in the synapse.

Examples of Direct Anticholinergics:

Natural Alkaloids –

- Atropine (Hyoscyamine)
- Scopolamine (Hyoscine)

Semi-Synthetic Derivatives –

- Homatropine
- Atropine methonitrate
- Hyoscine butyl bromide
- Ipratropium bromide
- Tiotropium bromide

Synthetic Compounds – Mydriatics:

- Cyclopentolate
- Tropicamide (fastest and shortest acting)

Antisecretory-antispasmodics:

- Quarternary ammonium compounds – Propantheline, Oxyphenonium, Clidinium, Pipenzolate methyl bromide, Isopropamide, Glycopyrrolate
- Tertiary amines – Dicyclomine, Valethamate, Pirenzepine

Vesicoselective:

- Oxybutynin
- Flavoxate
- Tolterodine

Antiparkinsonian:

- Trihexyphenidyl (Benzhexol)
- Procyclidine
- Biperidin

Comparison between atropine and hyoscine:

	Property	Atropine	Hyoscine
1	Source	Atropa belladonna, Datura stramonium	Hyoscyamys niger
2	Alkaloidal ester of tropic acid with	Tropine (base)	Scopine (base)
3	CNS effects	Low dose – Mild excitation High dose – Strong excitation	Low dose – Depression High dose – Excitation
4	Anticholinergic property	More potent on heart, bronchial muscle and intestine	More potent on eye and secretory glands
5	Duration of action	Longer	Shorter
6	Anti-motion sickness activity	++	+++

Comparison between Atropine and Glycopyrrolate:

Action	Atropine	Glycopyrrolate
Antisecretory	++	+++
Tachycardia	+++	++
CNS effects	+	-
Bronchodilation	++	++

ONE-LINERS:

- Atropine is a – Racemic mixture. But only L-atropine is biologically active.
- Scopolamine is – L-Hyoscine.
- Enzyme atropinase is present in – Rabbits
- Glycopyrrolate is **devoid of CNS effects** because – It is a quaternary ammonium compound; hence does not cross blood brain barrier. Thus why it is preferred pre anaesthetic agent
- Anticholinergic used to hasten dilation of cervix during labour – **Valethamate**

Examples of indirect anticholinergics:

	Drug	Mechanism of action
1	Hemicholinium	Inhibits choline uptake by pre-synaptic nerve terminals (rate-limiting step in ACh biosynthesis).
2	Vesamicol	Inhibits active uptake of ACh into synaptic vesicles.
3	Botulinum toxin	Inhibits release of ACh from pre-synaptic nerve terminals.
4	Black widow spider toxin	Causes massive release of ACh from pre-synaptic nerve terminals and hence, depletion of ACh.

Indications and Side Effects of Anticholinergics:

Receptor	Effect	Indication	Side effect
M_1	Decreased transmission of cholinergic impulse	Drug induced Parkinson's disease • Trihexyphenidyl (Benzhexol) (DOC) • Benztropine • Biperidin • H_1 antihistaminics Motion sickness • Hyoscine Cholinomimetic poisoning (Organophosphorus poisoning, Carbamate poisoning, Early type of mushroom poisoning) • Atropine	Excitement Psychotic behaviour Ataxia Delirium Dreadful visual hallucinations
	Decreased acid secretion	Peptic ulcer disease • Pirenzepine • Telenzepine	-
M_2	+ve chronotropic effect	-	Tachycardia
	+ve dromotropic effect	AV block Atropine	-
M_3	Decreased exocrine secretions	Pre-anesthetic medication to reduce secretions • Glycopyrrolate	Dry mouth Drying of secretions Difficulty in swallowing and talking Dry, flushed and hot skin
	Dilation of sphincter pupillae (Mydriasis)	Any ophthalmic procedure requiring papillary dilation • Atropine (longest acting) • Homatropine • Cyclopentolate • Tropicamide Anterior uveitis • Atropine	Photophobia Blurring of near vision
	Bronchodilation	Bronchial asthma and COPD • Ipratropium bromide • Tiotropium bromide • Oxytropium bromide	-



	Relaxation of GI musculature	As antispasmodics • Dicyclomine • Hyoscine butyl bromide	Constipation
	Decreased urinary voiding	Overactive bladder (Unstable bladder or Detrusor instability) • Oxybutynin (DOC) • Tolterodine • Trospium • Darifenacin • Solifenacin • Flavoxate • Imipramine	Urinary retention
N_M	Relaxation of skeletal muscle	Adjuvants to general anesthesia to facilitate intubation • d-Tubocurarine • Doxacurium • Pancuronium • Vecuronium • Rocuronium • Rapacurium • Atracurium • Cisatracurium • Mivacurium	Respiratory paralysis
N_N	Decreased ganglionic transmission	Hypertension (Ganglion blockers not used nowadays)	Bradycardia Hypotension

Antidotes for Mushroom Poisoning:

Type of mushroom poisoning	Antidote
Early mushroom (Muscarine) type INOCYBE toxin	Atropine
Late mushroom (Phalloidin) type A phalloids,	Thioctic acid
Hallucinogenic type ISOXAZOLE	No specific antidote

Indications of Botulinum Toxin (BoTox):

- Cosmetic – Masking of facial wrinkles
- Achalasia cardia
- Spasmodic dysphonia
- UMN lesions
- Long-standing back/neck pain
- Blepharospasm
- Strabismus

Valathamate - anti cholinergic- useful for cervical ripening (facilitating child birth)

- Ona botulinum toxin - useful in prophylaxis of chronic migraine, overactive bladder.

- Saxitoxin, Tetrodotoxin - origin from Dinoflagellates, blocks sodium channel- causing skeletal muscle paralysis.
- Alpha bungarotoxin = Component of venom of banded krait, Antagonist at N_M receptor, causing skeletal muscle paralysis, reversed by neostigmine.

Other Drugs Having Anticholinergic Properties

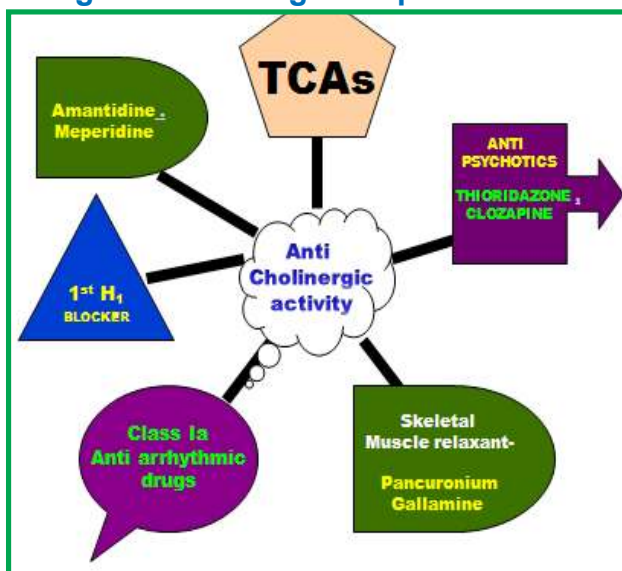


Fig. 2.14

Condition	Treatment
Motion sickness	Hyosine
Sea sickness	Meclizine (1 gen antihistamine)
Mountain sickness	Acetazolamide
Morning sickness	Doxylamine + Vit B6



Worksheet

To do

DRUG	USES
Pirenzepine, telenzepine	
Darifenacin	
Tolterodine	
Trospium	
Trihexyphenidyl	
Glycopyrrolate	
Valathamate	

Concept 2.6 : Adrenergic neurotransmission

Learning objective: To know the synthesis, storage, release & metabolism of NE

Time Needed

1 st reading	15 mins
2 nd reading	5 mins

Synthesis, Storage, Release and Metabolism of Noradrenaline

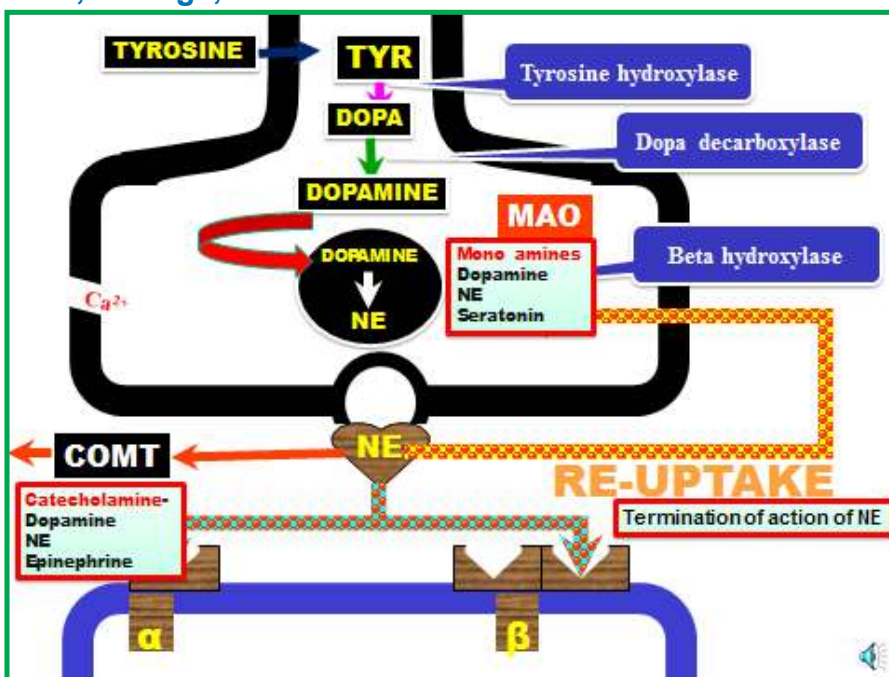


Fig. 2.15

Enzyme involved in synthesis of NE	Enzyme involved in metabolism of NE
Tyrosine hydroxylase (Rate limiting enzyme) -Dopa decarboxylase -Beta hydroxylase	Mono amino oxidase Catechol o methyl transferase

Co-factors for various enzymes involved in catecholamine synthesis:

Enzyme	Co-factor
Tyrosine hydroxylase	Tetrahydrobiopterin
Aromatic amino acid decarboxylase	Pyridoxal phosphate
Dopamine-β-hydroxylase	Ascorbate
Phenylethanolamine-N-methyltransferase	S-adenosylmethionine

Molecules released in the nerve endings along with catecholamines:

- ATP
- Dopamine- β -hydroxylase
- Chromogranin
- Enkephalin / Neuropeptide Y

ONE-LINERS:

- Rate limiting enzyme (tyrosine hydroxylase) blocked by- METYROSINE
- DOPA DECARBOXYLASE blocked by- carbidopa, ebzarside
- In the synaptic vesicles: NA is stored along with ATP in the ratio **4:1**, adsorbed on the protein **chromogranin**.
- Reserpine inhibits – **Vesicular monoamine transporter 2 (VMAT2)**.
- **Valbenazine – VMAT2 BLOCKER** – useful in treatment of Tardive dyskinesia.
- **Droxidopa** – Prodrug of NA/Adr; approved for neurogenic orthostatic hypotension, dialysis-induced hypotension and hypotension associated with fibromyalgia and chronic fatigue syndrome.
- **Reuptake of NE is blocked by** – TCA, SNRI, NDRI, cocaine.

Worksheet

To do

STEP	BLOCKED BY
Tyrosine hydroxylase	
Dopa decarboxylase	
Vesicular uptake of dopamine	
Beta hydroxylase	
Release of norepinephrine	
Reuptake of norepinephrine	

Concept 2.7 : Adrenergic receptors

Learning objective:

- To know the location & functions of adrenergic receptor
- To know the endogenous and exogenous catecholamines

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Receptors in the Sympathetic Nervous System:

Receptor	Location	Function	Selective agonist
α_1	Smooth muscle	Contraction E.g: • Dilator pupillae – Mydriasis • Urinary sphincter – Urinary retention • Vascular smooth muscle – Vasoconstriction and rise in BP	Nasal decongestant- Naphazoline Oxymetazoline Xylometazoline Mydriatic -Phenylephrine To treat postural hypotension- Methoxamine Mephenteramine Midodrine
	Central nervous system	Multiple unknown functions	
α_2	Central nervous system	Hypotension Analgesia Drying of secretions Sedation Anxiolysis (Mnemonic: HADSA)	Anti hypertensive- Clonidine. Methyl dopa. Guanafacine Guanabenz Moxonidine Rilmonidine Anti glaucoma- Apraclonidine Brimonidine Skeletal muscle relaxant= Tizanidine Preanaesthetic medication Dexmedetomidine

β_1	Heart	Increased heart rate (+ve chronotropic effect) Increased conduction (+ve dromotropic effect) Increased force of contraction (+ve inotropic effect) Increased excitability (+ve bathmotropic effect) Increased relaxation (+ve lusitropic effect)	Dobutamine
	Juxta-glomerular (JG) cells of kidney	Increased rennin release	
β_2	Smooth muscle	Relaxation E.g: Bronchial smooth muscle – Bronchodilation Uterus – Tocolysis Vascular smooth muscle – Vasodilation and fall in BP	Bronchodilators Salbutamol Terbutaline Salmeterol Formetrol Indacaterol Uterine relaxant Ritordine Isoxuprine
	Skeletal muscles	Tremors	
	Liver	Glycogenolysis	
β_3	Adipocytes	Lipolysis	Anti obesity SIBUTRAMINE(withdrawn because of cardiotoxicity) MIRABEGRON- beta 3 agonist Relax detrusor- overactive bladder

Classification of adrenergic

Examples of Endogenous Catecholamines:

Catecholamine	Action on receptors
Adrenaline (Epinephrine)	All α and β
Noradrenaline (Norepinephrine)	All α and β except β_2
Dopamine	D_1 , β_1 , α_1 $D_1 - D_4$

Indications of Adrenaline:

- Anaphylactic shock (DOC) – 0.5 mL of 1:1000 dilution IM.
- Cardio-pulmonary resuscitation.
- As an adjuvant to local anaesthetics.

Exogenous (Synthetic) catecholamines

Exogenous Catecholamine	Action on receptors
Isoprenaline	β_1 , β_2 , β_3
Dopexamine	D_1 , β_2
Dobutamine	β_1 $t_{1/2} = 2$ minutes
Fenoldopam	D_1 (IV infusion, HT emergency with renal impairment)

Effects of intravenous catecholamines on cardiovascular parameters:

Catecholamine	Heart rate	Systolic blood pressure	Diastolic blood pressure
Adrenaline	↑	↑	Low dose: ↓ High dose: ↑
Noradrenaline	↓ (Reflex vagal)	↑	↑
Isoproterenol (Isoprenaline)	↑	↑	↓

Dale's vasomotor reversal:

Adrenaline + α -blocker → Fall in BP (due to β_2 mediated vasodilation).

Re-reversal of Dale's vasomotor reversal:

Adrenaline + β -blocker → Rise in BP (due to α_1 mediated vasoconstriction).



Examples of indirect sympathomimetics (increase release of NA from adrenergic nerve endings):

- Tyramine (develops tachyphylaxis)
- Amphetamines

Examples of mixed action sympathomimetics (act both directly and indirectly):

- Ephedrine
- Dopamine
- Mephentermine



Worksheet

To do

Site	Receptor
Radial muscle of iris	
Myocardium	
Bronchial smooth muscle	
Adipose tissue	
Internal urethral sphincter	

Concept 2.8 : Alpha receptor agonist and antagonist

Learning objective:

- To know the alpha receptor agonist and antagonist
- To know the uses & adverse effects of alpha receptor agonist and antagonist

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

	Drug	Characteristics
1	Phenylephrine	• Used as a mydriatic • Devoid of cycloplegia • Hence, preferred in elderly
2	Phenylpropanolamine	• Vasoconstrictor • Approved for shock • Not used nowadays due to side effect of – Hemorrhagic stroke
3	Ephedrine	• Vasoconstrictor of choice in pregnancy • High incidence of tachyphylaxis as side effect
4	Xylometazoline	• Vasoconstrictors
5.	Oxymetazoline	• Decrease blood flow to nasal mucosa and relieve congestion • Hence, called nasal decongestants
6	Nafazoline	• Used for common cold and allergies • Long-term side effect – Rhinitis medicamentosa
7	Methylphenidate	• CNS stimulant • Amphetamine derivative but devoid of amphetanergic effects • Used for attention deficit hyperactivity disorder (ADHD)
8	Fenfluramine	• Suppress appetite in CNS
9	Dexfenfluramine	• Hence, called ‘anorectics’ • Cause weight loss
10	Sibutramine	• Approved for obesity
11	Phentermine	• Only phentermine used now; others have been withdrawn
12	Dexmedetomidine	• Central selective α_2 agonist • Approved for sedating critically ill patients in ICU
13	Clonidine	• Central selective α_2 agonist • Acts as a central sympatholytic • Used for – Hypertension, Opioid withdrawal and Autonomic diarrhea

14	Moxonidine Rilmonidine	- Central selective α_2 agonist - Acts as a central sympatholytic - Used for – Hypertension, Opioid
15	Guanafacine Guanabenz	- Central selective α_2 agonist - Acts as a central sympatholytic - Used for- ADHD, TICS
16	Apraclonidine Brimonidine	Used for GLAUCOMA
17	Tizanidine	Centrally acting skeletal muscle relaxant
18	LOFEXIDINE	Useful to control opioid withdrawal symptoms

Classification of α -blockers:

Non-selective		Selective	
Non-equilibrium type	Equilibrium type	α_1 selective	α_2 selective
Phenoxy-benzamine	- Phentolamine - Tolazoline	- Prazosin - Terazosin - Doxazosin - Tamsulosin - Alfuzosin - Silodosin - Bunazosin - Indoramine - Urapidil	- Yohimbine - Rauwolscine - Idazoxan

Alpha-receptor Antagonists:

	Drug	Characteristics
1	Phenoxybenzamine	- Irreversible blocker of α receptors - Long-acting - Gets accumulated in fat - Preferred for BP control in pheochromocytoma
2	Phentolamine	- Short-acting - DOC in cheese reaction - DOC in clonidine withdrawal - To control hypertension during pheochromocytoma surgery - To treat oxime induced hypertension

3	Tolazoline	- Vasodilator - Used in spasms of peripheral blood vessels - Antidote for vasoconstriction due to 5-HT_{2A} agonists
4	Prazosin	- Vasodilator - Approved for hypertension - Preferred in hypertension due to scorpion sting - Also preferred in hypertension + BPH - s/e – Postural (Orthstatic) hypotension - Significant after first dose; hence called 'first-dose effect' - Prevented by starting drug in low dose at bedtime - Starting dose – 0.5 mg OD HS - Gradually increased to full therapeutic dose – 1-4 mg BD/TDS
5	Tamsulosin	- Selective α_{1A} antagonist - Preferred for BPH - Symptomatic improvement by increasing urinary voiding
6	Yohimbine	- No approved indication - Used as aphrodisiac for recreational purposes

ONE-LINERS:

- α -blocker with additional phosphodiesterase inhibitory action – **Prazosin**.
- Retrograde ejaculation is a side effect of – **Tamsulosin**.
- Problem of floppy iris during cataract surgery is encountered with – **Tamsulosin**.
- Alpha blockers causing apoptosis of prostate – **Doxososin, Terazosin**.



Worksheet

To do

Drug	Receptor activity	Uses
Phenylephrine	Alpha 1 agonist	
xylometazoline		
Methyldopa		
Tizanidine		
Lofexidine		
Tamsulosin		
Phenoxybenzamine		

Concept 2.9 : Beta Receptor Agonist and Antagonist

Learning objective:

- To know the beta receptor agonist and antagonist
- To know the uses and adverse effects of beta agonists and blockers

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

β_1	Heart	Increased heart rate (+ve chronotropic effect) Increased conduction (+ve dromotropic effect) Increased force of contraction (+ve inotropic effect) Increased excitability (+ve bathmotropic effect) Increased relaxation (+ve lusitropic effect)	Dobutamine
	Juxta-glomerular (JG) cells of kidney	Increased rennin release	
β_2	Smooth muscle	Relaxation E.g: Bronchial smooth muscle – Bronchodilation Uterus – Tocolysis Vascular smooth muscle – Vasodilation and fall in BP	Bronchodilators Salbutamol Terbutaline Salmeterol Formoterol Indacaterol Uterine relaxant Ritordine Isoxuprine
	Skeletal muscles	Tremors	
	Liver	Glycogenolysis	
β_3	Adipocytes	Lipolysis	Anti obesity SIBUTRAMINE (withdrawn because of cardiotoxicity) MIRABEGRON- beta 3 agonist Relax detrusor-overactive bladder



Beta blockers

1 st generation	2 nd generation	3 rd generation
non selective beta blocker	cardioselective beta blockers	beta blockers with additional properties

Cardio selective beta blockers

Drug		Extra property
Nebivolol	(Most cardio selective)	releases NO
Betaxolol	useful in Glaucoma, safe in asthmatic.	
Bisoprolol	useful in CCF	
Atenolol	lipid insoluble	
Esmolol	Ultra short acting(~9Min) due to rapid metabolism by pseudocholinesterase, given IV, only for Emergency purpose	
Acebutolol	Partial agonistic action, membrane stabilizing action	
Metoprolol-	useful in HT. Angina, MI, CCF	
Celiprolol		Nitric oxide releasing property, weak beta 2 agonistic action

3rd generation beta blockers (beta blockers with additional properties)

Drug	Properties
Labetalol	β & α blocker, NERI USE- HT emergency in pregnancy ADR-Postural hypotension, hepatotoxic
Carvedilol	β & α Blocker Anti oxidant anti-inflammatory CCB USE- CCF, HT
Bucindolol Bevantolol Nipradilol	β & α Blocker
Nebivolol Nipradilol, Bopindolol Celiprolol, Carteolol	Release of NO
Tilisolol	Opening Of K ⁺ Channel
Sotalol	K ⁺ Channel Blocker
Carvedilol Betaxolol	Calcium channel blocker



Property	Drugs
Highest membrane stabilizing	propranolol
Highest lipid soluble-	propranolol
Lipid insoluble beta blockers	nadalol, atenolol, sotalol, celiprolol, bisoprolol, acebutolol-lipid insoluble beta blockers undergoes renal route of excretion- unsafe in renal failure
Highest intrinsic sympathomimetic	pindolol
Favourable effect on lipid profile	pindolol
Antidote for beta blocker poisoning	Glucagon

Condition	Recommended drugs
Beta blockers useful in CCF	Metoprolol, bisoprolol, carvedilol, nebivolol
Beta blockers useful in glaucoma	Timolol, betaxolol, levobunolol, carteolol, metipranolol
Pheochromocytoma	Phenoxybenzamine + beta blocker Beta blocker useful to control tachycardia, beta blockers are not first line choice
thyrotoxicosis	Propranolol Controls symptoms, inhibits peripheral conversion of T4 to T3
Hypertrophic obstructive cardiomyopathy	Beta blockers

ONE-LINERS:

- β -blockers with Additional β_2 **Receptor Agonism**-Celiprolol, Bopindolol, Carteolol
- β -blockers with Additional Calcium **Channel Blocking Property** Carvedilol, Betaxolol, Bevantolol.
- Shortest acting β -blocker – **Esmolol**, $T_{1/2}$: **8-10 min**. Duration of action: **15-20 min**
- Longest acting beta blocker- **nadolol** (>40hrs)
- BETA BLOCKERS USEFUL **IN LVF**- carvedilol, bisoprolol, metoprolol
- **Beta blockers useful in glaucoma** – **Timolol, Betaxolol (safe in asthmatic) Carteolol, Levobunolol, Metipranolol**

β -blockers that have Membrane Stabilizing (Local Anesthetic) Property:

Profound	Mild
• Propranolol. • Acebutolol. • Carvedilol.	• Pindolol. • Metoprolol. • Betaxolol. • Labetalol.

Drug interactions with β -blockers:

Interaction	Drug/s
Additive depression of SA node and AV conduction.	Digitalis. Verapamil.
Delayed recovery from hypoglycaemia	Insulin. Oral hypoglycaemic agents.
Marked rise in BP.	Phenylephrine. Ephedrine. Other α agonists.
Decreased antihypertensive action seen with	NSAIDs.
Propranolol metabolism inhibited by	Cimetidine.
Propranolol inhibits metabolism of.	Lignocaine.
Propranolol increases bioavailability of (by decreasing first pass metabolism)	Chlorpromazine

Worksheet

To do

Selective beta 1 agonist	
Very long acting beta 2 agonist	
B3 agonist useful for overactive bladder	
Cardio selective beta blocker having nitric oxide releasing property	
Safer beta blocker for glaucoma in bronchial asthmatic patient	
Beta blocker useful for hypertensive emergency	
Antidote for beta blockers overdose	

CONCEPTS

- ⇒ **Concept 3.1 Drugs useful in Bronchial Asthma**
- ⇒ **Concept 3.2 Drugs useful for cough**

Concept 3.1 : Drugs useful in Bronchial Asthma

Learning objectives:

- To know the classification of drugs useful for cough
- To know the actions uses adverse effects of beta 2 agonists in asthma
- To know the actions uses adverse effects of anticholinergic drugs in asthma
- To know the actions uses and adverse effects of methylxanthine
- To study the role steroids in asthma
- To know the importance of leukotriene antagonist & mast cell stabilizers in asthma
- To know the newer drugs useful in asthma

Time Needed

1 st reading	30 mins
2 nd reading	15 mins

Classification of Anti-Asthmatic

Drugs:

Direct bronchodilators	Anti-inflammatory
β_2 agonists Anti-cholinergics Selective PDE-4 inhibitors	Corticosteroids 5-LOX inhibitors LT receptor antagonists Mast cell stabilizers Monoclonal antibodies
Methylxanthines (both bronchodilator and anti-inflammatory)	

Treatment of acute severe asthma (status asthmaticus):

- Salbutamol / Terbutaline (Inhaled).
- Ipratropium (Inhaled).
- Hydrocortisone hemisuccinate (Intravenous).

Maintenance of Bronchial Asthma:

Gradation	Treatment
Mild intermittent	No daily medication needed (Inhaled β -agonist SOS)
Mild persistent	Low dose inhaled corticosteroid (ICS)
Moderate persistent	Low dose inhaled corticosteroid (ICS) + Inhaled β -agonist
Severe persistent	High dose inhaled corticosteroid (ICS) + Inhaled β -agonist

- Very long acting (acting for >24hrs)
- Indacaterol
- Vilanterol
- abedeterol

(A) β -receptor agonists:

Classification:

Short-acting ($T_{1/2}$: 2-4 hours)	Long-acting ($T_{1/2}$: 12-14 hours)
• Salbutamol (Albuterol) • Terbutaline	• Salmeterol • Formoterol • Bambuterol

Inhalational route of administration is preferred over oral route because of:

- Faster onset of action.
- Greater bioavailability (due to significant reduction in first pass metabolism).

ONE-LINERS:

- Bambuterol is a pro-drug of – Terbutaline.
- Indacaterol – ultra long acting β -agonist for COPD.
- β -agonist available as subcutaneous injection – Terbutaline.
- DOC for acute severe asthma (status asthmaticus) – Salbutamol.
- LABA(long acting beta agonist) effective in acute asthma – Formoterol (due to rapid onset of action).
- Most common adverse reaction to β -agonists – Muscle tremors.

(B) Anticholinergics (Also Known As Inhaled Anticholinergics):

Examples:

- Ipratropium bromide
- Tiotropium bromide
- Oxytropium bromide

Advantages of tiotropium bromide over ipratropium bromide:

	Property	Tiotropium bromide	Ipratropium bromide
1	Duration of action	Longer (24 h)	Shorter (4-6 h)
2	Frequency of administration	Once a day	Four times a day
3	Efficacy	More	Less
4	Antagonism of pre-synaptic M_2 autoreceptor	Less	More

ONE-LINERS:

- % of inhaled dose that is swallowed – **80-90%** (only 10-20% reaches the airways).
- DOC for COPD – **Tiotropium**.
- **Aclidinium** – long acting anticholinergic for COPD.
- Adverse effects due to inhaled anticholinergics are seen in – **20-30%**.

(C) Methylxanthines:

Alkaloid	Structure
Theophylline	1, 3-Dimethylxanthine
Theobromine	3, 7-Dimethylxanthine
Caffeine	1, 3, 7-Trimethylxanthine

Theophylline v/s Caffeine:

	Theophylline	Caffeine
CNS stimulation at low dose	++	+++
CNS toxicity	+++	++
Heart – stimulation	+++	++
Blood vessel relaxation	++	+
Bronchodilation	+++	+
Diuresis	++	+
Increased skeletal muscle contractility	++	+++
Gastric mucosal irritation	++	+
Phosphodiesterase inhibition	+++	++
Adenosine antagonism	+++	++

Mechanisms of Action of Theophylline:

- Phosphodiesterase (PDE) 3, 4 and 5 inhibition (**mainly PDE4**)
- Adenosine receptor antagonism
- Increased interleukin-10 release
- Histone deacetylase activation
- Inhibition of NF-κB
- Induction of apoptosis
- Increase release of calcium

Drug interactions of theophylline:

Metabolism increased by	Metabolism inhibited by
• Barbiturates • Barbecued / charcoal broiled meat • Rifampicin • Phenytoin • Smoking • Marijuana • Alcohol • High-protein, low- carbohydrate diet • Children (1-16 years of age)	• Allopurinol • Cimetidine • Ciprofloxacin • Erythromycin • Oral contraceptives • Fluvoxamine • Zileuton • Zafirlukast • Congestive heart failure Liver failure Pneumonia • Viral infection • Vaccination • High carbohydrate diet • Old age

Increased Dose of Theophylline with:

Phenytoin	1.5 times
Rifampicin	1.5 times
Cigarette smoking	1.6 times

Decreased dose of theophylline with:

Condition	Decrease in dose of theophylline
Allopurinol Cimetidine Ciprofloxacin Erythromycin Oral contraceptives	0.67 times (2/3rds of original dose)
Age >60 years	0.6 times
Congestive heart failure	0.6 times
Pneumonia	0.4 times
Liver failure	0.2 – 0.4 times

ONE-LINERS:

- Sustained release (SR) formulation of theophylline is preferred due to – **uniform absorption**.
- Theophylline is insoluble in water; cannot be given parenterally. Hence, parenteral salts are prepared –
- **Aminophylline** (Theophylline ethylenedia- mine; 85% theophylline).
- **Ethophylline** (Hydroxyethyl theophylline; 80% theophylline).
- V_d of theophylline = **0.5L/kg**.
- Plasma protein binding of theophylline – **50%**.
- Normal plasma levels of theophylline – **5-15 mg/L**.
- Theophylline is mainly metabolised mainly by – **CYP1A2**.

Concomitant administration of	Increase in dose of theophylline
Phenobarbitone	1.2 times
Charcoal boiled meat meal	1.3 times

Inhaled corticosteroids: Examples:

- Beclomethasone dipropionate
- Fluticasone propionate
- Triamcinolone acetonide
- Budesonide
- Ciclesonide
- Flunisolide

Measures to decrease the local tissue deposition of drugs:

- Use of spacer – directs the drug directly into the airways.
- Frequent mouth rinsing.
- Use of **soft steroids** – prodrugs that are activated by esterases in the lung alveoli (e.g.: beclomethasone dipropionate, ciclesonide).

ONE-LINERS:

- DOC for maintenance of BA – **In-haled steroids.**
- **Tipredane** – Inhalational corticosteroid, that was proved ineffective in clinical trials.
- Oral steroids are less preferred than inhalational steroids for maintenance of BA due to – **Increased adverse effects** and **increased requirement for tapering.**

(D) 5-Lipoxygenase inhibitors

E.g.:

- Zileuton
- Inhibits leukotriene synthesis.
- Purely anti-inflammatory action.
- Given orally.
- Only used for maintenance of BA.
- MC adverse effect – Sinusitis.
- Also hepatotoxic.

(E) Leukotriene receptor antagonists

E.g.:

- Montelukast.
- Zafirlukast.
- Pranlukast.

- Inhibit action of leukotrienes on cysteinyl-LT receptors.
- Purely anti-inflammatory action.
- Given orally.
- Only used for maintenance of BA.
- Rare adverse effect – Increased risk of Churg Strauss syndrome (headache, eosinophilia, vasculitis).
- **MEPOLIZUMAB-IL-5 Antagonist – useful for treating churgstrauss syndrome.**

ONE-LINERS:

- DOC for exercise induced asthma – **Inhaled short acting β -agonist(- SABA).**
- DOC for aspirin-induced asthma – **Leukotriene receptor antagonist.**
- T $\frac{1}{2}$ of Montelukast – **2-6 h.**
- T $\frac{1}{2}$ of Zafirlukast – **10 h.**

Mast cell stabilizers:

E.g.:

- Sodium cromoglycate (Cromolyn sodium).
- Nedocromil sodium.
- Ketotifen.

ONE-LINERS:

- Apart from mast cell stabilizer, ketotifen is also a **H₁ antagonist and LT receptor blocker.**
- Antihistamine having mast cell stabilizing property - **OLAPATA- DINE**

Examples of Selective PDE-4 Inhibitors:

Drug	Approved for
Roflumilast Cilomilast Tofamilast	Bronchial asthma
Drotaverine	Anti-spasmodic
Apremilast	Psoriatic arthritis

Examples of Monoclonal Antibodies Approved for Bronchial Asthma:

Drug	Target antigen
Omalizumab	IgE
Lebrikizumab	IL-13
Gomiliximab	CD23
Reslizumab	IL- 5

**ONE-LINERS:**

- Omalizumab is administered by – s.c. route.

Newer Drugs for Bronchial Asthma:

Benralizumab	Anti IL-5 receptor (CD125). Targets the α chain of the receptor.
Fevipirant	Oral selective Prostaglandin D ₂ receptor-2 antagonists.
Setipirant	
Seratrodast	Thromboxane A ₂ antagonist.
Pitrakinra	IL3 and IL4 antagonist.

ONE-LINERS:

- COPD patients are considered eligible for α 1-antitrypsin replacement therapy if the serum α 1-antitrypsin level is below – **50 mg/dL (11 μ M).**
- The only pharmacological therapy that has demonstrated an unequivocal decrease in mortality rates in COPD patients is – **Supplemental oxygen.**

Concept 3.2 : Drugs useful in cough

Learning objectives:

- To know the classification of drugs useful in cough
- To know the drugs useful for different types of cough

Time Needed

1 st reading	15 mins
2 nd reading	8 mins

Pharyngeal demulcents	Lozenges, cough drops and linctuses containing – Syrup Glycerine Liquorice
Expectorants (Mucokinetics)	
Bronchial secretion enhancers	Sodium citrate Potassium citrate Potassium iodide Guaiphenesin (Glycerylguaiacolate) Balsam of Tolu Vasaka Ammonium chloride
Mucolytics	Bromhexine Ambroxol N-acetylcysteine Carbocisteine
Antitussives	
Opioids	Codeine Ethylmorphine Pholcodeine
Non-opioids	Noscapine Dextromethorphan Chlophedianol
Antihistamines	Chlorpheniramine Diphenhydramine Promethazine
Peripherally acting	Prenoxdiazine
Adjuvant antitussives	Salbutamol Terbutaline

Indications of N-acetylcysteine:

- As a mucolytic.
 - Antidote in paracetamol poisoning.
 - Prevention of radiocontrast-induced nephropathy.
 - Prevention and treatment of hemorrhagic cystitis.
- Treatment of interstitial lung disease

ONE-LINERS:

- Bromhexine is a derivative of – alkaloid **Vasicine** obtained from **Adhatodavasica** (Vasaka).
- Prenoxdiazine was developed in – **Hungary**.

Treatment of Specific Types of Cough

Etiology	Treatment approach
Upper / Lower respiratory tract infection	Appropriate antibiotics
Smoking Chronic bronchitis Bronchiectasis	Cessation of smoking Avoidance of pollutants Steam inhalation Postural drainage
Pulmonary tuberculosis	Antitubercular drugs
Asthmatic cough	Inhaled β_2 agonists Ipratropium Inhaled corticosteroids
Cough in pulmonary eosinophilia	Diethyl carbazine citrate (DEC) Inhaled corticosteroids
Postnasal drip due to sinusitis	Antibiotics Nasal decongestants H_1 antihistamines
Postnasal drip due to allergic / perennial rhinitis	Avoidance of precipitating factors Corticosteroid nasal spray H_1 antihistamines
Gastroesophageal reflux	Bed head elevation Light dinner Diet modification H_2 antagonists PPIs Mosapride
ACE inhibitor-associated cough	Substitute ACE inhibitors with losartan NSAIDs may reduce cough
Post-viral cough	No specific treatment, subsides by itself

Recent FDA Approved Drugs:

	Drug	Mechanism	Indication
1	Grastek	Timothy grass pollen allergen extract	Grass-pollen induced allergic rhinitis
2	Oralair	Sweet Vernal, Orchard, Perennial Rye, Timothy and Kentucky Blue grass mixed pollens allergen extract	Grass-pollen induced allergic rhinitis with or without conjunctivitis
3	Ragwitek	Short ragweed pollen allergen extract	Short ragweed pollen-induced allergic rhinitis
4	Reslizumab	Monoclonal antibody targeting IL-5	Maintenance of asthma
5	Olodaterol	Long acting β -receptor agonist	Maintenance of COPD
6	Umeclidinium bromide	Anticholinergic – Blocks M_3 and M_2 receptors in bronchial smooth muscles	COPD
7	Lumacaftor	Prevents misfolding of the abnormal CFTR protein	Cystic fibrosis with $\Delta F508$ mutation
8	Ivacaftor	Potentiates the defective CFTR channel function	Cystic fibrosis with G551D mutation
9	Nintedanib	Inhibits VEGFR, FGFR and PDGFR tyrosine kinases	Idiopathic pulmonary fibrosis
10	Pirfenidone	Reduces fibrosis through anti-inflammatory action	Idiopathic pulmonary fibrosis

Worksheet

To do

Most common side effect of salbutamol	
Adverse effects of theophylline due to blockade of adenosine receptor	
Mast cell stabilizer having antihistamine activity	
Mucolytic useful in treating paracetamol poisoning	
Monoclonal antibodies useful in bronchial asthma	
PDE inhibitors useful in COPD	
Side effect of Chronic use of leukotriene antagonists	

CONCEPTS

- ⇒ **Concept 4.1 Anti histamines**
- ⇒ **Concept 4.2 Serotonin (5HT) modulators**
- ⇒ **Concept 4.3 Prostaglandin analogues**
- ⇒ **Concept 4.4 NSAIDs**
- ⇒ **Concept 4.5 Drugs useful for rheumatoid and gouty arthritis**
- ⇒ **Concept 4.6 Drugs useful in erectile dysfunction**
- ⇒ **Concept 4.7 Drugs useful pulmonary hypertension**

Concept 4.1 : Antihistamines

Learning objectives:

- To know the classification of antihistamines
- To know the action uses of antihistamines
- To know the drugs useful in vertigo

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

H₁ Antihistaminics:

Histamine Receptors:

	H ₁	H ₂	H ₃
Receptor type	Gq-protein coupled	Gs-protein coupled	Auto receptor Gi-protein coupled
Effector pathway	↑ IP ₃ /DAG and release of Ca ²⁺ from intracellular stores; Protein kinase C activation; NO release → ↑cGMP	Adenylyl cyclase activation → ↑cAMP	Restricting Ca ²⁺ influx; K ⁺ channel activation; ↓cAMP
Selective agonists	2-methylhistamine (8:1); 2-pyridylethylamine (30:1); 2-thiazolyl ethylamine (90:1)	Dimaprit (1:2000); Impromidine (1:10,000)	(R) α-methyl histamine (1:3000); Imetit
Selective antagonists	Mepyramine (6000:1); Chlorpheniramine (15,000:1)	Cimetidine (1:500); Ranitidine (1:>500)	Thioperamide (1:23,000); Impromidine; Clobenprofit Ciproxifan Proxifyfan Tiprolisant (or) pitolisant

Location and Actions of Histamine Receptors:

Receptor	Location	Function
H ₁	Smooth muscle	Contraction (e.g.: intestine, airways, uterus)
	Smooth muscles of larger vessels	Vasoconstriction
	Endothelium	Release of NO and PGI ₂ leading to vasodilation
	Afferent nerve endings	Stimulation
	Ganglionic cells	Stimulation
	Adrenal medulla	Release of catecholamines
	Brain	Neurotransmitter

H ₂	Gastric glands	Increased acid secretion
	Smooth muscles of blood vessels	Vasodilation
	Heart	Atria: +ve chronotropic effect; Ventricles: +ve inotropic effect
	Uterus (rat)	Relaxation
	Brain	Neurotransmitter
H ₃	Brain (presynaptic)	Inhibition of histamine release, leading to sedation
	Lung, spleen, skin, gastric mucosa	Decreased histamine release
	Ileum	Inhibition of Ach release from myenteric plexus neurons
	Certain blood vessels	Inhibit NA release, leading to vasodilation

ONE-LINERS:

H₄ receptor was identified in – 2001 on – eosinophils, mast cells and basophils.

Classification of 1st Generation H antihistaminics

Highly sedative	Diphenhydramine Dimenhydrinate Promethazine Hydroxyzine
Moderately sedative	Pheniramine Cyproheptadine Meclizine Buclizine Cinnarizine
Mildly sedative	Chlorpheniramine Dexchlorpheniramine Triprolidine Clemastine Cyclizine

ONE-LINERS:

- **Meclizine** useful in motion sickness.
- **Cyproheptadine**- has anti histaminic. Anti cholinergic and anti serotonin property- useful as an appetite stimulant, prophylaxis of chronic migraine and also for serotonin syndrome.
- **Hydroxyzine** has anti anxiety activity, produces active metabolite called cetirizine.
- **Doxepin** – a tricyclic antidepressant has high antihistamine property – useful in atopic dermatitis, lichen simplex to control pruritis.
- **Cinnarazine**- has anti histaminergic anti cholinergic and anti serotonin activity- useful for vertigo.

Anticholinergic Actions of H₁ antihistaminics:

High	Low	Minimal / Absent
• Promethazine • Diphenhydramine • Dimenhydrinate • Pheniramine	• Chlorpheniramine • Hydroxyzine • Triprolidine • Cyproheptadine	• Fexofenadine • Astemizole • Loratadine • Cetirizine • Mizolastine

Examples of Second Generation H₁ antihistaminics

• Terfenadine • Fexofenadine • Cetirizine • Levocetirizine • Loratadine • Desloratadine • Rupatadine	• Olopatadine • Azelastine • Mizolastine • Ebastine • Acrivastine • Astemizole
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ONE-LINERS:

- Fexofenadine is an active metabolite of – **Terfenadine**.
- Cetirizine is the metabolite of hydroxyzine.
- H₁ antihistaminic with additional PAF antagonistic properties – **Rupatadine**.
- A mast cell stabilizer has high antihistamine activity- ketotifen.
- Terfenadine and Astemizole have been withdrawn due to – **QT_c prolongation**.
- **Pitolisant / Tiprolisant** – Inverse agonist at H₃ receptor; useful for narcolepsy(FDA approved), under trial for **Schizophrenia and Parkinson's disease**.

Topical anti histamines

Azelastine	- Nasal spray
Olopatadine	- Nasal spray, Ophthalmic drop, oral - Mast cell stabilizing action
Alcaftadine	- Ophthalmic drop

Examples of Substances that Release Histamine from Mast Cells / Basophils

- Tissue damage due to trauma.
- Stings and venoms.
- Proteolytic enzymes.
- Phospholipase A.
- Antigen-antibody reaction involving
- IgE antibodies.
- Polymers – Dextran, Polyvinyl pyrrolidone (PVP).
- Basic drugs – d-Tubocurarine, Morphine, Atropine, Pentamidine, Polymyxin-B, Vancomycin, dexferioxamine.
- Some antihistaminics.

Surface acting agents – Tween 80, Compound 48/80.

Labyrinthine suppressants – Inhibit end-organ receptors or central cholinergic pathways in vestibular nuclei

1. Antihistaminics with anticholinergic action	• Cinnarizine • Dimenhydrinate • Diphenhydramine • Promethazine
2. Anticholinergics	• Atropine • Hyoscine
3. Antiemetic phenothiazines	• Prochlorperazine • Thiethylperazine
Vasodilators – Improve blood flow to labyrinth and brainstem	• Betahistine • Codergocrine • Nicotinic acid
Diuretics – Decrease labyrinthine fluid pressure	• Acetazolamide • Thiazides • Furosemide
Anxiolytics and antidepressants – Modify the sensation of vertigo	• Diazepam • Amitriptyline
Corticosteroids – Suppress labyrinthine edema	

ONE-LINERS:

- Most effective drug for violent vertigo and vomiting – **Parenteral prochlorperazine.**
- DOC for refractory pruritus – **Chlorpromazine.**
- DOC for refractory hiccups – **Chlorpromazine.**

Worksheet

To do

Anti histamine with highest anticholinergic action is	
Anti histamine useful in morning sickness	
Cetirizine is the metabolite of	
Anti histamine causing QT prolongation are	
Mast cell stabilizer having antihistamine activity	
Antihistamine having platelet activating factor (PAF) inhibitory action is	

Concept 4.2 : Serotonin (5HT) Modulators

Learning objectives:

- To know the drugs modifying actions of various serotonin receptors
- To the drugs useful in migraine

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Drugs Acting on Serotonin Receptors:

Serotonin (5-HT) Receptors:

Receptor	Location	Function
5-HT _{1A}	Raphe nuclei and hippocampus	Inhibits serotonergic activity
	GIT	Slow depolarization of enteric plexus neurons
5-HT _{1B/1D}	Cranial blood vessels	Vasoconstriction
5-HT _{2A}	Vascular smooth muscles	Vasoconstriction
	Visceral smooth muscles	Contraction
	Platelets	Aggregation
	Neurons (post-junctional)	Activation
5-HT ₃	GIT	Fast depolarization of enteric plexus neurons, Release of 5-HT from enterochromaffin cells, Emesis, ↑ peristalsis
	Heart	Bradycardia, transient hypotension
	Neurons	Apnea, pain, itch
5-HT ₄	GIT	↑ peristalsis, ↑ intestinal secretion
	Lower esophageal sphincter	Contraction
	Enteric plexus	Increased ACh release

ONE-LINERS:

- All 5-HT receptors are G-protein coupled except – **5-HT₃** (ion channel).

Drugs acting on 5-HT receptors:

Receptor	Agonist	Antagonist
5-HT _{1A}	(Partial agonists) • Buspirone • Gepirone • Ipsapirone	
5-HT _{1B/1D}	• Triptans • Ergot alkaloids	
5-HT _{2A/2C}		• Methysergide • Ketanserin • Lidanserin • Cyproheptadine • Atypical antipsychotics
5-HT ₃		• Ondansetron • Granisetron • Tropisetron • Palonosetron • Alosetron
5-HT ₄	• Cisapride • Mosapride • Renzapride • Tegaserod • Prucalopride	

ONE-LINERS:

- Methysergide is not used because of – **Retroperitoneal fibrosis.**
- DOC for chemotherapy-induced vomiting – **Ondansetron.**
- DOC for radiotherapy-induced vomiting – **Ondansetron.**
- 5-HT₃ antagonist with maximum receptor affinity – **Palonosteron.**
- Longest acting 5-HT₃ antagonist – **Palonosetron.**
- Alosteron is approved for – **Diarrhea-predominant IBS.**
- Cisapride has been withdrawn due to – **QTc prolongation.**
- Tegaserod has been withdrawn due to – **Increased risk of cardiovascular events.**
- Prucalopride has been approved for – **Constipation-predominant IBS.**

Pharmacokinetics of Triptans:

Characteristic	Suma	Frova	Riza	Nara	Zolmi
Oral bioavailability (%)	15	25	45	70	40
Tmax (h)	1.5-2	2-4	1-1.5	2-3	1.5-2
Plasma T _{1/2} (h)	~2	26	2-3	6	2-3
Initial oral dose (mg)	50-100	2.5	5-10	2.5	2.5
Maximum oral dose in 24 h (mg)	200	5-7.5	20	5	10

ONE-LINERS:

- DOC for acute migraine – **Sumatriptan**.
- Triptan with highest oral bioavailability – **Naratriptan** (70%).
- Longest acting triptan – **Frovatriptan** (duration of action: 26 h).
- Sumatriptan and ergot alkaloids should not be administered within – **24 hours of each other**.

Classification of Ergot Alkaloids:

NATURAL ERGOT ALKALOIDS	
Amine alkaloids	Ergometrine (Ergonovine)
Amino acid alkaloids	Ergotamine Ergotoxine
SEMISYNTHETIC ERGOT ALKALOIDS	
Dihydroergotamine	
Dihydroergotoxine (Codelergocrine)	
Bromocriptine	
Methysergide	

Drugs for Migraine:

Acute attack	Prophylaxis
• Triptans • Ergot alkaloids (except Methysergide) • NSAIDs	• Propranolol and other non-selective β-blockers • Amitriptyline / Other TCAs • Flunarizine, verapamil- CCB • Valproate • Topiramate • Gabapentin / Pregabalin • Methysergide • Cyproheptadine

Treatment of migraine:

Mild	Simple analgesics / NSAIDs or their combinations ± Antiemetics
Moderate	NSAID combinations / A triptan / An ergot alkaloid + Antiemetics
Severe	A triptan / An ergot alkaloid + Antiemetic + Prophylaxis

ONE-LINERS:

- Most commonly used drug for migraine prophylaxis – **Propranolol**
- Calcitonin gene-related peptide (**CGRP**)- **Olccegepant, Telcagepant—both are hepatotoxic.**
- **Lasmiditan- 5HT_{1F} agonist under clinical trial for migraine.**

MABs useful in migraine

- Erenumab, fremanezumab, galcanizumab

Worksheet

To do

Example for 5HT1A agonists	
Sumatriptan acts on which receptor	
Lorcaserin was withdrawn because of	
Name one female sexual stimulant useful for female with hypoactive sexual desire disorder	
Metoclopramide acts on which receptors	

Concept 4.3 : Prostaglandins

Learning objectives:

- To know the actions of various prostaglandins
- To know the actions uses and adverse effects of prostaglandin analogues

Time Needed

1 st reading	15 mins
2 nd reading	10 mins

	PGE ₂	PGF _{2α}	PGI ₂	TXA ₂
Blood vessels	Vasodilation, ↓ BP	Vasodilation mostly (larger veins constrict), little effect on BP	Vasodilation (marked and widespread), ↓↓ BP	Vaso-constriction, ↑ BP
Heart	Weak inotropic, reflex cardiac stimulation	Weak inotropic	-	-
Platelets	Variable effect	-	Anti-aggregatory	Aggregation and release reaction
Uterus	Contraction (in vivo), relaxes non-gravid human uterus in vitro , softening of cervix	Contraction (in vivo and in vitro), softening of cervix	-	-
Bronchi	Dilation, Inhibits histamine release	Constriction	Dilation (mild), Inhibits histamine release	Constriction
Stomach	↓ acid secretion, ↑ mucus production	-	↓ acid secretion (weak), mucosal vasodilation	-
Intestine	Contracts longitudinal and relaxes circular muscles, ↑ peristalsis, ↑ chloride and water secretion	Spasmogenic, ↑ fluid and electrolyte secretion (weak)	Weak spasmogenic, inhibits toxin-induced fluid secretion	Weak spasmogenic
Kidney	Natriuresis, ↓ chloride reabsorption, Inhibits ADH action, vasodilation, renin release	-	Natriuresis, vasodilation, renin release	Vaso-constriction
CNS	Pyrogenic	-	-	-
Release of NA	↑ or ↓	↑ or ↓	-	-

Afferent nerves	Sensitize to noxious stimuli → Tenderness	-	Sensitize to noxious stimuli → Tenderness	-
Endocrine system	Release of anterior pituitary hormones, steroids, insulin. TSH-like action	Release of gonadotropins and prolactin, luteolysis (in animals)	-	-
Metabolism	Anti-lipolytic, insulin-like action, mobilization of bone calcium	-	-	-

Prostaglandin Analogues:

Prostaglandin	Synthetic preparation / s	Indications
PGE ₁	Misoprostol	NSAID-induced ulcers 1 st trimester medical termination of pregnancy
	Alprostadil	Erectile dysfunction To maintain the patency of ductus arteriosus
PGE ₂	Dinoprostone	Cervical ripening
PGF _{2α}	Dinoprost	Induction of labour
	Carboprost tromethamine	Post-partum haemorrhage
	Latanoprost Isopropyl unoprostone Travoprost Bimatoprost Tafluprost	Glaucoma
PGI ₂	Epoprostenol Treprostinil Iloprost	Pulmonary arterial hypertension

ONE-LINERS:

- DOC for NSAID-induced ulcer – **PPIs**.
- Specific drug for NSAID-induced ulcer – **Misoprostol**.
- Misoprostal - teratogen - causing moebius syndrome.

Concept 4.4 : NSAIDs

Learning objectives:

- To know the classification of NSAIDs
- To know the clinical signs and symptoms of aspirin and paracetamol poisoning

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

A: Non-Selective Cox Inhibitors	
Salicylates	• Aspirin
Propionic acid derivatives	• Ibuprofen • Ketoprofen • Flurbiprofen • Naproxen
Fenamate (Anthranilic acid derivatives)	• Mefenamic acid
Enolic acid derivatives	• Piroxicam • Tenoxicam
Acetic acid derivatives	• Ketorolac • Indomethacin • Nabumetone
Pyrazolone derivatives	• Phenylbutazone • Oxyphenbutazone
B: Preferential Cox-2 Inhibitors	
• Nimesulide • Meloxicam • Diclofenac • Aceclofenac • Etodolac	
C: Selective Cox-2 Inhibitors	
• Celecoxib • Etoricoxib • Parecoxib • Lumiracoxib	
D: Analgesic-Antipyretic with Poor Anti-Inflammatory Action	
• Paracetamol • Metamizol • Propiphenazone • Nefopam	

Selective COX2 inhibitors causing side effects of thrombosis, myocardial infarction, heart failure and hypertension

Effects of Aspirin on Uric Acid

Dose	Effect
< 2g/d	Urate retention and antagonism of all other uricosuric agents
2-5 g/d	Variable effects, often no change
> 5g/d	Increased urate excretion

Acute Salicylate Poisoning:

- More common in children
- Fatal dose: **15-30 g** for adults, lower for children
- Serious toxicity seen at plasma levels > **50 mg/dL**

Signs and Symptoms:

- | | |
|---|--|
| <ul style="list-style-type: none"> • Vomiting • Dehydration • Electrolyte imbalances • Acidotic breathing • Hyper / hypo glycemia • Petechial hemorrhages • Restlessness | <ul style="list-style-type: none"> • Delirium • Hallucinations • Hyperpyrexia • Convulsions • Coma • Death due to respiratory failure • + Cardiovascular collapse |
|---|--|

Treatment

- Supportive and symptomatic.
- External cooling.
- IV fluid and electrolyte replacement.
- Glucose.
- Gastric lavage to remove the unabsorbed drug.
- Alkaline diuresis or hemodialysis to remove the absorbed drug.
- Blood transfusion and Vit. K if bleeding occurs.

Acute paracetamol poisoning:

- More common in small children.
- Toxicity seen at doses >150 mg/kg or > 10g in adults.
- Fatality seen at doses >250 mg/kg.
- Toxic metabolite is N acetyl benzoquinoneimine (hepato toxic).

Signs and Symptoms:

Early manifestations	• Nausea • Vomiting • Abdominal pain • Liver tenderness • No impairment of consciousness
After 12-18 hours	• Centrilobular hepatic necrosis • Renal tubular necrosis • Hypoglycemia • May progress to coma
After 2 days	• Jaundice

Fulminant hepatic failure and death due to paracetamol are more likely if plasma levels are above the line joining – **200 µg/mL at 4 h and 30 µg/mL at 15 h.**

Treatment

- Induction of vomiting.
- Gastric lavage.
- Activated charcoal.
- Antidote: **N-acetylcysteine**: 150 mg/kg IV infusion over 15 min f/b same dose over next 20 h OR 75 mg/kg oral every 4-6 h for 2-3 days .
- Alternative to N acetylcysteine I is methionine

ONE-LINERS:

- Reye's syndrome is most commonly associated with – **Aspirin**.
- Safest conventional NSAID / Conventional NSAID with least side effects – **Ibuprofen**.
- Most selective COX-2 inhibitor – **Lumiracoxib**.
- NSAID that inhibits COX-3 – **Paracetamol(Acetaminophen)**.
- The enzyme COX-3 has been so far identified and localized in – **Dog brain**.
- According to a recent FDA recommendation, the dosage of paracetamol in a single tablet should not exceed – **325 mg**.
- NSAID undergoing enterohepatic circulation – **Piroxicam**.
- NSAID with longest half-life – **Piroxicam** (~ 75 h).
- Single dose NSAID / NSAID that can be administered as OD dosing – **Piroxicam**.
- Phenylbutazone has been withdrawn due to – **Agranulocytosis**.
- Rofecoxib has been withdrawn due to increased risk of – **myocardial infarction**.
- Phenacetin has been withdrawn due to – **Analgesic nephropathy**.
- Nimesulide is contraindicated below – **12 years of age**.
- NSAIDs **decrease** the therapeutic effect of antihypertensives.
- Dual COX/LOX inhibitors – **Licofelone, Tepoxalin**.
- **Idroclamide** – Skeletal muscle relaxant and anti-inflammatory medication used as a topical cream to treat lumbago and other kinds of muscular pain.

Concept 4.5 : Drugs useful for Gout and Rheumatoid Arthritis

Learning objective:

- To know the drugs useful in treating acute and chronic gout
- To know the drugs useful for rhenumatoid arthritis

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Treatment of Gout:

Drugs for gout:

ACUTE GOUT	• NSAIDs . • Colchicine (disturbs microtubules, safe in renal failure, ADR- DIARRHOEA). • Steroids.
MAINTENANCE OF GOUT	
Xanthine oxidase inhibitors (Inhibit the synthesis of uric acid)	• Allopurinol –(ADR- allergy, SJS) • Febuxostat
Uricosuric drugs (increase urinary excretion of uric acid)	• Probenecid • Sulfinpyrazone • Benzbroomine • Lesinurod
Increase metabolism of uric acid to allantoin For rapid control of uric acid in case of tumor lysis syndrome	• Rasburicase (Recombinant uric acid oxidase) • Pegloticase (pegylated form of uric acid oxidase)
IL 1 blockers	• Anakinra, canakinumab, rilonacept

ONE-LINERS:

- DOC for acute gout – **NSAIDs**.
- NSAID not used in acute gout – **Aspirin** (due to its uric acid retention effects).
- Colchicine is derived from – *Colchicum autumnale*.
- DOC for Familial Mediterranean fever – **Colchicine**.
- Allopurinol is an analogue of – **Hypoxanthine**.
- Non-purine inhibitor of xanthine oxidase – **Febuxostat**.
- Most common side effect of probenecid is – **Dyspepsia** (25%).

Drug Interactions of Probenecid:

Decrease urinary excretion of	Decreases the biliary excretion of
• Penicillins • Cephalosporins • Sulfonamides • Methotrexate • Indomethacin • Nitrofurantoin	• Rifampicin

Standard Preventive Approach for Tumourlysis Syndrome

• Vigorous hydration. • Urinary alkalization. • Oral / IV allopurinol • Rasburicase (DOC).
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Drugs Causing Hyperuricemia as a Side Effect

• Thiazides. • Loop diuretics. • Pyrazinamide. • Ethambutol. • Levodopa.
--

Drugs useful for treating rheumatoid arthritis

conventional synthetic (cs) csDMARDs	biologic (b) bDMARDs
Methotrexate Azathioprine, Chloroquine Hydroxychloroquine Cyclophosphamide Cyclosporine Leflunomide (inhibits dihydro orotate dehydrogenase) Mycophenolate mofetil Sulfasalazine Tofacitinib (janus kinase 1, 3 blocker)	TNF- α -blocking agent IL-1 blockers (anakinra, rilonacept, canakinumab) IL-6 blockers (Tocilizumab) B-cell cytotoxic agent (Rituximab) T-cell-modulating biologic (Abatacept)

TNF- α blockers

- Infliximab
- Adalimumab
- Eterncept
- Golimumab
- certolizumab

TNF- α blockers side effects-

- Anti drug antibody formation, allergic reaction
- Activation of latent infection, hepatotoxicity, heart failure, secondary cancer

Contraindications to anti-TNF- α agents:

- Hepatitis B.
- HIV and other immunocompromised states.
- Active tuberculosis.
- Multiple sclerosis.
- Systemic lupus erythematosus.
- Pregnancy.
- Lactation.

Worksheet

To do

NSAIDs useful in acute gout	
Mechanism of action of colchicine	
Examples for uric acid metabolism enhancers	
Target of rituximab	
Mechanism of action of leflunomide	
IL 6 blocker useful in RA	

Concept 4.6 : Drugs useful in Erectile Dysfunction

Learning objectives: To know the drugs useful in erectile dysfunction

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

PDE V blockers	Local (intracavernous) injection into penis	Other drug useful in erectile dysfunction
- Sildenafil - Vardenafil - Tadalafil (longest acting) - Avanafil	- Alprostadil - Phentolamine - Papaverine	- Apomorphine - Trazadone - Avaptadil- VIP - Ketanserin - Naltrexone - Gingeng, Kava - Ginkgo

Sexual problem and treatment

Premature ejaculation	Delayed orgasm	Sexual stimulants
- SSRI - PDEV inhibitors	- Amantidine - Buspirone - Cyproheptadine	- Yohimbine - Zinc - Ginkgo biloba - ginseng

Worksheet

To do

Most common group of drug useful for erectile dysfunction	
Sildenafil should not be given with nitrate because of risk of	
Depoxetine is the SSRI approved for	
Alpha 2 agonist used as a sexual stimulant is	
Prostaglandin E1 analogue useful in erectile dysfunction is	

Concet 4.7 : Drugs useful in Primary Pulmonary Hypertension

Learning objectives: To know the various classes of drugs useful in pulmonary hypertension

Time Needed

1 st reading	15 mins
2 nd reading	7 mins

CCB	Nitric oxide	PDE V inhibitors	Endotheline receptor blockers	PGI 2 analogues	Rho kinase enzyme inhibitor	Direct guanyl cyclase activators	Newer drugs
Nifedipine	Inhalation nitric oxide	Sildenafil Tadalafil	Bosentan Ambresentan Macitentan	Epoprostenol- (IV- infusion) ADR- hypotension, headache, myalgia, jaw pain Treprostinil (IV/SC- infusion, inhalation, oral) Beroprost- oral (first PGI2 analogue), not approved Ilioprost- inhalation (first PGI2 analogue)	Fasudil	Riociguat, Cinociguat	Trebananib Aniopoietin 1 inhibitor Extra cellular elastase inhibitors- sivelestat, elafin

Worksheet

To do

Examples of direct guanyl cyclase activator useful in pulmonary hypertension	
What is fasudil	
Examples for endothelin receptor blockers useful in pulmonary hypertension	
Acute side effects of sildenafil	
What is Treprostinil	
Example for extra cellular elastase inhibitor	

CONCEPTS

- ⇒ **Concept 5.1 Anti anginal agents**
- ⇒ **Concept 5.2 Anti arrhythmic agents**
- ⇒ **Concept 5.3 Drugs useful for heart failure**
- ⇒ **Concept 5.4 Anti hypertensive agents**

Concept 5.1 : Anti Anginal Agents

Learning objectives:

- To know the classification of anti anginal agents
- To know the mechanism , uses and adverse effects of nitrates
- To know the role of beta blocker in angina and in myocardial infarction
- To know the uses and adverse effects of CCB
- To know the newer drugs useful in angina

Time Needed

1 st reading	45 mins
2 nd reading	20 mins

Classification

Vaso dilators	Cardia depressants	Vasodilatation & cardiac suppression	Pathway of fatty acid oxidation (pFOX)inhibitors (antioxidants)	Newer drug
Nitrates Nicorandil (potassium channel opener)	Beta blockers	Calcium channel blocker (diltiazem)	Trimetazidine Ranolazine	Ivabradine Molsidomine

Nitrates

Drugs	MOA	Uses	ADR	Drug interaction
Short acting- Glyceryl trinitrate (Nitroglycerine) Amyl nitrite Intermediate acting- Isosorbide dinitrate Long acting- Isosorbide mononitrate Longest acting- Pentaerythritol tetranitrate	Releases nitric oxide- activate guanyl cyclase- accumulates cGMP- causing smooth muscle relaxation	Cardiac- Angina, MI, CCF Non cardiac Achalasia cardia Biliary colic pain Cyanide poisoning	Headache Hypotension with reflex tachycardia Development of tolerance (Monday morning headache) Methemoglobinemia Skin rashes	With sildenafil – produces severe hypotension Beta blockers when added with nitrate controls the reflex tachycardia

ONE-LINERS:

- All nitrates are long-acting except – **Nitroglycerine.**
- Shortest-acting nitrate – **Nitroglycerine amyl nitrite.**
- Longest-acting nitrate – **Pentaerythritol tetranitrate.**
- All nitrates undergo extensive first-pass metabolism except – **Isosorbide mononitrate.**
- DOC for angina pectoris – **Nitrates.**
- Nitrate preferred in acute angina - **NTG, ISDM.**
- Sublingual for rapid action should be in lipid soluble and in non ionized form.
- Skin rashes are maximum with penta erythritol tetra nitrite.
- Nitrates should not be combined with – **Sildenafil.**
- Most common side effect of nitrate – **headache.**
- Nitrate causes hypotension with reflex tachycardia.
- Methemoglobinemia produced by nitrate useful in treating cyanide poisoning.
- Non cardiac use of nitrate - **biliary colic pain, achalasia cardia.**

Role of beta blockers

In angina	In MI
Reduces heart rate & force of contraction- Reducing work load- Reduces oxygen demand	Reduces work load- reduces oxygen demand Has anti arrhythmic action- so controls arrhythmia in MI, thereby prevents mortality

Classification of Calcium Channels:

	L-type (Long-lasting current)	T-type (Transient current)	N-type (Neuronal)
Conductance	25 pS	8 pS	12-20 pS
Activation threshold	High	Low	Medium
Inactivation rate	Slow	Fast	Medium
Location and function	Cardiac and smooth muscles – Excitation-contraction coupling; SA node and AV node – Conductivity; Endocrine cells – Hormone release; Neuro – transmitter release	SA node – Pacemaker activity; Thalamic and other neurons – T currents and repetitive spikes; Endocrine cells – Hormone release; Certain arteries – Constriction	Neurons in CNS, sympathetic and myenteric plexuses – Transmitter release
Antagonists	Verapamil Diltiazem Dihydropyridines	Mibefradil Flunarizine Ethosuximide Valproate Topiramate Gabapentin Pregabalin	ω -conotoxin

**ONE-LINERS:**

- CCB with –ve chronotropic action: **Verapamil > Diltiazem.**
- CCB with –ve dromotropic action: **Verapamil > Diltiazem.**
- CCB with –ve inotropic action: **Verapamil > Diltiazem.**

Pharmacokinetics of CCBs:

Drug	Oral bio-availability (%)	Vd (L/kg)	CL (L/h/kg)	Active metabolite	Elimination T _{1/2} (h)
Verapamil	15-30	5.0	0.9	Yes	4-6
Diltiazem	40-60	3.0	0.7	Yes	5-6
Nifedipine	30-60	0.8	0.42	Minor	2-5
Felodipine	15-25	10.0	1.0	None	12-18
Amlodipine	60-65	21.0	0.42	None	35-45

ONE-LINERS:

- Most popular DHP – **Amlodipine.**
- DHP supposed to be useful in CHF – **Amlodipine.**
- CCB used to treat cerebral vasospasm following sub arachnoid hemorrhage(SAH) – **Nimodipine** (due to excellent blood brain barrier penetration).
- FASUDIL(Rho kinase inhibitor is also useful in SAH.
- Long-acting DHPs – **Benidipine and Lercanidipine** (due to slow dissociation from the channels).
- CCB causing paradoxical angina – **Nifedipine.**
- CCBs should never be combined with – **Clarithromycin.**
- Clarithromycin inhibits CYP enzymes, leading to elevated CCB levels, leading to an **increased risk of hospitalization due to acute kidney injury.**
- **CCB useful in hypertensive emergency- Intravenous clevidipine, nicardipine.**
- **CCB useful for prophylaxis of chronic migraine- Flunarizine** (additionally has sodium channel blocking action and anti oxidant action), **verapamil.**
- **Verapamil is P-glycoprotein inhibitor.**
- **Class IV anti arrhythmic - verapamil, diltiazem.**

Potassium channel openers

Drug	Uses	ADR
Nicorandil (K channel opener + releases nitric oxide)	Angina	Hypotension, headache, aphthous ulcer
Hydralazine (K channel opener + releases nitric oxide)	Hypertension (preferred in pregnancy)	Hypotension, fluid retention, SLE
Minoxidil	Hypertension, alopecia	Hypotension, fluid retention, hirsutism
Diazoxide	Insulinoma, hypertension	Hyperglycemia, hyperuricemia
Adenosine	Supra ventricular arrhythmia	Bronchospasm

Anti oxidant

Drug name	ADR
Ranolazine (anti oxidant + sodium channel block)	QT prolongation
Trimetazidine	GI upset Thrombocytopenia Liver dysfunction Risk of movement disorders

ONE-LINERS:

- Anti-anginal with cardioprotective effects – **Ranolazine**.
- Anti-anginal **causing HbA1c reduction** – **Ranolazine**

Newer drugs

Drug	MOA	Action	Uses	ADR
Ivabradine	Funny current sodium channel blocker	Causing bradycardia	Angina, CCF	Bradycardia Luminous phenomena-PHOSPHENES (visual disturbance)
Molsidomine	Releases nitric oxide	Vasodilatation	Prophylaxis of angina	Headache

For stable angina

First line drugs	Second line drugs
Beta blockers CCB- Diltiazem Nitrates	pFOX inhibitors Trimetazidine , Ranolazine If Na channel blocker Ivabradine K channel opener- Nicorandil

Drugs for Myocardial Infarction

Pre-admission:

- Anti-platelet agent: ASPIRIN to arrest the progression of clot (**First drug to be given**).
- Opioid analgesic: **MORPHINE for pain relief** (Pethidine, Pentazocin Should be used as They Cause Tachycardia).
- 100% oxygen inhalation.

Prophylactic β -blocker to prevent arrhythmias

On admission:

- Thrombolysis (PTCA or Pharmacological)
- Nitrates
- ACEi / ARB
- β -blocker

Anticoagulant Heparin / LMWH followed by Warfarin

On discharge:

- Antiplatelet agents
- Statins

β -blocker / ACEi / ARB to prevent remodeling

Drugs useful in peripheral vascular disease

Drug	MOA	ADR
Pentoxifylline	non selective PDE inhibitor rheologic modifier (Flexibility of RBC)	Severe nausea & vomiting
Cilastazole	Selective PDE III inhibitor	Headache, arrhythmia,

Worksheet

To do

Nitrates acting by	
Nitrates should not used along with	
CCB safe in CCF	
CCB useful in SAH	
CCB useful for prophylaxis of chronic migraine	
CCB having anti arrhythmic action	
Potassium channel opener useful in angina	
Mechanism of action of ivabradine	
pFOX inhibitor useful in angina	
Mechanism of action of cilastazole	

Concept 5.2 : Anti-Arrhythmic Agents

Learning objectives:

- To know the classification of anti-arrhythmic agents
- To know the uses adverse effects of various classes of anti-arrhythmic agents
- To know the appropriate antiarrhythmic agents for appropriate arrhythmias

Time Needed

1 st reading	45 mins
2 nd reading	25 mins

Classification of anti-arrhythmic agents

Class I (Na ⁺ channel blockers)	
Class IA	• Quinidine • Procainamide • Disopyramide
Class IB	• Lignocaine • Phenytoin • Mexilitene • Tocainide
Class IC	• Flecainide • Encainide • Propafenone
Class II (β -blockers)	• Propranolol • Esmolol
Class III	• Amiodarone • Dronedarone • Bretylium • Sotalol • Ibutilide • Dofetilide • Vernakalant
Class IV (Calcium channel blockers)	• Verapamil • Diltiazem
Miscellaneous	• Digoxin • Adenosine • Magensium • Atropine

Class Ia drugs

- Quinidine may **cause thrombocytopenia** due to hypersensitivity reaction on bone marrow.
- The most common adverse reaction due to quinidine is – **Diarrhoea** (30-50%).
- Procainamide undergoes **acetylation** may cause SLE.
- Dispyramide has **highest antivagal action**

Class Ib drugs

- All **class Ib drugs useful only for** ventricular arrhythmia.
- Lignocaine causes **convulsion**(first sign of convulsion: nystagmus, first symptom of convulsion: circum oral paraesthesia).
- Mexilitine **off label use- diabetic neuropathy**.
- Phenytoin useful for treating digitalis induced VT (**drug of choice is lignocaine**).
- Tocainide causes **AGRANULOCYTOSIS**.

Class Ic drugs

- **Class Ic drugs are pro arrhythmic, but useful for WPW syndrome**
- **Propafenone** has additionally beta blocking property

Class II { beta blockers}- useful for both ventricular and atrial arrhythmias

Class III (potassium channel blockers)

AMIODARONE:

1. Whorl like pattern cornea (cornea verticillata)
2. Pulmonary fibrosis
3. Pseudo alcoholic liver injury
4. Hypo or hyper thyroidism

BRETYLIUM:

Chemical defibrillator

Drugs converting AF into NSR (given IV)

1. Amiodarone
 2. Ibutilide
 3. Vernakalant
- Dronedarone do not contain IODINE (no thyroid problem).
 - Bretylium – chemical defibrillator.
 - Sotalol- beta blocker with potassium channel blocking action.
 - Ibutilide useful for converting atrial fibrillation into normal sinus rhythm.

- Percentage of iodine in amiodarone – **33-37%**.
- Photosensitization and skin pigmentation seen with amiodarone.
- Pulmonary alveolitis and fibrosis is rare if daily dose of amiodarone is less than – **200 mg**.
- pseudo alcoholic liver injury, hypo or hyper thyroidism can be the adverse effect of amiodarone.

Class IV

Drugs	Uses
Verapamil & diltiazem	Mainly useful for atrial and supra ventricular arrhythmia

Miscellaneous drugs

- **ADENOSINE**- drug of choice for SVT, Suseful to produce Controlledhypotension, is also called Endogenousantiepileptic, its action antagonized by Methyl xanthane- theophylline, action potentiated by Dipyridamole.
- **IV. Magnesium sulphate (MgSO₄)**- drug of choice for long QT syndrome, Eclampsia.
- **Most common adverse effect of Mgso₄ is – Diminishing deep tendon reflex, most serious isRespiratory failure, antidote for MgSo₄ is - caclium gluconate.**
- Atropine- anticholinergic drugs, causing tachycardia, useful for symptomatic bradycardia, first degree heart block, Mobitz type I second degree heart block
- Digoxin- has vagomimetic property, acts on AV node, reducing conduction velocity, useful for atrial fibrillation and atrial flutter

ONE-LINERS:

- Classification of antiarrhythmics is given by – **Vaughan Williams and Singh**.
- DOC for congenital QT syndrome – β -blocker.
- DOC for acquired QT syndrome – **MgSO₄**.

Drugs that can Prolong QTC Interval

Anti-arrhythmics	• Class IA agents • Class IC agents • Class III agents
Antimalarials	• Quinine • Mefloquine • Artemisinin • Halofantrine
Antibacterials	• Sparfloxacin • Moxifloxacin • Gatifloxacin

Antihistaminics	• Terfenadine • Astemizole • Ebastine
Antidepressants	• TCAs
Antipsychotics	• Thioridazine • Pimozide • Aripiprazole • Ziprasidone

Clinical Classification of Antiarrhythmics:

Supraventricular arrhythmias only	Ventricular arrhythmias only	Supraventricular + Ventricular arrhythmias
Adenosine Verapamil Diltiazem Dronedarone Digoxin	Class IB agents	Amiodarone β -blockers Class IA agents Class IC agents

ONE-LINERS:

- DOC for supraventricular arrhythmias – **Verapamil**.
- DOC for PSVT – **Adenosine**.
- DOC for ventricular arrhythmias – **Lignocaine**.
- Management of choice for ventricular arrhythmias – **DC cardioversion / Defibrillation**.

Worksheet

To do

Class I drug with highest ant ivagal action	
Anti arrhythmic useful only for ventricular arrhythmia	
Antiarrhythmic causing pulmonary fibrosis	
Antiarrhythmic drug causing hypothyroidism	
CCB having antiarrhythmic property	
Anti arrhythmic drug useful for treating first degree heart block	
Antidote for MGSO ₄	
Antidote for digoxin	\

Concept 5.3 : Drug Useful For Heart Failure

Learning objectives:

- To know the classification of drugs useful in acute and chronic heart failure
- To know the action uses and adverse effects of drugs interfering RAAS system
- To know the natriuretic peptides and their analogues
- To know the role of inotropic agents in heart failure
- To know the newer drugs for heart failure

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Drugs useful for acute heart failure (only for symptom relief)	Drugs useful for controlling progression of heart failure (disease modifying drugs)- RAAS pathway inhibitors
Diuretics(furosemide) and natriuretics Vasodilators Inotropes (digoxin, dopamine, dobutamine, PDEIII inhibitors)	β blockers, ACEI, ARB & Mineralo corticoid receptor antagonist

Drugs Acting on Renin-angiotensin System:

Group	Mechanism	Examples
Renin release inhibitors	Inhibiting the release of renin from Gella of kidney	Beta blockers Clonidine, methyl dopa
Direct renin inhibitors	Inhibit conversion of angiotensinogen to AT-I	• Aliskiren • Remikiren • Enakiren
Angiotensin converting enzyme (ACE) inhibitors	Inhibit conversion of AT-I to AT-II	• Captopril • Enalapril • Lisinopril • Perindopril • Fosinopril • Ramipril • Quinapril • Trandolapril • Imidapril • Benazepril

Angiotensin receptor blockers (ARBs)	Prevent action of AT-II on the receptors	• Losartan • Candesartan • Irbesartan • Valsartan • Olmesartan • Telmisartan
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ONE-LINERS:

- Angiotensinogen is an – $\alpha 2$ globulin secreted by **Liver**.
- Angiotensin-I contains – **10 amino acids**.
- Angiotensin-II contains – **8 amino acids**.
- Angiotensin-III contains – **7 amino acids**.
- Angiotensin-IV contains – **6 amino acids**.
- Aliskiren has been approved for – **Hypertension**.

Structure and Pharmacokinetics of Ace Inhibitors:

Drug	Chemical nature	Time to peak action (h)	Elimination half-life (h)	Duration of action (h)	Dose (mg/d)
Captopril	Sulphydryl	1	2	6-12	25-150
Enalapril	Carboxyl	4-6	11	24	2.5-40
Lisinopril	Carboxyl	6-8	12	≥ 24	5-40
Fosinopril	Phosphinate	3-5	12	24	10-40
Perindopril	Carboxyl	6	25-30	> 24	2-8
Ramipril	Carboxyl	3-6	8-48	> 24	1.25-10

ONE-LINERS:

- All ACE inhibitors are **prodrugs except – Captopril and Lisinopril**.
- DOC for diabetic nephropathy – **ACE inhibitors**.
- DOC for scleroderma renal crisis – **ACE inhibitors**.
- All ACEI undergoes mainly renal route of excretion except- fosinopril (mainly bile, minimally through kidney).
- Bradykinin is the cause for dry cough and angioedema due to ACE INHIBITOR.
- BRADYKININ ANTAGONIST - **Icatibant**.
- ARB with maximum affinity for receptor – **Candesartan**.
- ARB with **anti-platelet** action and uricosuric action – **Losartan**.
- ARB with insulin sensitizing action – **Telmisartan** (due to PPAR- γ agonism).

Natriuretics:

Example	Mechanism	Current status
Nesiritide	Recombinant BNP analogue	Approved for acute decompensated heart failure Given IV, short acting(20minutes) Rapid metabolism by vasopeptidase
Carperitide	Recombinant ANP analogue	Not used much nowadays
Omapatrilat sampatrilat	Dual neutral endopeptidase (NEP) and ACE inhibitor	Withdrawn due to high incidence of angioedema
Sacubitril Ecadotril	Neutral endopeptidase (NEP) inhibitor	Recently approved for CHF

ONE-LINERS:

- Nesiritide is administered as an – **IV infusion** (due to short T_{1/2})

BETA BLOCKERS USEFUL **IN LVF**- carvedilol, bisoprolol, metoprolol

Mineralocorticoid antagonist useful in LVF- spironolactone, eplerenone

Inotropes:**ONE-LINERS:**

Digoxin is derived from – Digitalis lanata. Digitoxin is derived from – Digitalis purpurea. Ouabain is derived from – Strophanthus gratus. Cardiac glycosides were first used for dropsy by –

William Withering.

Cardiac glycoside found in toad skin – **Bufotoxin.**

Digoxin

MOA	Uses	Non Cardiac Adr	Cardiac Adr	Treatment for Digoxin Induced Arrhythmia	Antidote for Digoxin
Inhibits Na ⁺ K ⁺ ATPase pump, thereby increases intracellular Calcium, increase the force of contraction	CCF atrial fibrillation and atrial flutter.	Nausea, Vomiting (most common) DNS depression Yellow vision defect (Xanthopsia) Gynecomastia	Atrial arrhythmia AV block Ventricular arrhythmia	Correction of electrolyte imbalance (K, Mg) Propranolol for atrial arrhythmia Atropine for AV block Lignocaine for ventricular arrhythmia	Digibind (anti-digoxin Fab fragments)
Vagomimetic property- reduces AV conduction- controls heart rate					

Digoxin

Effects of digoxin on cardiac action potential	Effects of digoxin on ECG	Factor aggravating digoxin toxicity
- Resting membrane potential is progressively decreased . - Rate of phase 0 depolarization is reduced resulting in slowing of conduction. - Slope of phase 4 depolarization is increased in the Purkinje fibres. Ectopic automaticity is enhanced and latent pacemakers become overt at high doses producing extrasystoles. - Action potential duration is reduced and amplitude of the action potential is diminished	- Decreased amplitude or inversion of the T wave . - Increased PR interval. - Shortening of QT interval. - Depression of ST segment. - Inverted tick mark sign.	- Hypokalemia - Hypercalcemia - Hypomagnesemia - Severe renal disease - Severe hepatic disease - Myocardial ischemia - Thyrototoxicosis - Myxedema - Ventricular tachycardia - Partial AV block - Acute myocarditis - Wolff-Parkinson White (WPW) syndrome

Pharmacokinetic Properties of Digoxin

Oral bioavailability	~ 75%
Plasma protein binding	20 – 30%
Therapeutic serum levels	0.5 – 2 ng/mL
T _{1/2}	~ 36 h

ONE-LINERS:

- Digoxin undergoes renal route of excretion (glomerular filtration), digitoxin undergoes hepatic route of excretion**
- Most characteristic feature of digitalis-induced arrhythmias – **Atrial tachycardia with AV block.**
- Most common arrhythmia produced by digoxin- ventricular bigemini.**
- DC cardioversion is contraindicated in – Patients with elevated digitalis levels** (It may precipitate ventricular fibrillation).
- No role for hemodialysis in digoxin poisoning (due large volume of distribution).

Effects of Dopamine at Various Doses

Dose ($\mu\text{g/kg/ min}$)	Receptor	Primary action
Low (< 2)	Δ_1	Renal and splanchnic vasodilation
Intermediate (2-10)	β_1	Inotropic effect
High (> 10)	α_1	Vasoconstriction

Examples of Calcium Sensitizers

- Levosimendan (also has PDE iii blocking property). Pimobendan.

Examples of PDE3 Inhibitors

- Inamirinone – most common side effect is Thrombocytopenia.
- Milrinone – most common side effect is arrhythmia.
- Vesnarinone.
- Enoximone

Newer drugs:

ISTAROXIME- inhibits Na + K +ATPase pump- inotropic action.

Omecamtiv mecarbil (Direct myosin activator)-positive inotropic action.

Reduce Mortality in CCF:

Beta Blockers (carvedilol, bisoprolol, metoprolol).

A.C.E. Inhibitors.

Angiotensin Receptor Blockers. Spironolactone.

ISDN + HYDRALAZINE.

Sacubitril + valsartan

Worksheet

To do

Diuretic of choice for rapid symptom relief in CCF	
Disease modifying diuretic in CCF	
All ACEIs are prodrug except	
ARB having PPAR gamma agonistic action is	
BNP analogue useful for CCF	
Selective neprilysin inhibitors	
Calcium sensitizer examples	
Uses of ivabradine	

Concept 5.4 : Anti-Hypertensive Drugs

Learning objectives:

- To know the classification of antihypertensive drugs
- To know selection of appropriate antihypertensive drug for appropriate person

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Drugs for Hypertension:

1	Diuretics	
2	Drugs inhibiting RAS	• ACE inhibitors • ARBs • Aliskiren (Direct renin inhibitor / DRI)
3	Peripheral sympatholytics	• α-blockers • β-blockers
4	Central sympatholytics	• Clonidine • α-methyldopa • Rilmenidine • Moxonidine
5	Calcium channel blockers	
6	Potassium channel openers	hydralazine, minoxidil, diazoxide
7	Nitrates	
8	Other vasodilators	• Sodium nitroprusside • Hydralazine • Fenoldopam • Fasudil

ONE-LINERS:

- **Coombs' test** is positive upon administration of – α -methyldopa.
- **Cyanide poisoning** can occur due to overdose of – **Sodium nitroprusside- unsafe in pregnancy.**
- ACE inhibitor used in hypertensive emergencies – **Enalaprilat** (IV route).
- Fenoldopam acts as a – **D1 receptor partial agonist.**
- **Fasudil** acts as a – **Rho kinase inhibitor- useful for primary pulmonary hypertension, subarachnoid hemorrhage, angina pectoris.**

Selection of 1st Line Antihypertensive Agents:

Compelling indications	Suitable for	To be avoided in
Diuretics		
- Heart failure - High CAD risk - Recurrent stroke prevention - Diabetes mellitus	- Older patients - Isolated systolic hypertension - Obese with volume overload - Low cost therapy	- Gout or family history of gout - Abnormal lipid profile - Pregnancy induced hypertension
ACE inhibitors / ARBs		
- Heart failure - Post-MI - High CAD risk - Diabetes mellitus - Chronic kidney disease - Recurrent stroke prevention	- Relatively young patients - Patients with left ventricular hypertrophy - Gout - Peripheral vascular disease - Dyslipidemic patients	- Bilateral renal artery stenosis - Pregnancy - Hyperkalemia - Pre-existing dry cough (ARBs can be given)
β-blockers		
- Stable heart failure - Post-MI - High CAD risk	- Co-existing anxiety or tachycardia - Relatively young patient - Migraine patients - Low cost therapy	- Asthma - COPD - Bradycardia - Conduction disease - Decompensated heart failure - Peripheral vascular disease - Abnormal lipid profile
Calcium channel blockers		
Recurrent stroke prevention	- Older patients with poor arterial wall compliance - Isolated systolic hypertension - Asthma - COPD - Peripheral vascular disease - Pregnant hypertensive - Diabetes mellitus	- Verapamil and Diltiazem – - Myocardial inadequacy - CHF - Conduction defects - Sick sinus syndrome - Receiving β-blockers - DHPs – - Ischemic heart disease - Post-MI - Males with prostate enlargement - Gastroesophageal reflux

Condition	Preferred drug	Better avoid
HT with DM	ACEIs, ARBs or CCBs	Beta blockers, thiazide diuretics
HT with dyslipidaemia	ACEIs, ARBs, alpha blockers, CCB	Beta blockers, thiazide diuretics
HT with CAHD	Beta blockers, CCB (diltiazem)	Arteriolar dilators
HT with CCF	ACEIs, ARBs, diuretics	Beta blockers, CCB (diltiazem, verapamil)
HT with chronic renal failure	Diuretics, CCB	
HT with Bronchial asthma	CCB, diuretic, ACEi, ARB	Non selective beta blockers
HT due to pheochromocytoma	Non selective alpha blockers	Don't use beta blocker as a first line choice
Hypertensive emergency	<u>Hydralazine</u> <u>Esmolol</u> <u>Labetalol</u> <u>Enalaprilat</u> Nitroprusside Nicardipine, clevidipine Fenoldopam	
HT with pregnancy	Alpha methyl dopa Labetalol, hydralazine	ACEIs, ARBs

Worksheet

To do

Anti hypertensive causing side effect of SLE	
DOC hypertensive emergency in pregnancy	
First line drugs for treating hypertension are	
Hypertension with BPH preferred anti hypertensive drug is	
Adverse effects of sodium nitroprusside	
CCB useful for hypertensive emergency	
D1 agonist useful for hypertensive emergency is	
Preferred antihypertensive medication of young patient with chronic migraine problem is	

CONCEPTS

- ⇒ **Concept 6.1 Thrombolytic agents**
- ⇒ **Concept 6.2 Anti coagulants**
- ⇒ **Concept 6.3 Anti platelets**
- ⇒ **Concept 6.4 Hypolipidemic agents**

Concept 6.1 : Thrombolytic Agents

Learning objectives:

- To know the examples for thrombolytic agents
- To know the uses , contra indications of thrombolytic agents

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Examples of Fibrinolytics:

Drug	Source	T½
Streptokinase	Obtained from β -hemolytic Group C streptococci	60 – 80 min
Urokinase	Commercially prepared from cultured human kidney cells	10 – 15 min
Alteplase (rt-pA)	Recombinant tissue plasminogen activator (tPA)	4 – 8 min
Retepase	Modified longer acting form of rt-PA	13 – 15 min
Tenecteplase	Genetically engineered substitution mutant of native t-PA	13 – 15 min

Newer Drugs

- **ALFIMEPRASE-** obtained from copperhead snake (venom).
- **DESMOTEPLASE-** obtained from Vampire bat (saliva).

ONE-LINERS:

- Thrombolytic drugs acts by activation of plasminogen.
- First fibrinolytic to be clinically used – **Streptokinase**.
- Fibrinolytic resistant to inhibition by tissue plasminogen activator inhibitor-1 – **Tenecteplase**.
- Fibrinolytic that can be administered as a single IV bolus dose over 10 seconds – **Tenecteplase**.
- Thrombolytic agents are very useful for ST elevation MI, and for acute pulmonary embolism

Contraindications to Pharmacological Thrombolysis:

- h/o intracranial hemorrhage
- h/o ischemic stroke in the past 3 months
- h/o head injury in the past 3 months
- Intracranial tumour / vascular abnormality / aneurysms
- Active bleeding / bleeding disorders
- Peptic ulcer
- Esophageal varices
- Any wound or recent fracture or tooth extraction
- h/o major surgery within 3 weeks
- Uncontrolled hypertension
- Pregnancy

Anti fibrinolytics (antidote for thrombolytic agents)

- Epsilon-aminocaproic acid (EACA)
- Tranexemic acid
- Aprotinin

ONE-LINERS:

- Tranexamic acid is – **7 times** more potent than EACA

Concept 6.2 : Anticoagulants

Learning objectives:

- To know the classification of anticoagulants
- To know the various types of injectable anticoagulants and their uses
- To know the various types of oral anticoagulants and their uses

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Injectable Anticoagulants:

Unfractionated Heparin (UFH) v/s Low Molecular Weight Heparins:

	UFH	LMWH
Inhibits	Xa = IIa	Xa > IIa
Route of administration	IV, s.c.	Only s.c.
Subcutaneous bioavailability	Poor	Good
Plasma and tissue protein binding	Extensive	Negligible
Anticoagulant effect	Less consistent or less predictable	More consistent or more predictable
Regular monitoring	Required with aPTT	Not required May be needed in severe obese patients and in renal failure patients (monitor anti factor Xa)
Metabolism	Metabolized by heparinase enzyme in plasma	Mainly excreted unchanged by kidney
Safe in renal failure	Yes	No
Adverse effects	More likely	Less likely
Heparin-induced thrombocytopenia	Yes	No
Antidote	Protamine sulphate	Not available

Examples of Low Molecular Weight Heparin (LMWH) Preparations:

• Enoxaparin • Dalteparin • Tinzaparin • Ardeparin • Bemiparin	• Nadroparin • Reviparin • Pamparin • Certoparin • Semuloparin
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Examples of synthetic pentasaccharides (acting via antithrombin III – inhibits mainly Xa).

- Fondaparinux
- Idraparinux
- Idrabiotaparinux

ONE-LINERS:

- T_{1/2} of Fondaparinux – **17 h**
- T_{1/2} of Idraparinux – **80 h**.
- Idraparinux is administered as – **weekly subcutaneous injections**

Examples of Parenteral Direct Coagulation Factor Inhibitors:

Factor	Inhibitors
Ila (Thrombin)	Lepirudin Desirudin Bivalirudin Pegmusirudin Argatroban Flovagatran
Va	ART123
VIIa	Tifacogin
Xa	Otamixaban
XIa	Clavatadine
Dual Ila/Xa	Tanogitrin

ONE-LINERS:

- Drug used in heparin-induced thrombocytopenia – **Lepirudin**
- Injectable DTI safe in renal failure- argatroban

Oral anticoagulants

Examples of Coumarin Derivatives:

- Bishydroxycoumarin
- Warfarin
- Acenocoumarol
- Ethylbiscoumacetate
- Phenindione

Factors Affecting Warfarin Action:

Enhance warfarin action	Reduce warfarin action
• Debility • Malnutrition • Malabsorption • Liver disease • Chronic alcoholism • Hyperthyroidism • Newborns • Broad spectrum antibiotics • Drugs causing hypoprothrombinemia – Ceftriaxone, Cefoperazone • Aspirin • Drugs that displace warfarin from protein binding sites – Long-acting sulfonamides, Indomethacin, Phenytoin, Probenecid • Drugs that inhibit metabolism – Chloramphenicol, Erythromycin, Celecoxib, Cimetidine, Allopurinol, Amiodarone, Metronidazole, Tolbutamide, Phenytoin • Liquid paraffin	• Pregnancy • Nephrotic syndrome • Genetic warfarin resistance • Drugs that induce warfarin metabolism – Barbiturates, Carbamazepine, Rifampicin, Griseofulvin • Oral contraceptives

ONE-LINERS:

- Name 'warfarin' is derived from 'Wisconsin Alumni Research Foundation coumarin derivative'.
- Dose of warfarin – **2-10 mg/d.**
- Warfarin sensitivity – **< 1.5 mg/d** of warfarin is needed.
- Warfarin resistance – **> 20 mg/d** of warfarin is needed.
- Regular monitoring for warfarin is done with – **PT-INR.**
- PT-INR for warfarin is done – **once a week.**

Recommended INR for Various Indications of Warfarin:

• 2.0 – 2.5	• Prophylaxis of deep vein thrombosis
• 2.0 – 3.0	• Treatment of deep vein thrombosis • Pulmonary embolism • Transient ischemic attacks • Prior to hip surgery
• 3.0 – 3.5	• Recurrent thromboembolism • Arterial disease (MI) • Prosthetic heart valves

Clinical Features of Fetal Warfarin Syndrome (CONTRADI SYNDROME):

- Hypoplasia of nose, eye sockets and hand bones
- Growth retardation
- CNS defects
- Fetal hemorrhages

Anticoagulation During Pregnancy:

• 1-12 weeks	• Unfractionated heparin
• 12-36 weeks	• Warfarin
• 36 weeks – Delivery	• Unfractionated heparin
• Post-partum	• Warfarin

Oral Direct Thrombin(IIa) Inhibitors (DTIs):

- Ximelagatran (withdrawn due to hepatotoxicity)
- Dabigatran

Oral Factor Xa Inhibitors:

- Rivaroxaban
- Apixaban
- Edoxaban
- Batruxaban

Adverse effects of heparin and warfarin (ABOUT is the pneumonic)

Heparin	Warfarin
A- alopecia B-bleeding O-osteoporosis U-utricaria(Hypersensitivity) T- thrombocytopenia Hyperkalemia	A- alopecia B- bleeding O- oral(Gi intolerance) U- dermatitis T- teratogenicity Warfarin may also causes hypercoagulation, dermal necrosis, purple toe syndrome due to inhibition of protein C

Antidotes for Various Anticoagulants:

Anticoagulant	Antidote
Heparin	Protamine sulphate
Warfarin	Vitamin K1(Phytonadione)
Idrabiotaparinux	Avidin
Oral direct thrombin inhibitor(dabigatran)	Idarucizumab
Oral direct Xa inhibitors	Andexanet alfa-
Common antidote for both oral IIa and Xa inhibitor is	CIRAPARANTAG

ONE-LINERS:

- Protamine sulphate is obtained from – **sperm**
- **of certain fish.**
- 1 mg of protamine sulphate neutralizes – **100 U of heparin**
- Protamine antagonizing heparin is example for- chemical antagonism

Concept 6.3 : Antiplatelet Drugs

Learning objective: To know the various types of antiplatelet drugs and their mechanism of action

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

TXA2 Antagonist	P2Y ₁₂ (ADP) Antagonist	GP2B3A ANT Agonist	PAR-1 Antagonist
Txa synthesis inhibitors Low dose aspirin (inhibits irreversibly COX 1 enzyme) Dezoxiben- inhibits thromboxane synthase enzyme TXA2 receptor blockers Ifetroban Sultroban Daltroban Losartan Vapiprost	Ticlopidine Clopidogrel Prasugrel Ticagrelor Cangrelor	Abciximab Eptifibatide Tirofiban	Vorapaxar Atopaxar
	Avoid omeprazole with clopidogrel Safer PPI that donot have drug interaction with clopidogrel are pantoprazole and rabeprazole		

Monitoring:

- **Anti platelet drugs-** prolongs bleeding time
- **Heparin** (intrinsic pathway)- prolong aPTT
- **Warfarin** (extrinsic pathway)- prolongs PT
- LMWH- monitor anti factor Xa (renalfailure, obese)

Drugs Employed in Drug Eluting Stents:

- Paclitaxel
- Sirolimus (Rapamycin)
- Temsirolimus
- Everolimus
- Zotarolimus

Drugs to be Stopped before Surgery:

• ACEi	• On day of surgery
• ARBs	• On day of surgery
• Lithium	• 2 days
• Sulfonylureas	• 2-4 days
• Metformin	• 2-4 days
• Oral anticoagulants	• 4 days
• Estrogen containing OCPs	• 2-4 weeks

Worksheet

To do

Example for oral direct thrombin inhibitor	
Antidote for oral direct Xa inhibitor	
Antidote for thrombolytic agents	
Clot specific thrombolytic are	
Antiplatelet drug having PAR antagonistic action are	
DOC for treating patient with heparin induced thrombocytopenia	
Clopidogrel and omeprazole drug interaction is due to which microsomal enzyme	

Concept 6.4 : Hypolipidemic Agents

Learning objectives:

- To know the different group of cholesterol lower agents
- To know the action, uses and adverse effects of statins
- To know the mechanism of action adverse effects of fibrates
- To know the actions of niacin, bile acid binding resins
- To know the newer cholesterol lowering agents

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

Statins

Drugs	Mechanism of action a	Adverse effects
Lovastatin Simvastatin Pravastatin Fluvastatin Atorvastatin Cerivastatin Rosuvastatin Pitavastatin	Inhibits HMG CoA reductase Upregulate LDL receptors	Myopathy Myalgia Rhabdomyolysis Hepato toxic C/I- pregnancy
Prodrug	Lovastatin , Simvastatin	
Long acting	Atorvastatin, Rosuvastatin	
Least myopathy	Pravastatin Rosuvastatin	

Statin Metabolism

CYP3A4, 3A5	CYP2C9	No hepatic metabolism
Atorvastatin Lovastatin Simvastatin	Fluvastatin	Pravastatin Least drug interaction Least myopathy Less entry to CNS ↓ Sr. fibrogen
Preferred in Renal Dysfunction		Preferred in Liver Dysfunction
Atorvastatin		Pravastatin, Rosuvastatin

11yr & older children	8yr & older children
Atorvastatin Lovastatin Simvastatin	Pravastatin

ONE-LINERS:

- Statin may increase risk of developing diabetes
- When statin dose are doubled there will be addition reduction 6% LDL-C from the base line value (rule of six)
- Cholesterol lowering effect of statin achieved within 7-10days of therapy
- Statin not giving benefit in patients with Homozygous Familial Hypercholesterolemia
- PLEOTROPIC EFFECT- Anti inflammatory action, Reduces endothelial dysfunction
- Anti oxidant action

Fibrates

Drug	Mechanism of action	ADR
Clofibrate Fenofibrate Bezofibrate Gemfibrozil	PPAR alpha stimulation-activating lipoprotein lipase – reducing triglycerides	Myopathy, hepatotoxic

Niacin (vit B3)

Mechanism of action	Actions	Adverse effects
Inhibits lipolysis in adipose tissue	Reduces LDL, Lp(a) Increases HDL	Cutaneous Flushing Dyspepsia Hyperuricemia Diabetes Hepatotoxicity Myopathy

Drugs Inhibiting Dietary Cholesterol Absorption

Drug	Mechanism of action	Adverse effect
Ezetimibe	Inhibits NPC1L1 (Niemann Pick C1 Like)	steatorrhea

Bile Acid Sequestrants

Drugs	Mechanism of action	Adverse effects
Cholestyramine Colestipol Colesevelam-	Binds with bile salt loss more bile acid synthesis taking place from cholesterol- leads to depletion cholesterol Compensatory increase in LDL receptor in liver –leads to reduction in LDL	Increases the activity of hepatic HMG coA reductase- poor long term efficacy

Newer Drugs

PCSK9 inhibitor-	Cholesterol ester transport protein inhibitor	Microsomal triglyceride transport protein inhibitor	Inhibits conversion Cholesterol to cholesterol ester ACAT- 1 inhibitor	Inhibits ApoB100 Reduces VLDL
Alirocumab Evolocumab	Torcetrapib Dalcetrapib Evacetrapib Anacetrapib	Lomitapide	Avasimibe	Mipomersen

Worksheet

To do

Statin preferred in renal disease	
Cholesterol lowering agent safe in pregnancy	
PPAR alpha agonists are	
Adverse effects of niacin	
Examples for MAB useful for lowering cholesterol	
CETP inhibitors useful for lowering cholesterol	

CONCEPTS

⇒ **Concept 7.1 Diuretic & Anti Diuretics**

Concept 7.1 : Diuretic & Anti Diuretics

Learning objectives:

- To know the site of action and mechanism of actions of various diuretics
- To know the uses and adverse effects of various class of diuretics
- To know the uses and adverse effects of antidiuretic

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

Sites and Mechanisms of Actions of Diuretics:

Diuretic	Site of Action	Mechanism of Action
Carbonic anhydrase inhibitors	Proximal convoluted tubule	Inhibit Na^+ and HCO_3^- absorption due to carbonic anhydrase enzyme
Osmotic diuretics	PCT, Mainly in the loop of Henle	Draw water by osmosis
Loop diuretics	Thick ascending limb of loop of Henle	Inhibit $\text{Na}^+ - \text{K}^+ - 2\text{Cl}^-$ symport
Thiazides	Early distal convoluted tubule	Inhibit $\text{Na}^+ - \text{Cl}^-$ symport
Aldosterone antagonists	Late distal convoluted tubule	Inhibit Na^+ and H_2O absorption by aldosterone
Amiloride	Late distal convoluted tubule	Inhibits epithelial sodium channel (ENaC)
Triamterene	Late distal convoluted tubule	Inhibits epithelial sodium channel (ENaC)

Effect of Diuretics on Urine Electrolytes:

Drug	Urine electrolyte excretion				Max % of filtered load excreted	Efficacy
	Na^+	K^+	Cl^-	HCO_3^-		
Furosemide	↑↑↑	↑	↑↑	↑, -	25	High
Thiazide	↑↑	↑	↑	↑	8	Intermediate
Acetazolamide	↑	↑↑	↑, -	↑↑	5	Mild
Spironolactone	↑	↓	↑	-, ↑	3	Low
Amiloride	↑	↓	↑	-, ↑	3	Low
Mannitol	↑↑	↑	↑	↑	20	High

ONE-LINERS:

- All diuretics acting proximal to late DCT cause – **Potassium loss** and **hypokalemia**.
- All diuretics acting at or distal to late DCT cause – **Potassium retention** and **hyperkalemia**.
- Maximum hypokalemia caused by – **Carbonic anhydrase inhibitors** (due to most proximal action).
- All diuretics require access to tubular lumen except – **Aldosterone antagonists**.

Carbonic anhydrase enzyme inhibitors

Drugs	Uses	Adverse effects	Contraindications
• Acetazolamide • Dorzolamide • Brinzolamide	• Acetazolamide- • Mountain sickness • Periodic paralysis • Absence seizure and catamenial seizure • To control intra cranial tension • For making urine alkalization • For glaucoma • Oral – acetazolamide • Topical – dorzolamide, brinzolamide	• Metabolic acidosis • Hypokalemia • Hypersensitivity reaction (bone marrow suppression)	• Hepatic failure • Metabolic acidosis

Examples of Loop Diuretics:

• Furosemide. • Bumetanide. • Torsemide. • Ethacrynic acid.	• Axosemide. • Piretanide. • Tripamide.
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ONE-LINERS:

- Synergistic diuretic activity seen with the combination of – **Loop diuretic + Thiazide diuretic**.
- Loop diuretics are – **High ceiling diuretics**.
- All loop diuretics are well absorbed orally except – **Axosemide**.
- All loop diuretics are mainly eliminated by kidney except – **Torsemide**.
- Most potent loop diuretic – **Bumetanide**.
- Bumetanide is – **40 times** more potent than furosemide.
- Torsemide (**longest acting**) is – **3 times** more potent than furosemide.

Drugs that Increase the Risk of Ototoxicity with Loop Diuretics:

- | | |
|--|---|
| <ul style="list-style-type: none"> • Aminoglycosides. • Carboplatin. | <ul style="list-style-type: none"> • Paclitaxel. • Vancomycin . |
|--|---|
- Highly ototoxic loop diuretics is **Ethacrynicid**.
 - **Loop diuretic aggravates the adverse effects of digoxin by causing hypokalemia and hypomagnesemia.**
- **Diuretic of choice in the presence of Renal failure is loop diuretics, thiazides are ineffective except metolazone.**
 - **Loop diuretic relieve only the symptoms of LVF not modify the disease.**

Examples for thiazide diuretics

- | | |
|---|---|
| <ul style="list-style-type: none"> • Hydrochlorthiazide. • Hydroflumethiazide. • Benzthiazide . • Bendroflumethiazide. • Chlorthalidone. | <ul style="list-style-type: none"> • Metolazone. • Quinethazone. • Xipamide. • Indapamide. • Clopamide |
|---|---|

Thiazides:

Drug	Daily dose (mg)	Duration of action (h)	Carbonic anhydrase inhibition
Hydrochlorthiazide	12.5-100	6-12	+
Chlorthalidone	50-100	48	++
Metolazone	5-20	12-24	+
Xipamide	20-40	12	+
Indapamide	2.5-5	12-24	-
Clopamide	10-60	12-18	±

ONE-LINERS:

- Thiazides are not effective in renal failure except – **Metolazone**.
- **Thiazide and loop having opposite action on calcium (loop loses calcium in urine, thiazide retains calcium from urine).**
- **Thiazide have anti diuretic action (DOC- nephrogenic DI).**
- **Thiazide are first line therapy for hypertension.**
- **JNC VIII (first line therapy for HT) are- Thiazides, ACEI, ARBs and CCB.**
- **Thiazides because of inhibiting calcium excretion useful in Idiopathic hypercalciuria (William's syndrome), Treatment for Calcium Nephrolithiasis**

Comparing to loop diuretic thiazide diuretic causes maximum metabolic side effects like- hyperuricemia, hyperglycemia and hyperlipidemia

Thiazide diuretic may cause erectile dysfunction which may be aggravated when given with beta blockers

Potassium Sparing Diuretics:

Aldosterone antagonist	Epithelial sodium channel (ENaC) blockers
Spironolactone Canrenone (active metabolite of spironolactone) Eplerenone (no problem of gynecomastia) Drospirenone (progesterone)	Amiloride Triamterene Anti microbial having ENaC blocking property- Pentamidine Trimethoprim

Potassium sparing diuretics:

Drug	Action	Uses	Adverse effects
Spironolactone	Acting from interstitium against aldosterone receptor	- Primary Hyperaldosteronism (Conn's syndrome). - Edema of Liver Cirrhosis. - Heart Failure (disease modifying drug)	Hyperkalemia Metabolic acidosis Impotence Gynecomastia
Amiloride	Acting from lumen against ENaC	- Lithium-induced diabetes insipidus (DI). - Cystic fibrosis. - Treatment for Liddle's Syndrome (↑ENaC).	Hyperkalemia Metabolic acidosis

ONE-LINERS:

- DOC for Lithium-induced DI – **Amiloride**.
- DOC for nephrogenic DI- **thiazides**
- DOC for cranial DI- **DESMOPRESSIN**
- DOC for cirrhotic edema – **Spironolactone**.

Osmotic Diuretics

Drug	Site of action	Uses	Adverse effects	Contra indication
Mannitol Isosorbide Urea Glycerol	LOH PCT	Glaucoma Cerebral edema Cisplatin toxicity Dialysis disequilibrium syndrome	ADR- Hyponatremia Headache	C/I- Pul edema- LVF Cerebral Hge

Natriuretics:

Example	Mechanism	Current status
Nesiritide	Recombinant BNP analogue	Approved for acute decompensated heart failure Given IV, short acting (20minutes) Rapid metabolism by vasopeptidase
Carperitide	Recombinant ANP analogue	Not used much nowadays
Omapatrilat sampatrilat	Dual neutral endopeptidase (NEP) and ACE inhibitor	Withdrawn due to high incidence of angioedema
Sacubitril Ecadotril	Neutral endopeptidase (NEP) inhibitor	Recently approved for CHF

ONE-LINERS:

- Nesiritide is administered as an – **IV infusion** (due to short T_{1/2})

Anti-diuretics (Vasopressin)

	Name	Uses
Selective v2 analogue	Desmopressin	Cranial di Nocturnal enuresis Hemophilia Bleeding due to- Def vwf
Selective v2 antagonist	Lixivaptan, Mozavaptan, Tolavaptan.	(drug of choice for SIADH)
Selective v1 analogues	terlipressin, felypressin, lypressin	esophageal varices, but drug of choice is octreotide
Selective v1 antagonist	Relcovaptan	Hypertension
Non selective V1 and V2 blocker	Conivaptan, maximally blocks V2 than V1	in SIADH, given IV

ONE-LINERS:

Drugs causing SIADH- Chlorpropamide. MAO inhibitor, Carbamazepine, Phenothiazine, Oxytocin high dose SSRI, TCA, Vasopressin, Vincristine, Nicotine

Worksheet

To do

Diuretic of choice for rapid relief of symptom of LVF	
Diuretic causing metabolic acidosis	
Diuretic contra indicated in liver failure	
Diuretic useful in nephrogenic DI	
Diuretic cause adverse effect of gynecomastia	
Drug of choice for nocturnal enuresis	
Example for selective V1 antagonist	
Antimicrobial having ENac blocking property	

CONCEPTS

- ⇒ Concept 8.1 Sedative & Hypnotics
- ⇒ Concept 8.2 Ethyl & Methyl Alcohol
- ⇒ Concept 8.3 Anti Epileptics
- ⇒ Concept 8.4 Anti Parkinsonian Drugs
- ⇒ Concept 8.5 Anti Psychotics
- ⇒ Concept 8.6 Anti Manic Drugs
- ⇒ Concept 8.7 Anti Depressants
- ⇒ Concept 8.8 Anti Anxiety Agents
- ⇒ Concept 8.9 Opioids
- ⇒ Concept 8.10 Newer Drugs In Cns

Concept 8.1 : Sedative & Hypnotics

Learning objectives:

- To know synthesis, storage release and metabolism of neurotransmitter GABA
- To know drugs acting via GABA receptors
- To the various uses of BZD
- To know the drugs useful for treating BZD poisoning
- To know the general properties of barbiturates
- To know "Z" compounds
- To know the GABA analogues
- To know melatonin analogues

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Synthesis, storage, release and metabolism of GABA

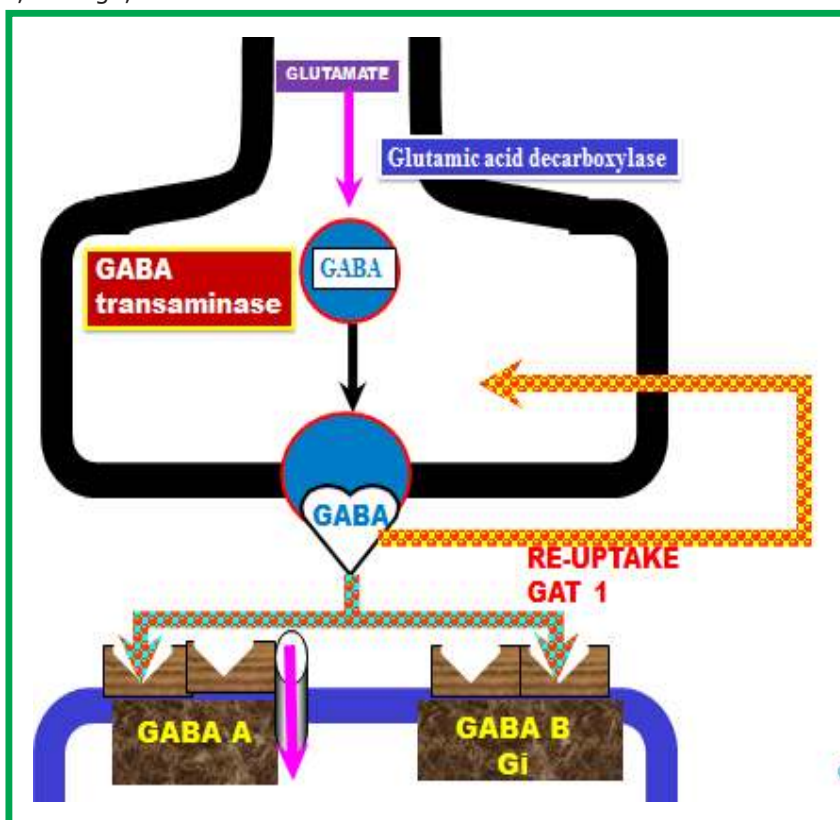


Fig. 8.1

GABA

Enzyme involved in synthesis	Glutamic acid decarboxylase
Drugs activating glutamic acid decarboxylase	Sodium valproate,
Enzyme involved in metabolism of GABA	GABA transaminase
Drugs inhibiting GABA transaminase	Vigabatrine, sodium valproate
GABA undergoes reuptake by	GABA transporter (GAT)
GABA reuptake blocked by	Tiagabine

Drugs acting on GABA_A receptor:

	Site	Agonist	Antagonist
1	• GABA	• GABA • Muscimol	• Bicuculline
2	• Alpha, gamma unit of GABA _A receptor	• Benzodiazepines	• Flumazenil • DMCM / β -carboline (Inverse agonist)
3	• Other miscellaneous sites	• Mucimol, Alphaxolone • Alcohol, Propofol	• Antagonist- • Bicuculline • Picrotoxin- Cl channel
4	• Barbiturates	• Ivermectin • Alpha, beta units of GABA _A receptor	• No specific antidote

Uses of BZD

Benzodiazepines used as anti-anxiety drugs	Benzodiazepines used as Anticonvulsants	Benzodiazepines used as Pre-Anaesthetic Medications
Chlordiazepoxide. Diazepam. Oxazepam. Lorazepam. Alprazolam.	Diazepam. Lorazepam. Clonazepam. Clobazam.	Diazepam. Lorazepam Midazolam. Remimazolam

Benzodiazepine	Uses
Diazepam	DOC-Acute febrile seizure Status epilepsy delirium tremors
Lorazepam	DOC- Status epilepsy
Alprazolam	Insomnia, Anxiety Disorder
chlordiazepoxid	DOC- alcohol withdrawal (delirium tremors)



ONE-LINERS:

- Specific antidote for benzodiazepine poisoning / overdose – **Flumazenil**.
- Flumazenil is administered as – **IV infusion**.
- T_½ of flumazenil – **~ 1 hr**.
- Duration of action of flumazenil – **30 to 60 min**.
- The maximum dose of flumazenil to be administered in a case of suspected benzodiazepine overdose is – **5 mg**.
- Competitive antagonist of BZD – **FLUMAZENIL**.
- Non competitive antagonist of BZD – **BICUCULINE**.
- Competitive antagonist of GABA – **BICUCULINE**.
- Direct chloride channel blocker – **PICROTOXIN**.
- Inverse agonist of benzodiazepine – **DMCM (β-carboline)**.
- BZD-Safe in Liver Failure:
 - Temazepam, Lorazepam, Oxazepam:
 - Date Raping Drug - Flunitrazepam (Anterograde amnesia)
 - Ketamine (general anaesthetic agent also used as date raping agent)

Barbiturates:

- Thiopentone sodium- intravenous induction GA, ultra short acting (due to rapid redistribution)
- Methohexitone- useful in electro convulsive therapy
- Phenobarbitone is the active metabolite of primidone
- Classical symptoms of barbiturate poisoning Flabby, Comatose, shallow and failing Respiration, bullous eruptions.
- Barbiturate are microsomal enzyme inducer- C/I in acute intermittent porphyria

Non-benzodiazepine Sedative Hypnotics (Z compounds):

- **“Z” compounds (useful in sleep onset insomnia)**
- Zolpidem (most commonly used drug)
- Zaleplone (shortest acting)
- Zopiclone
- Eszopiclone

GABA Analogues:

	Drug	Mechanism of action	Remarks
1	Tiagabine	• GABA reuptake inhibitor	
2	Valproate	• GABA Transaminase Inhibitor	• DOC - infantile spasm with tuberous sclerosis • ADR - bilateral visual field defect, psychosis
3	Ganaxalone	• Neurosteroid • Direct chloride channel opener	• Useful in catamenial seizure • Absence seizure
4	Levetiracetam	• Ligand for SV2A protein (which modify release of GABA & Glutamate)	• Anti epileptic
5	Valproate	• Inhibits GABA transaminase • Stimulate glutamic acid decarboxylase	• Broad spectrum anti epileptic • Also useful in prophylaxis of chronic migraine, Bipolar disorder
6	Gabapentin Pregabalin	• Promotes synthesis and release of GABA • Binds to alpha 2 delta subunit of voltage gated sodium channel, inhibits voltage gated calcium current and decreases synaptic transmission	• DM neuropathy, post herpetic neuralgia

GABA B (GPCR)**Agonist:**

Baclofen (central acting skeletal muscle relaxant, useful in Hiccough, controls Craving of alcohol).

Antagonist: Saclofen**Melatonin analogues-**

- **Ramelteon** – synthetic melatonin analogue.
- Mechanism of ramelteon – **Agonist at MLT-1 and MLT-2 receptors.**
- Oral bioavailability of ramelteon – **< 2 h.**
- $T_{1/2}$ of ramelteon – **~ 2 h.**
- Plasma protein binding of ramelteon – **~ 80%.**
- **Tasimelteon** – Congener of ramelteon. Melatonin MLT-1 and MLT-2 agonist for treatment of non-24-hour sleep-wake disorder in totally blind people.
- **AGOMELATINE**- Agonist on MT1/ MT2, Antagonist on 5-HT_{2C} – useful as an anti depressant.

Newer drug for insomnia

SUVOREXANT (FDA approved)- Non- selective **OREXIN** receptor antagonist useful in treating **Insomnia**.

Worksheet

To do

GABA transaminase inhibitors	
GABA reuptake inhibitor	
DOC of status epilepticus	
Antidote for BZD	
BZD safe in liver failure	
Melatonin analogue useful in insomnia	
Orexin receptor antagonist useful in insomnia	
Examples for Z compounds	

Concept 8.2 : Ethyl & Methyl Alcohol

Learning objectives:

- To know the Percentages of Alcohol in Various Alcoholic Beverages
- To know drugs useful for deaddiction and preventing craving for alcohol
- To know the drugs useful for treating alcohol withdrawal symptoms
- To know the drugs useful for treating methyl alcohol poisoning

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

Percentages of Alcohol in Various Alcoholic Beverages:

Alcoholic beverage	% of alcohol
Malted liquors (Stout, Beer)	3 – 6
Strong beer	Up to 10
Light wines (Claret, Cidar)	9 – 12 (does not exceed 15)
Fortified wines (Port, Sherry)	16 – 22
Effervescent wines (Champagne)	12 – 16
Spirits (Rum, Gin, Whiskey, Brandy, Vodka)	40 – 55
Rectified spirit	90
Absolute alcohol	99

ONE-LINERS:

- Denatured / Methylated spirit: **5 parts of methyl alcohol(wood naphtha) + 95 parts of rectified spirit.**

Drugs useful for deaddiction & prevention of craving against alcohol:

Deaddiction	Anti-craving agents
• Disulfiram (aldehyde dehydrogenase enzyme inhibitor)	• Naltrexone (opioid antagonist) • Acamprosate (NMDA blocker with GABA agonist action) • Topiramate (anti epileptic) • Baclofen (GABA B agonist) • Ondansetron • Fluoxetine

Drugs causing disulfiram like reaction:

Deaddiction	Drugs causing disulfiram like reactions
Disulfiram	- Chlorpropamide, Cefoperazone - Metronidazole - Procarbazine - Griseofulvin - Tinidazole - Nitrofurantoin

Drugs useful for treating alcohol withdrawal symptoms

- DOC for uncomplicated alcohol withdrawal (delirium and Tremor) – **Chlordiazepoxide**.
- DOC for complicated alcohol withdrawal –
- Diazepam**.

ONE-LINERS:

- Disulfiram inhibits – **Aldehyde dehydrogenase enzyme**.
- Chronic alcoholism causes **thiamine deficiency**.
- Chronic alcoholism will induce CYP2E1 enzyme.
- Alcohol undergoes **zero order kinetic of excretion**.

Methanol Poisoning:

- Etiology: **Consumption of illicit / adulterated liquor**.
- Toxic level: **60-100 mL**.
- Primary toxic metabolite: **Formic acid**.
- Formic acid mainly accumulates in: **Retina and optic nerve**.
- Clinical features: **Obscured, snowy vision that ultimately progresses to complete blindness**.
- Antidotes: **Ethanol and Fomepizole**.
- Ethanol: **Saturates** alcohol dehydrogenase enzyme.
- Fomepizole: **competitively inhibits** alcohol dehydrogenase enzyme.
- Calcium leucovorin** and **hemodialysis** are also effective.



Worksheet

To do

Drug inhibiting alcohol dehydrogenase	
Drug inhibiting aldehyde dehydrogenase	
Mechanism of action of acamprosate	
Wernicke encephalopathy is due to deficiency of	
Microsomal enzyme induced by chronic alcoholism is	
Antifungal drug causing disulfiram like reaction is	

Concept 8.3 : Anti Epileptic Drugs

Learning objectives:

- To know the classification of antiepileptic drugs
- To know the uses and adverse effects of anti epileptics
- To know newer antiepileptic drugs

Time Needed

1 st reading	40 mins
2 nd reading	25 mins

Classification of antiepileptic drugs:

Predom- inant Mechanism of Action	Examples
Sodium channel blockers	• Phenytoin and Fosphenyt- oin(prodrug, water solu- ble) • Carbamazepine, Oxcarba- zepine and Eslicarbazepine • Valproate and Divalproex • Lamotrigine • Topiramate • Rufinamide (approved for • lennox gestaut syndrome) • Lacosamide
GABA enhancers	• Progabide (prodrug, gets converted to GABA) • Vigabatrin (Inhibits GABA transaminase en- zyme) • Tiagabine (Inhibits GABA transporter-1 mediated re- uptake of GABA) Direct: • Barbiturates (Increase duration of chloride chan- nel opening) • Benzodiazepines (In- crease frequency of chlo- ride channel opening)
T-type calcium channel blockers	• Ethosuximide, Valproate • Zonisamide • Gabapentin • Pregabalin

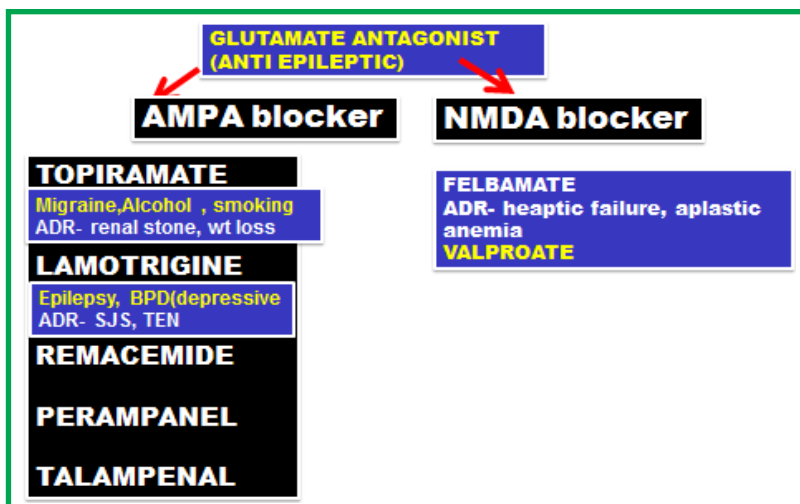


Fig. 8.2

Other NMDA blockers:

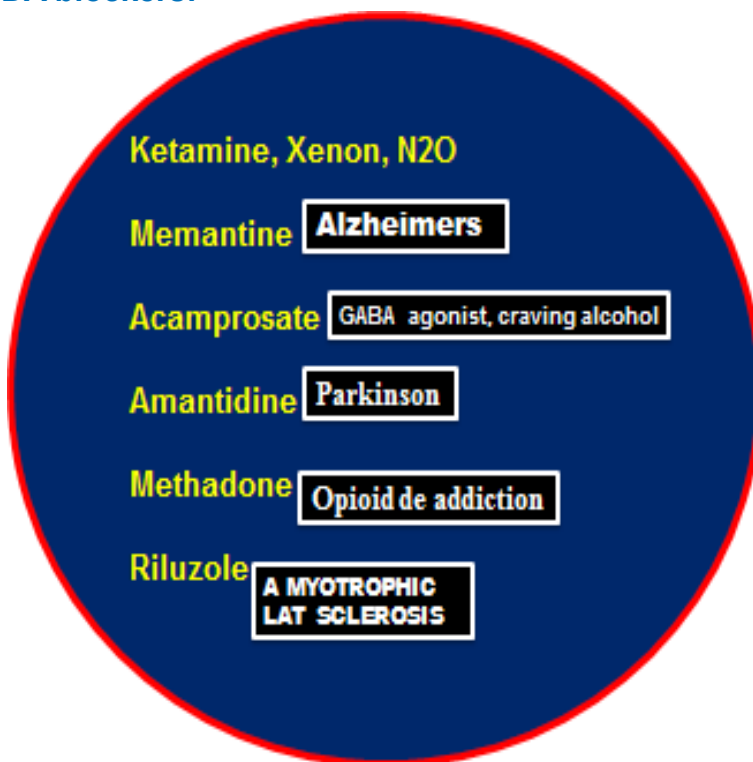


Fig. 8.3

Anti epileptic drugs uses and adverse effects

Generic Name	Trade Name	Principal Uses	Typical Dose: Dose Interval	Half-Life	Therapeutic Range	Adverse Effects		Drug Interactions ^a
						Neurologic	Systemic	
Brivaracetam	Bnviact	FocaFonset	100-200 mg/d; bid	7-10 h	Not established	Fatigue Dizziness Weakness Ataxia Mood changes	Gastrointestinal irritation	May increase carbamazepine- epoxide causing decreased tolerability May increase phenytoin
Carbamazepine	Tegretol [®]	Tonic-c Ionic Focal-onset	600-1800 mg/d (15-35 mg/ kg, child 1; bid (capsules or tablets), tid-qid (oral suspension)	10-17 h (variable due to autoinduction: complete 3-5 wk after initiation)	4-12 µg/mL	Ataxia Dizziness Diplopia Vertigo	Aplastic anemia Leukopenia Gastrointestinal irritation Hepatotoxicity Hyponatremia	Level decreased by enzyme- inducing drugs ^b Level increased by erythromycin, propoxyphene, isoniazid, cimetidine, fluoxetine
Clobazam	Onfi	Lennox Gastaut syndrome	40-50 mg/d (5-20 mg/d for patients <30 kg body weight); bid	36-42 h (71-82 h for less active metabolite)	Not established	Fatigue Sedation Ataxia Aggression Insomnia	Constipation Anorexia Skin rash	Level increased by CYP2C19 inhibitors
Clonazepam	Klonopin	Absence Atypical absence Myoclonic	1-12 mg/d; qd-tid	24-48 h	10-70 ng/mL	Ataxia Sedation Lethargy	Anorexia	Level decreased by enzyme- inducing drugs ^b
Ethosuximide	Zarontin	Absence	750-1250 mg/d (20-40 me/kg); qd-bid	60 h, adult 30 h, child	40-100 µg/mL	Ataxia Lethargy Headache	Gastrointestinal irritation Skin rash Bone marrow suppression	Level decreased by enzyme- inducing drugs ^b Level increased by valproic acid
Felbamate	Felbatol	Focal-onset Lennox- Gastaut syndrome Tonic-clonic	2400-3600 mg/d, tid-qid	16-22 h	30-60 µg/mL	Insomnia Dizziness Sedation Headache	Aplastic anemia Hepatic failure Weight loss Gastrointestinal irritation	Increases phenytoin, valproic acid, active carbamazepine metabolite

Generic Name	Trade Name	Principal Uses	Typical Dose: Dose Interval	Half-Life	Therapeutic Range	Adverse Effects		Drug Interactions ^a
						Neurologic	Systemic	
Gabapentin	Neurontin	Focal-onset	900-2400 mg/d; bid-qid	6-9 h	2-20 µg/mL	Sedation Dizziness Ataxia Fatigue	Gastrointestinal irritation Weight gain Edema	No known significant interactions
LaCoSamide	Vnimpat	FocaMwset	200-400 mg/d; bid	13 h	Not established	Dizziness Ataxia Diplopia Vertigo	Gastrointestinal irritation Cardiac conduction (PR interval prolongation)	Level decreased by enzyme- inducing drugs ^b .
Lamotrigine	Lamictal ^c	FocaFonset Tonic-clonic Abyptcaf absence Myoclonic Lennox- Gastaut syndrome	150-500 mg/d; bid (immediate release), daily (extended release) (lower daily dose for refrimms with valproic acid: higher daily dose for regimens with an enzyme inducer)	25 h 14 h {with enzyme- inducers}. 59 h (with valproic acid)	2.5-20 µg/mL	Dizziness Diplopia Sedation Ataxia Headache	Skin rash Stevens- John son syndrome	Level decreased by enzyme- inducing drugs ^b and oral contraceptives Level increased by valproic acid
Levetiracetam	Keppra ^c	FocaFonset	1000-3000 mg/d: bid (immediate release), daily (extended release)	6-8 h	5-45 µg/mL	Sedation Fatigue Incoordination Mood changes	Anemia Leukopenia	No known significant interactions
Oxcarbazepine:	Trileptal	Focal-onset Tonic-clonic	900-2400 mg/d (30-45 mg/kg; child): bid	10-17 h (for active metabolite)	10-35 µg/mL	Fatigue Ataxia Dizziness Diplopia Vertigo Headache	See carbamazepine	Level decreased by enzyme- inducing drugs ^b May increase phenytoin

Generic Name	Trade Name	Principal Uses	Typical Dose: Dose Interval	Half-Life	Therapeutic Range	Adverse Effects		Drug Interactions ^a
						Neurologic	Systemic	
Phenobarbital	Luminal	Tonic-clonic Focal-onset	60-180 mg/d qHd	90h	10-40 µg/mL	Sedation Ataxia Confusion Dizziness Decreased libido Depression	Skin rash	Level increased by valproic acid, ptietyoin
Phenytoin (diphenyl hydantoin)	Dilantin	Tonic-clonic Focal-onset	300-400 mg/d (3-6 mg/kg. adult 4-8mg/kg. child); qd-tid	24 h (wide variation, dose- dependent)	10-20 µg/mL	Dizziness Diplopia Ataxia Incoordination Confusion	Gingival hyperplasia LymphadenofHtiy Hirsutism Osteomalacia Facial coarsening Skin rash	Level increased by isoniazid, sulfonamides, fluoxetine Level decreased by enzyme- inducing drugs ^b Altered folate metabolism
Primidone	Mysoline	Tonic-clonic Focal-onset	750-1000 mg/d; bid-tid	Primidone. 8-15h Phenobarbital, 90 h	Primidone, 4-12 µg/mL Phenobarbital, 10-40 µg/mL	Same as phenobarbital		Level increased by valproic acid Level decreased by phenytoin (increased conversion to phenobarbital)
Rufinamide	Banzal	Lennox- Gastaut syndrome	3200 mg/d (45 mg/kg, child); bid	6-10 h	Not established	Sedation Fatigue Dizziness Ataxia Headache Diplopia	Gastrointestinal irritation Leukopenia: Cardiac conduction (QT interval shortening)	Level decreased by enzyme- inducing drugs ^b Level increased by valproic acid May increase phenytoin



Generic Name	Trade Name	Principal Uses	Typical Dose: Dose Interval	Half-Life	Therapeutic Range	Adverse Effects		Drug Interactions ^a
						Neurologic	Systemic	
Tiagabine	Gabitril	Focal-onset	33-56 mg/d; bid-bid (as adjunct to enzyme- inducing antiepileptic Aug teg; men)	2-5 h (w/iti enzyme inducer) ¹ , 7-9 h (without enzyme inducer)	Net established	Confusion Sedation Depression Dizziness Speech or language pract) Ferns Paresthesias Psychosis	Gastrointestinal irritation	Level decreased by enzyme- inducing drugs ¹
Topiramate	Topamax	Focal-onset Tonic-clonic Lennox-Gastaut synd<ictiroe	200-400 mg/d; bid (immediate release), daily (extended release) please)	20 fl I immediate releases 30 fl (extended release*)	2-20 µg/mL	Psychomotor slowing Sedation Speech or language problems Fatigue Paresthesias	Renal stones (avoid use with other cardan ic anhydrase inhibitors) Glaucoma weight loss Hyohurosis	Level decreased coenzyme- inducing drugs ^{2*}
valproic acid (valproate sodium, divalproex sodium)	Depakene Depalote ^c	Tonic-clonic Absence Atypical absence Myoclonic Focal-onset Atonic	760-2000 mg/d (20-60 mg/ kg); bid-qid (immediate and delayed release, daily (extended release)	15 h	50-125 µg/mL	Ataxia Sedation Tremor	Hepatotomy Thrombocytopenia Gastrointestinal Irritation weight gam Transient alopecia Hyperammonemia	Level decreased by enzyme- Inducing drugs ²
Zonisamide	Zonegran	Focal-onset Tonic-clonic	200-400 mg/d; qd-bid	50-68 h	10-40 µg/mL	Sedation Dizziness Confusion Headache Psychosis	Anorexia Renal stones Hypohidrosis	Level decreased by enzyme- inducing drugs ^b

NON EPILEPTIC USES OF PHENYTOIN-

Trigeminal neuralgia, Digoxin INDUCED VT, Wound healing

Carbamazepine- First line DOC for temporal lobe epilepsy, Trigeminal neuralgia. Carbamazepine has SIADH adverse effect.

Broad spectrum antiepileptic – sodium valproate (multiple mechanism of action- GABA agonism, Anti glutamate NMDA blocker, Na channel block, T-CCB).

Valproate is DOC for- **Myoclonic/ atonic/ clonic and tonic Absence /Lennox gestaut synd.**

Valproate useful for prophylaxis of chronic migraine, Bipolar disorder, very effective in rapid cyclers (>4 episodes of mania and depression /year).

Valproate most teratogenic, highly hepatotoxic.

- Phenytoin follows zero order kinetics at plasma levels – **>10 µg/mL.**
- Phenytoin toxicity occurs at plasma levels – **>20 µg/mL.**
- DOC for Rolandic epilepsy – **Carbamazepine.**
- DOC for trigeminal neuralgia – **Oxcarbazepine.**
- Antiepileptic drugs that can aggravate juvenile myoclonic epilepsy – **Phenytoin and Carbamazepine** (hence, contraindicated).
- All anti-epileptics are enzyme inducers except – **Valproate.**
- Anti-epileptic causing weight loss – **Topiramate**, zonisamide, felbamate.
- Anti epileptic causing weight gain- valproate, gabapentin.
- DOC for infantile spasms – **ACTH.**
- DOC for infantile spasms associated with tuberous sclerosis – **Vigabatrin.**
- Visual field contraction is a side effect of – **Vigabatrin.**
- DOC for status epilepticus – **Lorazepam.**
- DOC for emergency control of seizures – **Lorazepam.**
- Anti-epileptics contraindicated in porphyria – **Barbiturates.**
- Felbamate is on the verge of withdrawal due to – **Aplastic anemia and Hepatotoxicity.**
- Acetazolamide is effective in – Catamenial epilepsy at a dose of **10 mg/kg/d** (Maximum: 1000 mg/d).

Newer antiepileptic drugs

Stiripentol – GABA transmission enhancer for severe myoclonic epilepsy in infants (SMEI) / Dravet's syndrome.

Retigabine (ezogabine)- potassium channel opener- useful in partial seizure.

Anti epileptics causing:

Weight gain	Weight loss	Eye problem
Sod valproate, gabapentin	Topiramate, zonisamide, felbamate	Vigabatrine- visual field defect Topiramate- angle closure glaucoma ezogabine – retinal degeneration

Anti epileptic useful

Prophylaxis of chronic migraine	Bipolar disorder
Valproate Topiramate Gabapentin	Carbamazepine Valproate Lamotrigine

Worksheet

To do

Drug	Mechanism of Action	Uses	ADR
Topiramate			
Retigabine			
Vigabatrin			
Ethosuximide			
Fosphenytoin			
Oxcarbazepine			
Acetazolamide			

Concept 8.4 : Drugs Useful In Parkinson's Disease

Learning objectives:

- To know the classification of anti Parkinson drugs
- To know the drugs useful for other movement disorder

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Classification of Antiparkinson Drugs:

	Mechanism	Examples
1	Dopamine precursors	• Levo dopa
2	Dopamine receptor agonists	• ERGOT • Bromocriptine • Cabergoline • Pergolide • NON ERGOT- • Ropinirole • Pramipexole • Rotigotine • (trans dermal patch) • Apomorphine (D4 agonist)
3	Peripheral dopa decarboxylase inhibitors	• Carbidopa • Benserazide
4	COMT inhibitors	• Tolcapone (central + peripheral action, highly hepato toxic, causing rhabdomyolysis and severe diarrhea) • Entacapone (only peripheral)
5	MAO-B inhibitors	• Selegiline • Rasagiline • Safenamide
6	Dopamine facilitator	• Amantadine
7	Central anticholinergics	• Trihexyphenidyl (Benzhexol) • Benztropine • Biperiden • Procyclidine • Promethazine • (H1 antihistaminic) • Orphenadrine • (H1 antihistaminic)

ONE-LINERS:

- Single most effective drug for PD – **Levodopa**.
- Percentage of administered levodopa penetrating the blood-brain barrier – **1-3%**.
- Oral bioavailability of levodopa can be decreased by – **Pyridoxine**.
- Drug used to treat vomiting due to levodopa- Domperidone.
- Somnolence is a side effect of – **Dopamine receptor agonists**.
- **Chronic therapy of levodopa may cause on and off phenomina.**
- **Apomorphine (D4 agonist) useful in treatment of of phenomina.**
- **Levodopa contra indicated in malignant melanoma (levodopa is precursor or melanin).**
- Peripheral vasospasm can occur (**ERYTHROMELALGIA**) as a side effect of – **Bromocriptine**.
- Dopamine receptor agonist withdrawn due to **cardiac valvular fibrosis** – **Pergolide**.
- Drug available as a transdermal patch applied over cranium – **Rotigotine**.
- Vomiting due to apomorphine (useful in off phenomina) can be prevented or treated by – **Trimethobenzamide**.
- Orange discolouration of urine can be seen with – **COMT inhibitors**.
- Livedo reticularis is a side effect of – **Amantadine and Bromocriptine**.
- Ropinirole useful in restless leg syndrome.

Drugs useful for on and off phenomina

- Entacapone (COMT inhibitor)
- Selegiline (MAO-B inhibitor)
- Istradefylline – adenosine (A2) block
- Rescue therapy – S/C APOMORPHINE (D4 agonist)

Newer Drug

Pemavanserine - 5HT_{2A} inverse agonist useful for treatment of Parkinson disease psychosis

Drugs for Multiple Sclerosis (MS):

	Drug	Mechanism of action
1	Interferons (IFN β -1a, IFN β -1b)	Bind to VLA-4 on T cells ↓ Interfere with adhesion of T cells to endothelium and decrease T cell activation
2	Glatiramer acetate	Induces Th2 cells ↓ Mediates bystander suppression of inflammation

3	Mitoxantrone (antibiotic anti cancer drug)	Intercalates DNA and suppresses humoral and cell-mediated responses
4	Natalizumab ADR- Progressive multifocal leuco encephalopathy	Monoclonal antibody against $\alpha_4\beta_1$ integrin ↓ Inhibits T cell trafficking into CNS
5	Fingolimod ADR- bradycardia	Sphingosine-1-phosphate receptor agonist ↓ Sequesters lymphocytes in lymph nodes
6	Teriflunomide (active metabolite of leflunomide)	Dihydroorotate dehydrogenase inhibitor ↓ Inhibits de novo pyrimidine synthesis
7	Dalfampridine	Kv1 potassium channel blocker ↓ Increases duration of neurotransmitter release at the neuromuscular junction
8	Dimethyl fumarate	Activates Nrf-2 pathway

ONE-LINERS:

- Only drug useful in secondary progressive multiple sclerosis (SPMS) – **Mitoxantrone**.
- Drug found to improve walking in patients with multiple sclerosis – **Dalfampridine**.
- Decreased progression of disease on EDSS seen with – **Mitoxantrone and Natalizumab**.
- Natalizumab given once in a month, not more than 18 months (due to risk of progressive leuco encephalopathy).
- Drug useful for **Huntington's chorea and Tourette syndrome** – **Tetrabenazine (dopamine depletory)**. **Chlorpromazine, Haloperidol (D2 antagonists)**.

Huntington's chorea

Treatment D2 blockers	DOC
Chlorpromazine Haloperidol	Tetrabenazine Deutetrabenazine (dopamine depletor)

Tourette syndrome Associated Tics

Treatment Alpha 2 agonist-	Non Responders
Clonidine Guanafacine	Atypical antipsychotics (Aripipraole)

Drugs useful for Amyotrophic lateral sclerosis

Riluzole- NMDA blocker

Edaravone- anti oxidant

New Drugs under Investigation for Alzheimer's Disease (AD):

	Drug	Mechanism of action
1	Etazolate	Positive allosteric modulator of the GABA _A receptor at the barbiturate binding site; Adenosine receptor antagonist; Selective PDE4 inhibitor.
2	Semagacestat	γ -secretase inhibitor (Trials have been abandoned due to lack of efficacy).
3	Bapineuzumab	Antibody against A β ; Binds to A β and helps in its clearance (Trials have been abandoned again due to lack of efficacy).
4	Gantenerumab	Also an antibody against A β ; Binds to A β and helps in its clearance.
5	Aducanumab	Antibody against aggregated forms of β -amyloid; helps in amyloid clearance.



Worksheet

To do

Drug useful for Parkinson disease psychosis	
Drugs useful for restless leg syndrome	
Drugs useful for on and off phenomena	
Anti viral drug useful in Parkinson disease	
Antioxidant useful in ALS	
MABs useful in multiple sclerosis	

Concept 8.5 : Antipsychotic Drugs

Learning objectives:

- To know the classification of typical antipsychotic drugs
- To know the actions and adverse effects of typical antipsychotic agent
- To know the examples for atypical antipsychotic agents
- To know the uses and adverse effects of atypical antipsychotic agents
- To know the comparison between typical and atypical antipsychotic agents

Time Needed

1 st reading	40 mins
2 nd reading	25 mins

Examples of Typical Antipsychotics (Neuroleptics):

1. Phenothiazines	
Aliphatics	• Chlorpromazine • Trifluorpromazine
Piperidines	• Thioridazine • Mesoridazine
Piperazines	• Trifluoperazine • Prochlorperazine • Fluphenazine
2. Butyrophenones	• Haloperidol • Droperidol • Trifluoperidol
3. Diphenylbutylpiperidines	• Penfluridol • Pimozide
4. Thioxanthenes	• Chlorprothixene • Thiothixene • Flupenthixol • Zuclopenthixol
5. Dihydroindoles	• Molindone
6. Dibenzoxapines	• Loxapine

Typical antipsychotic drugs actions

D2 block	Anti histaminergic action	Anticholinergic action	Alpha blocking action
Anti psychotic action, anti emetic action Side effects EPS galactorrhoea	Side effects Sedation Increase in appetite, causing weight gain Lowering seizure threshold, causing seizure	Side effects like Dryness of mouth, constipation, urinary retention	Side effects like Hypotension

ONE-LINERS:

- Antipsychotic used in Gilles de la Tourette syndrome – **Haloperidol**.
- Neurolept anesthesia – **Droperidol + Fentanyl**.
- Autonomic side effects are maximum with – **Chlorpromazine** (due to maximum anticholinergic action).
- Autonomic side effects are minimum with – **Haloperidol** (due to minimum anticholinergic action).
- Most potent D2 blocking antipsychotics are- butyrophenones (haloperidol).
- Mydriasis is maximum with – **Thioridazine**.
- Impaired ejaculation is seen with – **Thioridazine**.
- Ocular side effects maximum with – **Thioridazine**.
- Sedation is maximum with – **Chlorpromazine**.
- Seizures is maximum with – **Chlorpromazine**.
- Cholestatic jaundice side effect of – **chlorpromazine**.
- Blue-gray metallic discolouration of skin is seen with – **Chlorpromazine**.

Examples of Extrapyramidal Reactions:

- Parkinsonian features (Drug-induced Parkinsonism).
- Akathisia (most common type of EPS).
- Rabbit syndrome (peri-oral tremors).
- Acute dystonia.
- Tardive dyskinesia.
- Neuroleptic malignant syndrome.

ONE-LINERS:

- Extrapyramidal side effects maximum with – **Haloperidol** (due to minimum anticholinergic action).
- Extrapyramidal reaction which may worsen with drug discontinuation – **Tardive dyskinesia**.
- Most fatal of the extrapyramidal reactions – **Neuroleptic malignant syndrome**.
- Specific treatment of drug- induced parkinsonism – **Central anticholinergics**.
- Specific treatment of akathisia –
- **Propranolol**.
- Specific treatment of neuroleptic malignant syndrome – **Dantrolene and Bromocriptine**.
- **VALBENAZINE (VMAT inhibitor)- approved for Tardive dyskinesia**.

Examples of Atypical Antipsychotics:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Clozapine. • Olanzapine. • Risperidone. • Paliperidone. • Sulpride. | <ul style="list-style-type: none"> • Amisulpride. • Levosulpride. • Quetiapine. • Ziprasidone. • Aripiprazole. |
|---|---|

Asenapine

Paliperidone – active metabolite of risperidone, Approved for schizoaffective disorder

Atypical antipsychotic drug causing metabolic syndrome	Atypical antipsychotic drug no risk of metabolic syndrome
• Clozapine • Olanzapine • Quetiapine	• Ziprasidone • Aripiprazole • Asenapine

Long-acting injectable preparations

(risperidone, paliperidone, olanzapine, aripiprazole)

Drug	Adverse effects
Clozapine	Granulocytosis, seizure, cataract, myocarditis, metabolic syndrome
Quetiapine	Cataract, priapism, metabolic syndrome
Olanzapine	Metabolic syndrome, EPS
Risperidone	EPS
Ziprasidone	QT prolongation

Worksheet

To do

Most potent typical antipsychotic	
Most oculo toxic antipsychotic drug	
Atypical antipsychotic drug causing metabolic syndrome	
Atypical antipsychotic causing long QT	
Antipsychotic drug given sublingually	
Atypical antipsychotic drug useful in schizoaffective disorder	

Concept 8.6 : Anti Manic Drugs

Learning objective

- To know the drugs useful for bipolar disorder
- To know the adverse effects and contra indications of lithium

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Drug useful for bipolar disorder:

Drugs useful for prophylaxis (maintenance)	Drugs for mania	Drugs for depression
Lithium (has anti suicidal activity)	valproate, Carbamazepine, Haloperidol, olanzapine, quetiapine, ziprasidone, risperidone, Aripiprazole, diazepam, Endoxifen	Lamotrigine

Therapeutic Plasma Levels of Lithium:

Maintenance of bipolar disorder	0.5 – 0.8 mEq/L
Acute mania	0.8 – 1.2 mEq/L
Toxicity	> 1.5 mEq/L
Hemodialysis is effective at	> 4 mEq/L

Adverse effects	Contra indications	Drug interactions
Leucocytosis Tremor (most common) Hypothyroidism (inhibits release of T3 & T4 from thyroid follicle) Increase urination(treated by amiloride a Enac inhibitor) Teratogenicity (Epstein anomalies) Cardio toxicity Aggravation of psoriasis	Pregnancy Sick sinus syndrome	NSAID, DIURETICS aggravates lithium toxicity Lithium aggravate the action of non depolarising SMR

ONE-LINERS:

- DOC for bipolar disorder – **Lithium**.
- DOC for rapid cyclers of bipolar disorder – **Valproate**.
- Lithium is used in Felty syndrome as it – **Increases neutrophil count**.
- **Acute mania**- valproate, **Carbamazepine**, haloperidol, olanzapine, aripiprazole, diazepam.
- **Depressive phase** - lamotrigine.
- **Half life of lithium is 20** - 24hrs, lithium should be stopped 2 days before surgical procedure

Incidences of Various Side Effects of Lithium:

1	Overt hypothyroidism	7 – 10%
2	Reversible T wave flattening	~ 20%
3	Polyuria / Diabetes insipidus	~ 60%
4	Fine tremors	15 – 70%

- Amiloride is the DOC for treating
- lithium induced DI.
- Lithium contraindicated in pregnancy and in sick sinus syndrome.

ONE-LINERS:

- Frequency of fine tremors due to lithium – **8-12 Hz**.
- Teratogenic effect of lithium – **Ebstein's anomaly**.

Worksheet

To do

Half life of lithium	
Anti epileptic drugs useful for mania	
Most common side effect of lithium	
Drug of choice for lithium induced DI	
Contra indication for lithium	
Toxic plasma level of lithium	
Drug aggravating lithium toxicity	

Concept 8.7 : Antidepressants

Learning objectives:

- To the classification of antidepressants
- To know the uses and adverse effects of SSRI, TCA and other reuptake inhibitors
- To know the uses and adverse effects of MAO inhibitors
- To know the newer antidepressants

Time Needed

1 st reading	40 mins
2 nd reading	25 mins

Classification

1	Tricyclic antidepressants	• Imipramine (useful in nocturnal enuresis, DOC-desmopressin). • Trimipramine. • Clomipramine –approved for OCD. • Desipramine. • Amitriptyline- useful in prophylaxis of chronic migraine, peripheral neuropathy. • Nortriptyline. • Protriptyline. • Doxepin- has high antihistamine action useful to control itching in atopic dermatitis, lichen simplex. • Dothiepin.
2	Tetracyclic antidepressants	• Mianserin. • Maprotiline. • Amoxapine.
3	Bicyclic antidepressants	• Viloxazine.
4	Selective serotonin reuptake inhibitors (SSRIs)	• Fluoxetine- longest acting. • Fluvoxamine. • Paroxetine. • Sertraline. • Citalopram. • Escitalopram- highly selective SSRI. • Dapoxetine.
5	Serotonin norepinephrine reuptake inhibitors (SNRI)	• Venlafaxine (side effect- sustained hypertension) • Milnacipram. • Duloxetine (useful in stress incontinence, fibromyalgia, diabetic neuropathy pain). • Levo-milnacipram, vilazodone. • vortioxetine.

6	Norepinephrine serotonin reuptake enhancer (NSRE)	• Tianeptine.
7	Norepinephrine dopamine reuptake inhibitor (NDRI)	• Bupropion (useful in smoking control, causing weight loss- useful in obesity, important adverse effect- seizure).
8	Noradrenergic and specific serotonergic antidepressant (NaSSA)	• Mirtazapine.
9	Serotonin antagonists and reuptake inhibitor (SARI)	• Trazodone. • Nafazodone.
	Serotonin antagonists and reuptake enhancer	• Tianeptine. • Amineptine.
10	Norepinephrine antagonists and reuptake inhibitors (NARI)	• Reboxetine.
11	Monoamine oxidase (MAO- A) inhibitors	• Moclobemide. • Clorgiline.
12	Herbal product (St John's wort) Active compound – Hyperphorin	• Inhibits the reuptake of NE, 5HT.

SSRI-

SSRI	USES	Adverse Effects	Drug interaction
- Fluoxetine- longest acting (least chance of causing withdrawal manifestation) - Fluoxetine- shortest acting - Paroxetine- may cause weight gain, approved for post menopausal tension syndrome, teratogen - Citalopram- may cause QT prolongation - Escitalopram- highly selective SSRI - Sertraline- safe in elderly person and cardiac patients - Depoxetine – approved for treating premature ejaculation	- Depression - OCD - Anxiety - Post traumatic stress disorder - Premature ejaculation - Bulimia nervosa - Post menopausal stress syndrome	- Nausea, vomiting, diarrhoea - Insomnia - Anxiety - Delay in orgasm - bleeding	- Risk of causing serotonin syndrome when given with MAO inhibitor - Fluoxetine is the CYP2D6 inhibitor, thereby interfere the activation of tamoxifen

TCA-

TCA	MOA & other actions	Uses	Adverse effects	TCA poisoning symptoms	TCA poisoning treatment
Clomipramine Imipramine Doxepine Amitriptyline Nortriptyline Amoxapine Desipramine Meproteline Reboxetine	MOA- inhibits reuptake of both NE & 5HT Other action Alpha blocking, anti cholinergic, and anti histamergic action	Depression Nocturnal enuresis (imipramine) Lichen simplex, atopidermatitis-doxepine Diabetic neuropathy, migraine prophylaxis-amitriptyline Psychosis-amoxapine	Sedation Weight gain Seizure Cardiotoxicity Dryness of mouth, constipation, urinary retention Postural hypotension	Altered level of consciousness Cardiotoxicity Convulsion Metabolic acidosis	Sodium bicarbonate Lignocaine or bretylim for treating arrhythmia (don't use class Ia drug for treating TCA induced arrhythmia) Diazepam for treating seizure No role for dialysis No specific antidote for TCA poisoning

SNRI

SNRI	USES
Venlafaxine- Duloxetine, Milnacipram Levo-milnacipram, vilazodone, vortioxetine	Depression Duloxetine- Neuropathy pain Fibromyalgia Stress incontinence

NDRI

NDRI	USES
Bupropion	Depression ADHD Obesity Smoking control
Solriamfetol	Narcolepsy

Other Anti Depressants

Serotonin reuptake enhancer	Presynaptic alpha 2 antagonist	Presynaptic alpha 2 & 5HT1 antagonist	Herbal medicine
Tianeptine, amineptine	Mianserine	Mirtazapine	At John's wort

ONE-LINERS:

- Longest acting SSRI – **Fluoxetine** (due to formation of an active metabolite).
- SSRI having **least chance of producing discontinuation** symptom on withdrawal-fluoxetine(due to longer duration of action).
- SSRI least likely to cause withdrawal symptoms – **Fluoxetine** (due to formation of an active metabolite).
- Antidepressant with antipsychotic properties – **Amoxapine**.
- Antidepressant used for aggression in elderly – **Trazodone**.
- DOC for endogenous depression – **SSRI**.
- DOC for neurotic disorders – **SSRI**.
- SSRI are drug of choice for- **OCD, PTSD, anxiety with panic attack**.
- DOC for acute exacerbations of neurotic diseases – **Benzodiazepines**.
- Avoid class Ia anti arrhythmic drug for treating TCA induced cardiotoxicity.
- FDA approved TCA for OCD is **clomipramine** but DOC is SSRI.

Drugs that can precipitate serotonin syndrome:

Drugs that increase serotonin synthesis	Drugs that increase serotonin release	Drugs inhibit reuptake of 5HT	Drugs inhibit metabolism of 5HT	Drugs that act as serotonin agonists
Tryptophan	Amphetamines Cocaine Ecstasy (Methylenedioxy methamphetamine acid) Sibutramine	Dextromethorphan Pentazocine Pethidine SSRIs SNRIs TCAs SARIs Tramadol	MAO inhibitors	Buspirone Tryptans Ergot alkaloids Lithium LSD Piperazine

MAO inhibitors

Selective MAO A inhibitors	Selective MAO-B inhibitors	Nonselective MAO inhibitors
Moclobamide Clorgiline	Selegiline Rasagiline Safinamide	Phenelzine Tranylcypromine Isocarboxazid

ONE-LINERS:

- **Selective MAO B inhibitors useful for Parkinson disease**
- **MAO inhibitors aggravates the adverse effects of pethidine**

Drugs that can precipitate cheese reactions with MAOi

- | | |
|--|--|
| <ul style="list-style-type: none">• Amphetamines• Ephedrine• Pseudoephedrine• Phenylephrine | <ul style="list-style-type: none">• Phenylpropanolamine• Tyramine• Linezolid• INH |
|--|--|

Worksheet

To do

Longest acting SSRI	
SSRI approved for premature ejaculation	
TCA approved for OCD	
TCA approved for nocturnal enuresis	
Uses of duloxetine	
Antidepressant useful for controlling smoking	
Most anxiogenic antidepressant	
MAO inhibitor useful for treating depression	

Concept 8.8 : Anti anxiety drugs

Learning objectives:

- To know the drugs useful for various type of anxiety disorder

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Examples of Anti-anxiety Drugs:

1	SSRIs- Fluoxetine, paroxetine, and sertraline
2	SNRIs- Venlafaxine
3	Benzodiazepines- Clonazepam /alprazolam/lorazepam
4	Azapirones: 5HT _{1A} AGONIST- non sedative, non habit forming
	• Buspirone • Gepirone • Ipsapirone
5	H ₁ antihistaminics
	• Hydroxyzine (cetirizine is the active metabolite)
6	β-blockers
	• Propranolol (DOC for performance anxiety)

ONE-LINERS:

- In general **Anti depressant are the – FIRST LINE drugs useful for treating anxiety disorder**

Worksheet

To do

I general first line drugs useful for anxiety disorder are	
Preferred drug for performance anxiety	
Anti histamine having anti anxiety activity	
5HT1A agonist useful for anxiety disorder are	

Concept 8.9 Opioids

Learning objective:

- To know the action of opioid based on opioid receptors
- To know the classification of opioid drugs
- To know the drugs useful for opioid poisoning and opioid deaddiction
- To know newer opioids

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Actions of various opioid receptors:

μ	κ	δ
• Physical dependence (morphine type) • Miosis • Constipation (μ_2) • Analgesia • Supraspinal – μ_1 • Spinal – μ_2 • Respiratory depression (μ_2) • Euphoria • Sedation	• Physical dependence (nalorphine type) • Miosis (lower ceiling) • Constipation • Analgesia • Supraspinal – κ_3 • Spinal – κ_1 • Respiratory depression (lower ceiling) • Dysphoria • Sedation • Diuresis	• Analgesia (spinal + affective supraspinal component) • Respiratory depression • Affective behavior • Reinforcing actions • Constipation

All the actions of morphine may develop tolerance on repeated administration except- **constriction of pupil (miosis), constipation, convulsion.**

Morphine causes convulsion due to formation of metabolite **M3G (morphine 3 glucuronide)**.

Pethidine causes convulsion due to formation of **NORPETHIDINE**.

Morphine has **histamine releasing property** – causing vasodilatation – useful in LVE.

Classification

Pure agonist

Natural	Semi synthetic	Synthetic
• Morphine. • Codeine. • Thebaine. • Papaverine. • Noscipine.	• Heroin (diacetyl morphine). • Pholcodeine. • Levorphanol.	• Pethidine (Meperidine). • Methadone (longest acting). • Fentanyl. • Dextropropoxyphene. • Tramadol. • Tapentadol.

Opioid agonists and antagonists:

Drug	μ	κ	δ
Morphine	Agonist (Strong)	Agonist (Weak)	Agonist (Weak)
Nalorphine	Antagonist (Strong)	Agonist (Moderate)	-
Pentazocine	Partial ago- nist (Weak)	Agonist (Moderate)	-
Butorph- anol	Partial ago- nist (Weak)	Agonist (Strong)	-
Buprenor- phine	Partial agonist	Antagonist (Moderate)	-
Naloxone	Antagonist (Strong)	Antagonist (Moderate)	Antagonist (Weak)
Naltrexone	Antagonist (Strong)	Antagonist (Strong)	Antagonist (Weak)

Peripheral opioid antagonist

- Methyl Naltrexone - Alvemopan	- Naloxegol - Naldemedine
--	--

Potency comparison of opioids:

Morphine	1
Codeine	0.1
Pethidine	0.1
Heroin	3
Fentanyl	80-100
sufentanyl	1000 times > than morphine

Half-lives of various opioids:

Drug	Half-life
Remifentanyl (rapid metabolism by pseudocholinesterase- useful in day care surgery)	8-20 min
Alfentanil	1-2 h
Morphine	3-4 h
Fentanyl	4-5 h
Sufentanil	4-5 h
Methadone	24-36 h

Drugs for Opioid Deaddiction:

Deaddiction	To treat withdrawal symptom	To prevent relapse
- Methadone - buprenorphine	- Clonidine - Lofexidine	Naltrexone

ONE-LINERS:

- **ALVIMOPEN**- peripheral opioid antagonist- useful for treating opioid induced constipation and for treatment of Post op ileus.

Fentanyl + Droperidol = Neuroleptic analgesia

Fentanyl + Droperidol + N2O = Neuroleptic anaesthesia post op truncal rigidity (max alfentanil)

Thorax muscle rigidity- wooden chest syndrome

Newer opioids

- Peripheral κ - antagonist **ASIMADOLINE** – useful in IBS.
- Peripheral μ and κ - agonist, delta antagonist- **Eluxadoline** -useful in IBS
- Peripheral κ - antagonist-**NALFURAFINE** – Anti pruritic- CKD.

Worksheet

To do

Opioid contra indicated for treating MI pain	
Example for mu agonist kappa antagonist	
Day analgesic useful in surgery	
Peripheral opioid antagonists	
Drugs useful for deaddiction of opioid	
Ant diarrhoeal opioid	
Cough suppressant	

Concept 8.10 Newer drugs:**Time Needed**

1 st reading	10 mins
2 nd reading	5 mins

	Drug	Mechanism	Indication
1	Topiramate	Predominantly blocks sodium channels	Partial onset and primary generalized tonic-clonic seizures; Lennox-Gastaut syndrome.
2	Brivaracetam	Binds to synaptic vesicle 2A (SV2A) glycoprotein	Partial seizures in patients >16 years of age.
3	Brexipiprazole	Atypical antipsychotic – SDAM (Serotonin-dopamine activity modulator)	Schizophrenia; Depression.
4	Pimavanserin	Inverse agonist at 5-HT _{2A} > 5-HT _{2C}	Hallucinations and delusions associated with Parkinson's disease.
5	Suvorexant	Blocks both orexin receptors (OX ₁ and OX ₂)	Insomnia.
6	Tasimelteon	Melatonin (MLT-1 and MLT-2) receptor agonist	Non 24-hour sleep-wake disorder in totally blind.
7	Alemtuzumab	Monoclonal antibody against CD52	Relapsing multiple sclerosis.
8	Daclizumab	Monoclonal antibody against CD25	Relapsing multiple sclerosis.
9	Droxidopa	Synthetic amino acid precursor (metabolized to norepinephrine and epinephrine)	Neurogenic orthostatic hypotension; Hemodialysis-induced hypotension.
10	Eteplirsen	Triggers excision of exon 51 of dystrophin mRNA, restoring the reading frame of dystrophin	Duchenne muscular dystrophy.

CONCEPTS

- ⇒ **Concept 9.1 Pituitary hormones**
- ⇒ **Concept 9.2 Antithyroid hormones**
- ⇒ **Concept 9.3 Antidiabetic drugs**
- ⇒ **Concept 9.4 Corticosteroids**
- ⇒ **Concept 9.5 Sex hormones**
- ⇒ **Concept 9.6 Drugs useful for bone health**

Concept 9.1 : Pituitary Hormones

Learning objectives:

- To know the GnRh analogue and antagonist
- To know the drugs useful for acromegaly and dwarfism

Time Needed

1 st reading	20 mins
2 nd reading	10 mins

GnRH Agonists:	Gn RH Antagonists	ADR
• Leuprolide. • Goserelin. • Busorelin. • Naferelin. • Desorelin. • Histrelin. • Tritprelin.	• Ganirelix. • Cetrorelix. • Abarelix. • Degarelix	Hot flush, loss of libido, impotence, sarcopenia (reduced muscle mass), osteoporosis

Drugs used for treating acromegaly

Growth hormone release inhibitors (S/C)	Octreotide Lanreotide Vapreotide Seglitide
Growth hormone receptor blocker (S/C)	Pegvisomant
Dopamine analogues (oral)	Bromocriptine Cabergoline

Drugs useful in treating dwarfism:

Growth hormone release factors	Sermorelin Hexarelin Tesamorelin
Recombinant GH-	Somatrem Somatropin
recombinant human IGF-1 + IGF binding protein 3	Mecasermin

Tesamorelin: GHRE, Reduces abdominal fatin HIV with lipodystrophy

- **Atosiban**- oxytocin receptor antagonist
- **Ergometrine** – useful in PPH
- **Carboprost** PGF@α- useful in PPH

**ONE-LINERS:**

- DOC for central DI – **Desmopressin**.
- Treatment of nephrogenic DI – **Thiazides and NSAIDs**.
- DOC for SIAD – **Vaptans > Demeclocycline**.
- **Mecasermin** – recombinant IGF-1 approved for growth failure in children with severe primary IGF-1 deficiency (administered subcutaneously).

Concept 9.2 Anti-Thyroid Drugs

Learning objective:

- To know the classification of antithyroid drugs
- To know the drugs useful for thyrotoxicosis

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Classification of anti-thyroid medications

1	Thyroid hormone synthesis inhibitors (Thioamides)
	Propylthiouracil Carbimazole Methimazole
2	Iodide trapping inhibitors
	Thiocyanates Perchlorates Nitrates Fluoroborates
3	Thyroid hormone release inhibitors
	Iodides
4.	Peripheral T ₄ to T ₃ conversion inhibitors
	propranolol, amiodarone, propyl thiouracil, dexamethasone, ipodate
5.	Radioactive iodine

Propylthiouracil v/s Carbimazole / Methimazole:

		Propylthiouracil	Carbimazole / Methimazole
1	Plasma protein binding	High (75%)	Absent
2	V _d	~ 20 L	~ 40 L
3	T _{1/2}	~ 75 min	~ 4 – 6 h
4	Dosing frequency	1 – 4 times a day	1 – 2 times a day
5	Efficacy	Lower	Higher
6	Inhibition of peripheral T ₄ → T ₃ conversion	Present	Absent
7	Use in thyroid storm	Effective	Not effective
8	Side effects	Higher	Lower

ONE-LINERS:

- DOC for Graves' disease – **Methimazole**.
- DOC for hyperthyroidism / thyrotoxicosis in 1st trimester of pregnancy – **Propylthiouracil**.
- DOC for hyperthyroidism / thyrotoxicosis in 2nd / 3rd trimester of pregnancy – **Methimazole > Carbimazole**.
- Most common side effect of antithyroid drugs – **Urticarial papular rash**.
- Most serious side effect of antithyroid drugs – **Agranulocytosis**.

Half-lives of radioactive iodine isotopes:

Isotope	Half-life
¹²³ I	8 days
¹²⁵ I	13 hours
¹³¹ I	60 days

ONE-LINERS:

- Radioactive iodine is contraindicated in – **Pregnancy**.
- Thyroid constipation is seen with – **Iodides**.

Drugs for thyroid storm (Acute thyrotoxic crisis):

- Propylthiouracil.
- Propranolol.
- Hydrocortisone.
- Saturated solution of potassium iodide (SSKI).

ONE-LINERS:

- DOC for thyroid storm (Acute thyrotoxic crisis) – **Propylthiouracil**.

Concept 9.3 : Drugs for Diabetes Mellitus:

Learning objective:

- To know the classification of anti diabetic dugs
- To know the classification insulin analogue
- To know the action and adverse effects of oral antidiabetic drugs
- To know the action adverse effects of newer antidiabetic drugs

Time Needed

1 st reading	1 hr
2 nd reading	30 mins

Classification of anti-diabetic drugs:

Parenteral

1. Insulin	
2. GLP-1 analogues	• Exenatide • Liraglutide
3. Amylin analogues	• Pramlintide

Oral antidiabetic drugs

Sulfonylureas	
Meglitinide analogues	• Repaglinide • Nateglinide
Thiazolidinediones	• Pioglitazone
α -glucosidase inhibitors	• Acarbose • Miglitol • Voglibose
Biguanides (AMPK activators)	• Metformin
DPP-IV inhibitors	• Sitagliptin • Vildagliptin • Saxagliptin • Alogliptin

Classifications of Insulins:

Type	Onset (h)	Peak (h)	Duration (h)	Can be mixed with
A: RAPID / ULTRA-SHORT ACTING				
Insulin lispro	0.2 – 0.3	1 – 1.5	3 – 5	Regular, NPH
Insulin aspart	0.2 – 0.3	1 – 1.5	3 – 5	Regular, NPH
Insulin glulisine	0.2 – 0.4	1 – 2	3 – 5	Regular, NPH



B: SHORT-ACTING				
Regular (soluble) insulin	0.5 – 1	2 – 3	6 – 8	All except insulin glargine / detemir
C: INTERMEDIATE ACTING				
Insulin zinc suspension (Lente)	1 – 2	8 – 10	20 – 24	Regular
Neutral protamine hagedorn (NPH)	1 – 2	8 – 10	20 – 24	Regular
D: LONG ACTING				
Insulin glargine	2 – 4	-	24	None
Insulin detemir	1 – 4	-	24	None

ONE-LINERS:

- 1U insulin activity reduces blood glucose of a fasting rabbit to – **45 mg/dL**
- 1 mg of International Standard of Insulin = **28U**
- **Degludec - longest acting insulin**

- Lente = 7:3 mixture of ultralente (crystalline) and semilente (amorphous) insulin zinc suspension.
- Mixtard (30/70) = 30% regular insulin + 70% isophane insulin.
- Mixtard (50/50) = 50% regular insulin + 50% isophane insulin.
- Actrapid = Regular insulin.

ONE-LINERS:

- DOC for T1DM – **Insulin**.
- All insulin preparations are clear except –
- **Lente and NPH** (which are cloudy).
- Peakless absorption insulins – **Glargine and Detemir**.
- Insulin whose absorption is not affected by exercise – **Glargine**.
- Insulins that cannot be mixed with others –
- **Glargine and Detemir**.
- **Insulin that is stable at acidic pH – Glargine**
- **Fastest subcutaneous absorption of insulin occurs at – Anterior abdominal wall.**
- **Absorption is usually most rapid from the abdominal wall, followed by the arm, buttock, and thigh. If a patient is willing to inject into the abdomen.**
- Most commonly insulin is injected subcutaneously over the – **Thigh** (due to lipodystrophy at the injection site).
- Inhalational insulin – **AFREZZA** was approved in June-July 2014.
- Earlier inhalational preparation Exubera has been withdrawn due to – **Lack of cost- effectiveness.**
- Possible risk of inhalational insulin – **Lung cancer.**

Amino acid modification in various insulin

Insulin	Position	Original amino acid	Substituent amino acid
Lispro	B28	Proline	Lysine
	B29	Lysine	Proline
Aspart	B28	Proline	Aspartate
Glulisine	B3	Aspartate	Lysine
	B29	Lysine	Glutamate
Glargine	A21	Asparagine	Glycine
	B chain		Two additional arginine residues
Detemir	B29	Lysine	Additional myristic acid
Degludec	B29	Lysine	Additional hexadecanedioic acid
	B30	Threonine	Deleted

ONE-LINERS:

- Lispro, aspart, glulisine **are monomer rapid** in onset of action.
- Regular insulin **hexamer**, take 30-40 minutes to become monomer.
- Detemir – Myristic acid is added to insulin → **Increases binding to albumin and hence, duration of action.**
- Duration of action of insulin degludec – **42 hours.**

Drug interactions of insulin:

Action antagonized by (Decreased effect of insulin seen with)	Increased risk of hypoglycemia with
• Thiazides • Loop diuretics • Corticosteroids • Oral contraceptives • β-agonists	• Sulfonylureas • β-blockers • Salicylates • Lithium • Theophylline

ONE-LINERS:

- Most common side effect of insulins – **Hypoglycemia**
- Non-diabetic use of insulin – **Hyperkalemia** (IV insulin + Glucose) → Causes shift of potassium into the cells

Sulfonylureas

Classification:

1 st Generation	2 nd Generation
Longer acting	Shorter acting
Decrease both basal and post-prandial hyperglycemia	Decrease only post-prandial hyperglycemia
Usually given once or twice a day	Usually given before each meal
Greater incidence of hypoglycemia	Lesser incidence of hypoglycemia
E.g: • Tolbutamide • Tolazamide • Chlorpropamide	E.g: • Glibenclamide (Glyburide) • Glimepiride • Glipizide • Gliclazide

ONE-LINERS:

- Longest acting sulfonylurea – **Chlorpropamide**.
- Most common side effect of sulfonylureas – **Hypoglycemia**.
- Sulfonylurea with maximum risk of hypoglycemia – **Chlorpropamide**.
- **Chlorpropamide- cause cholestatic jaundice, SIADH.**
- **Glyburide can be used in GDM**

Pharmacokinetic properties of sulfonylureas

Drug	Plasma T _{1/2} (h)	Duration of action (h)	Daily dose	No. of doses per day
Tolbutamide	6	6-8	0.5-3 g	2-3
Glibenclamide	2-4	24	2.5-15 mg	1-2
Glipizide	3-5	12	5-20 mg	1-2
Gliclazide	8-20	12-24	40-240 mg	1-2
Glimepiride	5-7	24	1-6 mg	1-2

Additional properties of chlorpropamide

- Syndrome of inappropriate antidiuresis (SIAD)
- Agranulocytosis
- Cholestatic jaundice
- Disulfiram-like reaction

Meglitinides

Pharmacokinetics of meglitinide analogues:

Drug	Plasma T _{1/2} (h)	Duration of action (h)	Daily dose	No. of doses per day
Repaglinide	≤1	3-5	1-8 mg	3-4
Nateglinide	1.5	2-4	180-480 mg	3-4

ONE-LINERS:

- Binding site of meglitinide analogues on SUR – **Kir6.1**
- Flu-like symptoms is a side effect of – **Nateglinide**

Thiazolidinediones

Thiazolidinediones withdrawn:

	Drugs withdrawn	Reason
1	Troglitazone	Hepatotoxicity (Acute fulminant hepatitis)
2	Rosiglitazone	Increased risk of precipitation of congestive heart failure

ONE-LINERS:

- Pioglitazone has also been withdrawn in most of the Western countries due to – **increased risk of urinary bladder cancer**

Pharmacokinetics

Drug	Plasma T _{1/2} (h)	Duration of action (h)	Daily dose	No. of doses per day
Pioglitazone	3-5	24	15-45 mg	1

ONE-LINERS:

- Onset of action of thiazolidinediones – **2-3 months.**
- Weight gain is maximum with – **Thiazolidinediones** (due to fluid retention and edema).
- **Thiazolidinediones- activates PPAR γ .**
- **TELMISARTAN – PPAR γ agonism property.**
- Increased risk of pathological fractures is seen with – **Thiazolidinediones.**

Alpha-Glucosidase Inhibitors

- Absorption of miglitol from GIT – **50%**.
- **Only oral antihyperglycemic effective in T1DM – α -glucosidase inhibitors.**
- **But FDA approved drugs for type 1 & 2 DM are- Insulin and pramlintide.**
- Antidiabetic that can reduce fibrinogen levels – **Acarbose.**
- Abdominal distension and flatulence is a side effect of – α -glucosidase inhibitors

Biguanides

Pharmacokinetics of metformin

Drug	Plasma T _{1/2} (h)	Duration of action (h)	Daily dose	No. of doses per day
Metformin	1.5-3	6-8	0.5-2.5 g	1-2

Contraindications to metformin: (factors precipitating lactic acidosis)

- Renal failure (S. creatinine > 1.5 mg/dL for males, 1.4 mg/dL for females).
- Severe liver disease.
- Severe pulmonary disease, including COPD.
- Decompensated heart failure.
- Chronic alcohol abuse.

Metformin to be discontinued if patient is:

- Severely ill.
- Not taking anything orally.
- **Requiring administration of radiological contrast media (Metformin increases the nephrotoxicity of radiological contrast agents).**
- One day before and after the radiological procedure metformin should be stopped.

ONE-LINERS:

- Maximum daily dose of metformin – **2250 mg.**
- **Antidiabetic effective in polycystic ovarian disease – Metformin (It can both regularize menses and increase chances of ovulation).**
- Antidiabetic that may be effective in cancers – **Metformin** (It is under evaluation for breast and colon cancers).
- Metformin is a AMPK activator.
- Megaloblastic anemia is a side effect of – **Metformin.**
- Phenformin has been withdrawn due to – **Lactic acidosis.**

Mechanisms of GLP-1 based action

- Increased insulin secretion from pancreas (glucose-dependent).
- Decreased glucagon secretion from pancreas (glucose-dependent).
- Suppression of appetite.
- Delayed gastric emptying.

ONE-LINERS:

- Exenatide is – **Synthetic exendin-4** derived from the saliva of Gila monster (**Heloderma suspectum**).
- Exenatide shares ~ **50%** homology with human GLP-1.
- **Liraglutide is – recombinant human GLP-1**
- Liraglutide shares ~**97%** homology with human GLP-1.
- GLP-1 analogues cause nausea as a side effect in – **40-50%** of cases at the initiation of therapy.
- Pancreatitis is a side effect of – **GLP-1 analogues**.
- **Increased risk of medullary thyroid carcinoma is seen with – Liraglutide.**
- **GLP=1 analogues will cause weight loss, Liraglutide is FDA approved drug of obesity.**

Pharmacokinetics of DPP-IV inhibitors:

Drug	Plasma T½ (h)	Dura- tion of action (h)	Daily dose
Sitagliptin	~12	24	100 mg
Vildagliptin	2-4	12-24	50-100 mg

ONE-LINERS:

- Reversible DPP-IV inhibitors – **Sitagliptin and Alogliptin.**
- Irreversible DPP-IV inhibitors – **Vildagliptin and Saxagliptin.**
- All DPP-IV inhibitors are metabolized by CYP enzymes in liver except – **Saxagliptin.**
- Recent side effect reported with DPP-IV inhibitors – **Joint pain.**
- **All DPP- IV inhibitors are unsafe in kidney failure except- LINAGLIPTIN.**
- **VILDAGLIPTIN- cause hepatotoxicity.**
- **Most common side effect of DPP-IV inhibitors will be upper respiratory infection.**

Mechanism of action of pramlintide

- Decreased glucagon secretion from pancreas (glucose-dependent).
- Suppression of appetite.
- Delayed gastric emptying.
- Pramlintide acts on amylin receptors in – **Hind brain.**
- Pramlintide is stable at pH of – **4.**
- Pramlintide is approved for – **both T1 and T2 DM**; only in combination with insulin.

Drugs affecting basal and post-prandial hyperglycemia

Both basal and post prandial	Only post-prandial
- Intermediate acting insulins - Long acting insulins 1st generation sulfonylureas - Thiazolidinediones - Biguanides - GLP-1 analogues - DPP-IV inhibitors	- Ultra-short acting insulins - Short-acting insulins 2nd generation sulfonylureas - Meglitinide analogues α-glucosidase inhibitors

Newer antidiabetics

	Class (Mechanism)	Examples
1	Sodium glucose translocase-2 (SGLT-2) inhibitors (Decrease the proximal tubular reabsorption of glucose)	- Canagliflozin - Dapagliflozin - Empagliflozin - Sergliflozin
2	Bile acid binding resins (Decrease the absorption of glucose from the GIT)	Colesevelam
3	D ₂ receptor agonist (Resets the hypothalamic glucose control mechanism and reduce plasma glucose levels)	Bromocriptine
4	Aldose reductase inhibitor (Reduces the conversion of glucose to sorbitol; hence, decreases the occurrence of diabetic complications)	Epalrestat
5	Dual PPAR agonists (Reduce both lipid and glucose levels; hence effective in diabetic dyslipidemia)	Saroglitazar
6	Newer GLP-1 analogues	- Albiglutide - Semaglutide - Taspoglutide - Lixisenatide

ONE-LINERS:

- Recent side effect with SGLT2 inhibitors – **Diabetic ketoacidosis.**
- **Empagliflozin has been specifically approved for – Diabetic neuropathy.**
- **Saroglitazar has been specifically approved for – Diabetic dyslipidemia.**
- Saroglitazar was approved on – **5th June, 2013**
- **Ruboxistaurin** – Protein kinase C- β inhibitor under investigation for diabetic peripheral retinopathy.

Effect of drugs on blood glucose

Drugs causing hyperglycemia	Drugs causing hypoglycemia
• Glucocorticoids • Antipsychotics • Protease inhibitors • β-agonists • Thiazides • Loop diuretics • Phenytoin • Opioids • Diazoxide • Nicotinic acid • Pentamidine • Epinephrine • Interferons • Amphotericin B • L-asparaginase • Acamprosate • Basiliximab • Thyroid hormones	• β-blockers • Ethanol • NSAIDs • Pentamidine • ACE inhibitors • Lithium • Theophylline • Bromocriptine • Mebendazole

Indications of glucagon:

- Hypoglycemia.
- Cardiogenic shock due to β -blockers (stimulates the heart).
- To facilitate radiographic examination of upper / lower GIT (by relaxing the stomach and intestines).

Worksheet

To do

Longest acting insulin analogue	
Insulin prepared in acidic pH	
Safer DPP4 inhibitor safe in renal failure	
Most common side effect of SGLT 2 inhibitor	
Most common side effect of alpha glucosidase inhibitor	
Antidiabetic drug approved for treating obesity	

Concepts 9.4 : Corticosteroids

Learning objective:

- To know the classification of steroids
- To know the side effects and contra indications of steroids

Time Needed

1 st reading	30 mins
2 nd reading	15 mins

Potency Gradation of Systemic Steroids

Classification	Compound	Gluco-corticoid potency	Mineralo-corticoid potency	Equivalent dose (mg)
Short-acting ($T_{1/2} < 12$ h)	Hydrocortisone (Cortisol)	1	1	20
	Cortisone	0.8	0.8	25
Intermediate-acting ($T_{1/2}$ 12-36 h)	Deflazacort	3 – 4	0	6
	Prednisone	4	0.8	5
	Prednisolone	4	0.8	5
	Methylpredni-solone	5	0.5	4
	Triamcinolone	5	0	4
Long-acting ($T_{1/2} > 36$ h)	Dexamethasone	25	0	0.75
	Betamethasone	25	0	0.75
Mineralo-corticoids	Desoxy-corticosterone acetate (DOCA)	0	100	2.5
	Fludricortisone	10	150	0.2
	Aldosterone	0.3	3000	Not used clinically

ONE-LINERS:

- DOCA is administered – **Sublingually.**
- Most potent mineralocorticoid – **Aldosterone.**
- Most potent clinically used mineralocorticoid – **Fludricortisone.**

Potency gradation of topical steroids

Class	Compound and formulation
1 (Most potent)	- Betamethasone dipropionate (cream, ointment: 0.05%). - Clobetasol propionate (cream, ointment: 0.05%). - Halobetasol propionate (ointment: 0.05%). - Diflorasone diacetate (ointment: 0.05%).
2	- Desoximetasone (cream, ointment: 0.25%, gel: 0.05%). - Amcinonide (ointment: 0.1%). - Fluocinonide (cream, ointment, gel: 0.05%). - Halcinonide (cream, ointment 0.1%).
3	- Betamethasone valerate (ointment: 0.1%). - Diflorasone diacetate (cream: 0.05%). - Triamcinolone acetonide (ointment: 0.1%, cream: 0.5%).
4	- Amcinonide (cream: 0.1%). - Desoximetasone (cream: 0.05%). - Fluocinolone acetonide (cream: 0.2%, ointment 0.025%). - Flurandrenolide (ointment: 0.05%, tape 4 µg/cm²). - Hydrocortisone valerate (ointment: 0.2%). - Triamcinolone acetonide (ointment: 0.1%). - Mometasone furoate (cream, ointment: 0.1%).
5	- Betamethasone dipropionate (lotion: 0.05%). - Betamethasone valerate (cream, lotion: 0.1%). - Fluocinolone acetonide (cream: 0.025%). - Flurandrenolide (cream: 0.05%). - Hydrocortisone butyrate (cream: 0.1%). - Hydrocortisone valerate (cream: 0.2%). - Trimacinalone acetnoide (cream, lotion: 0.1%).
6	- Alclometasone dipropionate (cream, ointment: 0.05%). - Desonide (cream: 0.05%). - Fluocinolone acetonide (cream, solution: 0.01%).
7 (Least potent)	- Dexamethasone sodium phosphate (cream: 0.1%). - Hydrocortisone (cream, ointment, lotion: 0.5%, 1%, 2%).

ONE-LINERS:

- Topical steroid is most potent as – **Ointment**.
- For exuding lesions – **Cream** is preferred, as it allows evaporation of the exudates and dries the lesion.
- For dry and scaly lesions – **Ointment** is preferred, as it forms an occlusive cover around the lesion and retains moisture.
- Steroid topical spray is preferred on – **Hairy regions**.

Effects of glucocorticoids on blood cells

Increased levels of	Decreased levels
• RBCs • Platelets • Neutrophils	• Eosinophils • Basophils • Lymphocytes

Side effects of steroids

- Hypothalamo-pituitary adrenal axis suppression
- Cushingoid features – Moon face, Buffalo hump, Lemon-on-sticks (or orange-on sticks) appearance due to lipid redistribution.
- Hyperglycemia.
- Myopathy.
- Osteoporosis.
- Thinning of skin.
- Easy bruisability.
- Telangiectasiae.
- Hypertension.
- Immunosuppression and increased risk of opportunistic infections.
- Peptic ulcer disease.
- Euphoria and psychosis.
- Glaucoma and cataract.
- Topical steroids in eye cause – **Glaucoma**.
- Systemic steroids in eye cause – **Cataract**.

Contra indication of steroids

- Peptic ulcers.
- Diabetes mellitus.
- Hypertension.
- Viral and fungal infections.
- Intestinal tuberculosis.
- Osteoporosis.
- Herpes simplex keratitis.
- Psychosis.
- Epilepsy.
- Congestive heart failure.
- Renal failure.
- Steroids are contraindicated in – **Intestinal tuberculosis**, as they increase the risk of perforation.

Inhibitors of steroid synthesis

- Metyrapone – Inhibits 11- β -hydroxylase.
- Aminoglutethimide.
- Trilostane.
- mitotane.
- High doses of ketoconazole.
- Pesireotide.
- Etomidate.

Maximum glucocorticoid action	Dexamethasone
Maximum mineralocorticoid action	Aldosterone
Glucocorticoid with Maximum Mineralocorticoid action	Hydrocortisone
Least potent Glucocorticoid action	Cortisone
Most potent Glucocorticoid action	Betamethasone
Maximum topical action	Triamcinolone

ONE-LINERS:

- ACTH- drug of choice for Infantile spasm
- Vigabatrin- drug of choice for infantile spasm with tuberous sclerosis

Worksheet

To do

Most potent glucocorticosteroid	
Glucocorticoid having highest mineralocorticoid action	
Example for mineralocorticoid antagonists	
Drugs inhibiting glucocorticoid synthesis	
Example for glucocorticoid receptor blocker	

Concept 9.5 : Gonadal Steroids:

Learning objectives:

- To know the estrogen and progesterone preparations
- To know the different methods of contraception
- To know the uses of selective estrogen receptor modulator & selective progesterone receptor modulators

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Examples of androgens:

Natural (Endogenous)	Synthetic
• Testosterone • Dihydrotestosterone • Dehydroepiandrosterone • Androstenedione • Androsterone	• Methyltestosterone • Fluoxymesterone • Testosterone undecanoate • Mesterolone

Examples of estrogens:

Natural (Endogenous)	Synthetic
• Estradiol • Estriol • Estrone	Steroidal – • Ethinyl estradiol • Mestranol • Tibolone Non-steroidal – • Diethylstilbestrol (Stilbestrol) • Hexestrol • Dienestrol

Examples of progestogens:

Progesterone derivatives	19-nortestosterone derivatives
Older compounds – • Medroxyprogesterone acetate • Hydroxyprogesterone caproate • Megesterol acetate • Dydrogestrone Newer compounds – • Nomegestrol acetate	Older compounds – • Norethisterone (Norethindrone) • Lynestrenol (Ethinyl estrenol) • Allylestranol • Levonorgestrel (It is also included under gonanes) Newer compounds (Gonanes) – • Desogestrel • Norgestimate • Gestodene

ONE-LINERS:

- Gestodene inhibits ovulation at doses as low as – **40µg/d.**
- Examples of pure progesterone antagonists – **Onapristone and Gestinone.**

Examples of oral contraceptive preparations:

Progestin	Estrogen		Trade Name	
Combined Pills				
Norgestrel	300µg	Ethinyl estradiol	30µg	Mala-D (21 tabs + 7 tabs of ferrous sulphate 60mg)
Norgestrel	500µg	Ethinyl estradiol	50µg	Ovral-G
Levonorgestrel	250µg	Ethinyl estradiol	50µg	Ovral, Duoluton-L
Levonorgestrel	150µg	Ethinyl estradiol	30µg	Ovral-L, Ovipauz
Levonorgestrel	100µg	Ethinyl estradiol	20µg	Loette, Ovilow, Combee
Desogestrel	150µg	Ethinyl estradiol	30µg	Novelon
Desogestrel	150µg	Ethinyl estradiol	20µg	Femilon
Phased Pills				
Levonorgestrel	50-75-125µg	Ethinyl estradiol	30-40-30µg	Triquilar (6+5+10)
Norethindrone	500-750-1000µg	Ethinyl estradiol	35-35-35µg	Orthonovum 7/7/7 (7+7+7)
Post-Coital Pills				
Levonorgestrel	250µg	Ethinyl estradiol	50µg	Ovral, Duoluton-L (2+2 tablets)
Levonorgestrel	750µg, 1500µg	-	-	Norlevo, Ecee2 (1+1 tablets) iPill, Nofear-72, Oh God (1 tablet)
Mini Pills (Progesterone-Only Pills)				
Norethindrone	350µg	-	-	Micronor, Nor-QD
Norgestrel	75µg	-	-	Ovrette

Effects of different forms of contraception:

Effect	Combined E+P	Minipill (Only P)	Post-coital pill (Only P)	Injections (Only P)
FSH inhibition	++	-	-	+
LH inhibition	+++	+	+	+++
Anti-ovulatory effect	+++	+	+, -	++
Hostile cervical mucus	+++	+++	-	+++
Endometrium	Hyper-secretory	Out of phase	Unfavour-able	Atrophic
Failure rate (pregnancy per 100 women years)	0.1-0.3	2-3	2-4	< 0.5
Contraceptive efficacy	++++	+++	++	++++

ONE-LINERS:

- **Gossypol** – Non-steroidal male contraceptive obtained from cotton seed, studied in China.
- **Ulipristal** – Selective progesterone receptor modulator (SPRM) approved as an emergency contraceptive – 30 mg taken within 72 hours of intercourse.
- **Bazedoxifene** – Third generation SERM approved in combination with estrogen for prevention and treatment of postmenopausal osteoporosis.

- SERM for breast cancer- **Tamoxifen, Toremifen, Doloxifen, Raloxifene.**
- SERD for breast cancer- **Fulvestrant.**
- SERM for DUB- **Ormeloxifene.**
- **Centchromin**- Contraception (Twice in wk with the gap of four days- first 3 months, Latter - once in a wk).
- **SERMs For Dyspareunia-Ospemifene.**
- **Ulipristal.**
- **Asoprisnil.**
- **Telapristone**- uterine fibroid, endometriosis.

- **SPRM.**
- **Ulipristal**- emergency contraceptive,
- effective upto five days.
- **Asoprisnil.**
- **Telapristone**- uterine fibroid, endometriosis.

Aromatase Inhibitors:

Aminoglutethimide, Formestane

Exemestane

Vorozole

Fadrozole

Letrozole

Anastrozole

Worksheet

To do

SERM useful for dyspareunia	
SERM useful for ovulation induction	
What is ulipristal	
Adverse effects of tamoxifen	
Mechanism of action finasteride	
Examples for aromatase inhibitors	

Concept 9.6 : Drugs useful for maintaining bone health

Learning objectives:

- To know the drugs useful in osteoporosis
- To know action, uses and adverse effects of bisphosphonates
- To know the vit D analogues and their uses
- To know the parathormone analogues
- To know the drugs useful for hypercalcemia

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Inhibiting osteoclast	Promoting osteoblast	Dual action(promoting osteoblast & inhibiting osteoclast)
Bisphosphonates Estrogen & Serm Cinacalcet Calcitonin Thiazide Diuretics Denosumab- Rank L antibody	Calcitriol Calcium Parathormone - teriparatide, Abaloparatide Anti sclerotic antibody Romosozumab Blosozumab	Strontium ranelate

Bisphosphonates

Genera rations	Examples
First generation-	• Medronate • Etidronate • Clodronate • Tiludronate
Sec generation-	• Alendronate • Pamidronate (IV) • Ibadronate
Third generation-	• Risedronate • Zoledronate (IV) (most potent)

Mechanism of action

- Accelerating apoptosis of osteoclast reducing their number
- Detect osteoclast precursors and inhibit their differentiation (by inhibiting IL-6)
- Inhibit G protein involved in activation of osteoclast

Uses	ADR
Osteoporosis Hypercalcemia Paget's Disease Antitumor action (breast cancer)	Erosive esophagitis Flu like symptoms Renal toxicity Osteonecrosis of jaw stress fractures in the lateral cortex of the femoral shaft

Cinacalcet, Etelcalcetide

- Activates the calcium sensing receptor in the parathyroid inhibits PTH secretion indication
- USES-**
- Sec hyper parathyroidism Hypercalcemia

Teriparatide, Abaloparatide

Mechanism of action	ADR	AVOID IN
Recombinant PTH Approved for treatment of severe osteoporosis Low & pulsatile dose (s/c) Stimulate Bone Formation Improves bone density in most of bones except – distal radius	Paresthesia Hypercalcemia Nephrolithiasis Risk of osteosarcoma	Pagets disease Hypercalcemia Thiazides, steroids Risk of osteosarcoma

Vitamin D Analogs

Dihydratachysterol	Hypoparathyroidism
Ergocalciferol (calciferol)	vitamin D deficiency treatment of familial hypophosphatemia, hypoparathyroidism vitamin D-resistant rickets type II
Alpha calcidol	Renal osteodystrophy
Calcipotriol (Calcipotriene)	Psoriasis (topical)
Paricalcitol Doxercalciferol (1α-hydroxyvitamin D2) Maxacalcitol	Sec.Hyperparathyroidism

Uses of vit D

- Hypocalcemia
- CRF, intestinal osteodystrophy
- Nutritional Rickets&osteomalacia
- Hypoparathyroidism
- Osteoporosis
- Fanconi syndrome
- PSORIASIS

Calcitonin

Uses	ADR
Pagets disease (S/C-Salcatonin) Hypercalcemia Osteoporosis (Intra nasal- Salcatonin) Diagnostic marker in medullary ca of thyroid	nausea, flushing, altered taste sensation

ONE-LINERS:

- **BUROSUMAB - Acting against FGF23**
- **USE - X linked hypophosphataemia**

Drug Therapy for Hyper Phosphatemia	Drug Therapy for Hypercalcemia
Sevelamer hydrochloride Sevelamer carbonate Lanthanum carbonate Sucroferric oxyhydroxide Ferric citrate	Hydration with saline Forced diuresis (saline + loop diuretic) Corticosteroids calcitonin Plicamycin IV bisphosphonates (pamidronate, zoledronate) dialysis

Uses of calcium

- hypocalcemic tetany
- Hyperkalemia
- black widow spider envenomation
- MgSO₄ toxicity

Denosumab

Target	uses	Avoid in
Against RANKL	Osteoporosis Giant cell tumor	

Worksheet

To do

Mechanism of action of bisphosphonates	
Target for denosumab	
Analogues PTH	
Diuretic useful for treating hypercalcemia	
Uses of calcitonin	
Burosumab useful in	
Topical vit d useful in psoriasis	

CONCEPTS

- ⇒ **Concept 10.1 Drugs for Peptic Ulcer Disease**
- ⇒ **Concept 10.2 Antiemetics**
- ⇒ **Concept 10.3 Drugs useful for constipation, diarrhea and IBS**

Concept 10.1 : Drugs useful for Peptic Ulcer Disease

Learning objective:

- To know the classification of drugs useful in APD
- To know the action uses and adverse effects of PPIs
- To know the actions adverse effects of H₂ blockers
- To know the drugs useful in H pylori disease

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

1	Proton pump inhibitors (PPIs)	• Omeprazole • Esomeprazole (Esmoprazole) • Rabeprazole • Dexrabeprazole • Lansoprazole • Dexlansoprazole • Pantoprazole • Iloprazole
2	H ₂ receptor antagonists	• Cimetidine • Ranitidine • Famotidine • Roxatidine • Nizatidine
3	Prostaglandin (PGE ₁) analogues	• Misoprostol
4	Anticholinergics	• Pirenzepine • Telenzepine
5	Antacids	• Sodium bicarbonate • Magnesium hydroxide • Magnesium trisilicate • Aluminium hydroxide • Hydroxy magnesium aluminate (Magaldrate) • Calcium carbonate
6	Ulcer protectives	• Sucralfate • Colloidal bismuth subcitrate • Dimethylpolysiloxane • Dimethicone • Simethicone

7	Drugs for <i>H. pylori</i> eradication	• Amoxicillin • Tetracyclines • Clarithromycin • Metronidazole • Tinidazole • Omeprazole • Lansoprazole • Bismuth subcitrate
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Proton Pump Inhibitors (PPIs):

Drug	Dose (mg/d)
Iloprazole	5
Dexrabeprazole	10
Lansoprazole	15-30
Omeprazole	20
Esomeprazole	20
Rabeprazole	20
Dexlansoprazole	30
Pantoprazole	40

ONE-LINERS:

- PPIs are available as enteric coated formulations – **To prevent degradation from gastric acid.**
- Active metabolite of PPIs – **Sulfenamide**
- $T_{1/2}$ of PPIs – **1 – 2 h.**
- Duration of action of PPIs – **1 – 2 days**
- **PPIs are - irreversible inhibition of proton pump - Hit and run drug.**
- Most potent PPI – **Iloprazole.**
- Least potent PPI – **Pantoprazole.**
- Fastest acting PPI – **Rabeprazole.**
- DOC for NSAID-induced ulcer – **PPIs.**
- Specific drug for NSAID-induced ulcer – **Misoprostol.**
- Teratogenic effect associated with misoprostol – **Mobius syndrome.**

H₂ Receptor Antagonists: Intravenous doses of H₂ Receptor Antagonists

	Cimetidine	Ranitidine	Famotidine
Intermittent bolus	300 mg every 6-8 h	50 mg every 6-8 h	20 mg every 12 h
Continuous infusion	37.5-100 mg/h	6.25-12.5 mg/h	1.7-2.1 mg/h

ONE-LINERS:

- Ranitidine is – **5 times** more potent than cimetidine.
- Famotidine is – **5-8 times** more potent than ranitidine.
- Ranitidine absorption is reduced by – **Antacids and Sucralfate**.
- DOC for stress ulcers – **Ranitidine**.
- Drug interactions among H₂ blockers are maximum with – **Cimetidine**.
- Cimetidine – **microsomal enzyme inhibitor**.
- Cimetidine – **causes gynecomastia**.
- **H₂ antagonist maximally reduces nocturnal gastric acid output (better given at night time).**
- **H₂ antagonist undergoes renal route of excretion.**

Antacids: 1g of Antacid Neutralizes How Much of mEq of HCl

Antacid (1g)	Neutralizes HCl (mEq)
Magnesium hydroxide	30
Magaldrate	28
Calcium carbonate	20
Sodium bicarbonate	12
Sodium citrate	10
Magnesium trisilicate	10
Aluminium hydroxide	1 – 2.5

ONE-LINERS:

- Encephalopathy is a side effect of – **Aluminium salts**.
- Milk alkali syndrome can occur due to excessive ingestion of – **Calcium carbonate**.
- **Ulcer Protective-Sucralfate (Acts only in acid medium pH below 4), BISMUTH (can cause black stool, tongue)**
- **Ulcer Healingng- Carbenoxolone- Epithelisation of ulcer without- decreasing acid production.**

Drugs Active Against *H.pylori*

• Amoxicillin • Tetracyclines • Clarithromycin • Levofloxacin • Metronidazole	• Tinidazole • Omeprazole • Esomeprazole • Lansoprazole • Colloidal bismuth subcitrate (CBS)
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Regimens Against *H.pylori*:

1st line: PPI-based Triple Therapy × 10-14 days:

PPI – Standard dose OD/BD
Clarithromycin – 500 mg BD
Amoxicillin – 1 g BD OR Metronidazole 500 mg BD

OR

Bismuth-based Quadruple Therapy × 10-14 days:

PPI – Standard dose OD/BD OR H ₂ receptor antagonist – Standard dose OD/BD
Bismuth subcitrate – 525 mg QID
Metronidazole – 250-500 mg QID
Tetracycline – 500 mg QID

OR

Sequential Therapy × 10 days

Days 1 – 5:
PPI – Standard dose OD/BD + Amoxicillin – 1 g BD
Days 6 – 10:
PPI – Standard dose OD/BD + Metronidazole – 250-500 mg BD + Clarithromycin – 250-500 mg BD

2nd line:

If Amoxicillin is used in 1st line → **Bismuth- based quadruple therapy × 10-14 days** (same as above)

If metronidazole is used in 1st line → **Levofloxacinbased triple therapy × 10-14 days**

PPI – Standard dose OD/BD
Amoxicillin – 1 g BD
Levofloxacin 250 mg BD



Worksheet

To do

PPI having interaction with clopidogrel	
Adverse effects of PPIs	
DOC for stress ulcer	
H2 blocker having microsomal enzyme inhibitory action	
Anti ulcer drug causing black colour stool is	

Concept 10.2 : Antiemetics

Learning objective:

- To know the classification of anti emetics
- To know the examples for prokinetic agents

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Classification of Antiemetics

1	5-HT ₃ antagonists	• Ondansetron • Granisetron • Tropisetron • Dolasetron • Palonosetron • Ramosetron
2	D ₂ antagonists	• Metoclopramide • Domperidone • Itopride
3	Anticholinergics	• Hyoscine (Scopol- amine) • Dicyclomine
4	Antihistaminics	• Promethazine • Diphenhydramine • Dimenhydrinate • Doxylamine • Cyclizine • Meclizine • Cinnarizine
5	Neuroleptics	• Chlorpromazine • Trifluopromazine • Prochlorperazine • Haloperidol
6	NK-1 receptor antagonists	• Aprepitant • Fosaprepitant • Netupitant
7	CB-1 receptor agonists	• Dronabinol • Nabilone

8	5-HT ₄ agonists	• Cisapride • Mosapride • Tegaserod
9	Miscellaneous	• Dexamethasone • Benzodiazepines

ONE-LINERS:

- Highly selective 5HT₃ antagonist – **Palonosetran**.
- Most common side effect of ondansetran – **Headache**.
- 5-HT₃ antagonists are **not** effective in – **Apomorphine-induced vomiting and Motion sickness**.
- Metoclopramide has additional – **5-HT₄ agonistic and 5-HT₃ antagonistic actions**.
- Most common drug for treating anticancer drug induced vomiting – **ondansetran + dexamethasone**.
- Anti emetic used for treating delayed vomiting due to anticancer drug – **aprepitant**.

Prokinetics: Examples of Prokinetics:

1	D ₂ antagonists	• Metoclopramide • Domperidone • Itopride
2	5-HT ₄ agonists	• Cisapride • Mosapride • Tegaserod • Prucalopride
3	5-HT ₃ antagonists	• Ondansetron • Granisetron • Tropisetron • Dolasetron • Palonosetron • Ramosetron
4	Miscellaneous Motilin receptor stimulant	• Erythromycin

Worksheet

To do

5HT3 antagonist having longer duration	
NK1 blockers useful for treating anticancer drug induced vomiting are	
Adverse effect of metoclopramide	
Antibiotic having prokinetic activity	

Concept 10.3 : Drugs for Diarrhoea, Constipation and IBS

learning objective:

- To know the drugs useful in treating diarrhoe and constipation
- To know the drugs useful in IBS and IBD

Time Needed

1 st reading	30 mins
2 nd reading	30 mins

Drugs Useful for Diarrhoea

A: ANTISECRETORY AGENTS	
1	Racecadotril – For short-term management of acute secretory diarrhea.
2	5-ASA agents – For diarrhea due to IBD.
3	Bismuth subsalicylate – For Traveler's diarrhea.
4	Atropine and its derivatives – For nervousness or drug-induced diarrhea.
5	Octotide – For secretory diarrheas.
B: ANTIMOTILITY AGENTS	
1	Loperamide (Opioid μ receptor agonist)
2	Diphenoxylate + Atropine
3	Difenoxin (Metabolite of diphenoxylate)
4	Codeine – For Traveler's diarrhea
5	Paregoric – Camphorated opium tincture for diarrhea in children.
C: ADSORBENTS	
1	Bile acid sequestrants Cholestyramine Colestipol Colesevelam
2	Bulk-forming laxatives Bran Psyllium (Plantago) Methylcellulose Calcium polycarbophil Isapghol

New -WHO Formula ORS (Oral Rehydration Salt):

Content		Concentrations	
NaCl	2.6 g	Na ⁺	75 mM
KCl	1.5 g	K ⁺	20 mM
Trisodium citrate	2.9 g	Cl ⁻	65 mM
Water	13.5 g	Citrate	10 mM
Water	1 L	Glucose	75 mM

ONE-LINERS:

- Atropine is added to diphenoxylate – **To discourage addiction.**

Drugs for Constipation

1	Bulk-forming laxatives	• Bran • Psyllium • Methylcellulose • Calcium polycarbophil • Isapghol
2	Surfactant laxatives	• Docusates • Poloxamers • Lactulose
3	Stimulant laxatives	• Phenolphthalein • Bisacodyl • Sodium picosulfate • Oxyphenisatin
4	Osmotic laxatives	• Magnesium sulfate • Magnesium citrate • Magnesium hydroxide • Milk of Magnesia • Sodium sulfate • Sodium phosphate • Sodium potassium tartarate • Castor oil
5	Anthraquinone derivatives	• Senna • Cascara sagrada

Types of Stools and Latency of Action of Purgatives in Usually Recommended Doses

Soft, formed faeces (takes 1-3 days)	Semifluid stools (takes 6-8 hours)	Watery evacuation (within 1-3 hours)
Bulk-form- ing laxa- tives Docusates Liquid paraffin Lactulose	Phenol- phthalein Bisacodyl Sodium picosulfate Senna	Saline pur- gatives Castor oil

ONE-LINERS:

- Laxative useful in hepatic encephalopathy – **Lactulose**
- Phenolphthalein has been withdrawn due to – **Potential carcinogenicity**
- Oxyphenisatin has been withdrawn due to – **Hepatotoxicity**
- Castor oil is derived from seeds of – **Ricinus communis**

Drugs for Inflammatory Bowel**Disease (IBD):**

1	Glucocorticoids	
2	5-ASA agents	• Sulfasalazine • Mesalazine • Olsalazine • Balsalazide
3	Immunosuppressants	• Azathioprine • 6-Mercaptopurine • Methotrexate • Cyclosporine • Tacrolimus
4	Biologicals (Anti TNF- α)	• Infliximab • Adalimumab • Certolizumabpegol
5	Antibiotics	• Metronidazole • Ciprofloxacin

ONE-LINERS:

- DOC for acute exacerbation of IBD – **Glucocorticoids**.
- Only drug more effective in ulcerative colitis than Crohn's disease – **Sulfasalazine**.
- Active metabolite of sulfasalazine – **5-ASA** (Acts locally).
- **Sulfasalazine** – 5-ASA + Sulfapyridine
- **Mesalazine** – Delayed release preparation of 5-ASA coated with a pH sensitive acrylic polymer.
- **Olsalazine** – 2 molecules of 5-ASA joined by an azo bond
- **Balsalazide** – 5-ASA linked to 4-aminobenzyl- β -alanine

Drugs for Irritable Bowel Syndrome:

A: CONSTIPATION-PREDOMINANT IBS		
1	Bulk-forming laxatives	• Bran • Psyllium • Methylcellulose • Calcium polycar- bophil • Isapghol
2	Osmotic purgatives	• Lactulose syrup • Sorbitol • Polyethylene glycol • Magnesium hy- droxide
3	5-HT ₄ agonists	• Prucalopride
4	Lubiprostone	
B: DIARRHOEA-PREDOMINANT IBS		
1	Loperamide	
2	Cholestyramine	
3	5-HT ₃ antagonist	• Alosetron
C: ABDOMINAL PAIN-PREDOMINANT IBS		
1	Antispasmodics	• Dicyclomine • Hyoscine butyl bromide • Mebeverine
2	TCAs	
3	SSRIs	

ONE-LINERS:

- **Lubiprostone** – **CLC-2 (Type 2 chloride channel activator)**, Cyclic prostaglandin derivative
- **Linaclotide** – (guanylatecyclase-C activator) **Cystic fibrosis transmembrane conductance regulator (CFTR) activator**- increases chloride rich secretion- Increases bowl movement.
- **Crofelemer** - Inhibitor of CFTR, USE- HIV drug induced diarrhoea.

Miscellaneous:

Gallstone Dissolving Drugs:

Chenodeoxycholic acid (Chenodiol)	Ursodeoxycholic acid (Ursodiol)
Acts primarily by inhibiting cholesterol synthesis in the liver	Acts primarily by inhibiting intestinal cholesterol absorption
Raises plasma LDL cholesterol by reducing LDL receptors in the liver	Does not raise plasma LDL cholesterol
Reduces cholesterol secretion in bile after prolonged administration	Reduces cholesterol secretion in bile immediately upon administration
Generates a more litholytic bile acid pool	Itself lowers cholesterol saturation index of bile
Promotes micellarsolubilization of cholesterol	Promotes solubilization of cholesterol by liquid crystal formation

Prebiotics v/s Probiotics

Prebiotics	Probiotics
Special forms of dietary fibers that nourish and enhance the growth of beneficial bacteria in the gut	Live bacteria that are supplemented from outside
Not affected by heat, cold, acid or time	Can be destroyed by heat, cold, gastric acid or storage for prolonged periods

Indications of Prebiotics and Probiotics:

- Acute infectious diarrhea
- Antibiotic-associated diarrhea (Pseudomembranous enterocolitis)
- Neonatal necrotizing enterocolitis
- Lactase deficiency
- Irritable bowel syndrome
- Inflammatory bowel disease
- Celiac disease
- Food protein hypersensitivity
- *H. pylori* infection

Newer drugs

	Drug	Mechanism	Indication
1	Vono- prazan	Potassium-competitive acid blocker	Peptic ulcer disease and reflux esophagitis (approved only in Japan)
2	Vedoli- zumab	Monoclonal antibody against $\alpha_4\beta_7$ integrin	Inflammatory bowel disease in adults

3	Rifaxi- min	Inhibits bacterial RNA polymerase	Diarrhea-predomi- nant irritable bowel syndrome
4	Eluxad- oline	μ and κ opioid receptor agonist and δ -opioid receptor antag- onist	Diarrhea-predomi- nant irritable bowel syndrome
5	Naloxe- gol	Pegylatednal- oxol (opioid receptor antag- onist)	Opioid-induced con- stipation in adults
6	Obeti- cholic acid	Semi-synthetic bile acid ana- logue	Primary biliary cholangitis
7	Yosprala	Fixed drug combination (FDC) of Aspi- rin + Omepra- zole	Patients who require aspirin for second- ary prophylaxis of cardio-vascular and cerebro-vascular events and who are at risk for develop- ing aspirin-associat- ed gastric ulcers



Worksheet

To do

Examples for antimotility drugs	
Lubiprostone mechanism of action	
What is croflemmer	

CONCEPTS

- ⇒ **Concept 11.1** **General considerations**
- ⇒ **Concept 11.2** **Sulfonamides, Cotrimaxazole and Quinolones**
- ⇒ **Concept 11.3** **Cell wall inhibitors (beta lactum antibiotics, glycopeptides)**
- ⇒ **Concept 11.4** **Protein Synthesis Inhibitors
(Aminoglycosides, Macrolides,
Linocsamide)**
- ⇒ **Concept 11.5** **Broad spectrum antibiotics
(Tetracyclines & Chloramphenicol)**
- ⇒ **Concept 11.6** **Antitubercular Drugs**
- ⇒ **Concept 11.7** **Antileprotic Drugs**
- ⇒ **Concept 11.8** **Antifungal Drugs**
- ⇒ **Concept 11.9** **Antiviral Drugs (Retro & Non Retro Viral)**
- ⇒ **Concept 11.10** **Antimalaria Drugs**
- ⇒ **Concept 11.11** **Ant Amoebic Drugs**
- ⇒ **Concept 11.12** **Anthelminthic Drugs**

Concept 11.1 : General Considerations

Learning objectives: Contribution of scientist

- To know the classifications on antimicrobial agents
- To know the problems that arise with the use of antimicrobial agents
- To understand the concept antimicrobial agents resistance
- To know the factors governing selection of antimicrobial agents
- To know the categories of antimicrobial therapy

Time Needed

1 st reading	45 mins
2 nd reading	30 mins

1	Paul Ehrlich	Developed arsphenamine (Salvarsan) and neoarsphenamine for syphilis. Developed atoxyl for sleeping sickness. Coined the term ‘chemotherapy’.
2	Gerhard Domagk	Used prontosil (a dye) for treatment of streptococcal infections. Sulphonamides were later developed from these dyes. Was awarded the Nobel Prize for Medicine in 1938.
3	Louis Pasteur	Demonstrated the phenomenon of antibiosis by showing that the growth of anthrax bacilli in urine can be inhibited by exposure to air-borne bacteria.
4	Alexander Fleming	Discovered penicillin in 1928.
5	Howard Florey and Ernst Chain	Isolated penicillin. Shared the Nobel Prize with Alexander Fleming for development of penicillin.
6	Selman Waksman and colleagues	Isolated streptomycin from soil actinomycetes.
7	Brotzu	Isolated Cephalosporium fungus, the source of cephalosporins in 1948.

Classification of antimicrobial agents

Based on source:

	Drug	Microbial source
Drugs obtained from bacteria		
1	Polymyxin B	Bacillus polymyxa
2	Colistin	Bacillus (Aerobacillus) colistinus
3	Bacitracin	Bacillus subtilis (Tracy-I strain)
4	Tyrothricin	Bacillus bravis
5	Aztreonam	Chromobacterium violaceum

6	Vancomycin	<i>Streptococcus orientalis</i>
7	Teicoplanin	<i>Actinoplanes teichomycetius</i>
8	Mupirocin	<i>Pseudomonas fluorescens</i>
Drugs obtained from fungi		
1	Penicillins	Originally isolated from – <i>Penicillium notatum</i> Nowadays, commercially prepared from a mutant of – <i>Penicillium chrysogenum</i>
2	Cephalosporins	<i>Cephalosporium acremonium</i>
3	Griseofulvin	<i>Penicillium griseofulvum</i>
4	Caspofungin	<i>Glarea lozoyensis</i>
	Micafungin	<i>Coleophoma empedri</i>
	Anidulafungin	<i>Aspergillus nidulans</i>
Drugs obtained from actinomycetes		
1	Streptomycin	<i>Streptomyces griseus</i>
	Gentamicin, Netilmicin	<i>Micromonospora</i>
	Tobramycin	<i>Streptomyces tenebrarius</i>
	Remaining aminoglycosides are semi-synthetic derivatives	
2	Oxytetracycline	<i>Streptomyces rimosus</i>
	Demeclocycline	<i>Streptomyces aureofaciens</i>
	Remaining tetracyclines are semi-synthetic derivatives	
3	Chloramphenicol	<i>Streptomyces venezuelae</i>
4	Erythromycin	<i>Streptomyces erythreus</i>
	Remaining macrolides are semi-synthetic derivatives	
5	Amphotericin B	<i>Streptomyces nodosus</i>
	Nystatin	<i>Streptomyces noursei</i>
	Hamycin	<i>Streptomyces pimprina</i>
	Natamycin	<i>Streptomyces natalensis</i>
6	Imipenem	Derivative of thienamycin, obtained from <i>Streptomyces cattleya</i> .
7	Clavulanic acid	<i>Streptomyces clavuligerus</i>
8	Streptogramins (Quinupristin / Dalfepristin)	<i>Streptomyces pristinaespiralis</i>
9	Spectinomycin	<i>Streptomyces spectabilis</i>
10	Daptomycin	<i>Streptomyces roseosporus</i>
11	Rifampicin	Derivative of Rifamycin B obtained from <i>Streptomyces mediterranei</i>

Based on mechanism of action-

Cell wall inhibitors	Cell membrane inhibitors	Protein synthesis inhibitor	Nucleic acid synthesis inhibitor	Folic acid synthesis inhibitors	Microtubule inhibitors
Beta-lactam antibiotics Vancomycin / Teicoplanin Bacitracin (polypeptide) Cycloserine Fosfomycin	1. Polypeptide – Polymyxin-B – Polymyxin-E (Colistin) 2. Polyene Antifungals – Amphotericin B 3. Azole Antifungal – Ketoconazole 4. Allylamine – Terbinafine	By acting on 30s ribosomes Aminoglycoside Tetracycline By acting on 50s ribosomes Chloramphenicol Macrolides Lincosamide (clindamycin) Streptogramin Linezolid	Metronidazole Rifampicin Nitrofurantoin	Sulfonamide Trimethoprim Pyrimethamine	Griseofulvin

Bases on bacteriostatic or bactericidal

Bacteriostatic	Bactericidal
• Protein synthesis Inhibitors (except aminoglycosides, streptogramins) • Those affecting “Folate Pathway” (sulfonamides / trimethoprim) • Affecting DNA- nitrofurantoin, novobiocin	• Cell Wall Inhibitors • Cell membrane inhibitors • Those affecting nucleic acid synthesis • First line ATT (Except ethambutol) • Cotrimoxazole • Aminoglycosides • Streptogramins

Classification of Antimicrobials Based on their Action and Post-Antibiotic Effect:

Time-dependent (with minimal or no PAE)	Concentration-dependent (with prolonged PAE)	Time-dependent, concentration-enhanced (with PAE)
• β-lactams • Vancomycin	• Aminoglycosides • Daptomycin • Fluoroquinolones • Metronidazole • Azithromycin • Ketolides	• Clarithromycin • Clindamycin • Erythromycin • Linezolid • Streptogramins • Tetracyclines • Tigecycline

Adverse effects of antimicrobial agents (AMA)

General	Aminoglycoside	Tetracycline	Chloramphenicol	Vancomycin	Amphotericin B
Hypersensitivity Anaphylaxis	Ototoxic Nephrotoxic Skeletal muscle paralysis	Pseudotumor cerebri Outdated drugs -fnaconis syndrome Teratogenicity	Bone marrow suppression Grey baby syndrome	Redman syndrome Ototoxic Nephrotoxic	Allergy Nephrotoxic Anemia Neurological damage

Phototoxicity	Ototoxicity	nephrotoxicity	Disulfiram like reaction	Hemolysis in G6PD deficiency
Sulfonamides Tetracyclines Demeclocycline Fluroquinolone Sparfloxacin Azithromycin	Aminoglycoside Vancomycin Macrolides	Aminoglycosides Tetracyclines -outdated Demeclocycline Sulfonamides Penicillin (methicillin)	Cephalosporins Metronidazole Chloramphenicol Griseofulvin	Sulfonamide Primaquine, chloroquine, quinine Chloramphenicol Nitrofurantoin Fluroquinolones

Antimicrobial Resistance

Inability to reach target site	Insensitive target	Drug inactivation / Modification	Through alternative pathways
Decreased influx- Aminoglycosides, Tetracycline Increased efflux – tetracyc, Erythromycin, Quinolone	Quinolone Rifampicin penicillin	- β -lactam antibiotics (β lactamase) Aminoglycosides Chloramphenicol (acetyl transferase)	Sulphonamides, Trimethoprim

Choice of antimicrobial agents

Patient Factors	Drug Factor	Micro Organism Factors
Age- Gray baby syndrome.- CHLORAMPHENICOL Jaundice (kernicterus) - SULPHONAMIDES Bone & teeth defect- TETRACYCLINES Renal & hepatic functions H/O allergy to AMA host defence Local factors- Presence of pus and secretions, pH Supha&cotrimoxazole-Kernicterus Tetracycline-Teeth & bone defect Aminoglycoside –ototoxicity Quinolones-arthropathy Clarithromycin-teratogenic Chloramphenicol- grey baby	Spectrum of activity (BROAD/ NARROW) Type of activity (CIDAL/ STATIC) Sensitivity of the organism Relative toxicity Pharmacokinetic profile (CONCENTRATION/ TIME DEPENDENT) Cost	Bacteriological/diagnostic services are not available- EMPIRICAL THERAPY to cover all likely organisms with a broad-spectrum drug Bacteriological services are available & treatment can be delayed for a few days- It is BETTER TO WAIT FOR The culture & sensitivity report Bacteriological /diagnostic services are available. But treatment cannot be delayed: EMPIRICAL THERAPY started provisionally- AMA should be changed later in the light of bacteriological findings

Antimicrobials in Renal Disease:

Drugs to be completely avoided	Dose reduction even in mild failure	Dose reduction only in moderate-to-severe renal failure
- Nalidixic acid - Nitrofurantoin - Talampicillin - Tetracyclines (except doxycycline)	- Aminoglycosides - Amphotericin B - Cephalosporins - Ethambutol 5-Flucytosine - Vancomycin	- Aztreonam - Carbenicillin - Clarithromycin - Cotrimoxazole - Fluoroquinolones - Imipenem - Meropenem - Metronidazole

Antimicrobials in Liver Disease:

Drugs to be avoided	Dose reduction needed
- Erythromycin esteolate - Nalidixic acid - Pefloxacin - Pyrazinamide - Talampicillin - Tetracyclines	- Chloramphenicol - Clindamycin Isoniazid Metronidazole Rifampicin

Combination of Antimicrobial Agents

Advantages	Disadvantages
Synergism Prevention of resistance Broadening of spectrum Initial therapy for severe infection	Increased incidence ADR (vancomycin + tobramycin – more nephrotoxic) Increased risk of superinfections (pseudomembranous colitis) If inadequate doses- risk of resistance Higher cost

Examples of Antimicrobial Synergistic Combinations:

- Penicillin + Tetracycline for Pneumococci.
- Penicillin + Chloramphenicol for Pneumococci.
- Penicillin + Erythromycin for Group A Streptococcus.
- Nalidixic acid + Nitrofurantoin for E.coli.

Examples of Antimicrobial Antagonistic Combinations:

- **Penicillin / Ampicillin + Streptomycin / Gentamicin** for enterococcal SABE.
- **Vancomycin + Gentamicin** for enterococcal SABE.
- Carbenicillin / Ticarcillin + Gentamicin for Pseudomonas infections, especially in neutropenic patients.
- Ceftazidime + Ciprofloxacin for Pseudomonas infected orthopedic prosthesis.
- Penicillin + Sulfonamide for actinomycosis.
- Streptomycin + Tetracycline for brucellosis.
- Streptomycin + Chloramphenicol for K. pneumoniae infection.
- Rifampicin + Isoniazid for tuberculosis.
- Rifampicin + Dapsone for leprosy.

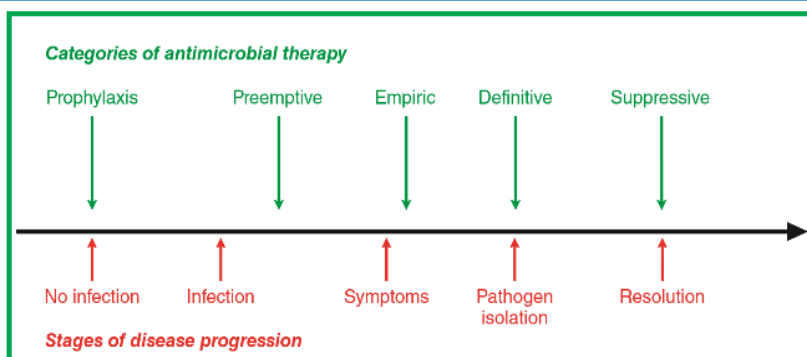


Fig. 11.1

Chemoprophylaxis

DISEASE	DRUGS
CHOLERA	Tetracycline/ doxycycline
Bacterial Conjunctivitis	Erythromycin ophthalmic ointment
Diphtheria	Erythromycin & 1st dose of vaccine
Influenza	Oseltamivir
Plaque	Tetracycline
Leprosy	Dapsone
Meningococcal meningitis	rifampin/ciprofloxacin/ ceftriaxone
Rheumatic fever	Benzathine penicillin
TB-	INH -6months
MAC	azithromycin/ clarithromycin
Gonorrhoea/syphilis	ampicillin/ceftriaxone
Rickettsial	tetracycline
Malaria	chloroquine/ mefloquine

Antimicrobials for Surgical Prophylaxis

Oral (single dose given 1 hour before procedure)

- Amoxicillin – 2g (50mg/kg)
- Cephalexin – 2g (50mg/kg)
- Cefadroxil – 2g (50mg/kg)
- Clindamycin – 600mg (20mg/kg) – for penicillin allergic patients
- Azithromycin – 500mg (15mg/kg) – for penicillin allergic patients
- Clarithromycin – 500mg (15mg/kg) – for penicillin allergic patients

Parenteral (single injection given just before procedure)

- Ampicillin – 2g (50mg/kg) IM/IV
- Cefazolin – 1g (25mg/kg) IV
- Vancomycin – 1g (20mg/kg) IV – in MRSA prevalent areas and/or penicillin allergic patients
- Clindamycin – 600mg (20mg/kg) IV – for penicillin allergic patients
- Cefuroxime – 1.5g (30mg/kg) IV + Metronidazole – 0.5g (10mg/kg) IV – for gut and biliary tract surgery
- Gentamicin – 160mg (3mg/kg) IV + Metronidazole – 0.5g (10mg/kg) IV – for gut and biliary tract surgery

Antimicrobials for Surgical Prophylaxis in Case of dirty Contaminated Wounds:

- **Cefazolin – 1g IV 8 hourly + Vancomycin – 1g IV 12 hourly.**
- Cefoxitin – 1g IV 6 hourly.
- Ceftizoxime – 1g IV 12 hourly.
- Clindamycin – 0.6g IV 8 hourly + Gentamicin – 80mg IV 8 hourly.
- Ampicillin – 2g IV 6 hourly / Vancomycin – 1g IV 12 hourly + Gentamicin – 80mg IV 8 hourly + Metronidazole – 0.5g IV 8 hourly.
- Amoxicillin – 1g + Clavulanate – 0.2g IV 12 hourly.

Duration of Surgical Prophylaxis:

Dirty contaminated wounds	5 days
All other cases	3 hours

Worksheet

To do

Antibiotic inhibiting cell wall synthesis	
Antibiotic following concentration dependent killing	
Mechanism of resistance of antimicrobial agents	
Antimicrobial to be avoided in renal failure	
antimicrobial to be avoided in liver failure	
Antibiotic causing red man syndrome	
Antimicrobials causing ototoxicity	
Antibiotics inhibiting protein synthesis	

Concept 11.2 : Sulfonamides, Cotrimaxazole and Quinolones

Learning objectives:

1. To study the mechanism of actions of sulfonamide and cotrimaxazole
2. To know the classification and clinical uses of sulfonamides
3. To the adverse effects of sulfonamides
4. To know the classification and clinical uses of quinolones
5. To know the adverse effects & withdrawn quinolones

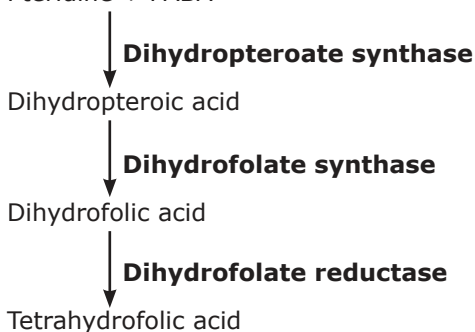
Time Needed

1 st reading	30 mins
2 nd reading	15 mins

Antimetabolites:

Pathway of folate metabolism:

Pteridine + PABA



Mechanism of action of antibacterial antimetabolites:

Sulfonamides Dapsone	Dihydropteroatesynthase inhibitors
Trimethoprim Pyrimethamine	Dihydrofolate reductase inhibitors

Sulfonamides:

Short-acting (4 – 8 h)	Sulfadiazine
Intermediate acting (8 – 12 h)	Sulfamethoxazole
Long acting (~ 7 days)	Sulfadoxine Sulfamethopyrazine
Topical	Sulfacetamide sodium Silver sulfadiazine Mafenide Sulfasalazine (eye)

**ONE-LINERS:**

- Sulfonamides and Dapsone are structural analogues of – **PABA**.
- Increased bacterial production of PABA
- can lead to resistance to – **Sulfonamides**.
- Sulfonamide which is a carbonic anhydrase inhibitor – **Mafenide**.

Adverse effects of sulfonamides – (Mnemonic: CAR-PHEK).

- | | |
|---|---|
| <ul style="list-style-type: none"> • Crystalluria. • Acute hemolysis. • Rashes. • Photosensitivity. | <ul style="list-style-type: none"> • Hypersensitivity reactions – SJS and TEN. • Erythema multiforme. • Kernicterus. |
|---|---|

ONE-LINERS:

- Sulfonamides cause kernicterus due to – **Displacement of bilirubin from plasma protein binding sites**.
- **Co trimaxazole Sulfamethaxazole 400mg + Trimethoprim 80mg, 5: 1 ratio** .
- **DOC- pneumocystis carinii Co trimaxazole DS**.
- Sulfasalazine- (**5ASA (ulcerative colitis) + Sulfapyridine (RA)**) .
- For Toxoplasmosis- **Sulfadiazine + Pyrimethamine + Folinic acid**.
- Toxoplasmosis in pregnancy- spiramycin (macrolide) .
- **Domagk (1938) - invented sulfonamide compound**.

Cotrimoxazole:

Trimethoprim + sulfamethoxazole in the ratio of –
 1:5 in the tablet
 1:20 in the tissues

ONE-LINERS:

- **Cotrimazine** – Trimethoprim + Sulfadiazine.

Cotrimoxazole is the DOC in:

- Nocardiosis
- Cryptosporidiasis
- Isosporiasis
- **Pneumocystis jiroveci pneumonia**

DNA Gyrase Inhibitors-

Examples for quinolones

Quinolones	Fluoroquinolones	
	1 st generation (Mnemonic: N-COP)	2 nd generation (Rest all)
Nalidixic acid	Norfloxacin	Levofloxacin
	Ciprofloxacin	Lomefloxacin
	Ofloxacin	Sparfloxacin
	Pefloxacin	Moxifloxacin
		Gatifloxacin
		Prulifloxacin

Fluoroquinolones that have been withdrawn:

Drug	Reason
Trovaflaxacin	Hepatotoxicity
Grepafloxacin	Cardiotoxicity (Adverse cardiac events)
Clinafloxacin	Phototoxicity
Temaflaxacin	Immune hemolytic anemia
Gatifloxacin (systemic preparation)	Pancreatic damage → Hypoglycemia f/b hyperglycemia

ONE-LINERS:

- Nalidixic acid was obtained as a by - product of – **Chloroquine synthesis**.
- Most potent 1st generation fluoroquinolone – **Ciprofloxacin**.

Pharmacokinetics of fluoroquinolones:

Drug	Oral F (%)	PPB (%)	% Metabolized
Norfloxacin	35–45	15	25
Ciprofloxacin	60–80	20–35	20
Ofloxacin	85–95	25	5–10
Pefloxacin	90–100	20–30	85
Levofloxacin	~ 100	25	5
Gemifloxacin	70	55–73	–
Moxifloxacin	85	40	70–80
Prulifloxacin	90	45	> 90

**ONE-LINERS:**

- Pefloxin injection should be diluted in – **Glucose** (as it precipitates in the presence of Cl⁻ ions).
- Pefloxin, gemifloxacin has high CNS penetration power.
- Antimicrobials that get concentrated in prostrate – **Fluoroquinolones and trimethoprim.**
- **Fluoroquinolones detected in ascitic fluid – Ofloxacin and Pefloxacin.**
- Fluoroquinolones detected in breast milk – **Ciprofloxacin, Ofloxacin and Pefloxacin.**
- Fluoroquinolones that do not require dose reduction in renal failure – **Pefloxacin, trovafloxacin and Moxifloxacin.**
- Trovafloxacin highly hepatotoxic.

Fluoroquinolones effective against tuberculosis:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Ciprofloxacin. • Ofloxacin. | <ul style="list-style-type: none"> • Levofloxacin. • Moxifloxacin. |
|--|--|

ONE-LINERS:

- Best fluoroquinolone against tuberculosis – **Moxifloxacin.**
- Fluoroquinolone preferred in bacterial meningitis – **Pefloxacin** (due to good BBB penetration).
- Strength of ciprofloxacin eye drops: **3 mg/mL.**
- Strength of ofloxacin eye drops: **0.3%.**
- Strength of levofloxacin eye drops: **0.5%.**
- Most important toxicity of nalidixic acid – **Neurotoxicity.**
- Nalidixic acid is contraindicated in **Infants** (due to greater risk of neurotoxicity) .
- Most common adverse reaction to fluoroquinolones – **GI symptoms.**
- Most common fluoroquinolone increasing the risk of **Cl. difficile** infection – **Ciprofloxacin.**
- Highest phototoxic quinolone- **sparfloxacin.**
- Risk of tendon damage due to fluoroquinolones is higher in patients – less than 6yr age and **above 60 years of age.**
- Most common tendon ruptured due to fluoroquinolones – **Achilles tendon.**
- QTc interval prolongation is seen with– **Sparfloxacin, Moxifloxacin and Gatifloxacin.**
- **Quinolone useful in TB Ciprofloxacin, Levofloxacin, Moxifloxacin.**
- **Purulifloxacin Ulifloxacin Broad spectrum.**
- **Quinolones are microsomal enzyme inhibitors.**

Achilles tendon rupture due to fluoroquinolones is more common in:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Age > 60 years. • Patients on corticosteroids. | <ul style="list-style-type: none"> • Solid organ transplant recipients. • Renal insufficiency. |
|--|--|

Worksheet

To do

Drugs useful for toxoplasmosis	
First line DOC for enteric fever	
Withdrawn quinolone & the reason for withdrawal	
Most common adverse effects of sulfonamide	
Cotrimaxazole is the DOC for	

Concept 11.3 : Cell wall Inhibitors (beta-lactum antibiotics, glycol petides)

Learning objectives:

1. To know the classification of penicillin
2. To know the spectrum and adverse effects of penicillin
3. To know the classification of cephalosporine
4. To know the spectrum and adverse effects of cephalosporines
5. To the uses and adverse effects of atypical beta lactam antibiotics
6. To know the uses and adverse effects of vancomycin
7. To know the drugs useful for treatment of MRSA & VRSA

Time Needed

1 st reading	45 mins
2 nd reading	20 mins

Examples of Bacterial Cell Wall Synthesis Inhibitors:

Enzyme	Drug Inhibiting
Alanin Ligase (Alanin Racemase)	Cycloserine
Enol Pyruate Transferase	Fosphomycin
Dephosphorylation of bactophrenol	Bacitracin
Trnasglycosilase	Glycopeptide (Vancomycin)
Transpeptidase	Beta lactum antibioitics

β-lactams (Bind to penicillin-binding proteins or PBPs) transpeptidase	Glycopeptides (Bind to D-Alanyl-D-Alanine) Transglycosylase	Miscellaneous
Penicillins Cephalosporins Carbapenems Monobactams	Vancomycin Teicoplanin	Bacitracin

Penicillins:

Classification according to Source:

Natural	Synthetic
• Penicillin G (also known as Benzyl penicillin / Crystalline penicillin / Aqueous penicillin) • Procaine penicillin G • Benzathine penicillin	Rest all

ONE-LINERS:

- Purpose of conjugating penicillin **with procaine** and benzathine – **Prolong half-life and hence, duration of action.**
- Longest acting penicillin – **Benzathine penicillin.**
- 1 U of crystalline penicillin (Penicillin G) sodium = **0.6 µg** of the standard preparation.
- Hence, 1 g of Penicillin G sodium = **1667 U.**
- However, 1 g of Penicillin G potassium = **1595 U.**

Route of administration:

Oral (acid resistant)	Parenteral (acid labile)
Mnemonic: VODKAA • Penicillin V (Phenoxymethyl penicillin) • Oxacillin • Dicloxacillin • C (K) loxacillin • Amoxicillin • Ampicillin	• Rest all

ONE-LINERS:

- **Tubular secretion of penicillins** and cephalosporins is inhibited by – **Probenecid.**
- Hence, it can **increasethe plasma levels** of penicillins and cephalosporins.

Spectrum of action of penicillins:

Extended-spectrum penicillins	Narrow-spectrum penicillins
Cover – Almost all Gram +ve and Gram –ve organisms	Cover – Only Gram +ve organisms, Neisseria and spirochaetes
Aminopenicillins: Becampicillin (prodrug) Amoxicillin Ampicillin	Rest all
Carboxypenicillins – Pseudomonal infection Carbenicillin Ticarcillin Ureidopenicillins – pseudomonal + klebsiella infection Piperacillin Azlocillin Mezlocillin	

ONE-LINERS:

- DOC for leptospirosis – **Penicillin G.**
- DOC for actinomycosis – **Penicillin G.**
- DOC for rat bite fever – **Penicillin G.**
- DOC for syphilis (except neurosyphilis) – **Benzathine penicillin.**
- Benzathine penicillin is contraindicated in neurosyphilis as – **it worsens the neurological features.**
- Treatment for neurosyphilis – **Penicillin G / Procaine penicillin.**
- **Jarisch herxheimer reaction**-side effect of penicillin in syphilitic patient.
- Amino penicillin (Ampicillin) **contra indicted in infectious mononeucleosis** (due to risk of skin allergy) .
- DOC for prophylaxis of rheumatic carditis – **Benzathine penicillin.**
- Dose of benzathine penicillin for prophylaxis of rheumatic carditis – **1.2MU (12 lac units) IM every 3-4 weeks.**
- DOC for listeriosis – **Ampicillin.**
- Anti-pseudomonal penicillin – (carboxypenicillin and urediopenicillin) **Piperacillin > Carbenicillin.**
- Most common mechanism of penicillin resistance – β -lactamase production.
- Most common mechanism of methicillin resistance – **Alterations in penicillin-binding proteins.**
- **METHICILLIN**- causes interstitial nephritis.
- **PROCAINE PENICILLIN**-causes seizure.
- **OXACILLIN**- side effects-hepatitis.
- **NAFCILLIN**- side effect-neutropenia.
- **CARBENICILLIN**- causes bleeding.
- Resistance to methicillin is primarily – **chromosome mediated.**

Susceptibility of Penicillins To β -Lactamase:

Penicillinase-resistant	Penicillinase-susceptible
• Methicillin • Nafcillin • Oxacillin • Cloxacillin • Dicloxacillin	• Rest all

Examples for beta lactamase inhibitors

- Sulbactam
- Clavulanic acid
- Tozabactam

ONE-LINERS:

- Usually **beta lactamase inhibitors** as such has no anti bacterial action.
- β -lactamase inhibitor exhibiting direct activity against **Acinetobacter** – **Sulbactam**
- The purpose of adding beta lactamase inhibitor with beta lactum antibiotic is to **prevent resistance**.
- β -lactams show cross-allergy with each other except – **Aztreonam**.
- Aztreonam shows cross-allergy only with – **Ceftazidime**.
- Aztreonam effective for aerobic gram negative infection.
- Aztreonam spectrum is simillr to aminoglycosides.

Cephalosporins:**Classification:**

Generation of cephalosporins	Examples	Route
1 st generation cephalosporins (Mnemonic: LO-ZO-LEX-RA-DROXIL)	Cefalothin	Parenteral
	Cefazolin	
	Cefalexin	Oral
	Cefradine Cefadroxil	
2 nd generation cephalosporins (Mnemonic: as a FACT, MAN kills FOX for FUR)	Cefaclor	Oral
	Cefamandole	Parenteral
	Cefoxitin Cefuroxime	
	Cefuroxime axetil	Oral
3 rd generation cephalosporins Mnemonics: TRI-ZI-PER-TAX-ZOX; FIX-PODO-MET-BUT-NIR)	Ceftriaxone	Parenteral
	Cetazidime	
	Cefoperazone	
	Cefotaxime Ceftizoxime	
	Cefixime	Oral
	Cefpodoxime	
	Ceftamet	
	Ceftibuten Cefdinir	
4 th generation cephalosporins	Cefepime Cefpirome Cefclidine	Parenteral
5 th generation cephalosporins	Ceftobipirole Ceftaroline	parenteral

Indications:

1 st generation cephalosporins	Prophylaxis of surgical infections
2 nd generation cephalosporins	Anaerobic bacterial infections
3 rd generation cephalosporins	Gram-ve bacterial infections Cefoperazone and ceftazidime- anti pseudomonal activity
4 th generation cephalosporins	Reserve drugs for nosocomial infections
5 th generation cephalosporins	Reserve drugs for nosocomial infections; also effective against MRSA

ONE-LINERS:

- DOC for gonorrhea – **Ceftriaxone**.
- DOC for meningococcal meningitis – **Ceftriaxone**.
- DOC for prophylaxis of meningococcal meningitis – **Ceftriaxone**.
- Most reliable drug for enteric fever – **Ceftriaxone**.
- Fastest acting bactericidal drug for typhoid fever – **Ceftriaxone**.
- DOC for **Pseudomonas** infection – **Ceftazidime**.
- Only cephalosporins with anti-MRSA activity – **5th generation cephalosporins**.
- Cephalosporins can cause bleeding due to – **Hypoprothrombinemia (Cefoperazone, Ceftriaxone, Cefotetan, Cefomondole, Moxolactam)**.
- Cefoperazone may cause disulfiram like reaction.
- Fifth generation cephalosporins useful-MRSA, Community acquired pneumonia.
- **Ceftazidime, Cefoperazone useful in pseudomonal infection.**

Atypical Beta Lactam Antibiotic (Carbapenem. Monobactam)**Carbapenems:**

• Imipenem. • Meropenem. • Doripenem.	• Ertapenem. • Faropenem.
---	--

ONE-LINERS:

- Orally active carbapenem – **Faropenem**.
- Imipenem is metabolized by – **dipeptidase enzyme** on the renal proximal tubular brush border; hence, combined with dipeptidase inhibitor – **Cilastatin**.
- β -lactam effective against ESBL producing strains – **Imipenem**.
- High doses of imipenem can cause – **Seizures**.
- Carbapenem resistance is seen due to production of – **metallo-beta lactamase (carbapenemase) enzyme**. E.g.: **NDM-1** (New Delhi metallo-beta lactamase-1) also known as 'superbug'.

Monobactams:

Aztreonam - Effective against aerobic Gr(-)ve infection.

- β -lactam used in **penicillin allergy cases**.

Glycopeptides:

Examples:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Vancomycin. • Teicoplanin. • Telavancin. | <ul style="list-style-type: none"> • Dalbavancin. • Oritavancin. |
|--|--|

ONE-LINERS:

- Glycopeptides are only effective against – **Gram+ve organisms** as D-Ala-D-Ala is absent in Gram-ve organisms.

Vancomycin:

One-liners:

ONE-LINERS:

- Vancomycin is orally – **not absorbed**; hence **given IV**.
- BBB penetration is – **Poor**.
- $T_{1/2}$: **~ 6 hours**.
- Dose: **500 mg QID**.
- **Use- orally for pseudo membranous colitis, intra venously for MRSA (DOC).**
- **Most common side effect – Red man syndrome (due to release of histamine).**
- Dose needs to be **reduced in renal failure**, as it is excreted unchanged by kidney.

Drugs Effective against MRSA:

- | | |
|--|---|
| <ul style="list-style-type: none"> • Glycopeptides. • Linezolid. • Quinupristin / Dalfepriatin (Streptogramins). • Daptomycin. • Mupirocin. | <ul style="list-style-type: none"> • 5th generation cephalosporins. • Tigecycline. • Cotrimoxazole. • Ciprofloxacin. |
|--|---|

Drugs effective against VRSA:

- **Linezolid**(common side effects- thrombocytopenia, neuropathy both peripheral ad optic, has MAO enzyme inhibitory action).
- Quinupristin / Dalfepriatin (**Streptogramins**) (common side effect- infusion reaction, arthralgia).
- **Daptomycin** (side effect- myopathy).
- **Tigecycline**(intravenous tetracycline, resistant to efflux pump, excreted via bile, safe in renal failiure).

**ONE-LINERS:**

- DOC for MRSA – **Vancomycin**.
- Route of vancomycin in pseudomembranous enterocolitis (drug-induced colitis) – **Oral**.
- Most common side effect due to vancomycin – **Maculopapular rash**.
- Red man syndrome is seen with – **Vancomycin**.

Red man syndrome:

- Infusion reaction due to vancomycin.
- Widespread histamine release due to non-specific mast cell degranulation.
- Appearance – 4–10 min after the commencement or soon after the completion of an infusion.
- Erythema of the peri-oral region and upper trunk.
- Prevention – Slow IV infusion of a diluted solution over 60 minutes.

Pseudomembranous Colitis:

- Caused by bacterium *Clostridium difficile*.
- Most commonly caused by 3rd gen cephalosporins.
- Treated by Metronidazole (firstline DOC), Vancomycin, Rifaximin, Fidaxomicin.

Worksheet

To do

Jarisch herxheimer reaction is the adverse effect of	
Cephalosporines causing disulfiram like reactions	
Redman syndrome is the adverse effects of	
Drugs useful for VRSA	
Antimicrobial causing cheese reaction	
Anti pseudomanla penicillins are	

Concept 11.4 : Protein Synthesis Inhibitors

Learning objectives:

- To know the various mechanism of actions of protein synthesis inhibitors
- To study the uses and adverse effects of aminoglycosides
- To know the uses and adverse effects of macrolides
- To know the uses and adverse effects of clindamycin (lincosanamide)

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

Mechanisms of Action of Protein Synthesis Inhibitors:

Aminoglycosides (binds with 30S ribosome)	• Prevent formation of polysomes • Promote disaggregation of already formed polysomes • Cause misreading of mRNA code ↓ Freeze the initiation of protein synthesis
Tetracyclines (binds with 30S ribosome)	Inhibit the binding to aminoacyl-tRNA to A site on ribosomes
Chloramphenicol (binds with 50S ribosome)	Inhibit peptidyltransferase enzyme
Macrolides Clindamycin Lincomycin (binds with 50S ribosome)	Inhibit the process of translocation

ONE-LINERS:

- **Puromycin** – Structural analogue of tyrosinyl-tRNA. It gets incorporated into the carboxy-terminal position of the peptide at the A site of ribosome and causes premature release of the peptide, thus prematurely terminating protein synthesis.
- **Retapamulin** – Topical antibiotic belonging to pleuromutilin class; approved for skin infections due to Gram+ve bacteria such as impetigo. Also binds to 50S and inhibits protein synthesis

Binding sites of Protein Synthesis Inhibitors:

30S	30S-50S interface	50S
Streptomycin Tetracyclines	All aminoglycosides except streptomycin	Rest all

Aminoglycosides:

• Streptomycin • Gentamicin • Kanamycin • Amikacin	• Capreomycin • Tobramycin • Netilmicin • Sisomicin	• Neomycin (topical) • Framycetin (topical) • Paromomycin (topical)
---	--	---

Nomenclature:

Source – Streptomyces	Source – Micromonospora
Add – _mycin E.g.: Streptomycin	Add – _micin E.g.: Gentamicin

ONE-LINERS:

- Aminoglycosides are not absorbed orally due to – **Polar nature**.
- BBB penetration of aminoglycosides – **Poor**.
- Aminoglycosides require **dose reduction in – Renal failure** as they are primarily excreted unchanged by the kidney.

Guidelines for dose adjustment of gentamicin in renal insufficiency:

Creatinine clearance (mL/min)	Dose adjustment needed
70	70 % of daily dose
50	50 % of daily dose
30	30 % of daily dose
20 – 30	80 % on alternate days
10 – 20	60 % on alternate days
< 10	40 % on alternate days

Aminoglycosides Effective in Tuberculosis:

• Streptomycin. • Amikacin.	• Capreomycin. • Kanamycin (Mnemonic: SACK) .
--	---

ONE-LINERS:

- Aminoglycoside used in hepatic encephalopathy – **Neomycin**.
- Intrinsic aminoglycoside resistance – **Anaerobes**, as penetration of aminoglycosides into bacterial cells is an aerobic process.
- Aminoglycoside showing least cross- resistance – **Amikacin** (effective even when organism is resistant to other aminoglycosides) .
- Most common cause for resistance of majority of aminoglycoside is due to **enzymatic degradation**.
- **Outstanding aminoglycosides** not yet developed resistance are amikacin and netilmicin.

Comparison of Toxicities of Aminoglycosides:

	Ototoxicity		Nephrotoxicity
	Vestibular	Cochlear	
Streptomycin	++	±	+
Gentamicin	++	+	++
Kanamycin	+	++	++
Amikacin	+	++	++
Tobramycin	+±	+	+
Sisomicin	++	+	++
Netilmicin	+±	+	++

ONE-LINERS:

- Hearing loss due to aminoglycosides is – **Irreversible**.
- Least ototoxic aminoglycoside – **Netilmicin**.
- Most vestibulotoxic aminoglycoside – **Streptomycin**.
- Least nephrotoxic aminoglycoside – **Streptomycin**.
- Most nephrotoxic aminoglycoside – **Neomycin**.
- Most skeletal muscle relaxing drug – **neomycin**

ONE-LINERS:

- Aminoglycoside follows **concentration dependent killing**.
- Aminoglycosides has long post antibiotic effect.
- Aminoglycoside effective for **kala azar – paramomycin**.
- Aminoglycosides effective for **pseudomonal infection – Toramycin, Amikacin, - gentamicin**.
- Gentamicin PMMA (poly methyl methacrylate) - given as an implantation into the bone in case of osteomyelitis.
- **DOC- plague- streptomycin (for mass prophylaxis- doxycycline)**.
- Aminoglycosides when combine with beta lactam antibiotic produces synergistic effect

Macrolides :

Examples of Macrolides:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Erythromycin. • Clarithromycin. • Azithromycin. | <ul style="list-style-type: none"> • Roxithromycin. • Spiramycin. • Tacrolimu. |
|---|---|

ONE-LINERS:

- Erythromycin was isolated in – **1952.**
- Telithromycin is a – **Ketolide.**

Azithromycin is Doc in:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Whooping cough. • Atypical pneumonia. • Trachoma. | <ul style="list-style-type: none"> • Chancroid. • Donovanosis. • Gonococcal and non gonococcal urethritis. |
|---|---|

Azithromycin is preferred over erythromycin today for most of the indications due to:

- Higher efficacy.
- Better gastric tolerance.
- Convenient once daily dosing regimen.

ONE-LINERS:

- DOC for prophylaxis of rheumatic carditis **in penicillin-allergic patients – Erythromycin.**
- **DOC for atypical** mycobacterial infections – **Clarithromycin.**
- DOC for toxoplasmosis in pregnancy – **Spiramycin.**
- Macrolides exhibit cross-resistance with – **Clindamycin and Lincomycin.**
- Biliary excretion is the major route for elimination of – **Erythromycin and Azithromycin.**
- Ototoxicity due to macrolides manifests as – **Transient auditory impairment / Reversible hearing loss.**
- Telithromycin is contraindicated in – **Myasthenia gravis** (due to exacerbation of symptoms).
- **Erythromycin estolate – unsafe in pregnancy (causing cholestatic jaundice to mother).**
- **Clarithromycin useful in Mycobacterium Avium Complex.**
- **Clarithromycin useful in H. pylori induced gastric ulcer.**
- **Azithromycin has large volume of distribution.**
- Azithromycin may cause QT prolongation.
- Erythromycin may aggravate congenital pyloric stenosis.
- Macrolide by acting on motilin receptor in GIT acting as prokinetic.

Lincosanamide (clindamycin)

- Useful In anaerobic infection
- Common side effect is pseudomembranous colitis

Worksheet

To do

Aminoglycosides useful In TB	
Most nephrotoxic aminoglycosides	
Antimicrobial group having prokinetic property	
Macrolide having immunosuppressant property	
Example for ketolide	

Concept 11.5 : Broad Spectrum Antibiotics (Tetracyclines & Chloramphenicol)

Learning objectives:

- To know the uses and adverse effects of tetracycline
- To know the uses and adverse effects of chloramphenicol

Time Needed

1 st reading	25 mins
2 nd reading	10 mins

Tetracyclines:

Examples of Tetracyclines:

- | | |
|--|---|
| <ul style="list-style-type: none"> • Tetracycline. • Chlortetracycline. • Oxytetracycline. • Methacycline. | <ul style="list-style-type: none"> • Demeclocycline. • Doxycycline. • Minocycline. |
|--|---|

ONE-LINERS:

- Oral absorption of tetracyclines is reduced by – **Food, milk and cations.**
- Tetracycline that can be given in full doses in renal failure – **Doxycycline, tigecycline (as it is excreted in bile).**

Doxycycline is DOC in:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Brucellosis (along with rifampicin) . • Cholera. • Rickettsial infections. • Lyme's disease. | <ul style="list-style-type: none"> • Relapsing fever. • LGV. • Prophylaxis of leptospirosis. |
|---|---|

Doxycycline Useful for Prophylaxis of:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Brucellosis (along with rifampicin) . • Cholera., Malaria | <ul style="list-style-type: none"> • Plague. • Prophylaxis of leptospirosis. |
|--|--|

ONE-LINERS:

- Most potent tetracycline – **Minocycline**.
- All tetracyclines are nephrotoxic except – **Doxycycline , tigecycline**.
- Tetracycline with highest phototoxicity – **Demeclocycline > Doxycycline**.
- Tetracycline exhibiting vestibular toxicity – **Minocycline**.
- Fanconi's syndrome is seen with – **Outdated / Expired tetracyclines**.
- Pseudo tumor cerebri (elevation intracranial pressure) – **side effect of tetracycline**.

Tigecycline:

- Glycylcycline.
- Tetracycline derivative.
- Given intravenously.
- Mechanism similar to tetracyclines.
- **Resistant to efflux pump**.
- Reserve drug for resistant infections including
- **MRSA**.
- Tetracycline useful in leprosy- **Minocycline**.
- **Demeclocycline** because of causing DI can be useful in SIADH.

Newer tetracyclines

Drug	Uses
Eravacycline	complicated intra abdominal infection
Omadacycline	Community-acquired bacterial pneumonia, skin and skin structure infections
Sarecycline	moderate to severe acne vulgaris

Chloramphenicol:

- **Bone marrow suppression** due to chloramphenicol occurs regularly if plasma concentrations are – $\geq 25 \mu\text{g/mL}$.
- **Grey baby syndrome** is a side effect of – **Chloramphenicol**.
- Grey baby syndrome is more common in – **Preterm infants**.
- Most common antibiotic **causing aplastic anemia** – **Chloramphenicol**.
- Most common reason for **resistance of chloramphenicol** is due to enzymatic **degradation (acetyl transferase)**.

Worksheet

To do

Doxycycline useful prophylactically for	
Safer tetracycline in renal failure includes	
Grey baby syndrome is the side effect of	
Fancony syndrome is the side effect of	
Tetracycline resistant to efflux pump is	
Most common cause for chloramphenicol resistant is	

Concept 11.6 : Antitubercular Drugs

Learning objectives:

- To know the first line and second lines drugs useful in TB
- To know the adverse effects of anti TB drugs
- To know the newer drugs for resistant TB

Time Needed

1 st reading	35 mins
2 nd reading	20 mins

Antimycobacterials:

Contributions of various scientists to treatment of tuberculosis:

1	Lehman	Developed para-amino salicylic acid (PAS)
2	Domagk	Developed thiacetazone
3	Kushner and colleagues	Developed pyrazinamide
4	Sensi and Margalith	Isolated rifampicin

ONE-LINERS:

- The first ever drug for TB was – **PAS** in 1943.

Conventional 1st line drugs for TB:

• Isoniazid. • Rifampicin. • Pyrazinamide.	• Ethambutol. • Streptomycin.
--	--

Chanisms of action of 1st line anti-TB drugs:

Cell wall synthesis inhibitors	1. Isoniazid 2. Pyrazinamide 3. Ethambutol	Inhibits mycolic acid synthesis Inhibits arabinogalactan synthesis
Protein synthesis inhibitors	1. Rifampicin 2. Streptomycin	Inhibits DNA-dependent RNA polymerase (Transcription inhibitor) Freezes initiation of protein synthesis

Mechanisms of resistance for 1st line anti-TB drugs:

Drug	Gene
Isoniazid	KatG > InhA > KasA
Rifampicin	rpoB
Pyrazinamide	pncA
Ethambutol	embB

Prevalence of drug resistance:

- | | |
|---------------------------------------|--|
| • Isoniazid: 1 in 106 bacilli. | • Rifampicin: 1 in 107 – 108 bacilli. |
|---------------------------------------|--|

ONE-LINERS:

- Resistance to anti-TB drugs is most commonly encountered with – **Isoniazid.**
- Overexpression of inhA may lead to cross-resistance of isoniazid with – **Ethionamide.**

Isoniazid (H) :

ONE-LINERS:

- Apart from tuberculosis, isoniazid is effective in – **M.kansasii infection.**
- **INH not effective in MAC infection.**

Peripheral neuropathy due to INH is more common in:

- | | |
|--|--|
| <ul style="list-style-type: none"> • Slow acetylators. • Diabetes mellitus. | <ul style="list-style-type: none"> • Poor nutrition. • Anemia. |
|--|--|

Indian population:

60 – 70 % - Slow acetylators of INH.

30 – 40% - Fast acetylators of INH.

ONE-LINERS:

- Peripheral neuropathy due to isoniazid is because of deficiency of – **Pyridoxine.**
- Dose of pyridoxine for prevention of peripheral neuropathy – **10 mg/d.**
- Dose of pyridoxine for treatment of peripheral neuropathy – **100 mg/d.**
- Antidote for isoniazid poisoning – **Pyridoxine.**
- Fast acetylators of isoniazid are supposed to be more prone to – **Hepatotoxicity (due to formation acety hydrazine hepatotoxic metabolite)** (not definitely proven) .
- INH may CNS side effects like memory loss, seizure, psychosis.
- Rarely causes shoulder hand syndrome.
- Safest anti-TB drug in pregnancy – **Isoniazid.**
- **Isoniazid safe in renal failure.**
- **Isoniazid is a microsomal enzyme inhibitor.**
- **Isoniazid is most effective against rapidly multiplying organism.**

Rifampicin:

Indications of rifampicin:

Therapeutic –

- All mycobacterial infections.
- Brucellosis.
- Legionellosis.
- Staphylococcal endocarditis and osteomyelitis.

Prophylactic –

- Meningococcal meningitis
- H.influenzae meningitis

ONE-LINERS:

- Anti-TB drug penetrating caseous material – **Rifampicin**
- Rifampicin penetrates BBB but – **effluxed out by pgp**. Hence, net rifampicin inside CNS is zero
- Anti-TB drug penetrating caseous material – **Rifampicin**.
- Rifampicin penetrates BBB but – **effluxed out by pgp**. Hence, net rifampicin inside CNS is zero.

Drugs causing interstitial nephritis as a side effect:

Antimicrobials	• β-lactams • Sulfonamides • Quinolones • Vancomycin • Minocycline • Rifampicin • Isoniazid* • Ethambutol • Acyclovir
Diuretics	• Thiazides • Loop diuretics • Triamterene
Anticonvulsants	• Phenytoin • Carbamazepine • Valproate • Phenobarbitone

Miscellaneous	• NSAIDs • PPIs • H₂ blockers • Captopril • Mesalazine • Indinavir • Allopurinol
---------------	---

Pyrazinamide:

- Pyrazinamide gets activated in – **Acidic environment.**
- Most hepatotoxic antitubercular – **Pyrazinamide.**
- Hyperuricemia and arthralgias are seen with – **Pyrazinamide.**
- Unsafe in renal failure.

Ethambutol:

ONE-LINERS:

- Ethambutol is effective in all mycobacterial infections excluding – **Leprosy.**
- Optic neuritis is seen with – **Ethambutol.**
- **Red-green colour blindness is seen with – Ethambutol.**
- **Optic neuritis due to ethambutol may be minimised by supplementing with Vit B12.**

Drugs causing colour vision abnormalities:

Ethambutol	Red-green colour blindness
Sildenafil	Blue-green colour blindness
Digoxin	Yellow vision
Thioridazine	Brown vision

Streptomycin – previously discussed under aminoglycosides

Use of 1st line antituberculars in renal and liver disease

How to use	Liver failure	Renal failure
Full doses	Ethambutol Streptomycin	Rifampicin
Reduced doses	Isoniazid Rifampicin	Isoniazid Pyrazin- amide
Avoided	Pyrazinamide	Ethambutol Strepto- mycin

**ONE-LINERS:**

- Rifabutin and Rifapentine have been re-grouped as – **Extended 1st line drugs**.
- Terizidone is a conjugate of – **2 molecules of cycloserine**.

Newer drugs for tuberculosis:

1	Bedaquiline	F ₀ /F ₁ ATP synthase inhibitor (inhibits aerobic respiration)
2	Delamanid	Inhibits mycolic acid synthesis; cell wall synthesis inhibitor.
3	Pretomanid	Inhibits both mycolic acid synthesis and F ₀ /F ₁ ATP synthase
4	Sutezolid	Congener of linezolid; protein synthesis inhibitor

Classification of antitubercular drugs into cidal and static drugs:

Tuberculocidal	Tuberculostatic
• Isoniazid • Rifamycins (Rifampicin, Rifabutin, Rifapentine) • Pyrazinamide • Aminoglycosides • Fluoroquinolones • Bedaquiline	• Rest all

ONE-LINERS:

- **Antitubercular drug causing hypothyroidism – Ethionamide, Para-amino salicylic acid.**
- **Anti TB drugs causing psychosis- INH, cycloserine.**
- **Anti TB drug causing uveitis- rifanutin.**
- Antitubercular absolutely contraindicated in immunocompromised – **Thiacetazone** (due to a high incidence of fatal hypersensitivity reactions) .
- **Anti TB drugs useful for MAC- rifabutin, ethambutol, clarithromycin (REC).**

Indications for chemoprophylaxis of TB

- Contacts of open TB with recent Mantoux conversion.
- Children with positive Mantoux test and a TB patient in the family.
- Patients of leukemia, diabetes or silicosis with a positive Mantoux test.
- Patients with old inactive disease with a recent Mantoux conversion.
- Neonates of tubercular mothers.

ONE-LINERS:

- Isoniazid monotherapy: **1020 mg/ kg/d x 6 months**.

Worksheet

To do

Anti TB drugs causing memory loss	
First line anti TB drug having microsomal enzyme induction property	
Non tuberculous uses of rifampicin	
Anti TB drug acting by inhibiting arabinogalaton synthesis	
Optic neuritis is the adverse effect	
First line Anti TB drug very safe in renal failure patient is	
Anti TB drugs causing hypothyroidism includeds	
Ant TB drug causing uveitis is	

Concept 11.7 : Antileprotic Drugs

Learning objectives:

- To know the drugs useful in paucibacillary and in multibacillary leprosy
- To know the drugs useful for treating lepra reaction
- To know the drugs useful in atypical mycobacterium infections

Time Needed

1 st reading	15 mins
2 nd reading	10 mins

Treatment of leprosy:

Paucibacillary (Duration: 6 months)		Multibacillary (Duration: 12 months)	
Drug	Dose	Drug	Dose
Rifampi- cin	600 mg on D1	Rifampi- cin	600 mg on D1
Dapsone	100 mg/d everyday	Dapsone	100 mg/d everyday
		Clofazi- mine	300 mg on D1 50 mg/d on re- maining days

- Anti TB drug useful in leprosy – **Rifampicin, ethionamide.**
- Antibiotics useful in leprosy – **Ofloxacin, Minocycline, Clarithromycin.**

- Most common adverse effect of dapsone – **skin rashes, hemolytic anemia.**
- INJ. Acadapsone – **given intra muscularly, long acting.**

ONE-LINERS:

- Dapsone is also useful in ulcers caused by – **Brown recluse spider.**
- Dapsone produces hemolysis in G6PD deficient cases at doses of – **> 50 mg/d.**
- Sulfone syndrome develops – **4-6 weeks after starting dapsone.**
- Main side effect of clofazimine – **Reddish black discolouration of skin and body secretions.**
- Clofazimine due to its anti inflammatory action useful in type 2 lepra reaction.
- Most potent drug against leprosy – **Rifampicin.**
- Most potent fluoroquinolone against **M. leprae** – **Moxifloxacin**

**ROM therapy:**

Rifampicin	600 mg single dose
Ofloxacin	400 mg single dose
Minocycline	100 mg single dose
Originally employed for single lesion leprosy Abandoned now	

Treatment of Lepra reactions:

Type 1	Steroids
Type 2 (Erythema nodosum leprosum)	Steroids
Recurrent and persistent cases of Type 2	Thalidomide, clofazimine, chloroquine

Treatment of atypical mycobacterial infections:

Clarithromycin + Azithromycin + Ethambutol + Rifampicin / Rifabutin

ONE-LINERS:

- DOC for atypical mycobacteria – **Clarithromycin**

Worksheet

To do

Quinolone useful in leprosy	
Drug useful for both type 1 and type 2 lepra reaction	
Anti TB drugs having antileprosy activity	
Most common adverse effect of clofazimine	

Concept 11.8 : Antifungal Drugs

Learning objectives:

- To know the classification and mechanism of antifungal agents
- To know the drugs useful for superficial and deep mycosis
- To know the uses and adverse effects of azole antifungal agent
- To know the uses and adverse effects of echinocandins

Time Needed

1 st reading	30 mins
2 nd reading	15 mins

1	Antibiotics		
	• Polyenes	• Echinocandins	• Heterocyclic benzofurans
	• Amphotericin B • Nystatin • Hamycin • Natamycin	• Caspofungin • Micafungin • Anidulafungin	• Griseofulvin
2	Antimetabolites		
	• 5-Flucytosine		
3	Azoles		
	• Imidazoles	• Triazoles	
	• Clotrimazole • Econazole • Miconazole • Oxiconazole • Butaconazole • Tioconazole • Sulconazole • Sertconazole • Ketoconazole	• Fluconazole • Itraconazole • Voriconazole • Posaconazole • Isavuconazole • Ternazole	
4	Allylamines		
	• Terbinafine • Butenafine • Naftifine		



5	Miscellaneous
	• Tolnaftate • Undecylenic acid • Benzoic acid • Quinodochlor • Ciclopirox olamine • Sodium thiosulphate • Haloprogin

ONE-LINERS:

- **Whitfield's ointment** – 3% Salicylic acid + 5% Benzoic acid (It is named after **Arthur Whitfield**, a British dermatologist) .

Mechanisms of action of antifungals:

Polyenes	Bind to ergosterol on the fungal cell membrane ↓ Disorientation of the cell mem- brane
Echinocan- dins	Inhibit the synthesis of 1,3-β-D-glucan ↓ Reduce the structural integrity of the fungal cell wall
Griseofulvin	Binds to β-tubulin ↓ Disorientation of microtubules
5- Flucyto- sine	Inhibits DNA and RNA synthesis
Azoles	Inhibit 14-α-sterol demethylase enzyme
Allylamines	Inhibit squalene epoxidase enzyme

All antifungals are fungicidal except:

- Griseofulvin.
- 5-Flucytosine.
- Azole antifungals.

Polyenes:

Examples:

- | | |
|--|--|
| <ul style="list-style-type: none"> Amphotericin B. Nystatin. | <ul style="list-style-type: none"> Hamycin. Natamycin. |
|--|--|

ONE-LINERS:

- Only polyene that can be used systemically – **Amphotericin B** (Others are too toxic, hence used only topically).

Special formulations of amphotericin B:

- Amphotericin B lipid complex (ABLC).
- Amphotericin B colloidal dispersion (ABCD) .
- Amphotericin B deoxycholate salt.
- Liposomal amphotericin B.

ONE-LINERS:

- Special formulations of amphotericin B are required because – **It is water insoluble.**
- Liposomal amphotericin B is usually preferred as it is – **Better tolerated or less toxic.**
- Amphotericin B is usually preferred for – **Disseminated fungal infections.**
- Only fungal infection that does not respond to Amphotericin B – **Maduramycosis.**
- DOC for Kala-azar – **Amphotericin B.**
- DOC for cryptococcal meningitis – **Amphotericin-B + 5-Flucytosine** (followed by long-term treatment with Fluconazole).
- Amphotericin B causes loss of – **Potassium and calcium ions.**
- Patient on amphotericin B requires – **Potassium supplementation.**
- Oral nystatin is used for – **Esophageal candidiasis.**
- DOC for **Fusarium solanii** keratitis – **Natamycin.**

Echinocandins:

Examples- Caspofungin, Micafungin, Anidulafungin

Mechanism of action- Beta-1,3-glucan synthase inhibition

Administered as IV infusion due to short half-lives

Approved for invasive candidiasis and invasive aspergillosis

Never the DOC due to high cost

Concomitant administration of cyclosporine / tacrolimus is contraindicated due to increase in levels

**ONE-LINERS:**

- Dose of caspofungin – **50 mg/d**
- Dose of micafungin and anidulafungin – **100 mg/d**

Griseofulvin:

Acts by binding to β -tubulin

Oral absorption is increased by fatty foods

Not used much due to high incidence of side effects and drug interactions

ONE-LINERS:

- Antifungal that gets deposited in keratin precursor cells – **Griseofulvin.**
- Only fungal infection in which Griseofulvin is effective – **Dermatophytosis.**
- Disulfiram-like reaction in caused by – **Griseofulvin.**

Azoles:**Fluconazole is DOC in:**

- | | |
|---|--|
| <ul style="list-style-type: none"> • Candidiasis • Esophageal candidiasis. • Localized (pulmonary) cryptococcosis. | <ul style="list-style-type: none"> • Coccidioidal meningitis • Cryptococcal meningitis |
|---|--|

Itraconazole is DOC in:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Histoplasmosis • Blastomycosis • Sporotrichosis | <ul style="list-style-type: none"> • Localized (pulmonary) aspergillosis. • Sporotrichosis. • Maduramycosis. |
|---|---|

Voriconazole is DOC in:

- | |
|---|
| <ul style="list-style-type: none"> • Invasive aspergillosis. • Persistent febrile neutropenia. • Side effects- phototoxicity, visual toxicity, QT prolongation |
|---|

- | |
|---|
| <ul style="list-style-type: none"> • Ketoconazole- useful for Cushing syndrome • Side effect- gynecomastia, |
|---|

ONE-LINERS:

- Only triazole antifungal that can be administered topically – **Fluconazole**.
- Triazoles effective in mucormycosis – **Posaconazole and Isavuconazole**.
- Posaconazole – **useful in mucormycosis**
- Isavuconazole – **useful for Candidiasis, aspergillus, mucormycosis**.

ONE-LINERS:

- Single agent 5-Flucytosine is used in – **Chromoblastomycosis**.

Worksheet

To do

Mechanism of action of griseofulvin	
Dose limiting adverse effect of amphotericin B	
Mechanism of action of azole antifungal agent is	
Anti fungal drug very useful for candidiasis	
Azole antifungal agent causing QT prolongation is	
Terbinafine acts by	
Antimetabolite having antifungal activity is	
Echinocandine useful for	

Concept 11.9 : Antiviral Drugs (Retro & Non Retro Viral)

Learning objectives: To know the classification of anti retroviral drugs based on their mechanism of action

Time Needed

1 st reading	45 mins
2 nd reading	25 mins

Classification of Antiretrovirals

1	Nucleoside reverse transcriptase inhibitors (NRTIs)	• Zidovudine • Lamivudine • Stavudine
		• Didanosine
		• Zalcitabine
		• Emtricitabine
		• Abacavir
		• Festinavir
2	Nucleotide reverse transcriptase inhibitor	• Tenofovir
3	Non-nucleoside reverse transcriptase inhibitors (NNRTIs)	• Nevirapine • Efavirenz • Etravirine
		• Delavirdine
		• Rilpivirine
		• Lersivirine
4.	Protease inhibitors	• Ritonavir • Saquinavir • Indinavir • Nelfinavir • Amprenavir • Fosamprenavir • Lopinavir • Atazanavir • Tipranavir • Darunavir
5.	Entry inhibitors / Fusion inhibitors	• Maraviroc- CCR 5 blocker • Enfuvirtide • Cenicriviroc • Ibalizumab



6.	Integrase inhibitors	• Raltegravir • Elvitegravir • Dolutegravir
7	Maturation inhibitors	• Bevirimat • Viveron

- Dose limiting side effect of zidovudine – **Myelosuppression.**
- Dose limiting side effect of didanosine – **Pancreatitis.**
- **Stavudine- causes severe Neuropathy, Lactic acidosis, Lipodystrophy.**
- Abacavir may cause SJS in apatient with HLA B 5701 allele, so genetic screening should be done before administration of abacavir.
- **Lamivudine,tenofovir useful in HBV infection.**

- **Common side effect of NNRTI – skin rashes.**
- **Specific side effect of nevirapine hepatotoxicity.**
- **Specific side effect of efavirenz psychosis.**
- **Indinavir- causes nephrolithiasis.**
- **Ritonavir – most potent microsomal enzyme inhibitor.**
- **Lipodystrophy least with atazanavir.**
- **Tipranavir- can cause intracranial hemorrhage.**
- **Common side effects of protease inhibitors - hyperglycemia, hyperlipidemia, fat redistribution .**

- **Tesamorelin- GHRF- Reduces abdominal fat in HIV patient with lipodystrophy.**
- **CROFELEMER-Inhibitor of CFTR, USE- HIV drug induced diarrhoea.**

ONE-LINERS:

- NRTIs are active against both – **HIV-1 and HIV-2.**
- Oldest antiretroviral drug – **Zidovudine.**
- Zidovudine should not be combined with – **Stavudine** as both compete for intracellular phosphorylation.
- **Stavudine should not be combined with – Didanosine due to increased risk of severe peripheral neuropathy and potentially fatal pancreatitis.**
- Peripheral neuropathy is maximum with – **Stavudine.**
- Pancreatitis is maximum with – **Didanosine.**
- Festinavir has been shown to be effective in – **Resistant infections.**
- NNRTIs are only active against – **HIV-1.**
- Teratogenic antiretroviral – **Efavirenz.**
- Lersivirine is effective against HIV with – **Y181C mutation.**
- 1st HIV protease inhibitor to be clinically approved – **Saquinavir.**
- Most potent protease inhibitor – **Ritonavir.**
- Least potent protease inhibitor – **Saquinavir.**

- **Boosted PI regimen:** PIs are combined with low dose (100 – 200 mg) ritonavir. Ritonavir inhibits metabolism of the PI, thus increasing its bioavailability.
- **PI not to be combined with ritonavir in boosted PI regimen – Nelfinavir (as it is metabolized by CYP2C19, which is not inhibited by ritonavir).**
- Renal stones are seen with – **Indinavir.**
- **Skin eruptions and hyperglycemia are seen with – Fosamprenavir.**
- **Enfuvirtide is always administered – subcutaneously.**
- Most prominent side effect with enfuvirtide – **Injection site reactions.**
- Antiretroviral drug that can bind to both CCR2 and CCR5 – **Cenicriviroc.**
- **Ibalizumab**– Monoclonal antibody against CD4.
- **Elvitegravir** is approved for initial treatment / naïve cases.
- **Dolutegravir** is approved for raltegravir-resistant infections.
- Maturation inhibitors bind to – **gag protein.**
- **Cobicistat- booster for elvitegravir.**

Prevention of parent-to-child transmission of HIV:

- Most effective drug for preventing parent-to-child transmission of HIV – **Zidovudine.**
- Drug most commonly used for preventing parent-to-child transmission of HIV in developing countries – **Nevirapine.**

Constituents of combined pill for HIV – STRIBILD

- | | |
|---|---|
| <ul style="list-style-type: none"> • Elvitegravir. • Tenofovir. • Emtricitabine. | <ul style="list-style-type: none"> • Cobicistat (acts as a bioenhancer; • inhibits the metabolism of elvitegravir) . |
|---|---|

Drugs for Herpes Virus Infections:

Drugs for human herpes virus infections:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Acyclovir. • Valacyclovir. • Famciclovir (prodrug, active form penciclovir) . • Penciclovir. • Ganciclovir. • Valganciclovir. | <ul style="list-style-type: none"> • Cidofovir. • Foscarnet. • Idoxuridine. • Trifluridine. • Docosanol (entry inhibitor, given topically) . • Fomivirsen. |
|---|---|

**ONE-LINERS:**

- Acyclovir is a – **Nucleoside analogue.**
- DOC for herpes simplex infections – **Acyclovir.**
- DOC for varicella zoster virus infections – **Acyclovir.**
- Dose of acyclovir for herpes simplex infections – **200 mg five times a day OR 400 mg three times a day.**
- Dose of acyclovir for varicella infections **800 mg five times a day.**
- Advantage of valacyclovir over acyclovir **Better oral bioavailability.**
- DOC for CMV infections – **Gancyclovir.**
- Dose limiting side effect of gancyclovir – **Myelosuppression.**
- **DOC for acyclovir / gancyclovir resistant infections – Foscarnet.**
- **Foscarnet- side effect – acute renal failure, external genitalia ulcer.**
- **Fomivirsen** – Antisense oligonucleotide against CMV mRNA.

Drugs for Hepatitis C:

	Targets	Drugs inhibiting
Viral Targets		
1	NS3-NS4a (protease)	• Telaprevir • Boceprevir • Danoprevir
2	NS5a (participates in formation of membrane-bound replication complex)	• Daclatasvir • Ladipasvir • Ombitasvir
3	NS5b (RNA polymerase)	• Sofosbuvir • Dasabuvir
Host Targets		
1	Cyclophilin A	• Alisporivir
2	miR-122 (micro- RNA 122)	• Miravirsen (antisense oligonucleotide)

ONE-LINERS:

- Drug whose approval has been kept on hold due to reports of severe pancreatitis – **Alisporivir**
- Ladipasvir is given – **Orally**
- **Viramidine (under trail)**

Drugs for Hepatitis B

- **Firstline:** Entecavir, Tenofovir
- **Secondline:** lamivudine Adefovir Telbivudine Injectable therapy- Interferons and Pegylated interferons

ONE-LINERS:

- Serum creatinine monitoring is required for – **Adefovir and Tenofovir** (due to nephrotoxicity) .
- **Drugs common to both HIV and HBV – Lamivudine and Tenofovir.**
- **Drug common to both HBV and HCV is INF- α .**
- Costliest treatment for Hep-B – **Pegylated interferons.**

Drugs for Viral Respiratory Infections:

Drugs for Influenza:

	Drug	Mechanism
1	Amantadine	Bind to M ₂ protein on influenza virus ↓ Prevent viral uncoating and replication
2	Rimantadine	
3	Oseltamivir	Bind to viral neuraminidase enzyme ↓ Prevent aggregation of virus at the cell surface ↓ Reduce the release of virus from infected cells
4	Zanamivir	
5	Peramivir	

ONE-LINERS:

- DOC for influenza – **Oseltamivir** (given orally).
- Route of administration of zanamivir - **intra nasal.**

Drugs for respiratory syncytial virus (RSV) :

- Ribavarin (aerosolized).
- Palivizumab (Once monthly IM injection) – Monoclonal antibody against an antigen on the F surface protein on RSV.

Worksheet

To do

NRTI causing myelosuppression is	
NRTI causing dose dependent pancreatitis	
NRTI safe in renal failure	
Anti retro viral having anti hepatitis activity includes	
Specific side effect of nevirapine	
Examples for integrase inhibitors	
Anti retro viral causing cerebral hemorrhage is	
Dose limiting side effect of acyclovir	
Common side effect of ganciclovir	
Sofosbuvir is useful in	
Anti viral drug useful in both influenza A and B virus	

Concept 11.10 : Antimalaria Drugs

Learning objectives

- To know the classification of antimalarial drug
- To know the actions and adverse effects of antimalarial drugs
- To the artemisinin combination therapy
- To know the chemoprophylaxis of malaria

Time Needed

1 st reading	000000 mins
2 nd reading	000000 mins

Classification of antimalarial drugs

4-aminoquinolines	• Chloroquine • Amodiaquine • Piperaquine
Quinoline-methanol	• Mefloquine
Cinchona alkaloids	• Quinine • Quinidine
Biguanide	• Proguanil (Chloroguanide)
Diaminopyrimidine	• Pyrimethamine
8-aminoquinolines	• Primaquine • Tafenoquine
Sulfonamides and sulfones	• Sulfadoxine • Sulfamethopyrazine • Dapsone
Antibiotics	• Tetracycline • Doxycycline • Clindamycin
Sesquiterpine lactones	• Artemisinins
Amino alcohols	• Halofantrine • Lumefantrine
Naphthyridine	• Pyronaridine
Naphthoquinone	• Atovaquone

Effects of antimalarials on various developmental stages of parasite:

Gp	Drug	Liver stages			Blood stages	
		Sporozoite	Primary	Hypnozoite	Asexual	Gametocyte
1	Artemisinins	-	-	-	+	+
	Chloroquine	-	-	-	+	±



	Mefloquine	-	-	-	+	-
	Quinine / Quinidine	-	-	-	+	±
	Pyrimethamine	-	-	-	+	-
	Sulfadoxine	-	-	-	+	-
	Tetracyclines	-	-	-	+	-
2	Atovaquone / Proguanil	-	+	-	+	±
3	Primaquine	-	+	+	-	+

ONE-LINERS:

- Quinine was isolated from – **Cinchona bark** in 1820
- Chloroquine was originally marketed as – **Resochin**
- Proguanil was introduced in – **1945**
- Fastest onset of action – **Artemisinin**
- Most toxic antimalarial – **Quinine**

Chloroquine:

Indications of chloroquine: (Mnemonic: RED LIP Mahatma Gandhi) :

- **R**heumatoid arthritis.
- **E**xtra-intestinal amoebiasis (Amoebic liver abscess) .
- **D**iscoid lupus erythematosus (DLE) .
- **L**epa reaction Type 2 (Erythema nodosum leprosum or ENL) .
- **I**nfectious mononucleosis.
- **P**hotodermatitis.
- **M**alaria.
- **G**iardiasis.

ONE-LINERS:

- MOA: **Binds to heme and prevents its degradation.**
- Plasma protein binding – **50%.**
- Therapeutic plasma levels – **1530 ng/mL.**
- $V_d = 1200 - 1500 \text{ L.}$
- Mainly gets concentrated in – **Liver and Retina.**
- Loading dose: **600 mg stat.**
- **Loading dose of chloroquine is needed because of – Large V_d .**
- Chloroquine may precipitate / exacerbate – **Porphyria and Psoriasis.**
- Bull's eye macula / Retinopathy is seen with – **Chloroquine.**
- Retinopathy due to chloroquine is – **Irreversible.**

Risk factors for retinopathy due to chloroquine:

- Age > 60 years.
- Duration of use > 5 years.
- Daily dose > 3 mg/kg for Chloroquine and > 6.5 mg/kg for Hydroxychloroquine.
- Cumulative dose > 100g.
- Renal disease.
- Liver disease.
- Pre-existing / Concomitant retinal disease.
- Obesity.

ONE-LINERS:

- Hydroxychloroquine is mainly used in – **RA and SLE**.

Artemisinin Derivatives:

Examples:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Dihydroartemisinin. • Artesunate. • Artemether. | <ul style="list-style-type: none"> • Arteether. • Arterolane. |
|---|---|

ONE-LINERS:

- Artemisinin is derived from – *Artemisia annua* (**Qinghasou**) mentioned in Chinese traditional medicine.
- Artemisinin cannot be directly used in humans as – It is **insoluble in both water and oil** (Hence, derivatives of artemisinin are used).
- Artemisinin derivative developed in India – **Arteether**.

Routes of administration:

Dihydroartemisinin	Oral
Artesunate	Oral, IM, IV, rectal
Artemether	Oral, IM
Arteether	IM
Arterolane	Oral

Artemisinin-Combination therapies (ACTs) :

- Artesunate + Mefloquine.
- Artesunate + Sulfadoxine / Pyrimethamine.
- Artesunate + Amodiaquine.
- Artesunate + Pyronaridine.
- **Artemether + Lumefantrine- available as FDC (fixed dose combination).**
- Arterolane + Piperaquine.
- Dihydroartemisinin + Piperaquine.

ONE-LINERS:

- Artemisinin monotherapy is contraindicated due to rapid development of – **Resistance and Recrudescence.**

Drugs of choice for malaria:

	Normal scenario	In pregnancy
DOC for uncomplicated malaria	Chloro - quine	Chloroquine
DOC for complicated malaria	ACT	1 st trimester – Quinine 2 nd / 3 rd trimester – ACT
DOC for chloro- quine-resistant malaria (both uncomplicated and complicated)	ACT	1 st trimester – Quinine 2 nd / 3 rd trimester – ACT

ONE-LINERS:

- Quinine is also useful in – **Nocturnal muscle cramps.**

Chemoprophylaxis of malaria:

1	CQ-sensitive area	Chloroquine: 300 mg or 5 mg/ kg weekly. Started 1 week before travel and continued up to 4 weeks after return from endemic area.
2	CQ-resistant area > 6 weeks stay	Mefloquine: 250 mg. Started 1 – 2 weeks before travel and continued up to 4 weeks after return from endemic area.
3	CQ-resistant area < 6 weeks stay	Doxycycline: 100 mg. Started 1 day before travel and continued up to 4 weeks after return from endemic area.
4	Alternative	Proguanil: 200 mg daily + Chloroquine 300 mg weekly.

ONE-LINERS:

- DOC for chemoprophylaxis – **Chloroquine**.
- DOC for chemoprophylaxis for chloroquine-resistant endemic area; Stay < 6 weeks – **Doxycycline**.
- DOC for chemoprophylaxis for chloroquine-resistant endemic area; Stay < 6 weeks – **Doxycycline**.
- DOC for chemoprophylaxis for chloroquine-resistant endemic area for a pregnant female – **Mefloquine**.
- Neuropsychiatric reactions are a side effect of – **Mefloquine**.

Mefloquine:

- Basic drug – binds with $\alpha 1$ acid glycoprotein.
- Long half life (20days).
- Side effect – Neuropsychosis.
- When given with Halofantrine, quinine – risk of QT prolongation.

Primaquine:

- Gametocidal.
- Vivax curative.
- In G6PD deficiency patients – causes hemolytic anemia.
- C/I= Pregnancy.

INTERMITTENT PREVENTIVE THERAPY FOR MALARIA

Single dose of Pyrimethamine (75 mg) / Sulfadoxine (1500 mg) each in 2nd and 3rd trimester (with a gap of not less than 1 month) is recommended for pregnant women in high *P. falciparum* endemic areas.

Worksheet

To do

Anti malarial drug used for radical cure of vivax infection	
Antimalarial drug causing neuropsychosis	
Indications of ACT	
DOC for cerebral malaria	
Drug effective only for extra intestinal amoebiasis	
Black water fever is the adverse effect of	

Concept 11.11 : Ant Amoebic Drugs

Learning objective : To know the drugs useful in amoebiasis and other protozoal infections

Time Needed

1 st reading	20 mins
2 nd reading	15 mins

Drugs for Amoebiasis:

Tissue Amoebicides		
1	Nitroimidazoles	• Metronidazole • Tinidazole • Ornidazole • Secnidazole • Satranidazole • Nimorazole
2	Alkaloids	• Emetine • Dehydroemetine
3	Chloroquine	
Luminal Amoebicides		
		• Paromomycin • Diloxanide furoate • Nitazoxanide • Quinidochlor (Clioquinol) • Diiodohydroxyquin (Iodoquinol) • Tetracyclines

ONE-LINERS:

- DOC for amoebiasis – **Metronidazole.**
- DOC for extra-intestinal amebiasis – **Metronidazole.**
- Chloroquine is effective only in – **Amoebic liver abscess.**
- Dose of metronidazole for intestinal amebiasis – **400 mg TDS × 5-7 days.**
- Dose of metronidazole for extra- intestinal amebiasis – **800 mg TDS × 7-10 days.**
- Luminal amebicide of choice / DOC to eradicate asymptomatic cyst passer state – **Paromomycin.**
- Dose of paromomycin as a luminal amebicide – **2535 mg/kg/d oral in three divided doses.**
- Subacute myelo-optic neuropathy (**SMON**) is a side effect of – **Clioquinol and Iodoquinol.**
- **Flatulence is th common side effect of Diloxanide furoate.**
- For Guinea worm infection – **NIRIDAZOLE is the Drug of choice.**

Other Protozoal Infections:

DOC for other protozoal infections:

Infection	DOC
Giardiasis	Tinidazole
Trichomoniasis	Metronidazole
Visceral leishmaniasis (Kala azar)	(Liposomal) Amphotericin B
Cutaneous leishmaniasis (Oriental sore)	Sodium stibogluconate
Chagas' disease (American sleeping sickness)	Benznidazole > Nifurtimox
Toxoplasmosis	Pyrimethamine/ Sulfadiazine + Folinic acid
Cryptosporidiasis	Nitazoxanide
Isosporiasis	Cotrimoxazole
Cyclosporiasis	Cotrimoxazole
Babesiosis	Clindamycin + Quinine
Balantidiasis	Tetracyclines
Naegleria	Amphotericin B (High dose) + Rifampicin

Drugs For African Sleeping Sickness:

Treatment of African sleeping sickness:

	Early stage (Haemo-lym- phatic stage / No CNS involvement)	Late stage (Neurolog- ical / CNS involvement)
West African (<i>T. brucei gambiense</i>)	Pentamidine	Eflornithine
East African (<i>T. brucei rhodesiense</i>)	Suramin	Melarsopol

Treatment of South American Trypanosomiasis (Chagas disease) - Benznidazole (DOC), Nifutrimox

ONE-LINERS:

- Dose of metronidazole for giardiasis and trichomoniasis – **400 mg TDS x 7-10 days.**
- DOC for Giardiasis in pregnancy – **Paromomycin.**



Worksheet

To do

Common side effect of metronidazole	
Aminoglycoside useful for amoebiasis	
Drug useful in toxoplasmosis in pregnancy	
Common side effect of diloxanide furoate	

Concept 11.12 : Anthelmintic Drugs

Learning objective:

- To know the examples for various anthelmintic agents
- To know appropriate choice of drugs for helminthic infections

Time Needed

1 st reading	30 mins
2 nd reading	15 mins

Examples for anthelmintic agents

- | | |
|---|---|
| <ul style="list-style-type: none"> • Mebendazole • Albendazole • Thiabendazole • Pyrantel pamoate • Piperazine | <ul style="list-style-type: none"> • Diethylcarbazine citrate (DEC) • Ivermectin • Levamisole • Niclosamide • Praziquantel |
|---|---|

ONE-LINERS:

- Ivermectin is a semi-synthetic derivative of active principle obtained from – **Streptomyces avermitilis**.

DOC for Helminthic Infestations:

Helminthic infestation	DOC	Other alternatives
Ascariasis (Round worm)	Albendazole	• Mebendazole • Pyrantel pamoate
Hookworm	Albendazole	Mebendazole
Enterobiasis (Pin worm or Seat worm)	Mebendazole	• Albendazole • Pyrantel pamoate
Trichuriasis (Whip worm)	Mebendazole	Albendazole
Trichinellosis	Albendazole + Steroids	Mebendazole + Steroids
Strongyloidiasis	Ivermectin	Albendazole
Lymphatic filariasis	DEC	Ivermectin
Loiasis	DEC	Ivermectin
River blindness (Onchocerciasis)	Ivermectin	
Dracunculosis (Guinea worm)	Metronidazole	Mebendazole
Visceral larva migrans	Albendazole	Steroids
Cutaneous larva migrans	Albendazole	• Thiabendazole • Mebendazole • Ivermectin

Taenia saginata (Beef tape- worm)	Praziquantel	Niclosamide
Taenia solium (Pork tapeworm) other than neurocysticercosis	Praziquantel	Niclosamide
Neurocysticercosis	Albendazole + Steroids	Praziquantel + Steroids
Diphyllobothriasis (Fish tape- worm)	Praziquantel	
Hymenolepis nana (Dwarf tapeworm)	Praziquantel	Albendazole
Hydatid disease	Albendazole	Praziquantel
Schistosomiasis	Praziquantel	• Oxamniquine • Artemether
Paragonimiasis	Praziquantel	Triclabendazole
Clonorchis sinensis	Praziquantel	
Opisthorchis viverrini	Praziquantel	
Fasciola hepatica (Liver fluke)	Triclabendazole	
Other intestinal nematodes	Praziquantel	

- Broad spectrum anti helminthic for Trematodes – Doc Praziquantel (Exception- Fasciola hepatica- Triclabendazole, Bithional) .
- Broad spectrum anti helminthic for Cystodes- – Doc Praziquantel (Exception Echi- nocoscosis- Neurocysticercosis- Albendazole.
- Broad spectrum anti helminthic for Nematodes- Doc Albendazole (Exception for Ochromocerca volvulus- Ivermectin, for W. bancrofti—DEC) .

ONE-LINERS:

- Pyrantel pamoate acts by causing – **Spastic paralysis.**
- Piperazine acts by causing – **Flaccid paralysis.**
- DOC for worm infestation in pregnancy – **Piperazine.**
- Dose of albendazole in worm infestation – **400 mg single oral dose for adults and children > 2 years; 200 mg oral single dose for children < 2 years.**
- Dose of mebendazole in pin worm infestation – **100 mg oral single dose.**
- Dose of mebendazole in trichinellosis – **200 mg BD x 4 days.**
- Dose of mebendazole in other worm infestations – **100 mg BD x 3 days.**
- Dose of ivermectin in strongyloidiasis and onchocerciasis – **150200 µg/kg single oral dose.**
- Dose of DEC in filariasis – **2 mg/kg TDS (6 mg/kg/d) x 21 days.**
- Dose of albendazole in neurocysticercosis – **400 mg BD x 8-30 days for adults; 15 mg/kg/d in 2 divided doses x 830 days for children.**
- Levamisole is now **only used as an anthelmintic;** not as an immunomodulator.



Scabicides:

Scabicide	Method of use
Permethrin (5%)	Single application
Benzene hexachloride (BHC)	Single application
Benzyl benzoate	3 applications at 12 hourly intervals
Crotamiton (10%)	2 applications daily for 14 days
Ivermectin	Single oral dose (200 µg/kg)

Drugs used for visceral leishmaniasis-

- **Amphotericin B (DOC) .**
- Pentamidine.
- **Paromomycin.**
- **Miltefosine.**
- **Sitamaquine.**

for all forms of leishmaniasis: Sodium stibogluconate.



Worksheet

To do

Anthelmintic drug causing spastic paralysis	
Uses of ivermectin	
Broad spectrum drug useful for majority of nematodes	
DOC for neurocysticercosis	

Recent FDA approved drugs:

	Drug	Mechanism	Indication
1	Ceftazidime + Avibactam	3 rd gen cephalosporin + Non- β -lactam β -lactamase inhibitor	Complicated intra-abdominal and urinary tract infections
2	Obiltoxaximab	Monoclonal antibody targeting Bacillus anthracis toxin	Inhalational anthrax
3	Dalbavancin	2 nd generation lipoglycopeptide; inhibits bacterial cell wall synthesis	Acute bacterial skin and skin structure infections
4	Oritavancin	Glycopeptide; inhibits bacterial cell wall synthesis	Acute bacterial skin and skin structure infections
5	Tedizolid	Oxazolidinone; inhibits bacterial protein synthesis	Acute bacterial skin and skin structure infections
6	Finafloxacin	Fluoroquinolone; inhibits bacterial DNA gyrase	Acute otitis externa
7	Atazanavir	HIV protease inhibitor	HIV-1 infection
8	Darunavir	HIV protease inhibitor	HIV-1 infection
9	Cobicistat	Inhibits the CYP metabolism of elvitegravir	HIV-1 infection
10	Elbasvir	Inhibits NS5a protein; prevents formation of membrane replication complex	HCV genotypes 1-4 infection
11	Velpatasvir	Inhibits NS5a protein; prevents formation of membrane replication complex	HCV genotypes 1-6 infection
12	Daclatasvir	Inhibits NS5a protein; prevents formation of membrane replication complex	HCV genotype 3 infection
13	Ombitasvir	Inhibits NS5a protein; prevents formation of membrane replication complex	HCV genotype 4 infection
14	Paritaprevir	NS3-NS4a protease inhibitor	HCV genotype 4 infection
15	Grazoprevir	NS3-NS4a protease inhibitor	HCV genotype 1-4 infection
16	Dasabuvir	NS5b RNA polymerase inhibitor	HCV infection
17	Peramivir	Neuraminidase inhibitor	Acute uncomplicated influenza in adults

18	Tavaborole	Inhibits leucyl-tRNA synthetase; inhibits fungal protein synthesis	Onychomycosis of toe nails
19	Efinaconazole	Inhibits lanosterol 14- α demethylase (14- α sterol demethylase) enzyme in fungi	Onychomycosis of toe nails
20	Isavuconazonium sulphate (prodrug of isavuconazole)	Inhibits lanosterol 14- α demethylase (14- α sterol demethylase) enzyme in fungi	Invasive aspergillosis; Invasive mucormycosis
21	Miltefosine	Disturbs alkyl-phospholipid metabolism and synthesis of glycolipids and glycoproteins in <i>Leishmania donovani</i> .	Leishmaniasis

CONCEPTS

- ⇒ **Concept 12.1 Classification of Cytotoxic Agents (Cell Cycle Specific and Cell Cycle Non Specific)**
- ⇒ **Concept 12.2 Antimetabolites**
- ⇒ **Concept 12.3 Antibiotic Anticancer Drugs**
- ⇒ **Concept 12.4 Alkylating and Related Anticancer Drugs**
- ⇒ **Concept 12.5 Vinca Alkaloids, Taxanes**
- ⇒ **Concept 12.6 Topoisomerase Inhibitors**
- ⇒ **Concept 12.7 Hormonal Agents Used in Cancer**
- ⇒ **Concept 12.8 Tyrosine Kinase & other Small Molecule Inhibitors**
- ⇒ **Concept 12.9 Monoclonal Antibodies**
- ⇒ **Concept 12.10 Chemotherapy**
- ⇒ **Concept 12.11 Anticancer Drugs ADR & Management of Adverse Drug Reaction**

Concept 12.1 : Classification of Cytotoxic Anticancer Drug

Learning objective : To know the classification anticancer drugs according to cell cycle specific or cell non specific

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

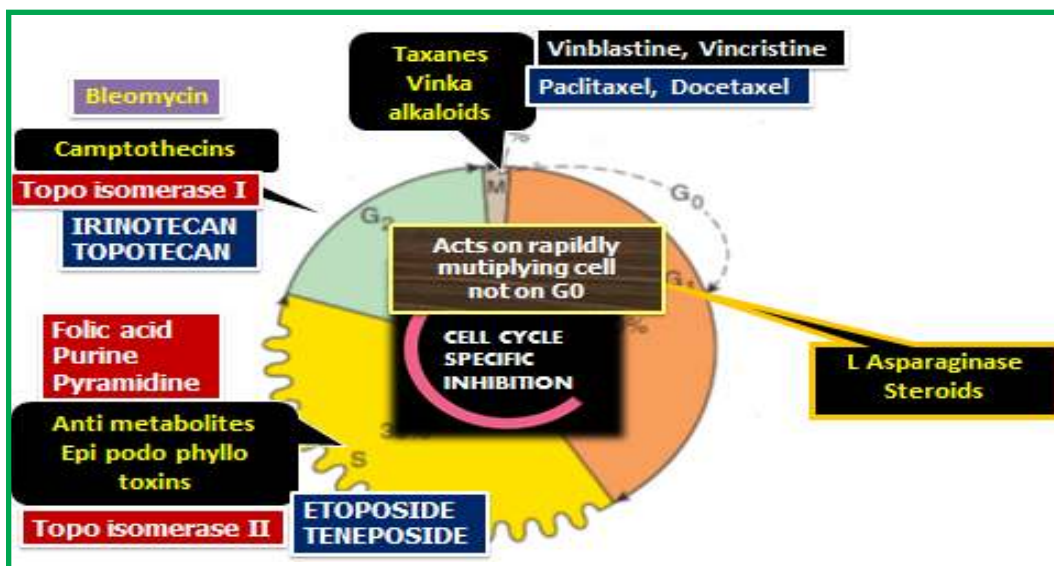


Fig. 12.1

Cell cycle non specific agents

Alkylating agents

Cisplatin

Antibiotic anticancer except Bleomycin

Natural Sources of Anti-Cancer Drugs:

	Drug	Natural source
A: DRUGS OBTAINED FROM PLANTS		
1	Vinca alkaloids	<i>Catharanthus roseus</i> / <i>Vinca rosea</i> (Madagascar periwinkle)
2	Taxanes	Bark of <i>Taxus brevifolia</i> (Western yew / Pacific yew tree)
3	Camptothecin analogues	<i>Camptotheca acuminata</i>
4	Epipodophyllotoxins	<i>Podophyllum peltatum</i> , (Mandrake plant / Mayapple)

B: DRUGS OBTAINED FROM MICRO-ORGANISMS		
1	Epothilones	<i>Sorangium cellulosum</i> , a mycobacterium
2	Actinomycin D	<i>Streptomyces</i> spp.
3	Anthracyclines	<i>Streptomyces peuceticus</i> var. <i>caesius</i>
4	Bleomycin	<i>Streptomyces verticillus</i>
5	Mitomycin C	<i>Streptococcus caespitosus</i>
6	Trabectedin	<i>Ecteinascidin turbinata</i> (Marine tunicate)
7	L-asparaginase	<i>Escherichia coli</i>

Worksheet

To do

Methotrexate acts on which phase of cell cycle	
Antibiotic anticancer drug acts on G2 phase of cell cycle is	
Paclitaxel acts on which phase of cell cycle	
Cyclophosphamide is cell cycle specific or cell cycle non specific	

Concept 12.2 : Antimetabolites

Learning objectives:

- To know the classification of antimetabolites
- To know the uses and adverse effects of antimetabolites

Time Needed

1 st reading	15 mins
2 nd reading	10 mins

Classification:

Folic acid / Folate antagonists	Methotrexate (Amethopterin) Pemetrexed Trimetrexate Pralatrexate Lometrexol Raltitrexed
Pyrimidine analogues	5-Fluorouracil (5-FU) Capecitabine Floxuridine Cytarabine (Ara-C) Azacytidine Decitabine Gemcitabine
Purine analogues	6-Mercaptopurine (6-MP) 6-Thioguanine (6-TG) Fludarabine Cladribine Clofarabine Nelarabine Pentostatin

ONE-LINERS:

- DMARD of choice for rheumatoid arthritis – **Methotrexate**.
- Starting dose of Methotrexate in RA – **7.5 – 15 mg oral once a week**.
- Dose of methotrexate for psoriasis – **2.5 mg/d x 5 days in a week OR 10 – 25 mg IV once a week**.
- Dose of intrathecal methotrexate for leukemic meningitis and meningeal carcinomatosis in < 3 years of age – **12 mg repeated every 4 days** until malignant cells are no longer evident in CSF.
- **Glucarpidase** – Recombinant carboxypeptidase G₂, that cleaves methotrexate. When given IV, methotrexate levels fall by ≥ 99% within 5 – 15 minutes.
- Erythematous pruritic rash is seen with – **Pemetrexed**.
- Pemetrexed toxicity can be ameliorated by – **Dexamethasone + Folic acid + Vitamin B₁₂**.
- Indication of trimetrexate – ***Pneumocystis jiroveci* pneumonia**.
- Genetic deficiency of dihydropyrimidine dehydrogenase (DPD) can lead to increased toxicity with – **5-FU**.
- **Capecitabine** – Prodrug of 5-FU.
- **Gemcitabine- DOC for pancreatic cancer**.
- **Gemcitabine is the radio sensitizer**.
- Most common drug useful for colorectal cancer – 5FU.
- Hand-foot syndrome is a side effect of – **Capecitabine > 5-FU**.
- Cerebellar **toxicity is a side effect of – Cytarabine**.
- Drugs that act by causing global hypomethylation of DNA – **Azacitidine and Decitabine**.
- While co-administering with allopurinol, the oral dose of 6-MP should be reduced by – **75%**.
- **DOC for hairy cell leukemia –Cladribine**.
- Capillary leak syndrome is a side effect of – **Clofarabine**.
- **Pentostatin inhibits – Adenosine deaminase (ADA)**.

Worksheet

To do

Antidote of methotrexate	
Cerebellar ataxia is the adverse effect of	
6MP inactivated by	
Drugs causing hand foot syndrome are	
Anticancer drugs causing radio sensitizer	
Pentostatin acts by inhibiting	
Drug of choice for pancreatic cancer is	

Concept 12. 3 : Antibiotic Anticancer Drugs

Learning objectives:

- To know the examples of anticancer antibiotics
- To know the uses and adverse effects of anticancer antibiotics

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Anthracyclines	• Doxorubicin (Adriamycin). • Donorubicin (Rubidomycin). • Epirubicin. • Idarubicin. • Valrubicin.
Mitoxantrone.	
Mitomycin C.	
Actinomycin D (Dactinomycin) .	
Bleomycin.	

ONE-LINERS:

- Pericarditis-myocarditis syndrome can be caused by – **Doxorubicin**.
- **Antidote for doxorubicin- DEXRAZOXANE+ Alpha tocopherol.**
- **Mithramycin (Plicamycin)-useful for treatment of hypercalcemia.**
- **Adriamycin flare** – Erythematous streaking near the site of doxorubicin infusion.
- Mitomycin was isolated by – **Wakaki and co-workers in 1958.**
- Mitomycin C can be used topically for – **Laryngotracheal stenosis.**
- Actinomycin D was first isolated by – **Selman Waksman and H. B. Woodruff in 1940.**
- DOC for Wilms' tumour – **Actinomycin D.**
- **Radiation recall phenomina side effect of Actinomycin D.**
- Bleomycin mainly accumulates in – **Lungs and Skin.**
- Flagellate dermatitis can be caused by – **Bleomycin.**
- Treatment of flagellate dermatitis – **Topical steroids.**
- Radio protector – **Amifostine.**
- Radio sensitizer – **Gemcitabine.**
- Radiation recall phenomina – **Dactinomycin.**

Worksheet

To do

Anti cancer drugs causing lung fibrosis	
Flagellate dermatitis can be caused by	
Anticancer drug useful for hypercalcemia	
Drug useful for laryngo treacheal stenosis is	
Radiation recall reactions are seen with	
Antidote for doxorubicin	

Concept 12.4 : Alkylating agents

Learning objective:

- To know the examples for alkylating anticancer drugs
- To know the general properties of alkylating agents
- To know the uses and adverse effects of alkylating agents

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Nitrogen mustards	• Mechlorethamine (Mustine HCl) • Cyclophosphamide • Ifosfamide • Glufosfamide • Melphalan • Chlorambucil • Bendamustine
Ethylenimines	• ThioTEPA • Altretamine
Alkyl sulfonates	• Busulfan
Nitrosoureas	• Carmustine (BCNU) • Lomustine (CCNU) • Semustine • Streptozotocin
Methylhydrazines	• Procarbazine
Triazines	• Dacarbazine • Temozolomide
Platinum coordination complexes	• Cisplatin (1st generation) • Carboplatin (2nd generation) • Oxaliplatin (3rd generation)

ONE-LINERS:

- The first clinically used nitrogen mustard Mustine HCl.
- Most reactive of the nitrogen mustards Mustine HCl.
- Vesication due to Mustine HCl can be treated with – **Sodium thiosulphate.**
- DOC for Wegener's granulomatosis Cyclophosphamide.
- DOC for steroid-dependent nephritic syndrome – **Cyclophosphamide.**
- MESNA – Mercaptoethane sulfonate sodium
- Most preferred route for administering Mesna – **Intravenous.**
- Glufosfamide – β -D-glucose + Ifosforamide mustard (A metabolite of ifosfamide).
- Myelosuppression due to nitrosoureas is Delayed (Onset around 6 weeks after institution) and prolonged.
- Anti-CA drug which is a MAO inhibitor – **Procarbazine.**
- Anti-CA drug causing disulfiram-like reactions – **Procarbazine.**
- Diuretic used with cisplatin – **Mannitol (to establish a chloride diuresis).**
- While administering cisplatin infusion, infusion equipment should not be made of aluminium (Since cisplatin reacts with Aluminium).
- Amifostine is the antidote for cisplatin.
- Dose limiting side effect of oxaliplatin is Peripheral neuropathy.
- Site of action of alkylating agents – **N7 GUANINE RESIDUE.**
- VENO OCCLUSIVE DISEASE OF LIVER side effects of alkylating agents, which can be minimized by (Budd-Chiari syndrome) – **DEFIBROTIDE.**
- Secondary cancer produced maximally by procarbazine.
- Chemical pancreatectomy – **Streptozocin.**

Cisplatin	• OTOTOXIC • NEPHROTOXIC- • NEUTORTOXIC • Highest emetogenic • Hypo Mg, Cal, K & • Hypophosphatemia
Carboplatin	• Myelosuppression
Oxaliplatin	• Neurotoxic • PharyngoLaryngeal • paraesthesias



Worksheet

To do

Antidote for cyclophosphamide	
Amifostine is the antidote for	
Chemical pancreatectomy agent is	
Antithrombotic drug useful for treating veno occlusive disease of liver is	
Most emetogenic anticancer drug is	
Anticancer drug having MAO enzyme inhibitory action is	

Concept 12.5 : Vinca Alkaloid & Taxanes

Learning objectives: To know the action and adverse effects of vinca alkaloid & taxanes

Time Needed

1 st reading	10 mins
2 nd reading	5 mins

Vinca alkaloids	• Vincristine (Oncovin) • Vinbastine • Vinorelbine
Taxanes	• Paclitaxel • Docetaxel
Epothilones	• Ixabepilone • Patupilone • Sagopilone • 21-aminoepothilone-B
Estramustine	

ONE-LINERS:

- Bone marrow sparing cytotoxic agent – **Vincristine**
- Taxol – **Brand name of Paclitaxel.**
- Stocking and glove neuropathy seen with – **Paclitaxel**
- Docetaxel is administered with – **Polysorbate 80**
- Ixabepilone is indicated in – **Metastatic breast cancer**
- Estramustine = Estradiol (Hormone) + Normustine (Nornitrogen mustard).
- Estramustine is indicated in – Hormone refractory prostate cancer



Worksheet

To do

Anticancer drugs acting on M phase	
Anticancer drugs causing skin vesicant	
Ixabepilone useful in	
Estramustine useful in	

Concept 12.6 : Drugs Inhibiting Topo isomerase Enzyme

Learning objective: To study the uses and adverse effects of topoisomerase inhibitors

Time Needed

1 st reading	00000 mins
2 nd reading	00000 mins

Classification:

Camptothecin analogues (Topoisomerase-I inhibitors)	• Topotecan • Irinotecan
Epipodophyllotoxins (Topoisomerase-II inhibitors)	• Etoposide • Teniposide

ONE-LINERS:

- Active metabolite of irinotecan – **SN-38**.
- Dose-limiting toxicity of irinotecan – **Delayed diarrhea**.
- Toxicity of irinotecan is increased in – **Gilbert's syndrome**.
- Irinotecan has anti cholinesterase property.

Enzymes

- L-asparaginase is synthesized from – **E.coli**.
- L-asparaginase is used in – **ALL**.
- Hypercoagulability is a side effect of – **L-asparaginase**.
- Pegasparginase is administered – **IM once every 14 days**.

Miscellaneous

Examples:

- Hydroxyurea.
- Trabectedin.
- Arsenic trioxide.
- Vorinostat.
- Mitotane.
- Tretinoin (all-trans retinoic acid).
- Romidepsin.

ONE-LINERS:

- Hydroxyurea inhibits – **Ribonucleoside diphosphate reductase**.
- Hydroxyurea can also be used as a – **Radiosensitizer and Inducer of fetal hemoglobin**.
- DOC for sickle cell anemia – **Hydroxyurea**.
- Mitotane is used for – **Palliation in inoperable adrenocortical carcinoma**.
- Tretinoin and arsenic trioxide are used in – **Acute promyelocytic leukemia**.
- Romidepsin and vorinostat inhibit – Histone deacetylase, Romidepsin and vorinostat are used in – **Cutaneous T cell lymphomas**.



Worksheet

To do

Concept 12.7 : Hormonal Therapy for Cancer

Learning objectives: To know the classification hormonal agents useful in cancer

Time Needed

1 st reading	25 mins
2 nd reading	15 mins

Selective Estrogen Receptor Modulators (SERMs)	• Tamoxifen • Toremifene • Raloxifene	• Breast cancer
Selective Estrogen Receptor Downregulator (SERDs)	• Fulvestrant	• Breast cancer
Aromatase inhibitors	• 1st generation: • Aminoglutethimide • 2nd generation: • Formestane • 3rd generation: • Letrozole • Anastrozole • Exemestane	• Breast cancer
GnRH Analogues	• Nafarelin • Buserelin • Goserelin • Histrelin • Deslorelin • Triptorelin • Leuprolide	
GnRH Antagonists	• Abarelix • Degarelix	
Antiandrogens	• Flutamide • Bicalutamide • Nilutamide • Enzalutamide	
Glucocorticoids		

Anti-androgens:

Steroidal	Non-steroidal
• Danazol • Cyproterone acetate	• Flutamide • Bicalutamide • Nilutamide • Enzalutamide
Not used nowadays	Used in CA prostate

Malignancies in which Glucocorticoids are Useful / Effective

• ALL • CLL • Hodgkin's disease	• Non-Hodgkin's lymphoma • Multiple myeloma
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Newer Drugs for Prostate Cancer

1	Abiraterone	17- α hydroxylase inhibitor
2	Galeterone	Androgen receptor inhibitor + 17- α hydroxylase inhibitor
3	Erismodegib	SMO antagonist (Hedgehog pathway antagonist)

ONE-LINERS:

- Advantage of raloxifene over tamoxifen – **Antagonist at uterus; hence minimal risk of endometrial carcinoma.**
- Ormeloxifene (Centchroman) – **non-steroidal SERM used as oral contraceptive (marketed as Saheli).**
- Steroidal inhibitors of aromatase – **Formestane and Exemestane (Rest are non-steroidal).**
- Irreversible inhibitors of aromatase – **Formestane and Exemestane.**
- Ganirelix and Cetrorelix are GnRH antagonists mainly for – **Ovarian hyperstimulation.**
- OSPEMIFENE – **is a SERM useful in DYSPAREUNIA.**

Worksheet

To do

Side effects of tamoxifen	
Advantage of raloxifene over tamoxifen	
Example for SERD	
Examples for steroidal aromatase inhibitors	
Androgen receptor blockers useful cancer prostate are	

Concept 12.8 : Tyrosine Kinase Inhibitors

Learning objectives: To know the uses of tyrosine kinase inhibitors

Time Needed

1 st reading	30 mins
2 nd reading	15 mins

PDGFR	Imatinib	- CM; - GIST
EGFR1 specific	- Erlotinib - Gefitinib - Afatinib	Useful in Metastatic non small cell lung cancer
VGEFR	- Sunitinib - Sorafenib	- Sunitinib- RCC, HCC - Sorafenib- RCC, GIST
Dual EGFR1 and EGFR2 specific	Lapatinib	Breast cancer

Indications of imatinib-

- CML (Almost all cases are bcr-abl +ve).
- ALL (bcr-abl +ve).
- CMML (EVT6-PDGFR +ve).
- GIST.
- Dermatofibrosarcoma protuberans

ONE-LINERS:

- The first tyrosine kinase inhibitor to be approved by FDI – **Imatinib**.
- DOC for CML – **Imatinib**.
- Dasatinib and Nilotinib**. - Treatment of imatinib-resistant CML – **Dasatinib**.
- Side effect of dasatinib – **Pleural effusion**.
- Treatment of dasatinib – **resistant CML Nilotinib**.
- Side effect of nilotinib – **QTc prolongation**.
- Erlotinib and gefitinib are approved for Refractory non-small cell lung cancer.
- Erlotinib- causesdysmorphic eyelashes.
- Bioavailability of erlotinib is – **Increased by food** (67% → 100%); hence **not combined** with food to prevent toxicity.
- Treatment of imatinib-resistant GIST – **Sunitinib**.
- Only drug approved for metastatic hepatocellular carcinoma – **Sorafenib**.
- The most common side effect of sunitinib and sorafenib – **Fatigue**.

Newer Tyrosine Kinase Inhibitors:

	Drug	Target	Indication
1	Cabozantinib	c-Met and VEGFR2 tyrosine kinases	Medullary thyroid cancer
2	Masitinib	Multiple tyrosine kinases (c-Kit, PDGFR and FGR tyrosine kinases)	Mast cell tumours, specifically in dogs
3	Semaxanib	VEGFR tyrosine kinase (antiangiogenic)	Advanced colorectal cancer (Trials discontinued due to lack of efficacy)
4	Ruxolitinib	Janus kinase	Myelofibrosis (Intermediate or high-risk)
5	Ibrutinib	Bruton's tyrosine kinase	Mantle cell lymphoma & Chronic lymphoid leukemia
6	Vemurafenib Dabrafenib dramtenib	B-raf enzyme	Late stage melanoma
7	Ponatinib	bcr-abl fusion protein	CML & Philadelphia chromosome +ve ALL
8	Ceritinib	ALK	ALK +ve metastatic non small cell lung cancer
9	Vandetanib	Multiple (VEGFR, EGFR, Ret) tyrosine kinase	Late stage medullary thyroid cancer
10	Regorafenib	Multiple	Metastatic colorectal cancer Advanced gastrointestinal stromal tumours

ONE-LINERS:

- Sales of ponatinib were temporarily suspended in 2013 due to – **the risk of life-threatening blood clots and severe narrowing of blood vessels.**
- 1st drug approved by FDA for late stage (metastatic) medullary thyroid cancer – **Vandetanib.**
- Vandetanib is contraindicated in – **Congenital long QT syndrome.**
- Regorafenib is anti-angiogenic due to – **Dual VEGFR2-TIE2 tyrosine kinase inhibition.**
- Side effect of regorafenib – **Hepatotoxicity.**
- Tofacitinib – janus kinase blocker – useful in rheumatoid arthritis.

Worksheet

To do

Doc for CML	
Examples for VGFR blocking tyrosine kinase inhibitors	
TKI useful in medullary cancer of thyroid	
Janus kinase inhibitor useful in myelofibrosis	
TKIs useful in breast cancer	

Concept 12.9 : Monoclonal Antibodies

Learning objectives: To know the MABs target and their indications

Time Needed

1 st reading	30 mins
2 nd reading	20 mins

Examples:

Target	Drug (Antibody)	Indication
CD 20	Rituximab	- B-cell neoplasms - Autoimmune disorders
	Ofatumumab	CLL (not responding to fludarabine and alemtuzumab)
EGFR1	Cetuximab	- Metastatic colorectal cancer - Palliative treatment of head and neck squamous cell cancer
	Panitumumab	Metastatic colorectal cancer
EGFR2 (HER2/neu)	Trastuzumab	HER2-neu +ve breast cancer
VEGF	Bevacizumab	- Renal cell carcinoma - Nitrosourea-resistant glioblastoma - Metastatic breast cancer - Metastatic colorectal cancer - Non small cell lung cancer - Wet Age-related macular degeneration (ARMD)
	Ranibizumab	Wet ARMD
TNF- α	Adalimumab Infliximab Golimumab Certolizumab pegol	- RA - Ankylosing spondylitis - Psoriatic arthritis - Crohn's disease
IL-6	Tocilizumab	RA
CD 25 (IL-2 receptor)	Basiliximab Daclizumab	Prophylaxis of acute transplant rejection
CD 52	Alemtuzumab	CLL not responding to fludarabine
RANKL	Denosumab	Osteoporosis
Proteasome	Bortezomib	- Multiple myeloma - Relapsed / Refractory mantle cell lymphoma

IL-12 and IL-23	Ustekinumab	Psoriasis
Lymphocyte function associated antigen-1 (LFA-1)	Efalizumab	Psoriasis
C5 complement component	Eculizumab	Paroxysmal nocturnal hemoglobinuria
CD38	Daratumumab	Multiple myeloma

Side Effects of Bevacizumab:

- Pulmonary hemorrhages.
- Arterial thromboembolic events.
- Hypertension. If poorly controlled → Reversible posterior leukoencephalopathy.
- GI perforation.
- Asymptomatic proteinuria.

ONE-LINERS:

- Autoimmune disease in which rituximab is not effective – **Paroxysmal nocturnal hemoglobinuria.**
- **Eculizumab- targeting against complement component C5- useful in paroxysmal nocturnal hemoglobinuria.**
- Rituximab may be associated with – **Reactivation of BK and JC virus infections.**
- **Ocrelizumab** – anti-CD 20 that entered trials for RA and SLE but was suspended due to increased risk of deaths due to opportunistic infections (presently under trial for multiple sclerosis).
- Bevacizumab may **restore hearing** in patients with – NF-2 related tumours.
- Asymptomatic proteinuria due to bevacizumab is more common in patients with – **ovarian cancer and primary colon cancer.**
- **Pegaptanib** – A pegylated single strand of nucleic acid that binds to 165 isoform of VEGF; approved for wet ARMD.
- **VEGFR blocking MABs useful in ARMD-** Bevacizumab, Ranizumab.
- **Etanercept** – Recombinant fusion protein containing two soluble TNF p75 receptor moieties linked to Fc portion of human IgG1.
- All anti-TNF- α are administered subcutaneously except – **Infliximab (which is administered IV).**
- Most common s/e of Bortezomib – **Thrombocytopenia.**
- Most chronic s/e of Bortezomib – **Peripheral neuropathy.**
- Bortezomib(proteasome inhibitor) – **DOC multiple myeloma.**
- **BELIMUMAB – useful in SLE** MABs for specific conditions

CONDITION	DRUG NAMES	TARGET
Mycosis Fungoides	Mogamulizumab	CCR- 4
Rheumatoid Arthritis	Infliximab, Adalimumab, Certolizumab, Golimumab	TNF - alpha
Rheumatoid Arthritis	Tocilizumab, Sarilumab	IL-6
Gout	Canakinumab	IL- 1
Osteoporosis	Denosumab	RANKL
Osteoporosis	Bolosozumab Romosozumab	Sclerostin
X Linked Hypophosphatemia	Burosumab	FGF 23

PLAQUE PSORIASIS

Brodalumab Secukinumab Ixekizumab	IL 17
Guselkumab Tildrakizumab	IL-23
Ustekinumab	IL 12, 23
Efalizumab	CD11a

Multiple sclerosis

Natalizumab	($\alpha 4\beta 1$ - integrin)
Ocrelizumab	CD20
Alentuzumab	CD 52

Bronchial Asthma

Omalizumab	Ig E
Reslizumab Mepolizumab Benralizumab	IL- 5
Tralokinumab Lebrikizumab	IL-13
Dupilumab	IL- 4, Atopic Dermatitis



Migraine

Erenumab Fremanezumab Galcanezumab	CGRP antagonist
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Programmed Death Receptor-1 (PD-1) Inhibitors

Malignant melanoma

Ipilimumab	Cytotoxic T Lymphocyte–Associated Protein 4 (CTLA- 4)
Nivolumab Pembrolizumab	immune check point inhibitor, Programmed Death-1 (PD-1)

Atezolizumab, Durvalumab – targeting Against programmed death ligand-1 (PDL1), useful in urothelial carcinoma

AVELUMAB - Programmed Death Ligand- 1 (PDL-1 useful in merkel cell carcinoma

BI PHASIC MABs

Catumaxomab	Epithelial Cell Adhesion Molecule (EpCAM) + CD 3	Malignant ascities
Ertumaxomab	HER 2 + CD 3	Breast Cancer
Blinatumomab	CD19 B CELL +CD3 T CELL	Philadelphia Chromosome Negative Refractory All
Solitumab	EpCAM + CD 3T CELL	Solid Tumors
Duligotuzumab	HER 1 + HER 3	Head And Neck Cancer
	TNF α + Human Serum Albumin	Rheumatoid Arthritis

Concept 12.10 : Chemotherapy

Learning objective : to know the various regimen useful for cancer chemotherapy

Time Needed

1 st reading	00000 mins
2 nd reading	00000 mins

MOPP	Mustine HCl. Oncovin (Vincristine). Procarbazine. Prednisolone.	Hodgkin's lymphoma (Not used nowadays).
ABVD	Adriamycin (Doxorubicin) Bleomycin Vinblastine Dacarbazine	Hodgkin's lymphoma
CHOP-R	Cyclophosphamide Hydroxydaunorubicin (Doxorubicin) Oncovin (Vincristine) Prednisolone Rituximab	Non- Hodgkin's lymphoma
BEP	Bleomycin Etoposide Platinum (Cisplatin)	Testicular cancers
FOLFOX	Folinic acid 5-Fluorouracil Oxaliplatin	Colorectal cancer
FOLFIRI	Folinic acid 5-Fluorouracil Irinotecan	Colorectal cancer
VAMP	Vincristine Amethopterin (Methotrexate) 6-MP Prednisolone	Acute leukemia

ONE-LINERS:

- T-10 protocol is used for – **Osteosarcoma.**

- **PD-L1 inhibitors** (programmed death-ligand).
- **Atezolizumab** –urothelial carcinoma. **Avelumab** – Merkel cell carcinoma. **Durvalumab**- urothelial bladder cancer.

Concept 12.11 : Adverse effects of anticancer drug

Learning objective:

- To know the common adverse effects and specific adverse effects of anti cancer drugs
- To know the treatment available for preventing and treating anticancer drug induced toxicities

Time Needed

1 st reading	40 mins
2 nd reading	20 mins

- **Myelosuppression** – Granulocytopenia, Agranulocytosis, Aplastic anemia, Thrombocytopenia (myelosuppression less with busulphan, vincristine, L- asparaginase).
- Lymphocytopenia and inhibition of lymphocyte function → Immunosuppression and increased risk of opportunistic infections.
- Oral mucositis and stomatitis.
- Diarrhoea.
- GI hemorrhages.
- Nausea and vomiting.
- Alopecia.
- Dermatitis.
- Oligospermia and impotence in males.
- Inhibition of ovulation and amenorrhea in females.
- Fetal death and abortions.
- Teratogenesis.
- Increased risk of secondary malignancies.
- Hyperuricemia and tumour lysis syndrome.

Emetogenic Potential of Cytotoxic Drugs:

High	Moderate	Mild
• Cisplatin(most emetogenic) • Carmustine • Cyclophosphamide • Actinomycin D • Dacarbazine • Lomustine	• Carboplatin • Cytarabine • Procarbazine • Vinblastin • Doxorubicin • Daunorubicin • Ifosfamide • 6-Mercaptopurine • Paclitaxel	• Bleomycin • Chlorambucil • Busulfan • 5-Fluorouracil • 6-Thioguanine • Hydroxyurea • Vincristine • Methotrexate • Etoposide • L-asparaginase

**ONE-LINERS:**

- Most serious side effect of cytotoxic drugs – **Myelosuppression.**
- Most emetogenic anti-cancer drug – **Cisplatin.**

Anti-Cancer Drugs Causing Cardiotoxicity as a Side Effect:

- | | |
|---|--|
| <ul style="list-style-type: none"> • Anthracycline antibiotics. • Mitoxantrone. • Trastuzumab. | <ul style="list-style-type: none"> • Cyclophosphamide. • 5-Fluorouracil. |
|---|--|

Anti-Cancer Drugs Causing Peripheral Neuropathy as a Side Effect:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Vinca alkaloids. • Taxanes. • Platinum co-ordination complexes. | <ul style="list-style-type: none"> • Thalidomide. • Bortezomib. |
|---|---|

Salvage Therapy:**Specific treatments for side effects of anticancer drugs**

Hemorrhagic cystitis due to cyclophosphamide and ifosfamide	Mesna
Cisplatin-induced nephrotoxicity	Amifostine
Methotrexate toxicity	Folinic Acid(Leucovorin)
Anthracycline- induced cardiotoxicity	Dexrazoxane (Iron Chelator)
Chemotherapy- induced hyperuricemia	Allopurinol
Hypercalcemia of malignancy	Bisphosphonates

Amifostine:

- | |
|---|
| <ul style="list-style-type: none"> • Chemoprotectant + Radioprotectant. • MOA – Free radical scavenger. • Short acting($T_{1/2} = 8$ min). • Prevents nephrotoxicity due to cisplatin. • Cannot protect CNS as it does not cross blood brain barrier. • Reduces the incidence of xerostomia. |
|---|

Doses for Treatment of Chemotherapy-Induced Neutropenia:

G-CSF (Filgrastim)	5 mg/kg/d s.c
GM-CSF (Sargramostim) Molgramstim lenogramstim	250 mg/m ² s.c
Pegfilgrastim	1 dose of 6 mg 24 hours after institution of chemotherapy
Continue till absolute neutrophil count is 10,000 / μ L	

- Recombinant- erythropoietin- Epoetin, Darbopoietin Oprelvekin (interleukin- 11analogue)- thrombopoietic factor.
- Other thrombopoietic factors - Romiplostim, Eltrombopag.

ONE-LINERS:

Most common manifestation of chemotherapy- induced myelosuppression – Neutropenia (since WBCs have the shortest life span) f/b Thrombocytopenia f/b Anemia.

Drugs used to Treat Chemotherapy Induced Nausea and Vomiting:

1	5-HT ₃ antagonists	• Ondansetron (DOC). • Granisetron. • Tropisetron. • Dolasetron. • Palonosetron- (longest acting, highly selective 5HT 3 blocker, can be used for delayed phase vomiting also).
2	NK-1 / Substance P antagonists	• Aprepitant (DOC for delayed vomiting). • fosaprepitant. • Netupitant.
3	D ₂ antagonists	• Metoclopramide • Promethazine • Haloperidol • Droperidol
4	Steroids	• Dexamethasone • Methylprednisolone
5	Cannabinoid agonists (anti emetic + appetite stimulant)	• Dronabinol • Nabilone
6.	Palifermin	• Mucosal ulcer

Worksheet

To do

Antidote for doxorubicin	
Anti emetic useful for anticancer drug induced vomiting	
Examples for granulocyte CSF	
Il 11 analogue useful for treating thrombocytopenia is	
Radioprotector	
MESNA useful for	

13

Immunomodulators

CONCEPTS

➤ **Concept 13.1 Immunomodulators**

Concept 13.1 : Immunomodulators

Learning objectives:

- To know the classification of immunosuppressants
- To know the uses and adverse effects of immunosuppressants
- To study the uses and adverse effects of immunostimulants
- To know the uses and adverse effects of INF

Time Needed

1 st reading	45 mins
2 nd reading	25 mins

Classification of Immunosuppressants

1	• Glucocorticoids
2	• Calcineurin Inhibitors
	• Cyclosporine • Tacrolimus (FK-506)
3	• Antiproliferative And Antimetabolic Drugs
	• Sirolimus • Temsirolimus • Everolimus • Azathioprine • Mycophenolate mofetil • Leflunomide • Fingolimod
4	• Biologicals (Monoclonal Antibodies and Receptor Fusion Proteins)
	• Muromonab CD3. • Antithymocyte globulin. • Anti-Rho immunoglobulin. • Infliximab. • Adalimumab. • Etanercept. • Canakinumab. • Daclizumab. • Basiliximab. • Inolimomab. • Alemtuzumab. • Efalizumab. • Alefacept. • Rilonacept. • Tofacitinib.

Side Effects of Calcineurin Inhibitors

Cyclosporine > Tacrolimus	Tacrolimus > Cyclosporine
Hirsutism. Hyperplasia of gums. Hypertension.	Nephrotoxicity. Neurotoxicity. Hepatotoxicity. Hyperglycemia and diabetes mellitus. Diarrhea. Alopecia.

ONE-LINERS:

- Tacrolimus is a – **Macrolide**.
- Cyclosporine binds to – **Cyclophilin**.
- Tacrolimus binds to – **FKBP (FK-506 binding protein)**.
- Calcineurin inhibitors inhibit the lymphocyte production of – **IL-2**.
- Steroids inhibits IL17 & IL6.
- **Voclosporine** – Congener of cyclosporine; evaluated for prevention of anterior uveitis in LUMINATE programme.
- **Pimecrolimus** – Congener of tacrolimus; approved for topical treatment of atopic dermatitis in children > 2 years of age.

Mechanisms of Antimetabolites:

1	Sirolimus (Rapamycin)	Inhibit mTOR (mammalian target of rapamycin)
2	Temsirolimus	
3	Everolimus	
4	Azathioprine	Inhibits T lymphocyte proliferation (Anti metabolites)
5	Mycophenolate mofetil	Inhibits inosine monophosphate dehydrogenase (IMP) dehydrogenase enzyme
6	Leflunomide	Inhibits dihydro-ototate dehydrogenase enzyme
7	Fingolimod	Activates sphingosine-1- phosphate receptors

Triple Therapy Regimen for Induction of Immunosuppression in Renal Transplant Recipients

- Cyclosporine
- Azathioprine
- Prednisolone (± Muromonab CD3)

Indications apart from Prophylaxis of Transplant Rejection:

1	Temsirolimus	Renal cell carcinoma.
2	Everolimus	Hepatocellular carcinoma. Mantle cell lymphoma.
3	Azathioprine	RA.
4	Mycophenolate mofetil	SLE.
5	Leflunomide	RA.
6	Fingolimod	Multiple sclerosis.

ONE-LINERS:

- DOC to prevent acute transplant rejection – Corticosteroids (IV methylprednisolone).
- Progressive multifocal leukoencephalopathy is a side effect of – Mycophenolate mofetil.
- Hepatotoxicity is a side effect of –
- Leflunomide.
- Anakinra – Anti IL-1 for RA.
- Abatacept – T-cell co-stimulation inhibitor (Anti CD80/CD86); approved for RA.
- Tofacitinib – Janus kinase inhibitor for RA (also under evaluation for psoriasis).
- Alefacept – LFA-3-IgG1 fusion protein targeting CD2; approved for psoriasis.

Examples of Immunostimulants:

• Thalidomide and its congeners • (Lenalidomide and Pomalidomide). • Levamisole. • BCG vaccine.	• Interferons. • IVIg. • Aldesleukin.
--	---

Mechanism of Action of Thalidomide at The Molecular Level

1	Promotes apoptosis by inhibiting anti-apoptotic proteins: • Inhibits NF-κB. • Inhibits A1/Bfl-1.
2	Inhibits angiogenesis: • Inhibits IL-6.
3	Inhibits tumour cell growth and survival: • Inhibits interaction of cells with adhesion molecules.
4	Enhances T-cell mediated cytotoxicity.

Peripheral neuropathy due to thalidomide:

- Most serious side effect of thalidomide.
- Occurs in 10-30% of cases.
- Predominantly sensory.
- Asymmetrical, painful, peripheral paresthesia with sensory loss.
- Commonly presents with numbness of toes and feet, muscle cramps, weakness, signs of pyramidal tract involvement and carpal tunnel syndrome.
- Increases with higher cumulative doses, especially in elderly.
- Symptoms improve upon discontinuation, but long-standing sensory loss may not reverse.

TNF- α blockers

- | | |
|---|---|
| <ul style="list-style-type: none"> • Infliximab • Adalimumab • Eterncept | <ul style="list-style-type: none"> • Golimumab • Certolizumab |
|---|---|

TNF- α blockers side effects-

Anti drug antibody formation, allergic reaction
Activation of latent infection, hepatotoxicity, heart failure, secondary cancer

Contraindications to anti-TNF- α agents:

- | | |
|--|---|
| <ul style="list-style-type: none"> • Hepatitis B. • HIV and other immunocompromised states. • Active tuberculosis. • Multiple sclerosis. | <ul style="list-style-type: none"> • Systemic lupus erythematosus. • Pregnancy. • Lactation. |
|--|---|

IL- modifiers:

Anakinra	Analogue of IL 1 receptor Antagonist	Useful In rheumatoid arthritis
Pitrakinra	IL 3 & 4 Antagonist	Under clinical trial for bronchial asthma
Aldeslukin	Analogue- IL 2	Useful In RCC, malignant melanoma
Basiliximab, Daclizumab	IL-2 receptor blockers	immunosuppressants
Denilukin diftotox	Combination of IL 2 + Difteria toxin-	Useful in Cutaneous T cell lymphoma
Reslizumab	IL5 blocker	Useful in bronchial asthma

Mepolizumab	IL 5 blocker	Hypereosinophic synd, churg- strauss syndrome
Tocilizumab	IL- 6 blocker	Useful in rheumatoid arthritis
Opelvikin	Analogue of IL-11	Useful for treatment of anticancer drug induced Thrombocytopenia
Ixekizumab Brodalumab	IL- 17 Blocker	Plaque psoriasis

ONE-LINERS:

- Most common side effect of thalidomide – **Sedation f/b Constipation.**
- Most serious side effect of thalidomide
- Peripheral neuropathy.
- Thalidomide: **Renantiomer** is responsible for **sedation**; S enantiomer for biological effects and teratogenicity (Mnemonic: **S is not S**).
- Teratogenic effects due to Thalidomide – **Phocomelia** and **Mobius syndrome.**
- Lenalidomide is indicated in – **Multiple myeloma and Myelodysplastic syndrome** (especially in patients with 5q deletion).
- Myelosuppression is a side effect of – **Lenalidomide.**
- Pomalidomide is approved in – **Multiple myeloma.**

Indications of interferons

IFN- α -2b	• Hairy cell leukemia. • Follicular lymphoma. • Malignant melanoma. • AIDS-related Kaposi's sarcoma. • Condyloma acuminata. • Chronic hepatitis B & C.
IFN- β -1a, IFN- β -1b	• Multiple sclerosis.
IFN- γ	• Chronic granulomatous disease.

Indications of Intravenous Immunoglobulin (IVIG)

- Kawasaki's disease.
- Idiopathic thrombocytopenic purpura (ITP).
- Guillain-Barre syndrome (GBS).
- Myasthenic crisis.
- Agammaglobulinemia (as replacement therapy)

ONE-LINERS:

- Fatal agranulocytosis is a side effect of
- – **Levamisole.**
- Imiquimod acts as a ligand for – **Toll- like receptor-7** (TLR-7).
- Imiquimod is used for – **Genital warts.**
- BCG vaccine is used for – **Superficial bladder cancer.**
- DOC for Kawasaki's disease – **IVIg.**
- **Aldesleukin** – Recombinant IL-2 for metastatic renal cell carcinoma and malignant melanoma.
- **Denileukin diftitox** – Recombinant IL-2 + Diphtheria toxin for cutaneous T cell lymphomas.

Newer drugs:

	Drug	Mechanism	Indication
1	Blinatumomab	Monoclonal antibody targeting CD19 and CD3	Philadelphia chromosome –ve relapsed or refractory B cell precursor ALL
2	Venetoclax	Inhibits Bcl-2 (anti-apoptotic protein)	Chronic lymphocytic leukemia (CLL)
3	Ibrutinib	Inhibits Bruton's tyrosine kinase	Chronic lymphocytic leukemia (CLL)
4	Idelalisib	Inhibits phosphoinositide-3-kinase- δ	Relapsed CLL
5	Belinostat	Inhibits histone deacetylase	Relapsed or refractory peripheral T cell lymphoma
6	Panobinostat	Inhibits histone deacetylase	Multiple myeloma
7	Siltuximab	Monoclonal antibody targeting IL-6	Multicentric Castleman's disease
8	Palbociclib	Inhibits CDK-4 and CDK-6	ER +ve and HER-2 –ve breast cancer
9	Lenvatinib	Inhibits VEGFR2 and VEGFR3 tyrosine kinases	Thyroid cancer resistant to radio-iodine
10	Nivolumab	Monoclonal antibody targeting programmed cell death-1 (PD-1) receptor on T cells	Metastatic squamous non-small cell carcinoma lung; Unresectable / metastatic melanoma
11	Ceritinib	Inhibits ALK (tyrosine kinase)	ALK +ve metastatic non-small cell lung cancer
12	Ramucirumab	Monoclonal antibody targeting VEGFR2	Gastric cancer
13	Dinutuximab	Monoclonal antibody targeting GD2 (glycolipid disialoganglioside)	High-risk neuroblastoma

14	Olaparib	Inhibits PARP (poly ADP ribose polymerase) enzyme involved in DNA repair	Previously treated BRCA mutated advanced ovarian cancer
15	Atezolizumab	Monoclonal antibody targeting PDL-1 (Programmed cell death ligand-1)	Locally advanced/metastatic urothelial carcinoma refractory to chemotherapy and / or radiotherapy
16	Fluciclovine	¹⁸ F labeled synthetic L-leucine analogue	Diagnostic agent for PET imaging for prostate cancer
17	Sonidegib	Inhibits smoothened receptor pathway (Hedgehog signalling pathway)	Locally advanced basal cell carcinoma
18	Pembrolizumab	Monoclonal antibody targeting programmed cell death-1 (PD-1) receptor on T cells	Unresectable / metastatic melanoma
19	Netupitant	Neurokinin-1 (NK-1) receptor antagonist / Substance P antagonist	Prevention of chemotherapy-induced nausea and vomiting
20	Apremilast	Selective PDE-4 inhibitor	Moderate – severe plaque psoriasis; Psoriatic arthritis
21	Secukinumab	Monoclonal antibody targeting IL-17A	Plaque psoriasis
22	Ixekizumab	Monoclonal antibody targeting IL-17A	Plaque psoriasis
23	Ivermectin	Anti-inflammatory and anti-parasitic action against Demodex mites	Inflammatory lesions of rosacea
24	Lifitegrast	Inhibits binding of lymphocyte function-associated antigen 1 (LFA-1) to intercellular adhesion molecule 1 (ICAM-1)	Keratoconjunctivitis sicca (dry eye disease)
25	Aflibercept	Inhibits VEGF	Wet macular degeneration

Worksheet

To do

Most common side effect of cyclosporine	
Mechanism of action of mycophenolate mofetil	
Examples for mTOR inhibitors	
Most common side effect of thalidomide	
Tyrosine kinase inhibitor useful for thyroid cancer	
VGFR blocking MABs example	
MABs useful in multiple sclerosis	
Examples for immunostimulants	

CONCEPTS

- ⇒ **Concept 14.1** Special Topics Drugs Useful for Obesity

Concept 14.1 : Special Topics Drugs Useful for Obesity

1. Drugs for Medical Management of Obesity:

	Drug	Mechanism of action
1	Orlistat	Inhibits pancreatic lipase (ADR- Steatorhoea)
	Olestra	Sucrose polyester, cooking medium
	Rimonabant	CB-1 antaonist (withdrawn due to psychiatry problem)
2	Lorcaserin	5-HT _{2C} agonist (Anorectic), withdrawn because of gastric cancer
	GLP-1-analogue	LIRAGLUTIDE
3	Phentermine + Topiramate	Phentermine – α_1 agonist; Topiramate – Anti-epileptic
	Bupropion + Naltrexone	Bupropion- NADRI Naltrexonep opioid antagonist Zonisamide-
	Bupropion + Zonisamide	antiepileptic

ONE-LINERS:

- Tesofensine – Serotonin-noradrenaline-dopamine reuptake inhibitor (SNDRI) for obesity.
- Cetilistat – Pancreatic lipase inhibitor (Similar to Orlistat) for obesity. Sibutramine has been withdrawn due to – Stroke
- Sibutramine has been withdrawn due to – Stroke

2.

Condition	Drug Useful
ADHD	• DOC for ADHD in children – Methylphenidate (metabolite name is Ritalinic acid) • DOC for ADHD in adults – Atomoxetine • Other drugs useful in ADHD are • Clonidine • Guafacine, guanabenz • Modafinil
Narcolepsy	• MODAFINIL • SOLRIAMFETOL- NDRI • PITOLISANT / TIPROLISANT - H3 Inverse agonist (orphan drug status)

Drug	Uses
Modafinil	• Narcolepsy • Shift workers • Obstructive sleep apnoea syndrome • ADHD (not approved by FDA)

3. Drugs useful for Glaucoma

Agents Decreasing Aqueous Secretion

Alpha agonist	Beta blockers	Carbonic anhydrase inhibitors
Selective α_2 Agonists Apraclonidine Brimonidine	Beta blockers- Timolol Betaxolol Carteolol Levobunolol Metipranolol	Brinzolamide Dorzolamide Oral- Acetazolamide

Agents Promoting Drainage of Aqueous

Trabecular outflow

Cholinomimetics	Rho kinase inhibitors
Pilocarpine (direct acting, short acting) Ecothiophate- long acting- ADR- Cataract	Netarsudil Repasudil –

Uveoscleral outflow

PGF ₂ (Agonist)	Non selective α agonists
- Latanoprost - Bimatoprost - Tafluprost - Travoprost - Unoprostone	- Epinephrine - Dipivefrine

Adverse effects of antiglaucoma drugs

Lid lag	Apraclonidine
Anterior uveitis	Brimonidine
Bronchospasm	Non selective beta blockers
Iris pigmentation	Latanoprost, bimatoprost
Hypertrichosis of eye lash	Bimatoprost
cataract	Ecothiophate

4. Antismoking drugs

First line therapy

- Varenicline- ($\alpha 4\beta 2$) nicotinic agonist
- Nicotine
- Bupropion - **NDRI**

Cystine

Second line therapy-

- Clonidine
- Nortriptyline

Miscellaneous-

- **Rimonabant**- IInverse agonist of Cannabinoid 1 receptor, eight [loss](#) (obesity), Prevents craving of alcohol, ADR- Psychiatry problems (withdrawn)
- Topiramate
- **Mecamylamine** – ganglionic blocker



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